NEW DEVELOPMENT OF ELECTROSPRAY IONIZATION MASS SPECTROMETRY

HU BIN

Ph.D

The Hong Kong Polytechnic University

2015

The Hong Kong Polytechnic University

Department of Applied Biology and Chemical Technology

New Development of Electrospray Ionization Mass Spectrometry

HU Bin

A thesis submitted in partial fulfilment of the requirements for the degree of doctor of philosophy

July 2015

Certificate of originality

I hereby declare that this thesis is my own work and that, to the best of my knowledge and belief, it reproduces no material previously published or written, nor material that has been accepted for the award of any other degree or diploma, except where due acknowledgement has been made in the text.

_________________________________ (Signed)

______HU Bin____________(Name of student)

Abstract

Electrospray ionization mass spectrometry (ESI-MS) is widely used in many fields including chemistry, biology, medicine and pharmacy, and is still under fast development. Conventional ESI uses a capillary for sample loading and ionization. Non-capillary ESI can avoid the clogging problem, allow easy sample loading, and create new analytical features. In this study, ESI on solid substrates has been developed and their analytical properties and mechanism have been investigated. We developed ESI on wooden tips (wooden toothpicks) and demonstrated that wooden-tip ESI-MS was applicable for qualitative and quantitative analysis of various samples, including samples that cannot be directly analyzed by conventional ESI techniques, e.g., slurry samples and powder samples. Wooden-tip ESI-MS with modified wooden-tip surfaces was found to allow selective and sensitive detection of analytes in mixtures, and direct coupling of ESI-MS with commercially available solid-phase micro-extraction (SPME) fibre was developed for enhanced detection of analytes in raw samples. Other materials, including bamboo, fabrics, sponge, polyester, and aluminium (Al) foil and thin-layer chromatography (TLC) plate have also been successfully employed as solid substrates for ESI-MS analysis. Particularly, Al-foil ESI-MS allowed on-target sample extraction, cleanup and heating, extending ESI device from usually only for sample loading and ionization to including sample processing; TLC-ESI-MS could be used for direct analysis of samples containing salts and detergents, and rapid detection and quantitation of target analytes in raw samples. Solid-substrate ESI-MS has been extended to direct ionization analysis of sample materials including plant tissue and animal tissue. Furthermore, by avoiding direct

connection of samples with high voltage, field-induced direct ionization was developed to much facilitate sample analysis and allow *in vivo* and real-time monitoring of secondary metabolites of living organisms by mass spectrometry.

In conclusion, replacement of capillary with solid substrates generates many new possibilities for ESI-MS and allows us to get more insight into the fundamentals of ESI-MS. The new techniques developed in this study enable direct analysis of raw samples with no or little sample pretreatment and can be applied in various sample analysis including bioanalysis, food quality control, pharmaceutical analysis, environmental analysis and forensic analysis. The motion of molecules on porous substrates under ESI conditions was found to be size and shape-dependent, suggesting that solid-substrate ESI-MS could be further developed for investigating molecular structures and protein conformations in solutions.

Research publications

Journal papers

1 **B Hu**#, P K So#, H W Chen, Z P Yao. Electrospray ionization using wooden tips. *Anal Chem* **2011**, *83*, 8201-8207. (#Equal contribution)

2 **B Hu**, Y H Lai, P K So, H W Chen, Z P Yao. Direct ionization of biological tissue for mass spectrometric analysis. *Analyst* **2012**, *137*, 3613-3619. **(Cover paper)**

3 **B Hu**#, L Wang#, W Ye, and Z P Yao. *In vivo* and real-time monitoring of secondary metabolites of living organisms by mass spectrometry. *Sci Rep* **2013**, *3*, 2104. (#Equal contribution)

4 **B Hu**, P K So, Z P Yao. Analytical properties of solid-substrate electrospray ionization mass spectrometry. *J Am Soc Mass Spectrom* **2013**, *24*, 57-65.

5 P K So, **B Hu**, Z P Yao. Mass spectrometry: towards *in vivo* analysis of biological systems. *Mol BioSys* **2013**, *9*, 915-929

6 P K So, T T Ng, H X Wang, **B Hu**, Z P Yao. Rapid detection and quantitation of ketamine and norketamine in urine and oral fluid by wooden-tip electrospray ionization mass spectrometry. *Analyst* **2013**, *138*, 2239-2243. **(Cover paper)**

7 **B Hu**, P K So, Z P Yao. Electrospray ionization with aluminum foil: a versatile mass spectrometric technique. *Anal Chim Acta* **2014**, *817*, 1-8. **(Cover paper)**

8 G Z Xin[#], **B Hu**[#], Z Q Shi, Y C Lam, T T Dong, P Li, Z P Yao, K W K Tsim. Rapid identification of species sources of plant materials by wooden-tip electrospray ionization mass spectrometry: a strategy to differentiate the bulbs of fritillaria. *Anal Chim Acta* **2014**, *820*, 84-91 ([#]Equal contributions)

9 P K So, **B Hu**, Z P Yao. Electrospray ionization on solid substrates. *Mass Spectrom* **2014**, *3*, S0028

10 **B Hu**[#], G Z Xin[#], P K So, Z P Yao. Thin layer chromatography coupled with electrospray ionization mass spectrometry for direct analysis of raw samples. *J Chromatogr A* **2015.** *1415*, 155-160**.** ([#]Equal contributions)

11 **B Hu**, Z P Yao, Mobility of proteins in porous substrates under electrospray ionization conditions. **Submitted.**

12 G Z Xin, **B Hu**, Z Shi, J Zheng, L Wang, W Chang, P Li, Z P Yao, L Liu. A direct ionization mass spectrometry-based approach for differentiation of medicinal ephedra species. **Submitted.**

13 L Wang, Z P Yao, S B Chen, P K So, Z Q Shi, **B Hu**, L F Liu. Global quantitation of non-targeted components of herbs based on multiple reaction monitoring without authentic standards. **Submitted.**

Conference papers

1. **B Hu**, Y Y Yang, J W Deng, Y C Choi, Z P Yao, Development of surface-enhanced wooden-tip electrospray ionization mass spectrometry. HKSMS Symposium 2015 & the 17th Annual General Meeting of The Hong Kong

Society of Mass Spectrometry. Hong Kong, 13 Jun, **2015. (Poster, Best Poster Award)**

2. **B Hu**, Y Y Yang, J W Deng, Z P Yao, Surface-modified wooden-tip electrospray ionization mass spectrometry for enhanced detection of analytes in complex samples: mechanism and applications. The 63rd ASMS Conference on Mass Spectrometry and Allied Topics. St. Louis, MO, USA, 30 May - 4 Jun, **2015**. **(Poster)**

3. **B Hu**, Y Y Yang, J W Deng, Y C Choi, Z P Yao, Surface-enhanced wooden-tip electrospray ionization mass spectrometry for improved detection of analytes in complex samples. The 22nd Symposium on Chemistry Postgraduate Research in Hong Kong. Hong Kong. 18 Apr, **2015. (Poster, Best Poster Award)**

4. **B Hu**, Z P Yao. Investigation on sequential motion of biomolecules by using solid-substrate electrospray ionization mass spectrometry with porous media. The 62nd ASMS Conference on Mass Spectrometry and Allied Topics. Baltimore, MD, USA, 15-19 Jun, **2014**. **(Poster)**

5. **B Hu**, Z P Yao. Investigation of molecular motions in porous media by solid-substrate electrospray ionization mass spectrometry. HKSMS Symposium 2014 & the 16th Annual General Meeting of The Hong Kong Society of Mass Spectrometry. Hong Kong, 7 Jun, **2014. (Poster, Best Poster Award)**

6. **B Hu**, P K So, Z P Yao. Investigation of molecular motion on porous

substrates under electrospray ionization conditions. The 21st Symposium on Chemistry Postgraduate Research in Hong Kong. Hong Kong. 12 Apr, **2014. (Poster, Best Poster Award)**

7. **B Hu**, P K So, Z P Yao. Electrospray ionization using aluminium foil. Young Scientist Forum of 4th Asia Oceania Mass Spectrometry Conference. Taipei, Taiwan. 8-10 Jul, **2013. (Oral presentation, Travel Award)**

8. **B Hu**, P K So, Z P Yao. Aluminium foil as substrate for electrospray ionization mass spectrometry. The 4th Asia Oceania Mass Spectrometry Conference. Taipei, Taiwan. 10-12 Jul, **2013. (Poster, HKSMS Conference Award)**

9. **B Hu**, Z P Yao. Solid-substrate for electrospray ionization mass spectrometry with thin layer chromatography plates. The 20th Symposium on Chemistry Postgraduate Research in Hong Kong. Hong Kong. 27 Apr, **2013. (Poster)**

10. **B Hu**, Z P Yao. Development of aluminium of foil/electrospray ionization mass spectrometry. HKSMS Symposium 2013 & the 15thAnnual General Meeting of The Hong Kong Society of Mass Spectrometry. Hong Kong, 22 Jun, **2013. (Poster, Best Poster Award)**

11. **B Hu**, Z P Yao. Tip electrospray ionization (TESI) mass spectrometry of thin layer chromatography (TLC) plate. HKSMS Symposium 2012 & the 14th Annual General Meeting of The Hong Kong Society of Mass Spectrometry. Hong Kong, 16 Jun, **2012. (Poster, Best Poster Award)**

Acknowledgements

I would like to sincerely express my gratitude to my supervisor, Dr. Zhongping Yao. Five years ago he accepted me as a research assistant, and after two years, he accepted me as a PhD student. In the five years, I have learned not only abundant professional knowledge, but also learn how to manage a scientific project. I am really grateful for his guidance, insightful comments, and the freedom he has given me in conducting my research.

I would like to acknowledge my co-supervisor, Dr. Jian-Yong Wu, who has taken the time to read this thesis and helped to improve it with his insightful comments and suggestions.

I would also like to thank the BoE chair, Dr. Yan-Xiang Zhao, and my thesis examiners, Prof. Zongwei Cai and Dr. Ivan Keung Chu.

I wish to acknowledge the help provided by Dr. Pui-Kin So (The Hong Kong Polytechnic University), Dr. Gui-Zhong Xin (The Hong Kong University of Science and Technology), Prof Wei-Cai Ye (Jinan University), Prof Yun-Yun Yang (China National Analytical Center Guangzhou), Prof. Janusz Pawliszyn (University of Waterloo), and the members at Dr. Yao's group: Dr. Lei Wang, Miss Ying-Han Lai, Dr. Bo Zheng, Miss Yi-Ching Choi, Mr. Tsz-tsun Ng, Mr. Ho-Yi Wong, Miss Jie-Wei Deng, Miss Li Wang, and Dr. Hai-Xing Wang; this project would not be able to have these results without their contributions.

It is a special pleasure to acknowledge Prof. Huan-Wen Chen and Prof. Jian-Hua Ding at East China Institute of Technology for their recommendations to The Hong Kong Polytechnic University for my PhD program.

Especially, I would like to thank Ms Josephine Leung and Miss Yi-Ching Choi for their proofreading of this thesis.

I would like to thank my friends and family for their encouragement and supports during my PhD study.

Supports by RGC (No. 5029/13P), NSFC (No. 81373369), the grant from MOST of China (No. 2014DFH30160), and the studentship from The Hong Kong Polytechnic University are gratefully acknowledged.

Table of contents

List of abbreviations

AFADESI	Air Flow-Assisted Desorption Electrospray Ionization
Al	Aluminium
Cp	Clobetasol Propionate
Da	Dalton
DAPCI	Desorption Atmospheric Pressure Chemical Ionization
DART	Direct Analysis In Real Time
DEMI	Desorption Electrospray/Metastable-Induced Ionization
DEFFI	Desorption Electro-Flow Focusing Ionization
DEP	Direct Electrospray Probe
DES	Droplet Electropsray
DESI	Desorption Electrospray Ionization
DGDG	Digalactosyl Diacylglycerol
DI	Direct Ionization

EADESI	Electrode-Assisted Desorption Electrospray Ionization
EASI	Easy Ambient Sonic-Spray Ionization
EESI	Extractive Electrospray Ionization
ELDI	Electrospray Laser Desorption Ionization
ESA-Py	Electrospray-Assisted Pyrolysis Ionization
ESI	Electrospray Ionization
FA	Formic Acid
FIDI	Field-Induced Droplet Ionization
GABA	γ-Aminobutyric Acid
GC	Gas Chromatography
GD	Gramicidin D
HV	High Voltage
Ket	Ketoconazole
LADESI	Laser Assisted Desorption Electrospray Ionization

LAESI	Laser Ablation Electrospray Ionization
LC	Liquid Chromatography
LDESI	Laser Desorption Electrospray Ionization
LDI	Laser Desorption/Ionization
LDSPI	Laser Desorption Spray Post-Ionization
LIAD/ESI	Laser-Induced Acoustic Desorption/Electrospray Ionization
LOD	Limit of Detection
LOQ	Limit of Quantitation
LS	Laser Electrospray
LSI	Laserspray Ionization
m/z	Mass-to-Charge Ratio
MAIV	Matrix Assisted Ionization Vacuum
MALDI	Matrix-assisted Laser Desorption/Ionization
MALDESI	Matrix-assisted Laser Desorption Electrospray Ionization

MC	Melamine Cyanurate
MeOH	Methanol
MGDG	Monogalactosyl Diacylglycerol
MS	Mass Spectrometry
MS/MS	Mass Spectrometry/Mass Spectrometry
NanoESI	Nanoelectrospray Ionization
OG	Octly β-D Glucopyranside
PC	Phosphatidylcholine
PESI	Probe Electrospray Ionization
QTOF	Quadrupole Time-of-Flight
RSD	Relative Standard Deviation
S/N	Signal-to-Noise Ratio
SCF	Fruit of *Schisandra Chinensis*
SDS	Sodium Dodecyl Sulfonate

SEM	Scanning Electron Microscope
SESI	Secondary Electrospray Ionization
SFF	Fruit of *Schisandra Sphenanthera*
SIC	Selected Ion Chromatogram
SIMS	Secondary Ion Mass Spectrometry
SPME	Solid-Phase Micro-Extraction
SRM	Selected Reaction Monitoring
SSI	Sonic Spray Ionization
TIC	Total Ion Chromatography
TM-DESI	Transmission Mode Desorption Electrospray Ionization
UV	Ultraviolet

List of figures

List of tables

Chapter 1: General Introduction

1.1 Mass spectrometry

MS is a technique used for determining the abundances and mass-to-charge ratios (*m/z*) of ions of analytes in the gas phase and providing important molecular information about analytes, including their masses, purities and compositions. A mass spectrometer consists of three components: (1) an ion source that ionizes analytes and transfers them into the gas phase, (2) a mass analyzer, a device that separates ions according to their mass-to-charge (*m/z*) values and (3) a detector that measures and amplifies the ion current of mass-resolved ions. Sample introduction systems such as chromatographic techniques and electrophoretic techniques are usually coupled to MS.

The first mass spectrometry instrumentation was developed in 1913 by Sir J. J. Thomson[1] built on the work of Eugen Goldstein and Wilhelm Wien in the late19th century, which showed that superimposed electric and magnetic fields could deflect positive ions.[2, 3] Commercial mass spectrometers were eventually developed in the 1940s.[4] Year by year, with the increasing demand for analysis of various samples from life science, biology, chemistry, environment, pharmacy, food, forensic science, and other disciplines, the high sensitivity, high resolution, and miniaturization of mass spectrometers are highly needed. Mass spectrometers with different mass analyzers, including time-of-flight, magnetic sector, quadruple, ion trap, Fourier transform ion cyclotron resonance, have been well developed.[5-11] Recently, the development of portable/miniature mass spectrometers has been introduced.[12-17]

On the other hand, various ionization techniques, such as electron ionization (EI), chemical ionization (CI), field ionization (FI), field desorption (FD), fast atom bombardment (FAB), matrix-assisted laser desorption/ionization (MALDI), electrospray ionization (ESI), and the recent ambient ionization, have been successfully developed for mass spectrometric analysis. Some of those early common ion sources are summarized in many reviews.[18-21]

Among these ionization techniques, ESI is one of the most powerful ionization techniques commonly used for ionization of various compounds. Considerable efforts have been made to further improve the sampling and ionization of ESI with different materials. The history, theory, evolution and recent development of ESI are briefly introduced in this chapter.

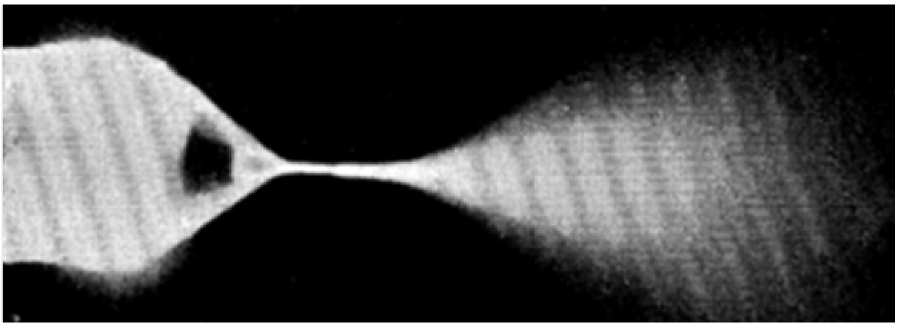

Figure 1-1 Early photograph of electrospray. (Reprinted from ref [22])

1.2 History, theory and evolution of electrospray ionization

The phenomenon of electrospray was first observed by John Zeleny In the early 20th century.[22] Figure 1-1 is an early photograph showing a conical spray from the tip of capillary. Following this work, the theory and experiments of electrospray were studied by Taylor, and the conical shape was thus named as Taylor cone.[23, 24]

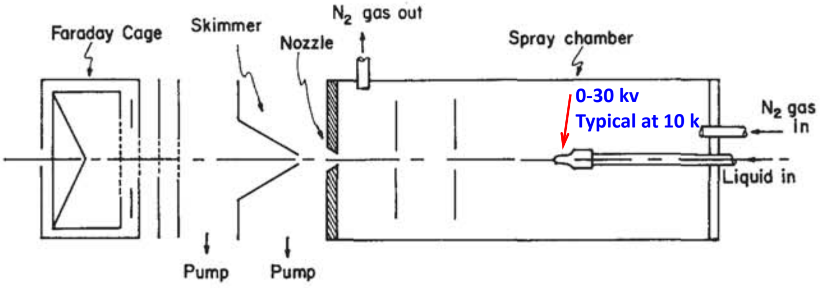

Figure 1-2 Electrospray as an ionization technique (Reprinted from ref[25]).

However, electrospray was not used as a ionization technique until 1968 by Dole and co-workers[25](Figure 1-2). Electrospray ionization (ESI) was named for the technique after the pioneer work by Fenn et al in the 1980s[26] to coupled ESI with mass spectrometer for analysis of biomacromolecules. Figure 1-3 shows a schematic diagram of the early coupling of ESI to MS.[27] Nowadays, ESI-MS is a basic tool used in biological, chemical and pharmaceutical studies worldwide. A lot of studies on the mechanisms and new developments of ESI have been carried out.[28-31]

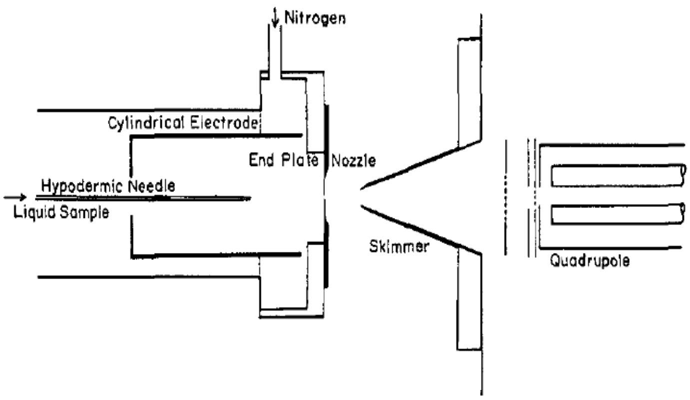

Figure 1-3 Coupling of electrospray ionization with mass spectrometer(Reprint from ref [27])

Briefly, in conventional ESI, charged droplets are generated by capillary sampling of solution sample under a strong electric field. The ESI process is described as break up of surface tension of the liquid by the action of electric forces. The scheme of evolution of charged droplets is described to be caused by solvent evaporation at constant charge and droplet fissions at the Rayleigh limit (Figure 1-4). The mechanism of ion generation in ESI was explained by two models, i.e., ion evaporation model and charged residue model.[32, 33] In ESI-MS, when a high voltage V is applied to a liquid on the tip of the capillary emitter, generation of electrospray is related to the following equation[34]:

$$E = \frac{2V}{r\ln(4d/r)}$$

where r is the radius of capillary, d is the distance between capillary emitter and the counter electrode, and E is the electrical field force. This field leads to liquid polarization under strong electric field: the negative/positive charge carriers leave the capillary exit when the high positive/negative voltage is applied and migrate toward the counter electrode located on the MS inlet.

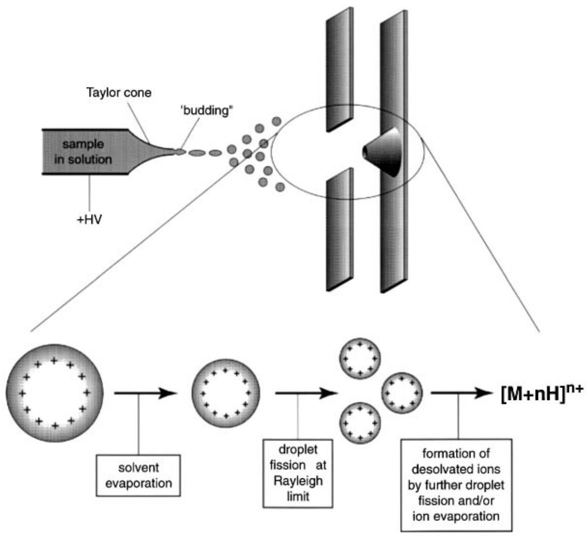

Figure 1-4 Charge droplet and ions production in the ESI (reprint form ref[35])

ESI has some unique features, and one of the most important is that ESI is a soft ionization technique which makes it especially useful in mass spectrometric analysis of small and large molecules at ambient conditions. ESI coupled to different commercial mass spectrometers have been developed with different designs.[36] Figure 1-5 shows three typical commercial ESI-MS, which are constructed by using three different designs: a) linear (on-axis/line-of-sight) b) orthogonal spray, c) z-spray. Recently, based on these mass spectrometers, many ESI-based ambient ionization techniques have been developed for direct analysis of complex sample.

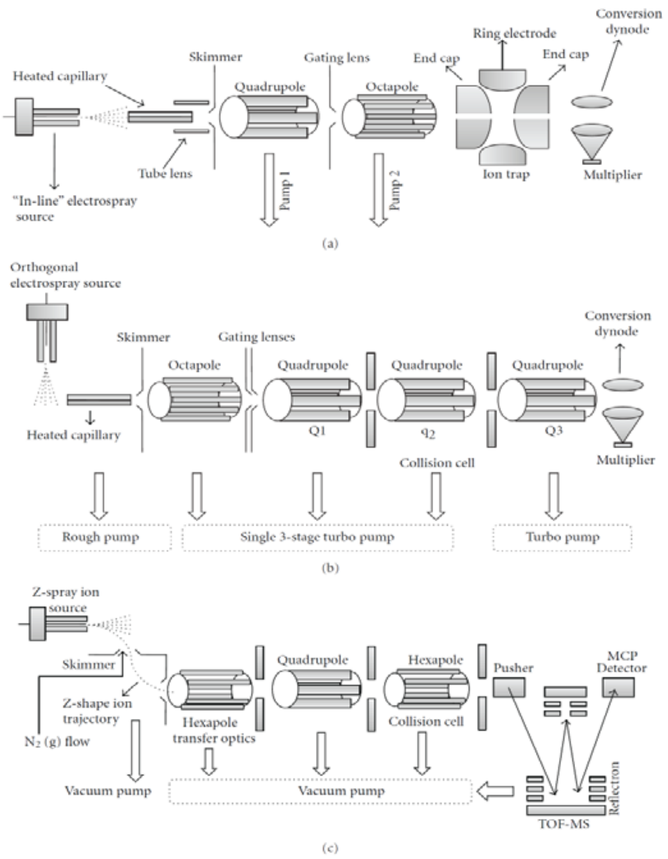

Figure 1-5 Schematic of a) Thermo-Finnigan on-axis spray coupledwithion trap mass spectrometer, b) Agilent off-axis/orthogonal spray coupledwithtriple Quadrupolemass spectrometer, and c)Waters z-spray couple withQ-ToF mass spectrometer. (reprint form ref[36])

1.3 Development of ESI-based ambient ionization techniques

1.3.1 ESI-based desorption/extraction ionization techniques

Desorption electrospray ionization (DESI), a typical ESI-based desorption/extraction ionization technique, was introduced in2004 and is currently the most commonly used ambient ionization technique.[37] In this technique, charged microdroplets are generated from an ESI spray, and directed towards analytes on the sample surface, leading to desorption/ionization of analytes as shown in Figure 1-6.

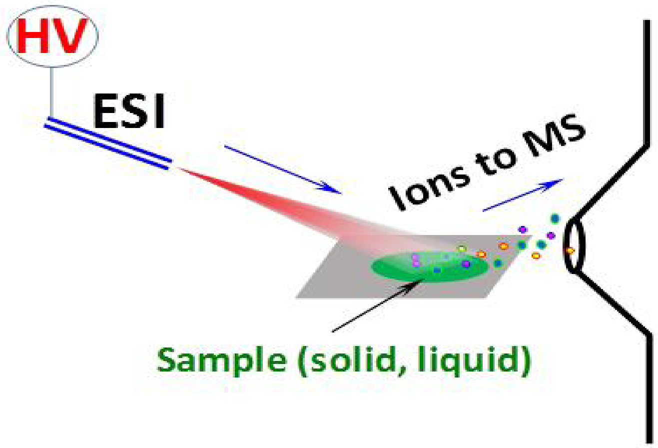

Figure 1-6 Schematic diagram of DESI source.

DESI has been applied for detection of analytes in complex samples in areas such as food safety,[38, 39] pharmaceutical quality control,[40, 41] environmental analysis,[42] forensic analysis,[43, 44] and metabolites analysis.[45] Moreover, DESI can be used for MS imaging of various samples, including biological tissues and even whole animals as well as fingerprinting of drugs, metabolites or lipids.[46-49]

Table 1-1 Typical ESI-based desorption/extraction ionization techniques.

Technique	**Acronym**	**Year of invention**	**Ref**
Desorption electrospray ionization	DESI	2004	37
Extractive electrospray ionization	EESI	2008	50
Transmission mode desorption electrospray ionization	TM-DESI	2008	51
Desorption electrospray/metastable-induced ionization	DEMI	2009	52
Electrode-assisted desorption electrospray ionization	EADESI	2010	53
Nanospray desorption electrospray ionization	nanoDESI	2010	54
Desorption electro-flow focusing ionization	DEFFI	2013	55
Air flow-assisted desorption electrospray ionization	AFADESI	2013	56

ESI is also used as a primary ion beam for extraction/ionization of analytes in gas or liquid samples (Figure 1-7). In 2000, the design of ESI for extraction/ionization of analytes was first established by Hill et al and the technique was named as secondary electrospray ionization (SESI).[57] Investigation of the ionization process of SESI revealed that gas-phase proton and charge-transfer reactions played an important role in SESI. In SESI, both liquid and gaseous samples can be introduced to be analyzed by MS.[58]

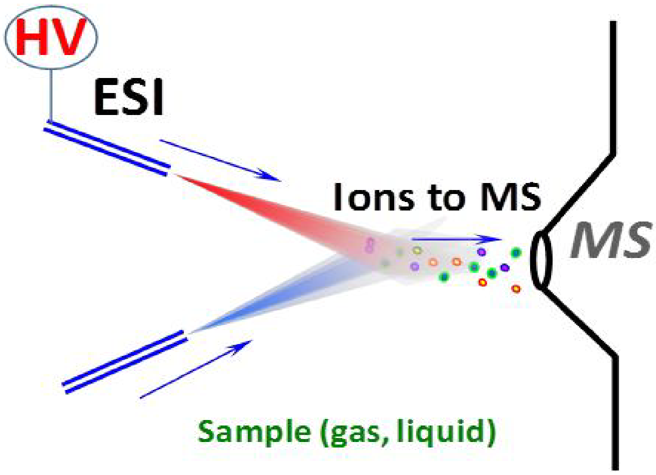

Figure 1-7 Schematic diagram of the of SESI, FD-ESI and EESI source.

A similar technique, fused-droplet electrospray ionization (FD-ESI), was developed for analysis of peptides and proteins in 2002.[59] It was found that FD-ESI-MS had advantages in providing salt tolerance for biological analysis.[59] Another similar technique, electrospray-assisted pyrolysis ionization-mass spectrometry (ESA-Py-MS), has been developed to ionize polymers. The pyrolytical products are produced by a commercial Curie point pyroprobe, and are conducted to a reaction chamber, where charged methanol spray is produced continuously by ESI. Polar components in the pyrolysates may interact with the polar methanol ions and thus are ionized for MS detection.[60] Moreover, the ESA-Py/MS also could be used for sensitive detection of trace polar compounds in nonpolar raw complex samples such as the crude oil, amber, humic substances, and rubber.[61] Polymer samples of different origins could be rapidly distinguished using ESA-Py in positive ion detection mode without separation or chemical pretreatment. Hydrocarbons are the major components in the raw samples. However, hydrocarbons are nonpolar and contain no functional group for accepting protons and thus produced no signals in the spectra, allowing the sensitive detection of trace polar components in the pyrolysates. In addition, extractive ESI,[50] another ESI-based extraction/ionization technique, which is proposed to have droplet-droplet extraction[62, 63] when a sample spray interacted with reagent electrospray, allowing continuous analysis of liquid and gas samples.[50, 64-68]

Since the introduction of DESI, many ambient ionization techniques have been developed, and ambient ionization can be found in several recent reviews.[69-73] Table 1-1 lists the representative ESI-based desorption/extraction ionization techniques that have been developed in recent years. In fact, these methods utilize ESI as the primary ion beam for desorption or/and extraction of analytes in complex samples.

1.3.2 ESI-assisted laser desorption/ionization

Laser is conventionally used in MADLI-MS for desorption/ionization of analytes on the surface. Taking its advantages in desorption/ionization of analytes, ESI was combined with laser for improved analysis of samples (Figure 1-8). ESI was used for production of primary ions under ambient conditions, and ultraviolet or infrared laser beam is often adopted as a light source for desorption/ionization of analytes. With the combination of laser desorption/ionization and ESI, both small and large molecules could be ionized at ambient conditions.

Particularly, due to the use of laser beam, ESI-based laser desorption/ionization techniques could also be used for imaging of solid samples, e.g., biological tissue. To improve desorption/ionization efficiency of analytes, matrix was often used.

Some of the representative ESI-assisted laser desorption/ionization techniques are listed in Table 1-2.

Table 1-2 Typical ESI-assisted laser desorption ionization techniques.

Technique	**Acronym**	**Year of invention**	**Refs**
Electrospray laser desorption ionization	ELDI	2005	74
Matrix-assisted laser desorption electrospray ionization	MALDESI	2006	75
Laser ablation electrospray ionization	LAESI	2008	76
Laser assisted desorption electrospray ionization	LADESI	2008	77
Laser desorption electrospray ionization	LDESI	2009	78
Laser-induced acoustic desorption/electrospray ionization	LIAD/ESI	2009	79
Laser desorption spray post-ionization	LDSPI	2010	80
Laser electrospray	LS	2010	81
Laserspray ionization	LSI	2010	82

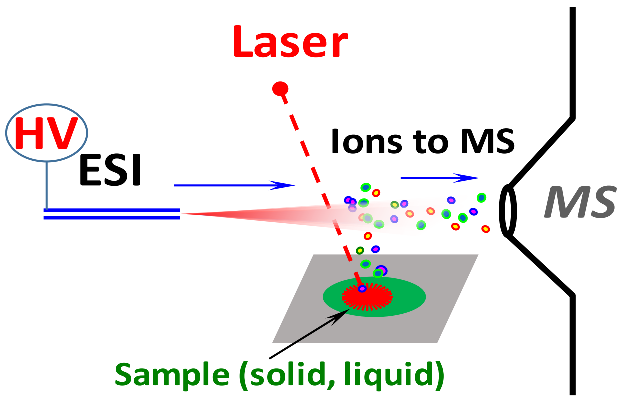

Figure 1-8 Schematic diagram of ESI-assisted laser desorption/ionization ion source

1.3.3 ESI with solid substrates

Conventional ESI uses capillary for sample loading and ionization. In recent years, the use of solid substrates for sample loading and ionization has attracted a lot of attentions. Solid-substrate ESI-MS avoids the clogging problem that can easily be encountered in capillary-based ESI-MS, allows more convenient sampling, and have ionization features that are different from those of conventional ESI-MS. In addition, solid-substrate ESI-MS allows direct analysis of complex samples, and requires no additional solvent beam, plasma or laser, different from other ambient ionization techniques discussed above.

Various solid substrates including wick element,[83] copper wire,[84] metal needles,[85] optical fibres,[86] glass rod,[87] tungsten oxide,[88] metal needles,[89] paper,[90] wooden tips[91] and aluminium foil[92] have been developed for mass spectrometric analysis of complex samples with little or no sample pre-treatment.

Unlike capillary-based ESI, in solid-substrate ESI, sample solution is loaded on the surface of a solid substrate for spray ionization. Initially, developments of these non-capillary ESI techniques were mainly for avoiding the clogging problem that could easily occur with capillary. In recent years, new features that are different from those of conventional ESI have been observed with solid-substrate ESI. For example, probe ESI (PESI), which uses metal needles as substrates for ESI analysis, was reported to be highly tolerant to salts and detergents[93, 94] and allow exhausted and sequential ionization of analytes;[95] direct analysis of raw samples could be achieved with paper spray,[96, 97] which uses paper as substrate for sample loading and ionization.

In 2011, we developed ESI on wooden tips (toothpicks).[91] Commercially available wooden toothpicks have a uniform o.d (2.0 mm),that is right compatible with typical nano-ESI sources, thereofore, no hardware modification is required to couple wooden-tip ESI tomass spectrometers. We demostrated that wooden-tip ESI-MS could be used for analysis of various samples, and was tolerant to impurity and matrix background and thus could be used for direct analysis of raw samples.[98-100] Our recent study revealed that this technique could be used for direct detection and quantitation of compounds in raw samples.

Moreover, following the studies with ESI on wooden tips, we further developed direct ionization of biological tissuefor direct analysis of plant tissue, animal tissue,[101, 102] and bulky solid samples such as mushroom and bone.[99] Other research groups also have developed similar techniques for direct analysis of plant samples, termed tissue spray[103] and leaf spray.[104, 105]

1.3.4 Other varieties of ESI technique

Direct current was directly applied to the ESI emitter in the previous experiments and theoretical studies of electrospray.[22, 25, 106, 107] Other electric modes, including friction electrification,[108] alternating current,[109] pulsed,[110] electrostatic,[111] and triboelectric ionization[112] have been successfully applied to ESI emitters for versatile applications. Interestingly, the voltage applied to ESI emitters does not have a strict limitation. For example, the voltage can be applied on a cylindrical electrode of ESI emitter,[27] and in some commercial ESI, the voltage is applied to a nozzle of ESI to avoid leakage.[113]

However, a high voltage could also be applied to an MS inlet[114] to induce electrospray ionization from a ESI emitter for MS analysis as well. In this way, the sample does not need any connection for ionization and analysis. This contactless feature of field-induced ESI mode has been largely ignored for a long time. Until in recent years, this feature was developed for monitoring chemical reaction and biomolecule analysis.[115-117] More recently, a similar field-induced ESI mode was introduced for *in vivo* study of living organisms by our group.[118,119] In field-induced ESI-MS, a sample is simply placed in front of the MS inlet for obtaining its signals, and the technique thus has great potential for various applications.

Voltage-free ESI techniques, including sonic-spray ionization (SSI),[120] easy ambient sonic-spray ionization (EASI),[121] matrix assisted ionization vacuum (MAIV),[122] and zero-volt paper spray ionization,[123] have also been developed. These voltage-free ionization techniques were proposed to have similar ion formation mechanism as ESI.

1.4 The objectives and outline of this thesis

Following the development of ESI on wooden tips and direct ionization of biological tissue, we aim to further develop solid-substrate ESI-MS and investigate its mechanism and analytical properties in this project. Particularly, we hope to address the following fundamental questions in solid-substrate ESI-MS:

1) How do the substrates affect the sampling and ionization during solid-substrate ESI-MS?

2) How to improve the sensitivity and selectivity in detection of analytes in complex samples?

3) How to achieve *in vivo* analysis and real-time monitoring of living organisms under ESI-MS conditions?

To this end, we have investigated various materials, factors and potential applications of solid-substrate ESI-MS in this study. The results are summarized in the different chapters in this thesis.

In Chapter 2, the analytical properties of ESI with different solid substrates were investigated. The sampling mechanism of wooden-tip ESI was explained. The results showed that wooden tips could serve as a chromatographic column for separation of sample components, indicating that the wooden surface could play an important role in sampling and ionization. Nonconductive materials that contain microchannels/pores could be used as tips for solid-substrate ESI-MS analysis with sample solutions loaded to the sharp-ends only, since rapid diffusion of sample solutions by capillary action would enable the tips to become conductive. The capacity of the technique for quantitative analysis of analytes in complex samples was also tested. Sequential and exhaustive ionization was observed for proteins and salts on wooden tips with salts ionized sooner and proteins later.

In Chapter 3, following the study in Chapter 2, effects of the wooden tip surface on ionization and sampling were further studied. Surfaces of wooden tips were modified with various functional groups and their analytical properties were investigated. It was found that the functional group on the surface of wooden tip could allow selective sampling and ionization of analytes during the WT-ESI-MS analysis. We demonstrated that improved detection of trace target compounds in complex samples could be achieved with surface-modified WT-ESI-MS.

In Chapter 4, mobility of proteins in porous materials substrates under ESI conditions was investigated. It was found that that under electrospray ionization (ESI) conditions, mobility of proteins in porous substrates, e.g., polyester tips and wooden tips, were size- and shape-dependent. A linear relationship was found between the relative integration area of the selected ion chromatogram and the cross section of the protein ion. Preparative separation of proteins could be achieved by collecting the proteins remained on the porous substrates.

In Chapter 5, a novel ESI technique based on household aluminium foil (Al foil) was developed. The technique was demonstrated to be applicable in analysis of a wide variety of samples, ranging from pure chemical and biological compounds to complex samples in liquid, semi-solid, and solid states. The inert, hydrophobic and impermeable surface of Al foil allowed convenient and effective on-target extraction of solid samples and on-target sample clean-up. Direct monitoring of thermal reactions could also be achieved with Al-foil ESI due to the heat conductivity and heat tolerance of Al foil.

In Chapter 6, the Al-foil ESI-MS technique was further extended to use Al-foil-based thin-layer chromatography (TLC) plate with silica coatings for loading and ionization of various samples. Direct analysis of samples containing high contents of salts, detergents and matrices, as well as quantitative measurements was demonstrated. Offline and online analyses of mixture samples were also attempted.

In Chapter 7, a commercial fibre with coatings, i.e., solid phase micro-extraction (SPME) fibre, was used as ESI emitter for enhanced detection of analytes in raw samples. The effects of various factors in the sampling and ionization process were systematically investigated. We demonstrated that detection of ketamine in urine could be significantly improved by SPME-ESI-MS with simple sample preparation. The analytical performances of SPME-ESI-MS, including the linear range, analytical speed, LOD and LOQ, were evaluated.

In Chapter 8, field-induced ESI-MS was developed. By avoiding direct connection of samples to a high voltage, much facilitated sample analysis could be achieved for routine sample analysis, various ambient techniques and direct ionization of biological tissues.

In Chapter 9, *in vivo* analysis and real-time monitoring of secondary metabolites released from living animals and plants were investigated with field-induced ESI-MS. Different sampling methods for living organisms, by detecting venoms secreted from living scorpion and toad upon attack and variation of alkaloids in living *Catharanthus* roseus upon stimulation, were introduced.

In Chapter 10, the study in this thesis was summarized and a perspective was given

Chapter 2: Analytical Properties of Solid-substrate ESI-MS with Various Materials

2.1 Introduction

In conventional ESI[124] or nano-ESI,[125] a sample solution is introduced into a capillary. Upon application of a high voltage to the capillary and usually with the assistance of gas, the sample solution would spray out of the capillary tip, leading to the formation of fine droplets and subsequently analyte ions. Non-capillary ESI techniques have also been developed. Materials such as wick element,[83] copper wire,[84] metal needle,[85, 93, 126] optical fibre wired with a metal coil,[86, 127] surface-modified glass rod,[87] nanosturctured tungsten oxide,[88] and more recently paper,[90, 128] and wood[91] have been used for sampling and ionization in ESI. Use of non-capillary emitters avoids the problem of clogging that could occur in capillary-based ESI and allows much easier sampling. Particularly, analysis of samples on surfaces or in difficult-to-access corners could be greatly facilitated by collecting sample using paper[90, 128] or wooden tips[91] followed by paper spray or wooden tip ESI analysis. More recently, non-capillary ESI techniques have been further extended to direct mass spectrometric analysis of biological tissue.[102, 105, 129]

Since the ionization conditions of non-capillary ESI such as the loading of sample solutions on surface and the absence of desolvation gas are different from those of capillary-based ESI, different features have been observed with non-capillary ESI. Some non-capillary tips such as stainless steel needle,[93, 94] paper,[97, 130-133] and wooden tips[91] enabled direct analysis of raw samples. Hiraoka and co-workers reported the sequential and exhaustive ionization of analytes on solid metal probe.[95] Separate ionization of proteins and peptides in a mixture sample was also observed with hydrophilic wooden tips.[91] The study on paper spray ionization of polar analytes using non-polar solvents indicated that solvents could transport insoluble analytes to the paper tip, and that field desorption and surface tension of solvents might play important roles in the ionization of analytes.[134] Further investigation on characteristics and emitters of non-capillary ESI is necessary for development and applications of this kind of techniques.

We recently reported the ESI ionization using wooden tips (toothpicks).[91] In this solid-substrate ESI technique, sample solution was loaded onto the sharp tip-end of a wooden tip. Upon application of a high voltage to the other end of the wooden tip, spray ionization was induced from the cusp and characteristic mass spectrum was observed. In the present study, the analytical properties and mechanism of Solid-substrate ESI was investigated and various tips were further developed for Solid-substrate ESI-MS analysis.

2.2 Experimental section

2.2.1 Materials

Wooden toothpicks used for wooden tips in this study were of the brand BEST buy from PARKnSHOP (Hong Kong). Bamboo toothpicks used for bamboo tips were of the brand Zhouzi™ from Zhennan Bamboo & Wood Co. Ltd (Hangzhou, China). Cloth pieces were of 35% cotton and 65% polyester and were cut from a lab coat from Guandong Tianyi Co. Ltd (Zhongshan, China). Cellulose sponge was of the brand Scothc-Brite™ from 3M (Hong Kong). PEEK tubing was from

Waters (UK). Stainless steel needles and silver needles were from Tongyong Office Co. Ltd (Shanghai, China) and Chow Sang Sang (Hong Kong) respectively. Fresh coriander leaves, spinach leaves, mushroom and ribs were purchased from supermarkets in Hong Kong. Reserpine, myoglobin (from horse heart), cytochrome *c* (from bovine heart), lysozyme (from chicken egg white), dimethoate, acetone, ligroin, ammonium acetate, sodium chloride and formic acid were purchased from Sigma (St. Louis, MO), and losartan from Gracure Pharmaceuticals Ltd. (New Delhi, India). Ketoconazole cream was purchased from Dihon Pharmaceutical Co. Ltd (Kunming, China). Green fluorescer was purchased from Zebra (USA). Methanol was purchased from Tedia (Fairfield, OH). Water used in this study was Milli-Q water. Extract of spinach leaf was prepared as described in the literature.[135] Wooden tips were prepared as described previously.[91] Other tips were prepared by modifying the materials to form sharp ends pointing to the MS inlet. Washing of tips with methanol before use could reduce contamination and background if necessary.

2.2.2 Solid-substrate ESI-MS measurements

Mass spectra were acquired on a Q-TOF2 mass spectrometer (Waters, Milford, MA, USA). Various tips were connected to the high voltage supply of the mass spectrometers via a chip except for toothpick tips that were via a nano-ESI ion source. The tips were perpendicular to the MS inlet with the tip-ends ~10 mm away from the inlet. The capillary voltage and cone voltage were set at 3.5 kV and 30 V respectively unless otherwise specified. The source temperature was set at 80ºC. Unless specified, mass spectra were obtained by accumulating the data acquired in 1 min. Other settings were similar as those previously described for the wooden tips.[91]

Resistances and voltages were measured using a multimeter from John Fluke MFG Co. Inc. (USA). Diffusion of the fluorescence solution on the wooden tips was visualized by an UV lamp from Macherey-Nagel (Düren, Germany) and recorded using a Sony T20 camera (Japan) with a speed of 28 frames/sec. Optical images of ESI plume was recorded using the same camera with the assistance of a KL 1500 LCD halogen cold light lamp (Schoott, Germany).

2.3 Results and discussion

2.3.1 Conductivity of wooden tips

To investigate the conductivity of wooden tips used in the study, the resistance of a typical wooden toothpick (length: 20 mm, root diameter: 2 mm) was measured. A resistance larger than the measurement limit of the multimeter was obtained when the wooden tip was dry. When 5 μL methanol was loaded onto the sharpend of the tip, immediate measurement revealed that the resistance of the tip became 5.0 ± 0.5 MΩ (3 independent measurements), indicating that the wooden tip became much more conductive even though the solvent was not applied to the entire tip. To further investigate how the conductivity was improved in this case, 5.0 μL of green fluorescer solution (1.0 mg/mL in methanol) was loaded onto the sharp end of a wooden tip, and an UV lamp and a camera were used to trace the movement of the solution.

It was found that the solution diffused rapidly and longitudinally along the microchannels (the vascular system) on the surface of the wooden tip by capillary action (Figure 2-1), and it took about 0.36 s (10 recorded frames) for the solution to travel 30 mm. This result suggested that for a wooden tip of 20 mm, a common length used in our studies, it took about 0.24 s for the solution to reach the other end to make the tip become conductive. It was noted that only a trace amount of solution had diffused along the tip, and only the sharp end of the tip onto where the sample solution had been loaded was apparently wet, while other parts of the tip still looked or felt dry after the tip had become conductive.

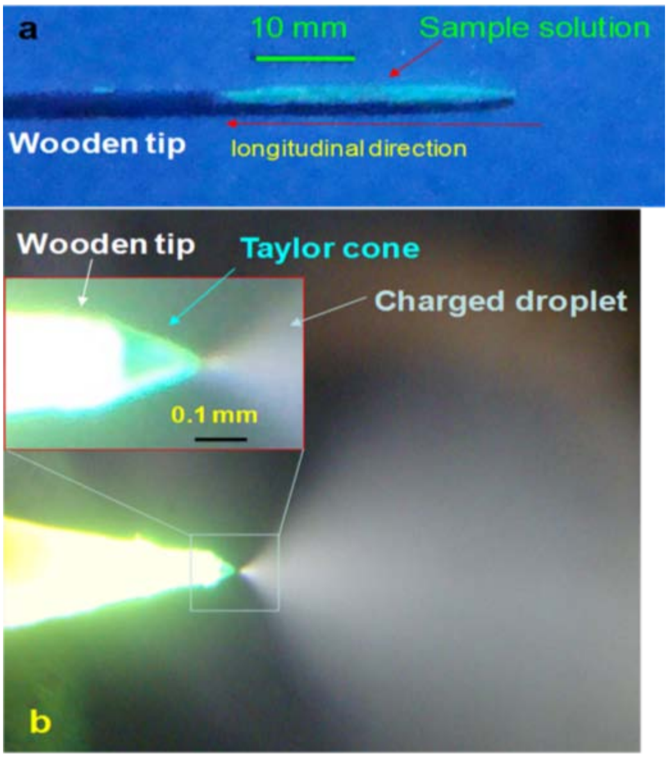

Figure 2-1 a) Monitoring the movement of the green fluorescer solution on a wooden tip revealed that the solution diffused longitudinally along the stripes; (b) optical image of the plume of methanol spray from the wooden tip at 3.5 kV. The inset shows the view of the Taylor cone in detail.

It was noted that the fluorescer solution only diffused longitudinally along the microchannels and almost no transverse diffusion was observed (Figure2-1). To further investigate the effects of microchannel direction on diffusion and spray ionization of sample solutions, wooden tips with longitudinal and transverse microchannels were made respectively (Figure 2-2). Investigation using the green fluorescer solution confirmed that when the solution was applied to the sharp tip-end, rapid solution diffusion to the other end occurred only on the tips with longitudinal microchannels but not on those with transverse microchannels. The resistance of the wooden tips with transverse microchannels remained larger than the measurement limit of the multimeter after sample loading to the sharp end, in contrast with the much improved conductivity of the wooden tips with longitudinal microchannels. Wooden tips with longitudinal microchannels and those with transverse microchannels were connected to a high voltage (3.5 kV) and they were compared for ESI analysis. When 5.0 μL reserpine solution (1.0 ppm in methanol) was applied to the sharp end of each tip, reserpine signals were immediately observed for the tips with longitudinal microchannels (Figure 2-2a), while no spray and no reserpine signals were observed for the tips with transverse microchannels (Figure 2-2b). When the tips with transverse microchannels got entirely wet with 20 μL methanol, reserpine signals were detected (Figure 2-2c, also see Figure 2-1b for an image of the plume of spray from a wooden tip). These results further confirmed that it was the diffusion of solution along the wooden tips that caused conductivity and subsequently spray ionization of the wooden tips. Commercially available wooden toothpicks usually have longitudinal microchannels and thus can be conveniently used for solid-substrate ESI analysis with sample solution loaded onto the sharp tip-end. The unnecessariness to get the entire tip wet allows less sample consumption and more convenient sample loading.

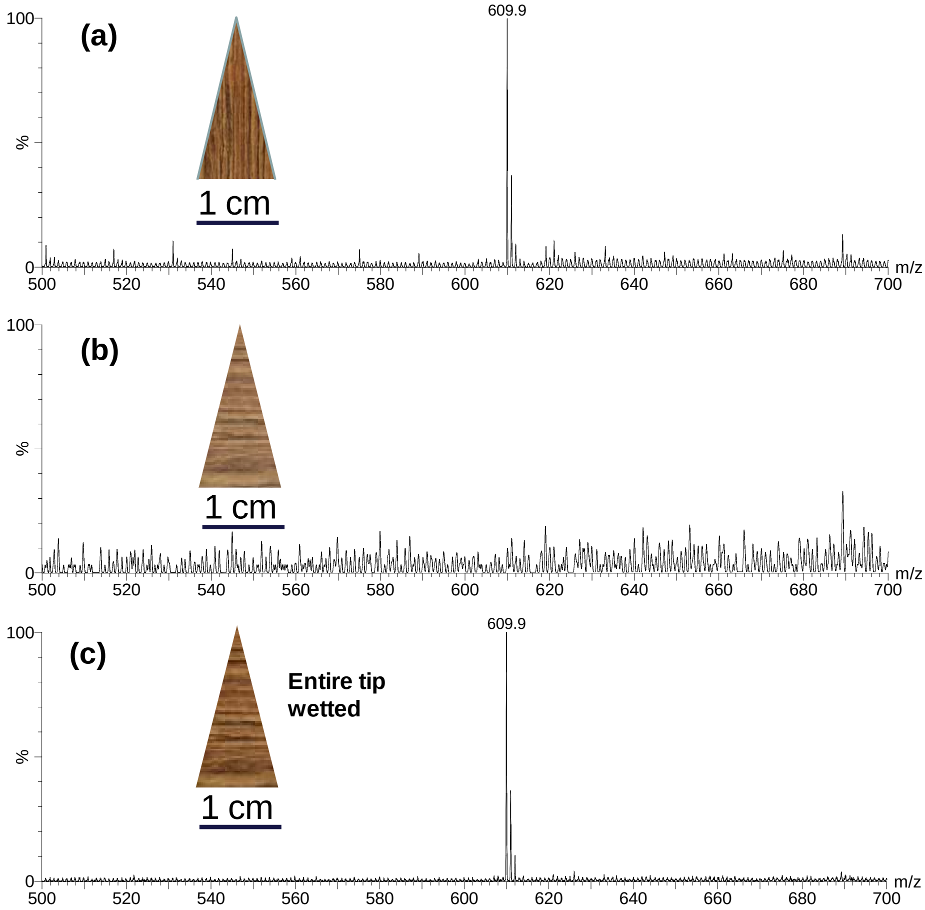

Figure 2-2 Mass spectrum obtained after loading 5 μL reserpine (1 ppm in methanol) onto the sharp end of a wooden tip. a) Reserpine signals were immediately obtained from the tip with longitudinal microchannels; b) no reserpine signals were obtained from the tip with transversal microchannels; c) reserpine signals were obtained from the tip with transversal microchannels after the entire tip was wetted by methanol.

2.3.2 Ionization on wooden tips

After loading a sample solution to the sharp tip-end and applying a high voltage to the wooden tip, it could be observed that the solution on the tip-end became elongated and then sprayed out to form microdroplets. An image of the plume of spray from a wooden tip is shown in Figure 2-1b. To better understand the ionization process of wooden tips, effects of major settings including tip size (size of the tip cusp), high voltage applied to the tip and tip angle were investigated.

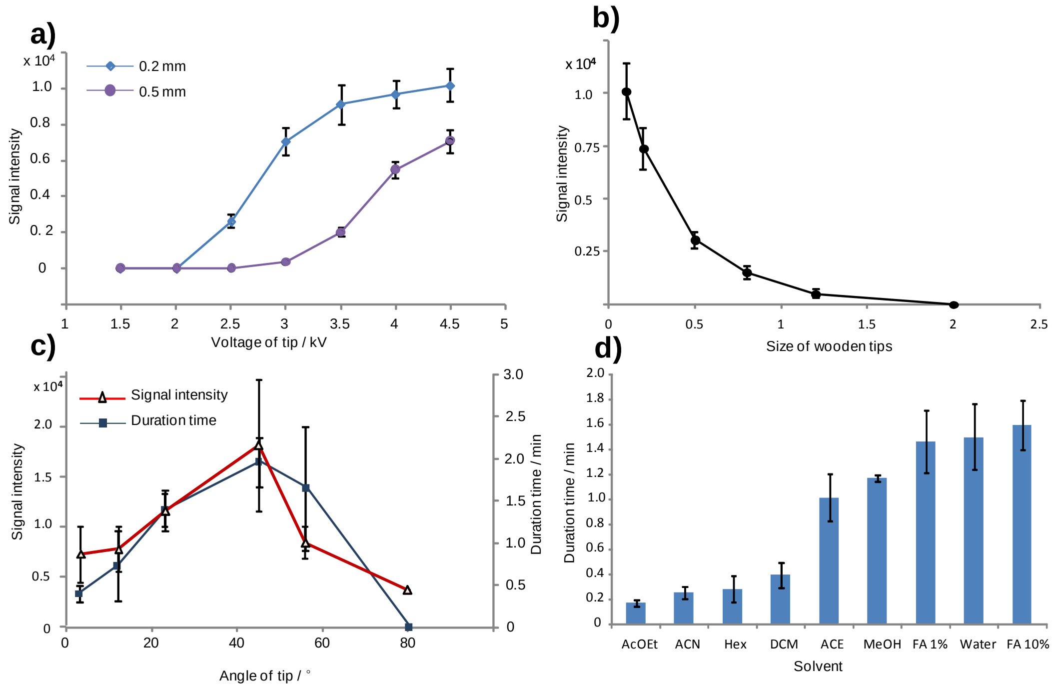

Figure 2-3 a) Effects of the voltage applied to the tip on signal intensity for tip sizes of 0.2 mm and 0.5 mm; b) effect of the tip size on signal intensity; c) effects of the tip angle on signal intensity and signal duration, data were obtained by detection of 5 μL reserpine (1 ppm in methanol) and each point was the average of three measurements; d) effects of solvent on signal duration.

For a wooden tip with a tip size of 0.5 mm, a common size for commercially available wooden toothpicks, no significant signals were detected when 5 μL reserpine solution (1.0 ppm in methanol) was analyzed with the applied high voltage of lower than 3.0 kV, and higher intensities of the signals were observed with higher voltages applied to the tip (Figure 2-3a). However, corona discharge could be easily induced at voltages higher than 4.0 kV. When the tip size was larger than 0.5 mm, with voltages lower than 4.0 kV it was difficult to generate spray from the tip end and accordingly no significant signals were observed. For a tip with a size of 0.2 mm, a threshold voltage of ~ 2.0 kV was obtained (Figure 2-3a). Much stronger signals were observed with the tip size of 0.2 mm than with the tip size of 0.5 mm with the same applied voltages. Other tip sizes were also assessed and the results revealed that the sharper the tip was, the stronger the signals were observed (Figure 2-3b). These results revealed that ionization on wooden tips followed a Taylor cone mode, with which appropriate voltage and tip size were required for formation of liquid jet and charged droplets from the tip. A tip size of 0.2 mm and a voltage of 3.5 kV were usually used in our studies. Although smaller tip sizes give better signals, their fabrication requires more delicate work. Solid-substrate ESI using wooden tips requires a higher voltage for ionization than nano-ESI.[128] This is believed to be due to the larger tip size and poorer conductivity of wood.

The effect of tip angle on solid-substrate ESI signals was also investigated. Wooden tips of the same tip size (0.2 mm) but different tip angles were prepared and used for analysis of 5 µL reserpine solution (1.0 ppm in methanol) with a high voltage of 3.5 kV. No spray ionization was observed with tip angles larger than 80º. Signals could be observed for wooden tips with tip angles smaller than 80º. The longest duration and strongest signals were obtained with a tip angle at around 45º (Figure 2-3c). It was observed that with smaller or larger tip angles, the tips had smaller surface areas of sharp-end parts and the sample solution would accumulate at the tip ends to form thicker solution layers, from which

Taylor cone was generated. Larger droplets were formed and sample solution was thus consumed quickly. Bulk solution movement was believed to be the major process for solution transfer during spray ionization in these cases. When the tip angle at around 45^{o}, the tip had larger surface area at the sharp-end apart and the solution was distributed to form a thinner solution layer. Taylor cone was generated from the tip cusp with formation of finer droplets, and signals with longer duration and stronger intensity were thus observed, however, the duration is different in different solvent (Figure 2-3d). Capillary action and electrophoresis were believed to play a more important role for solution transfer during spray ionization in this case.

2.3.3 Separation properties of wooden tips

Wooden tips have hydrophilic surface of cellulose structures. To test whether wooden tips have separation effect like paper chromatography and thin-layer chromatography (TLC), a 5 cm wooden tip was used and 5.0 µL extract of spinach leaf was loaded 4cm away from the tip cusp (Figure 2-4). After the solution was dried, acetone/ligroin (1/9, v/v), a solvent system used for TLC separation of the extract,[136] was added onto the coarse end of the tip to drive the components to the sharp end. The tip was cut into two halves after the tip became dry and each half was analyzed using the solid-substrate ESI technique with 5 µL methanol added to the sharp end. Pheophytin A and pheophytin a O-allomer,[137] compounds of lower polarity, were found at the front part (Figure 2-5a, compounds confirmed by MS/MS with data not shown), while polar compounds such as glucose were detected at the back part (Figure 2-5b). Some pigments such as chlorophylls were not observed probably due to their low signal responses under the solid-substrate ESI conditions.

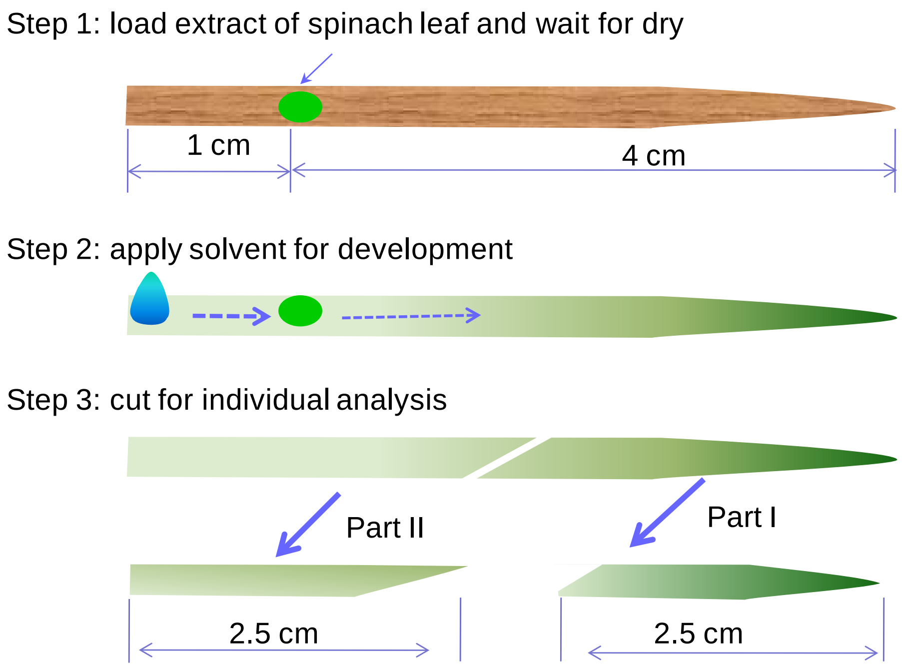

Figure 2-4 Procedures for testing the separation effect of the wooden tip. 5 μL extract of spinach leaf was loaded and acetone/ligroin (1/9, v/v) was used as the mobile phase for development.

These results revealed that the wooden tip could serve as chromatographic column to allow separation of components based on their polarities. The wooden tip loaded with extract of spinach leaf at the 4-cm position was also online analyzed with a high voltage applied to the tip and solvents of different polarities added to the coarse end of the tip (Figure 2-6). Low-polar components pheophytin A and pheophytin A O-allomer (Figure 2-6) and polar components such as glucose (Figure 2-6b) were predominantly detected when low-polar solvent acetone/ligroin (9/1, v/v) and polar solvent solvent/water (1/1, v/v) were respectively used as the mobile phase and ionizing solvents, confirming the separation effects of the wooden tips and the effects of running solvents.

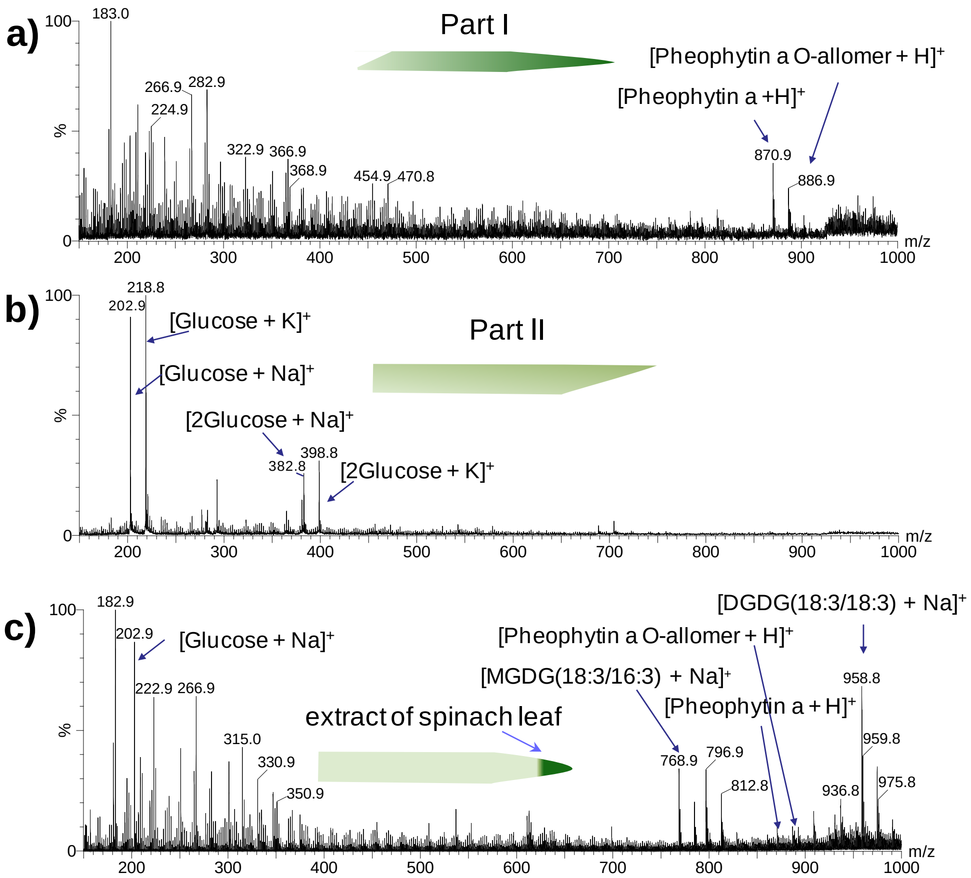

Figure 2-5 a) and b): mass spectra obtained by analyzing Part I and Part II in Fig. 1 respectively; c) mass spectrum obtained by analysis of the extract loaded to the sharp end of the tip.

For comparison, the extract was directly analyzed with the extract solution loaded to the sharp tip-end of the wooden tips. Both polar and low-polar compounds and compounds such as monogalactosyl diacylglycerol (MGDG) and digalactosyl diacylglycerol (DGDG) were detected in this case (Figure 2-5c), covering most components observed by LC-MS.[102] These components, however, could not be detected by analysis of the extract using nano-ESI.[102] These results indicated that solid-substrate ESI had lower signal suppression and could provide a more complete profile of the sample.

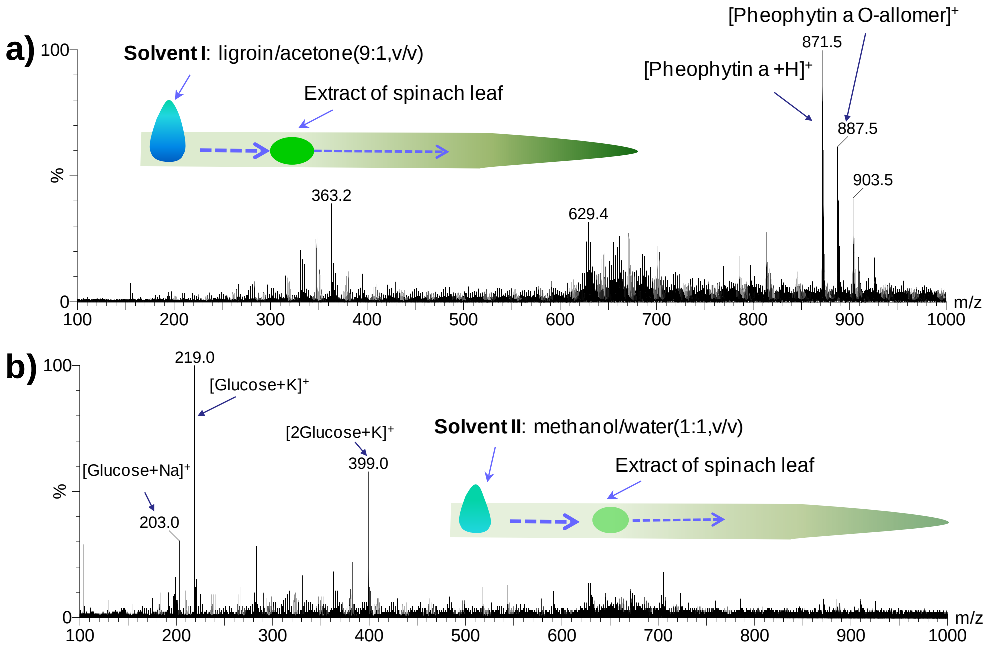

Figure 2-6 Mass spectra obtained by online wooden-tip ESI analysis of spot of extract of spinach leaf using a) ligroin/acetone (9/1, v/v) followed by b) methanol/water (1/1, v/v) as the driving and ionizing solvents.

The separation property of the wooden tips could be used for analysis of proteins containing high contents of salts. As shown in Figure 2-7a, 5.0 μL solution containing 10 μM lysozyme and 10 mM NaCl in 50% methanol was loaded 0.5 cm away from the cusp of a 2 cm wooden tip and left for dry. The spot was subjected to solid-substrate ESI-MS analysis with driving and ionizing solvents added to the coarse end of the tip. When water was used as the solvent, only cluster ions of NaCl and almost no protein signals were observed. When methanol, a solvent that favored elution of proteins, was used as the solvent, lysozyme signals was detected with few NaCl interferences (Figure 2-7b).

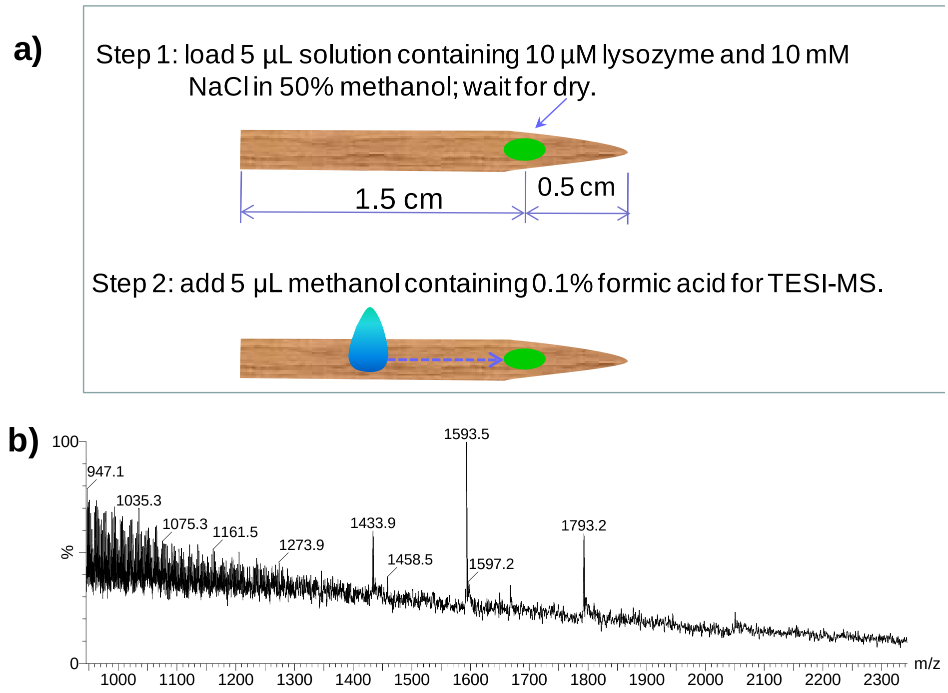

Figure 2-7 a) Procedures for analysis of spot of lysozyme containing NaCl, and b) the spectrum obtained.

However, such large molecules were found to have difficulty being carried through and detected if they were loaded too far away from the tip cusp, similar to the observation in TLC.[138, 139] When the lysozyme and NaCl solution was loaded to the sharp end of a wooden tip and directly analyzed by solid-substrate ESI-MS, the protein and salt were found to be ionized separately. As shown in Figure 2-8, NaCl was ionized at the first stage while lysozyme was detected at the last stage, showing an ionization order different from that obtained by Mandall *et al* [95] in the analysis of insulin and NaCl on metal needle. Mandall *et al* [95] recently reported the sequential and exhaustive ionization of analytes on metal needles. They concluded that analytes were ionized and detected in the order of their surface activities, i.e., compounds of higher surface activity were exhausted sooner than those of lower surface activity. However, there were also controversial cases in their studies.[95]

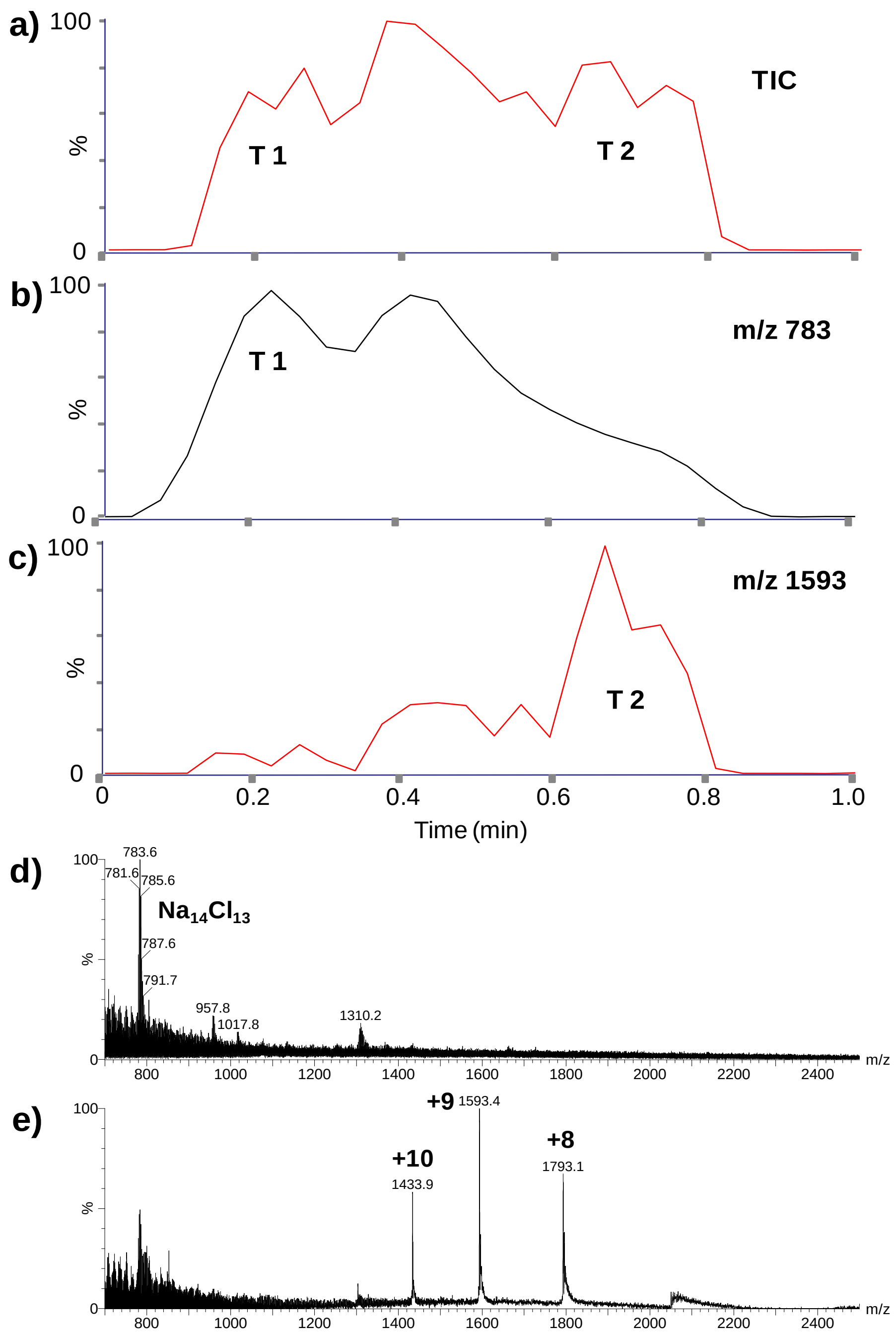

Figure 2-8 Sequential and exhaustive ionization on a wooden tip for 5 μL solution containing 10 μM lysozyme and 10 mM NaCl in 50% methanol: a)Total ion chromatogram (TIC); b) extracted ion chromatogram for *m/z* 783 ($Na_{14}Cl_{13}^{+}$); c) extracted ion chromatogram for *m/z* 1593 ($[\text{lysozyme} + 9H]^{9+}$); d) spectrum acquired at T1; e) spectrum acquired at T2.

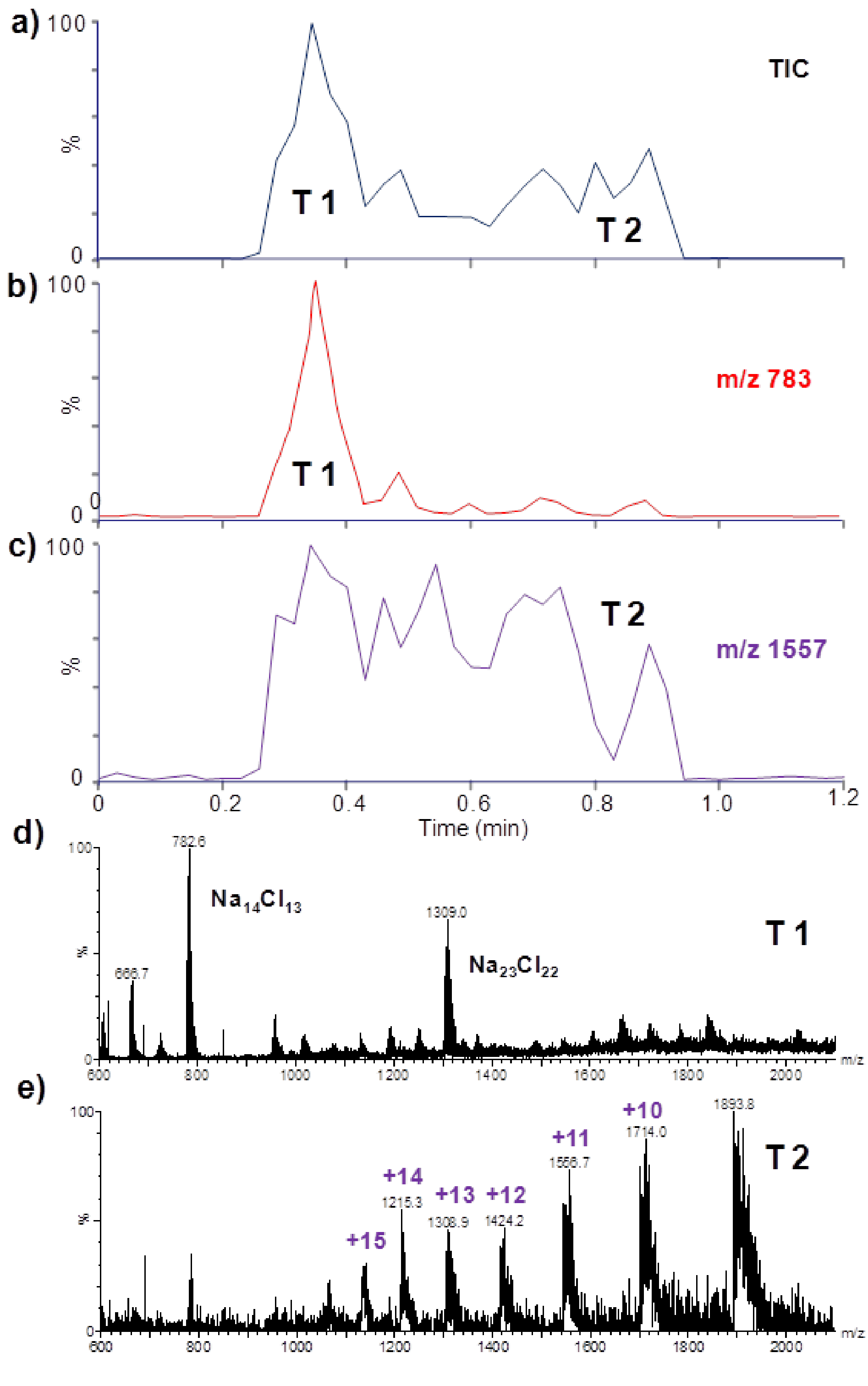

Figure 2-9 Sequential and exhaustive ionization on a wooden tip for 5 µL solution containing 10 µM myoglobin and 10 mM NaCl in 50% methanol: a) Total ion chromatogram (TIC); b) extracted ion chromatogram for *m/z* 783 ($Na_{14}Cl_{13}^{+}$); c) extracted ion chromatogram for *m/z* 1557 ($[\text{myoglobin}+11H]^{11+}$); d) spectrum acquired at T1; e) spectrum acquired at T2.

In the analysis of protein and peptide mixtures using wooden tips, we observed that protein signals were exhausted first while peptide signals lasted much longer.[91] In the present study, lysozyme and NaCl were found to be ionized in the order of increasing surface activity. This ionization sequence was further confirmed by analysis of a mixture of myoglobin and NaCl using wooden tips (Figure 2-9). Once again the salt was exhausted first and the protein signals were clearly observed at the late stage. Our results indicated that wooden tips could be used to directly analyze protein solutions with high salt concentrations. More investigations will be needed for a better understanding of the mechanism and order of the sequential and exhaustive ionization in solid-substrate ESI-MS.

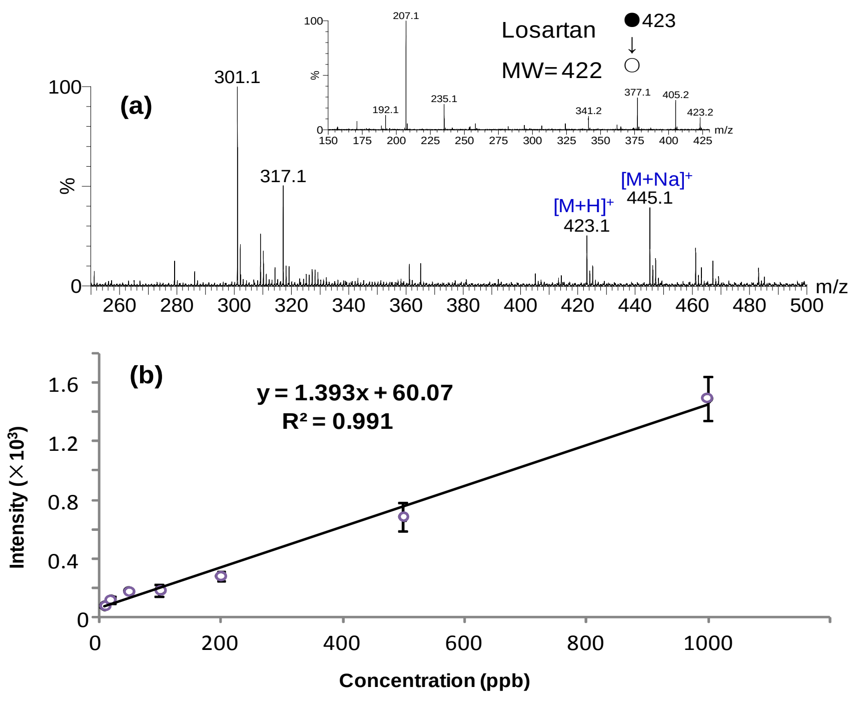

Figure 2-10 a) Mass spectrum obtained by analyzing 5 µL raw urine spiked with 0.2 ppm losartan. The inset MS/MS spectrum of protonated losartan confirmed the identity of the ion; b) linear relationship obtained by plotting intensity of SRM (*m/z* 423 > *m/z* 207) against the concentration of losarton in urea. Each point was measured 3 times.

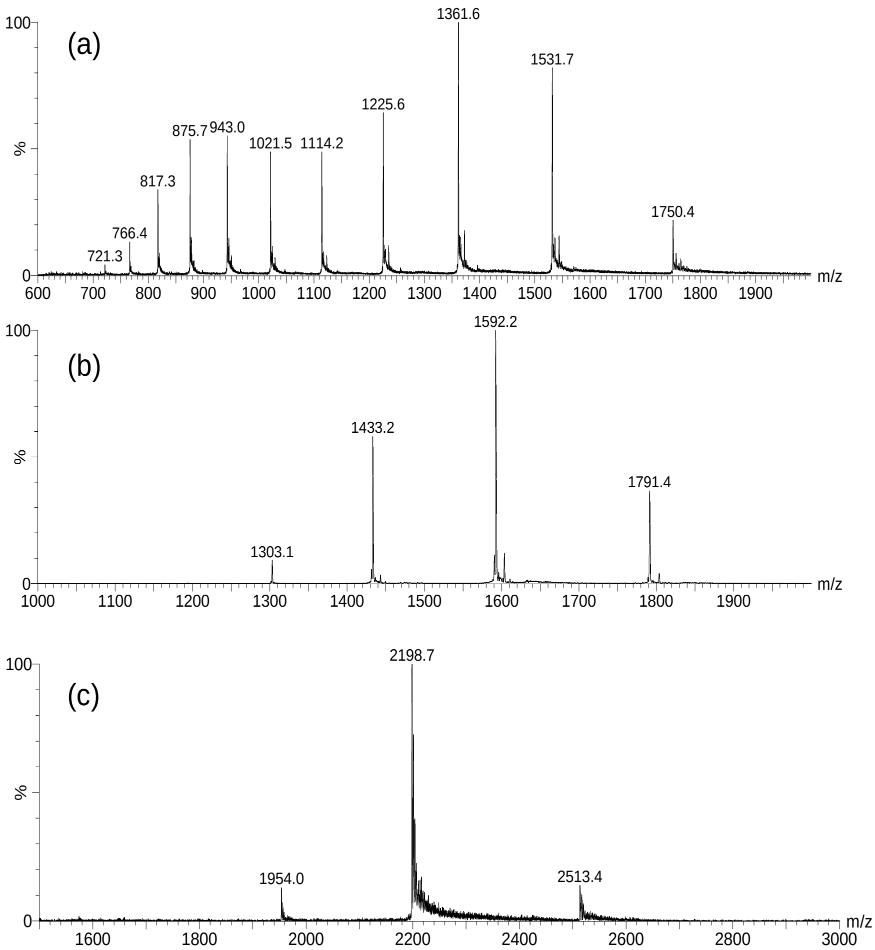

Figure 2-11 Solid-substrate ESI-MS spectra obtained with various tips, a) 5 µM cytochrome c in water/MeOH/ formic acid (50/50/0.05, v/v/v) with a bamboo tip; b) 5 µM lysozyme in water/MeOH/ formic acid (50/50/0.05, v/v/v) with a cloth (35 % cotton and 65 % polyester) tip; c) 10 µM myoglobin in 20 mM ammonium acetate with cellulose sponge tip.

2.3.4 Quantitative analysis using wooden-tip ESI-MS

Wooden tips can be used not only for qualitative analysis but also for quantitative analysis of raw samples. A wooden tip was used to directly analyze lorsartan, a medicine for hypertension, in raw urea (Figure 2-10a). As shown in figure 2-10b, SRM intensity shows a good linear relationship against the concentration of the compound in urea with an average RSD of 13.43 %, suggesting good reproducibility of the measurements and potential application of solid-substrate ESI-MS in quantitation.

2.3.5 Sampling and ionization with various tips

Conductivity is an important factor for selection and fabrication of ESI emitters. Our above results indicated that non-conductive materials containing microchannels or pores to allow longitudinal diffusion of solution could also be used as ESI emitters even the sample solution was loaded to the tip ends only. Rapid diffusion of the solution by capillary action would enable the materials to become conductive and ESI ionization could thus be induced. Various other materials were tested as ESI tips in this study.

Common non-conductive materials, including porous materials such as bamboo, cloth and sponge, and non-porous materials such as PEEK were attempted for sampling and ionization with the sample solutions loaded to the sharp-ends. No signals were observed with the non-porous tips due to the intrinsic insulating property of the materials, and the solution could not be efficiently diffused along the tips to improve conductivity. Spray ionization could be observed with the porous tips made of bamboo, cloth (35% cotton and 65% polyester) or cellulose sponge. With these tips, the diffusion of solutions along microchannels/pores enabled the applied voltage to reach the tip end to induce ionization. Some of the spectra obtained using these porous tips are shown in Figure 2-11. Bamboo is more rigid than wood and holds solution less efficiently (Figure 2-12). The signal span of bamboo tips was thus shorter than that of wooden tips for the same volume of the solutions. Shorter signal durations were also observed for the tips of paper and cloth, indicating that solutions on these planar materials of larger pores spread out and evaporated more easily. Sponge including cellulose sponge and other sponge could also be used as the ESI tips. Compact sponges with smaller pores tended to offer better signals and longer signal durations.

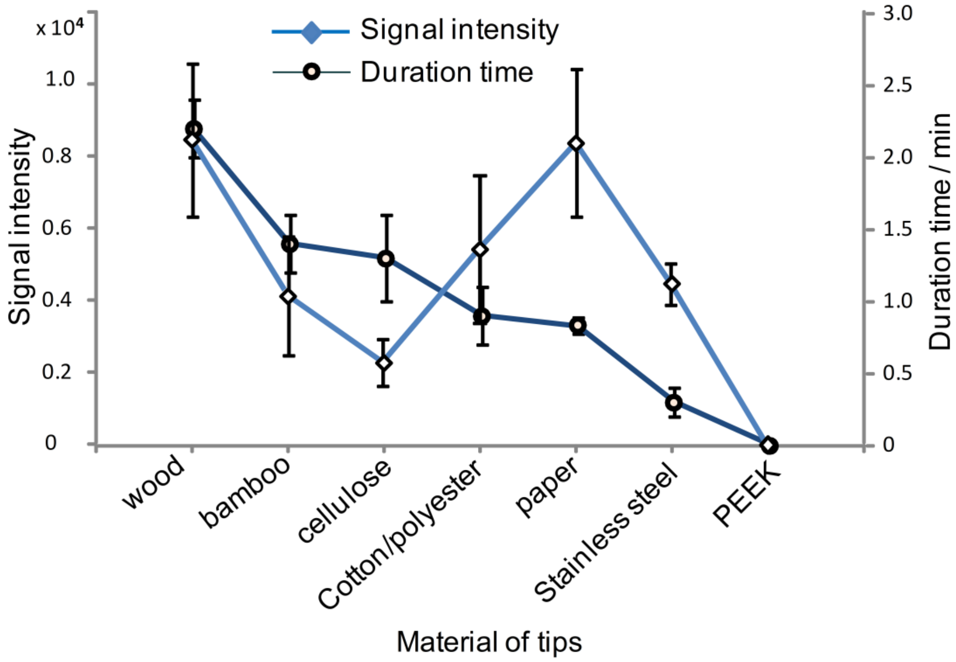

Figure 2-12 Effects of materials on signal intensity and duration

Metal tips such as silver and stainless steel needles were also examined as solid-substrate ESI tips in this study (Figure 2-13). These materials are intrinsically conductive and do not require solution diffusion to improve conductivity and induce spray ionization. Sample signals could be observed with these metal needles, but the signal durations were very short because the surface of these needles was smooth, hydrophobic and thus less capable of holding solutions. This result was consistent with the previous observation with metal needles.[85, 95] Although metal needles have short signal duration for analysis of solution samples, their good conductivity and relative inertness are good for direct analysis of some special samples. Analysis of viscous samples is a challenging task in mass spectrometry.[140] Metal needles could hold viscous samples well and thus could be used in their direct analysis.

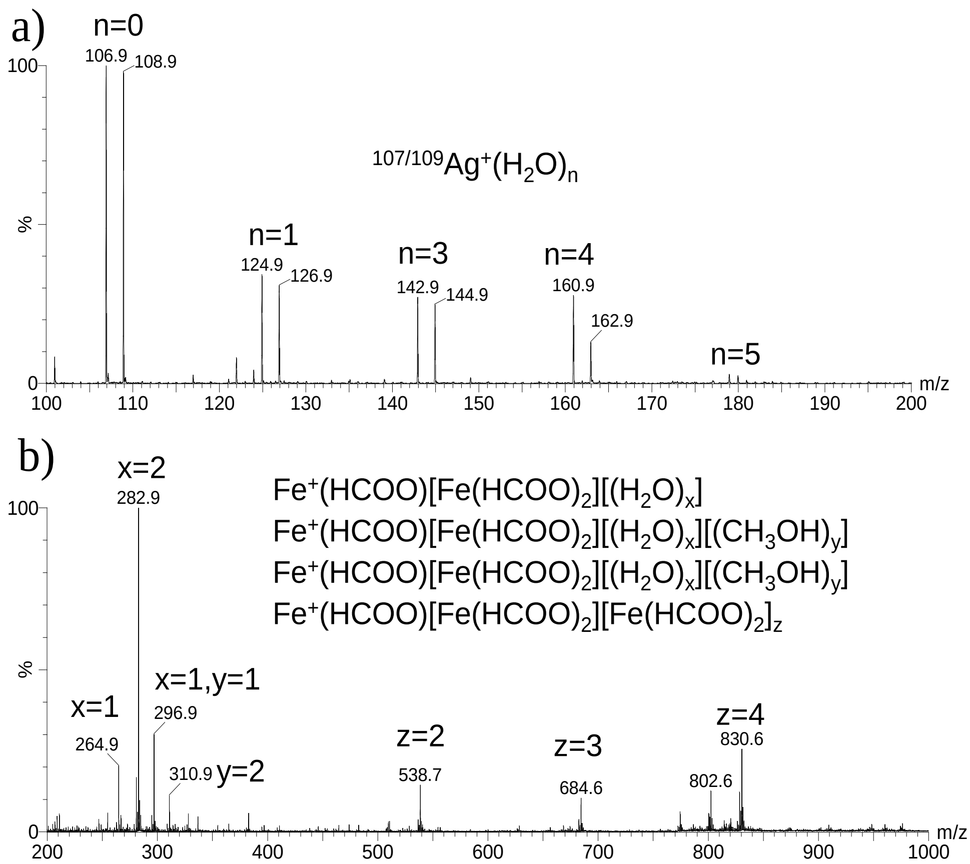

Figure 2-13 Direct analysis of a) silver and b) stainless steel, the HNO_3 (50% in water) and formic acid (1 %) were used as spray solvent, respectively.

Figure 2-14a is a spectrum for ketoconazole cream, a common medicine for fungal infections on skin,[141] obtained by direct analysis of the cream on a stainless steel needle without addition of solvents. Ketoconazole (Ket) and clobetasol propionate (CP), two major ingredients of the medicine, and the cream Peregal O were detected with excellent signal-to-noise ratios in the spectrum. With ~ 2 µL of the cream, the signals lasted for about 15 min. Metal needles can also be used to hold soft animal tissue for direct ionization analysis.[102]

Direct ionization could also be achieved with samples that could form tips to allow for MS analysis. These samples could be intrinsically conductive or become conductive with addition of solvents. Added solvents are also usually required to dissolve and facilitate ionization of chemicals from the samples. Figure 2-14b is a spectrum obtained by direct ESI-MS analysis of an intact fresh coriander (*coriandrum sativum*) leave previously spiked with 5 ng of dimethoate, a pesticide. After adding 5 μL methanol and applying a high voltage, a plume of spray could be observed from the leaf tip and protonated dimethoate (*m/z* 230) was predominantly detected in the spectrum. As no damage was done to the leaf and the waxy layer on the leaf surface prevented extraction of internal compounds by the solvent, only chemicals on the leaf surface were detected. This provides a simple and rapid method for analysis of substances on leaf surface. Leaves could also be cut into V-shape[102, 129] to allow their endogenous compounds to be analyzed by solid-substrate ESI-MS. Mushroom is a fruiting body of fungi. Figure 2-14c is a spectrum obtained by solid-substrate ESI-MS analysis of a piece of mushroom (~ 20 mg) with 5 μL methanol as the added solvent. Compounds such as arginine and sucrose could be observed. Animal bones could also be directly analyzed using the solid-substrate ESI-MS method. Figure 2-14d is a spectrum obtained for a small piece of fresh porcine rib (~ 5 mg). One side of this triangular rib was connected to a high voltage of 3.5 kV via a clip. Upon addition of 3 μL methanol containing 0.1% formic acid to the rib, a spectrum was obtained with ingredients of marrow such as lipids and hemoglobin detected.

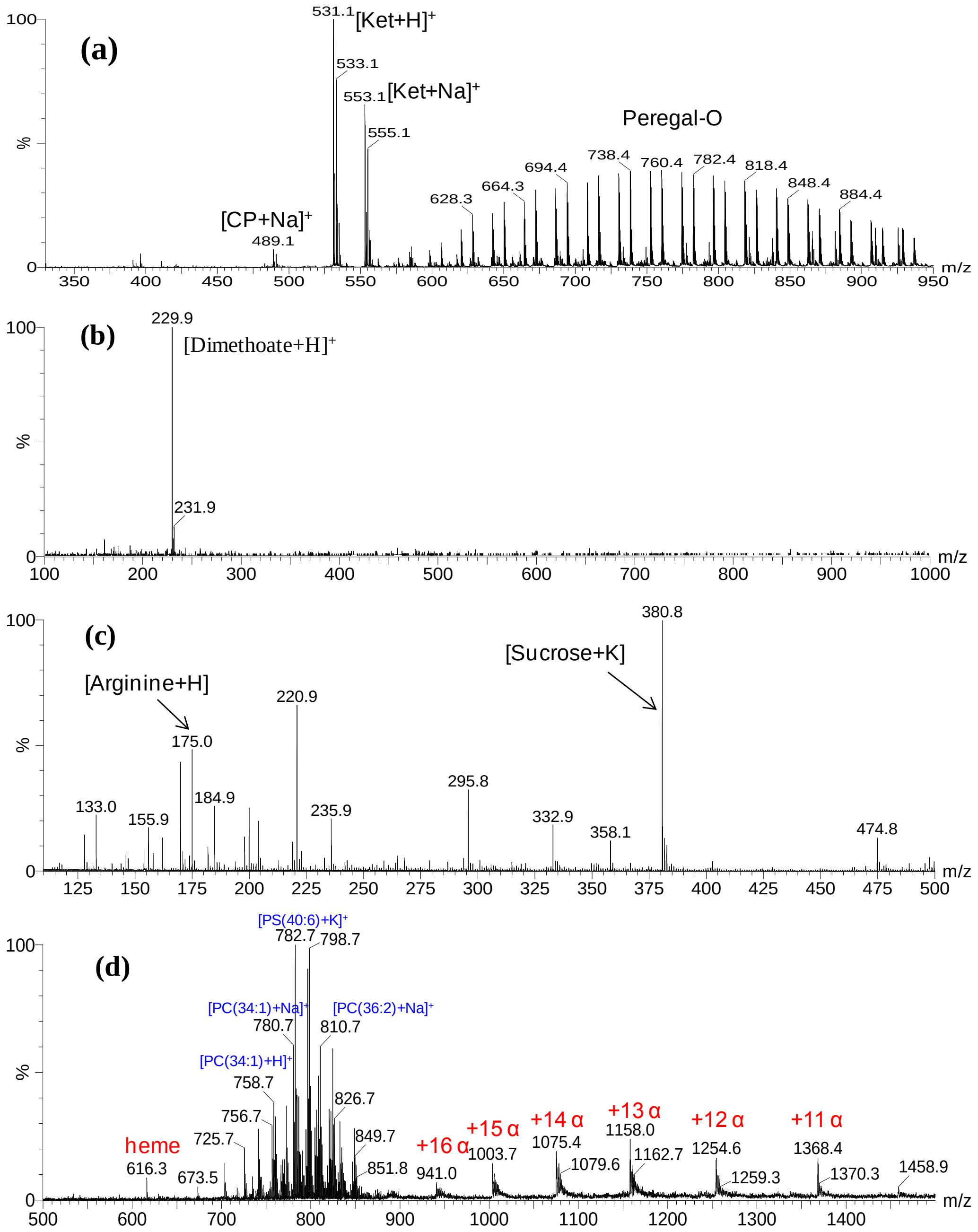

Figure 2-14 Mass spectrum obtained by analysis of a) ~2 µL ketoconazole cream loaded on a stainless steel needle without addition of solvents; b) intact fresh coriander leaf with 5 ng dimethoate spiked on the surface; c) a piece of mushroom with 5 µL methanol as the added solvent, and d) a piece of porcine rib with 5 µL methanol containing 0.1% formic acid as the added solvent.

2.4 Summary

The hydrophilic surface of wooden tips could serve as a chromatographic column for separation of sample components. Exhaustive and sequential ionization was observed for proteins and salts on wooden tips with salts ionized sooner and proteins later, allowing direct analysis of protein samples with high salt contents. Further investigation is required to understand the mechanism and order of exhaustive and sequential ionization on wooden and other tips. Our results also revealed that rapid longitudinal diffusion of solutions/solvents by capillary action enabled the wooden tips to become conductive and the high voltage applied to the tip could thus reach the tip cusp to induce spray ionization. Non-conductive porous materials could thus be used as tips for ESI analysis and ionization could be induced with sample solutions loaded to the sharp ends of the tips only. We demonstrated that in addition to wood, paper and metals, materials such as bamboo, cloth and sponge could also be used as tips for sample loading and ionization of solid-substrate ESI analysis, and ionization could be induced from tips of plant and animal tissues, mushroom and animal bones to allow them to be directly analyzed. These ESI ionization processes have a common feature: spray ionization is induced from tips with application of an electric field. In solid-substrate ESI-MS, tips of inert materials can be used for sample loading and ionization, and bulky samples can form tips to be directly analyzed. Searching for tips of higher efficiency for sample loading and ionization and extending solid-substrate ESI-MS for direct ionization analysis of a broader range of samples are under investigation.

Chapter 3: Surface-modified Wooden-tip ESI-MS for Enhanced Detection of Analytes in Complex Samples: Mechanisms and Applications

3.1 Introduction

In our previous work, we have developed ESI using disposable wooden tips (toothpicks) for direct analysis of various complex samples,[91] facilitating sampling and operation because there is no gas or laser beam required. To date, the WT-ESI-MS has been successfully applied in multiple disciplines such as food safety,[142] pharmaceutical analysis,[143] metabolites analysis,[144, 145] drug testing,[146] plant identification,[147] environmental analysis,[148] ionization mechanism study,[149, 150] explosive detection,[151] and development of new ionization technique.[152] However, it is noted that the sensitivity of direct measurement of drug-of-abuses in human body fluids using wooden-tip ESI-MS is lower than liquid chromatography coupling to conventional ESI-MS.[153] How to improve the WT-ESI-MS analysis of complex samples is thus an important issue for its further development and applications. Our previous work and other works on the ESI with solid substrates indicated that the surface of the solid substrate could play an important role in sampling and ionization.[92, 148-151, 154, 155] Therefore, understanding the mechanism of interaction between surface and analytes in sampling and ionization during solid-substrate ESI-MS analysis is critical for improving its sensitivity and selectivity.

In this study, the surfaces of wooden tips (toothpick) were modified with different functional groups (i.e., C_{18}, SO_3H and NH_2) to investigate the role of substrate surface in WT-ESI-MS analysis using positive and negative ion detection mode, respectively. The two sampling modes, extractive sampling and direct sample loading, were investigated in this study. Moreover, various applications for sensitive and selective detection of analytes in complex sample such as forensic sample, environmental sample, biological sample and juicy solid sample, have been demonstrated in this study.

3.2 Experimental section

3.2.1 Reagent and materials

Dimethyl formamide (DMF), octly β-D-glucopyranside (OG), sodium sinapine, hydroxide (NaOH), sodium pyrosulfite ($Na_2S_2O_5$), lysozyme, hexane, estrodiol, trimethoxyoctadecylsilane, sodium chloride (NaCl), 3-aminopropyltriethoxysilane, and trimethoxy(7-octen-1-yl)silane were purchased from Sigma (St. Louis, MO). Cocaine and cocaine-d3 were purchased from Alfasan (Woerden, Holland). Methanol was purchased from Tedia (Fairfield, OH). Formic acid was purchased from Fluka (Buchs, Germany). Phosphoeptide (AC-Ile-Tyr(PO_3H_2)-Gly-Glu-Phe-NH_2, PP) was purchased from Shanghai Apeptide Co., Ltd. (Shanghai, China). Gramincidin D was purchased from International Laboratory U.S.A. (South San Francisco, CA). Gly-Gly-His-Ala (GGHA) was purchased from Bachem Bioscience Inc (King of Prussia, PA). Wooden tips (BestBuy brand toothpick) were purchased from ParknShop (Hong Kong). Fresh water was supplied from HKPU (Hong Kong); other water used in this study was produced by the Milli-Q water-purification system (Milford, MA).

3.2.2 Surface modification of wooden tips

To prepare the modified wooden tips (e.g., C_{18}-WT, SO_3H-WT and NH_2-WT), 50 wooden toothpicks (tip-end size: 0.2 mm, length: 1.8 cm), whose major component was cellulose that was rich in hydroxyl groups (OH-WT), were dispersed in 100 mL of anhydrous DMF, followed by adding 2.5 mL of reagents (i.e., trimethoxyoctadecylsilane, trimethoxy(7-octen-1-yl) silane, and 3-aminopropyltriethoxysilane) under vigorous stirring (~ 700 rpm). The mixtures were heated to reflux at 120 °C for silanization for 12 hours under nitrogen atmosphere in a heat magnetic stirrier (DF-101SZ, Gongyi Kerui Co. Ltd. Zhengzhou, China). Thus the C_{18}-WT and NH_2-WT were prepared after silanization. For the SO_3H-WT, after the silanization, the surface modified wooden tips of trimethoxy(7-octen-1-yl) silane were soaked in 100 ml of 3 M $Na_2S_2O_5$ solution (pH = 6.5, adjusted by NaOH) for sulfonation for 24 hours under nitrogen atmosphere. All modified wooden tips were washed using DMF (100 ml), methanol (100 ml) and then water (100 ml) three times and then dried in the air before use.

3.2.3 Characterization of modified wooden tips

Scanning electron microscope (SEM) with energy dispersive X-ray spectroscopy (EDS) detector (JSM-6490, JEOL Ltd. Tokyo, Janpan), and X-ray photoelectron spectroscopy (XPS) (ESCA LAB 250 XPS instrument, Thermo Fisher Scientific, San Jose, CA) with monochromated Al Kα radiation (hυ = 1486.6 eV), 45° photoelectron takeoff angle, and a beam size of 500 μm, were used to characterize the modified wooden tips in this study.

3.2.4 Wooden tip sampling and MS analysis

Two sampling methods, extractive sampling and direct loading, were investigated using modified wooden tips and normal wooden tips. The procedures of wooden tip sampling and WT-ESI-MS analysis are shown in Figure 3-1. Briefly, in extractive sampling of liquid samples (Figure 3-1a), the wooden tip was soaked in the liquid (1.0 ml for all liquid sample except 0.5 ml for oral fluid) to extract the analytes for 5 minutes with vortex; in extractive sampling of the juicy solid samples (Figure 3-1a), the wooden tip was inserted in the sample (0.5 cm) to sample the juices for10 seconds. After sampling using these two methods, the wooden tips were washed twice for 10 seconds. Pure water was used as washing solvent in this study unless otherwise noted.

Then, the wooden tips were mounted on a commercial nanoESI device (Figure 3-1c) (Waters, Milford, MA) for performing of WT-ESI-MS (QToF2, Waters, Milford, MA) as previous methods,[91, 150, 153] In the direct ionization mass spectrometry (DI-MS) of potato tissue, the 0.5 cm (from the peel to inner tissue) of potato tissue were cut as a V-shaped tips and were placed on the front of MS inlet with a clip for MS analysis as previous work.[156] In the capillary ESI, the sample was introduced by the capillary for MS analysis. The distance between the tip-end of tissues or wooden tips and MS inlet is typically 0.8 cm. The ionization voltage and cone voltage were set at 3.5 kV and 30 V, respectively. 5.0 µl solvent was added for elution and spray ionization of the analyte of wooden tips. Methanol with 0.1 % formic acid and pure methanol was selected as elution/spray solvent in positive ion detection mode and negative ion detection mode, respectively. In direct loading, 5.0 µl liquid sample was directly added on the wooden tip for WT-ESI-MS analysis (Figure 3-1b).

Selected-reaction monitoring (SRM) mode in the triple quadrupole mass spectrometer (Quattro Ultima, Waters, Milford, MA) was operated to determinate the limit of detection (LOD), limit of quantitation (LOQ) and the quantification of cocaine in oral fluid and the LOD of β-estradiol in fresh water. The collision energy and dwell time were set at 30 eV and 0.2 sec, respectively.

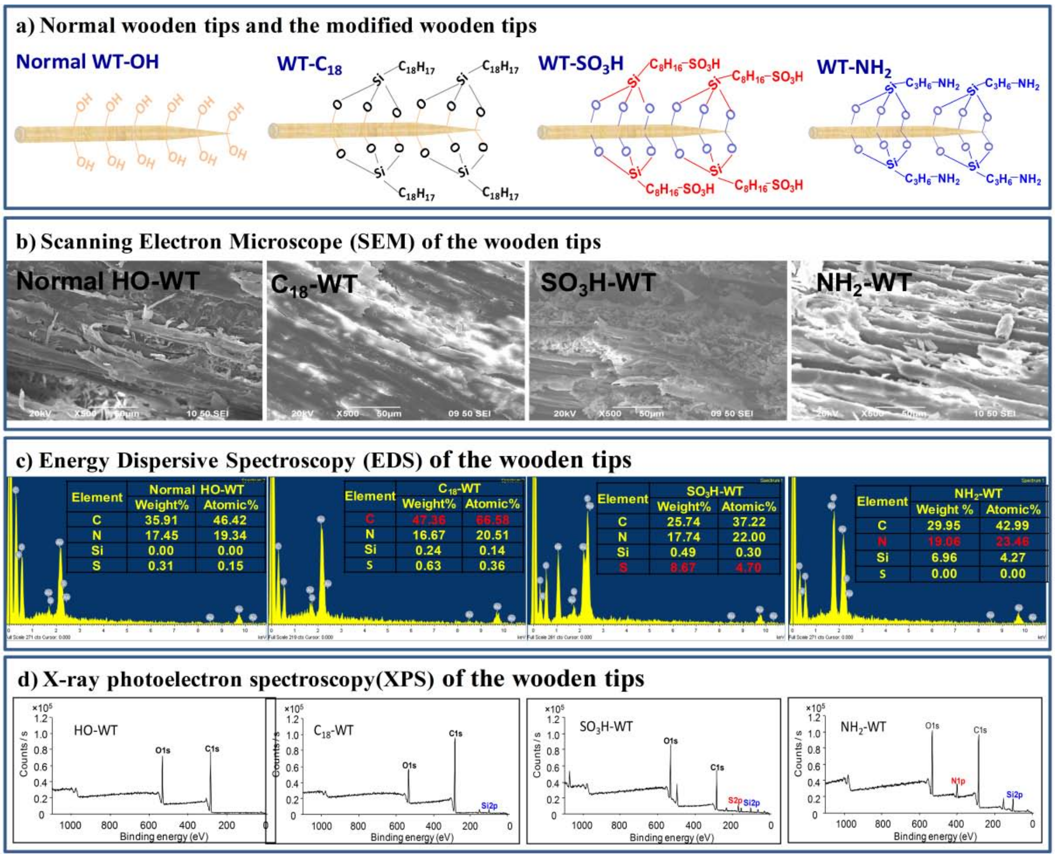

Figure 3-1 Characterization of wooden tips: a) the modification of wooden tips, b) SEM characterization of modified wooden tips; c) EDS characterization of modified wooden tips, insets tables show the elements compositions of surface of wooden tip; d) XPS characterization of modified wooden tips.

3.3 Results and discussion

3.3.1 Characterization of modified wooden tips

To characterize the target modifications of wooden tips (Figure 3-2a), the texture of surface of wooden tips was investigated using SEM. It is found that there was no significant change of surface texture after the modification processes (Figure 3-2b), probably because the modification of wooden tip is monomolecular layer level at porous surface of wooden tips and the wooden tip is inert to the conditions of the modification process of organic synthesis.

In the EDS spectra (Figure 3-2c), it was found that the content of the silicon (from silane-terminated) and the characteristic elements from the target groups (i.e., C, S and N) on the wooden tips significantly increased after modifications, indicating that Si-O is chemically bonded between the OH group of the wooden tip and the target groups since the surfaces had been carefully washed before EDS analysis.

To further confirm the modification of target groups, the XPS analysis was operated. In the normal wooden tip, the peaks of oxygen and carbon were observed only; the sulphur, nitrogen and silicon-related peaks, which are from the target function groups, were clearly observed in the XPS spectra (Figure 3-2d) in modified wooden tips, revealed that the wooden tip were successfully modified with the target groups.

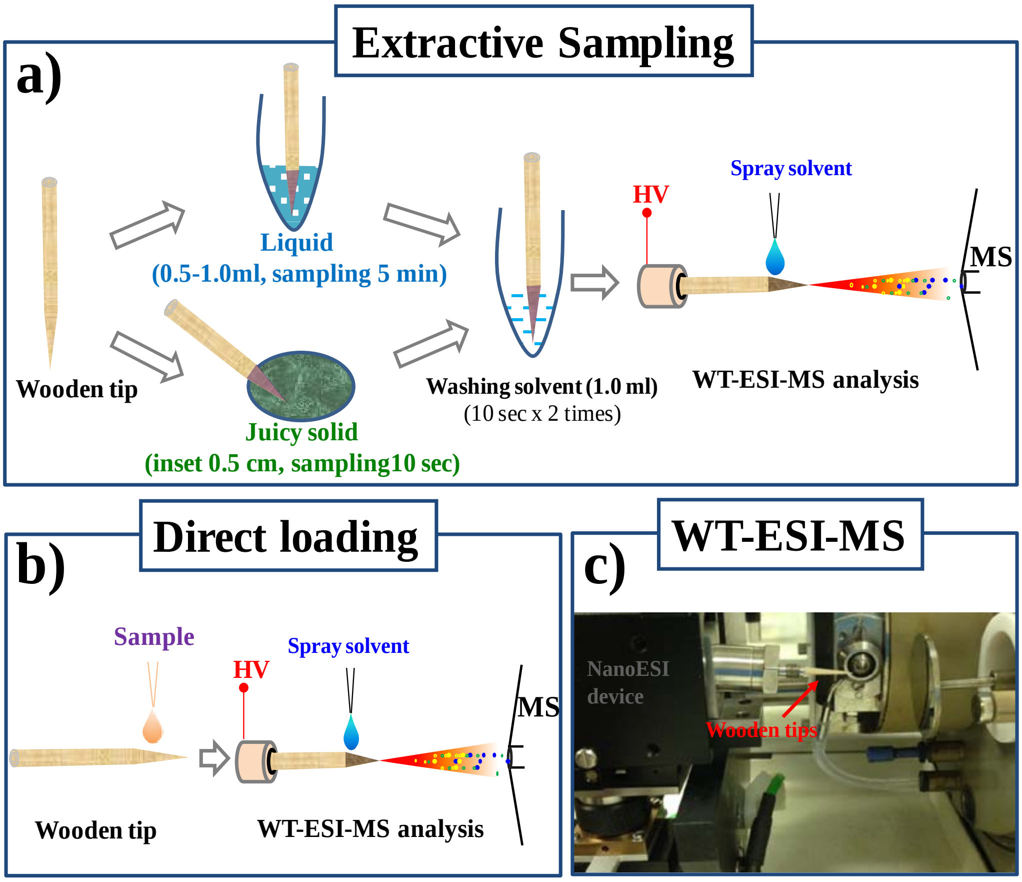

Figure 3-2 Wooden tip samplings and WT-ESI-MS: a) the procedures of extractive sampling of liquid sample and juicy solid sample, b) direct loading of liquid samples, c) a photo of WT-ESI-MS setup.

3.3.2 Analytical properties of modified wooden tips

To understand effects of substrate on sampling and ionization of analysts in WT-ESI-MS, the modified and normal wooden tips, the methods of extractive sampling and direct loading, the positive and negative ion detection mode, were investigated in this study. A mixtures of sinapine (a typical quaternary ammonium salt, MW: 310, 1.0 ppm) and a peptide (GGHA, MW: 340, 30.0 ppm) in methanol (Figure 3-3a) was prepared in positive ion detection mode. In the spectrum of this mixture with conventional capillary ESI-MS, the peak ratio of sinapine-to-GGHA was recorded at 1.0 (Figure 3-3b). The peak of sinapine was cation $[M]^+$ at *m/z* 310 and the peak of GGHA was protonated molecules $[M+H]^+$ at *m/z* 341. Interestingly, the responses of ratio of sinapine-to-GGHA were diversiform when the mixture was directly loaded on the different wooden tips for WT-ESI-MS analysis (Figure 3-3c). The peaks of sinapine were significantly reduced relative to GGHA in the HO-WT and SO_3H-WT, and signal ratio of sinapine is lower in the SO_3H-WT than HO-WT. It is believed that the cation ion (i.e., sinapine) was binding with the electronegative group on the substrate surface (i.e., OH, SO_3H). Obviously, the electronegativity of SO_3H group is stronger than OH and thus stronger interaction with cation ion than OH, resulting in a lower signal of sinapine. The reduction of signal of GGHA was also recorded in C_{18}-WT because the hydrophobic amino acid residue of GGHA could interact with the hydrophobic C_{18} group. In NH_2-WT, there were no significant changes in the sinapine-to-GGHA ratio since there was no special interaction between mixtures and NH_2 group. In extractive sampling, analytes were extracted based on the surface-analyst interaction mechanism of mentioned above, as shown in Figure 3-3d, the signal of sinapine was significantly improved in OH-WT and SO_3H-WT, and the GGHA was significantly improved in C_{18}-WT. Again, there were no significant changes in the sinapine-to-GGHA ratio in NH_2-WT.

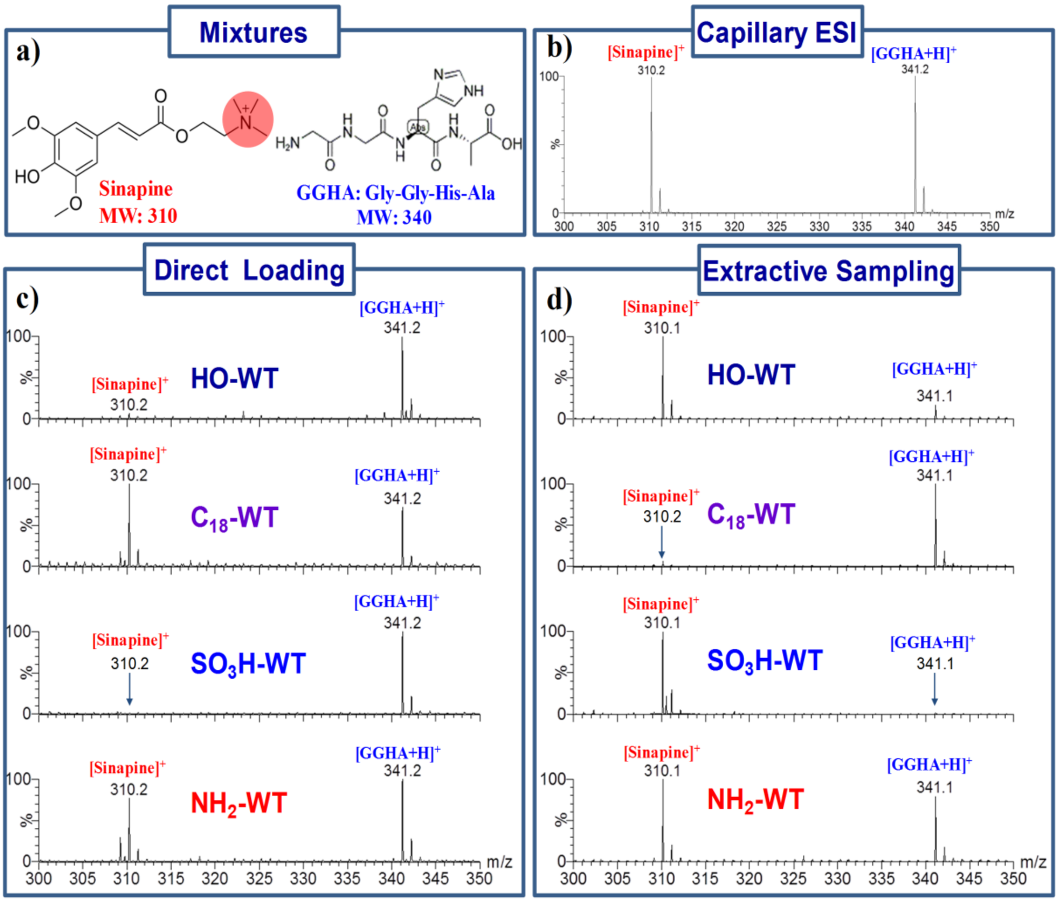

Figure 3-3 Analysis of mixture of sinapine (1.0 µg/ml) and GGHA (30.0 µg/ml) in positive ion detection mode: a) the molecular structures of the mixture, b) conventional capillary ESI-MS detection of the mixture, c) WT-ESI-MS detection of the mixture with direct loading method, d) WT-ESI-MS detection of the mixture with extractive sampling method.

A mixture of phosphoeptide (PP, MW: 748, 14.0 µg/ml) and the peptide (GGHA, MW: 340, 30 µg/ml) in methanol (Figure 3-4a) in the negative ion detection mode was prepared to investigate the mechanism of sampling and ionization of the modified wooden tips. In the spectrum of conventional capillary ESI-MS of the mixtures, the peak ratio of PP(*m/z* 747)-to-GGHA(*m/z*339) was recorded at 1.0; and the ions of the double charged of PP (*m/z* 372), dimmer of GGHA (*m/z* 679) and dephosphorylation of PP (*m/z* 667) were observed (Figure 3-4b). In the direct loading method (Figure 3-4c), it was found that the peak of PP was maintained as the base peak in HO-WT, C_{18}-WT and SO_3H-WT, but almost disappeared in NH_2-WT. It could be explained by the fact that acidic PP had very strong interaction with the basic NH_2 group but only weak interaction with other groups. In contrast, the signal of PP was improved in NH_2-WT extractive sampling due to the surface-analyte interaction. In extractive sampling with C_{18}-WT, the signal of dephosphorylation was improved since it is more hydrophobic after the loss of HPO_3 from PP and thus extracted by hydrophobic C_{18}-WT. Noted that a sodiated PP adduct was observed in SO_3H-WT probable because SO_3H not only adsorb the basic compound but also the free metal ion.

These results revealed that the mechanism of WT sampling is surface reversed-phase adsorption or/and acid-base coordination in surface-analyte interaction. In the direct loading sampling, the surface-analyte interactions could reduce the signal in ionization; while the interaction could be used for improving the detection of target analytes in the extractive sampling. Particularly, C_{18}-WT, SO_3H-WT and NH_2-WT are highly efficient in extracting hydrophobic, alkali-like, and acidic-like analytes, respectively. In this study, various important applications were verified the mechanisms of surface-analyte interactions.

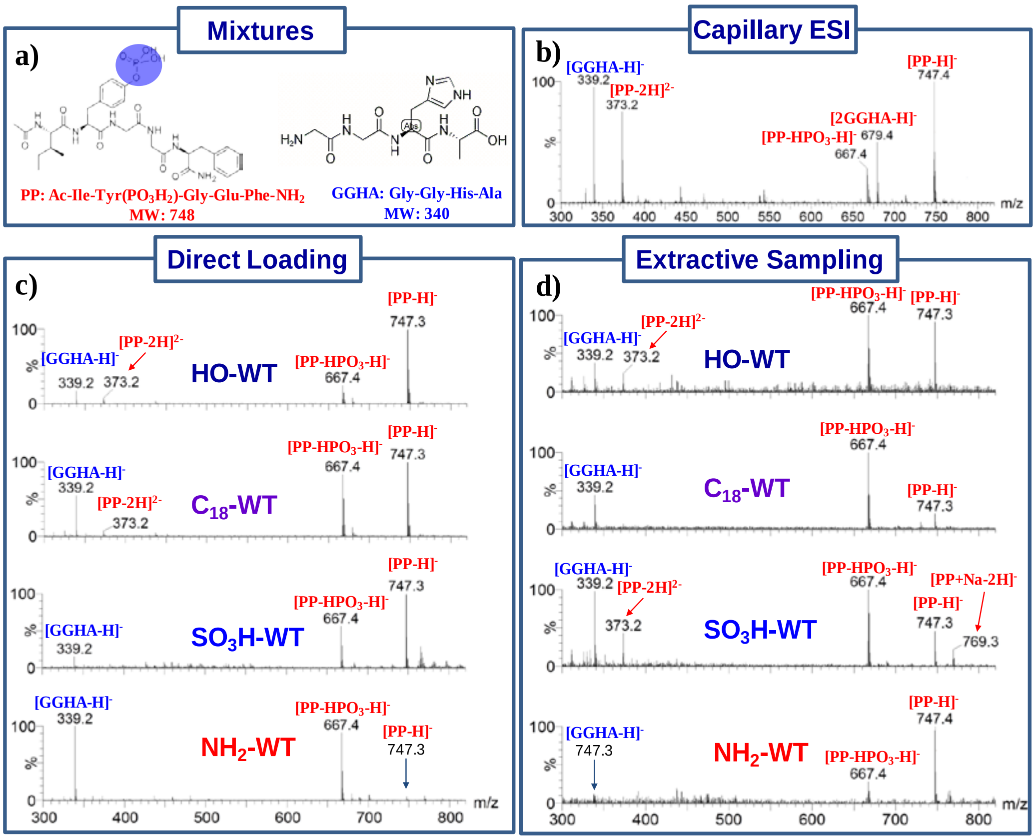

Figure 3-4 analysis of mixture of PP (14.0 μg/ml) and GGHA (30.0 μg/ml) in negative ion detection mode: a) the molecular structures of the mixture, b) conventional capillary ESI-MS detection the mixture, c) WT-ESI-MS detection of the mixture with direct loading method, d) WT-ESI-MS detection of the mixture with the extractive sampling method.

3.3.3 Detection of hydrophobic analytes

Characterization of proteins and peptides using ESI-MS analysis is highly susceptible to interferences from most buffer solutions used with salts and detergents.[157-159] Conventional methods such as liquid chromatography, solid-phase extraction, ultrafiltration, and ion exchange, which are usually time-consuming and laborious, are usually applied to remove any interfering compounds prior to ESI-MS analysis.[158, 160] Rapid, simple and efficient techniques for cleaning biological samples before ESI-MS analysis are needed. Herein, we demonstrated the use of modified wooden tips for extractive sampling of the protein from high concentration of slat solution. The mass spectral results obtained from the lysozyme (40 µM) solution containing a high concentration (100 mM) of NaCl with different wooden tips and capillary ESI (Figure 3-5). As expected, lysozyme was clearly observed in C_{18}-WT-ESI-MS (Figure 3-5c) since the hydrophobic protein was maintained on the hydrophobic C_{18}-WT and the hydrophilic salt was washed away in the washing process.

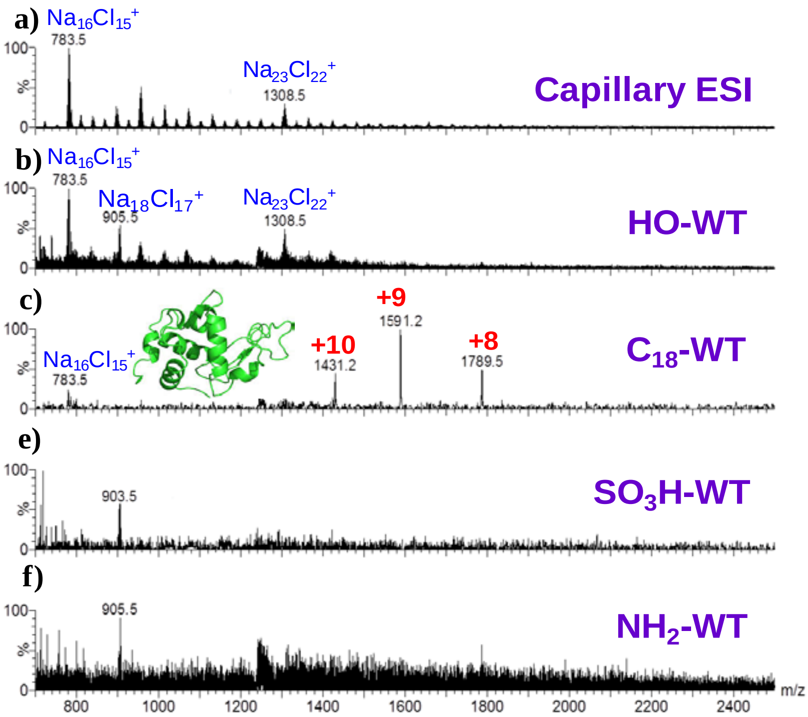

Figure 3-5 Analysis of lysozyme (40 µM) in salt solution (100 mM) using different methods: a) conventional capillary ESI-MS; b) normal WT-ESI-MS; c) C_{18}-WT-ESI-MS; d) NH_2-WT-ESI-MS; e) SO_3H-WT-ESI-MS.

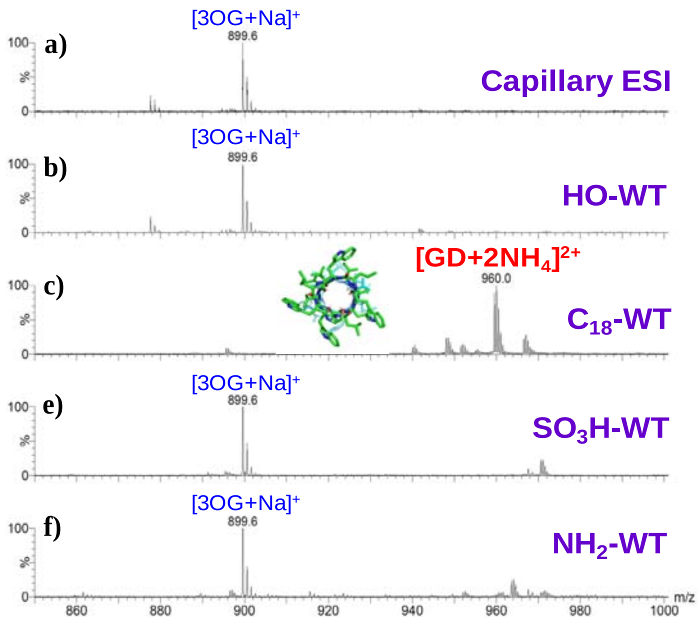

Figure 3-6 Analysis of gramincidin D (5 μM) in detergent solution (2.5 %, w/w) using different methods: a) conventional capillary ESI-MS; b) normal WT-ESI-MS; c) C_{18}-WT-ESI-MS; d) NH_2-WT-ESI-MS; e) SO_3H-WT-ESI-MS.

Moreover, the C_{18}-WT-ESI-MS was also demonstrated for removing detergents from biological samples, which is an important process for the characterization of membrane peptides and proteins.[161, 162] The mass spectra (Figure 3-6) were obtained with different WT-ESI-MS and conventional ESI-MS for detection of gramicidin D (a typical membrane peptide) solution containing 2.5% (w/w) octyl β-D glucopyranside (OG, a typical detergent). It was found that the peak of target hydrophobic peptide was predominated in the mass spectrum in the C_{18}-WT-ESI-MS as expected, while sodiated OG ions and no ion signal of the peptide could be clearly observed in the other modified WT-ESI-MS and conventional ESI-MS. This result was explainable because the OG easily dissolved in water therefore was washed away during the cleanup process, while gramicidin D was insoluble in water but easy to dissolve in methanol and thus was ionized. These results revealed that the unique modification of wooden tips with C_{18} group allows the extraction of hydrophobic analytes and removal of the hydrophilic matrix using water, showing an attractive method for the rapid detection of target compounds in mixtures.

3.3.4 Detection of acidic analytes

Nowadays, the public has more concerned water population of environmental estrogens (EEs) due to the industrial and agricultural activities, which leads to the serious causes of human reproductive failure.[163, 164] However, the determination of estrogens in natural water is difficult analytical task, because the available methods including mass spectrometry, immunoassay, liquid chromatography and gas chromatography suffer from low sensitivity or requiring complicated, time consuming extraction and purification processes.[164] A simple, rapid, and sensitive method for the determination of estrogens is highly considerable.

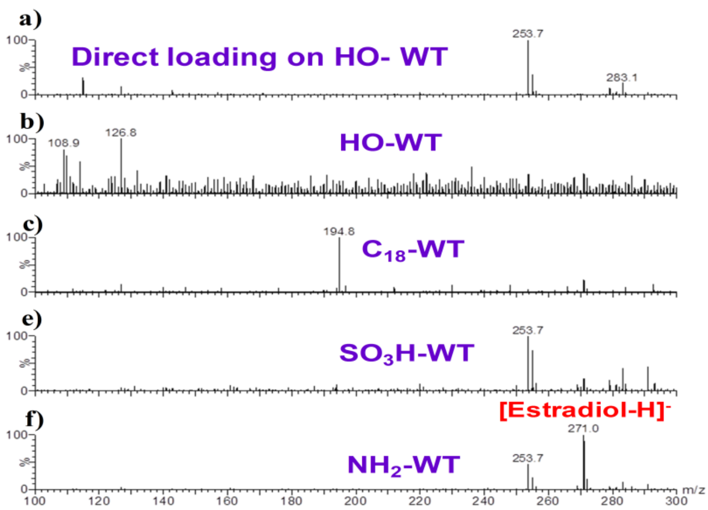

Figure 3-7 Analysis of fresh water spiked with β-estradiol (0.1 µg/mL) in the negative ion detection mode: a) direct loading on normal wooden tip; b). C_{18}-WT extractive sampling; d) NH_2-WT extractive sampling; e) SO_3H-WT extractive sampling.

Detection of estrogen residues in fresh water by modified WT-ESI-MS was investigated. In the negative ion detection mode, mass spectra were collected (Figure 3-7) from fresh water sample spiked with β-estradiol (0.1 µg/mL) using different WT-ESI-MS. As expected, deprotonated β-estradiol at *m/z* 271 was observed in NH_2-WT-ESI-MS, and it was confirmed upon MS/MS experiment (Figure 3-8). This is because the basic NH_2 group easily interacted with the acidulous β-estradiol in the extractive sampling process. The limit of detection (LOD) for the β-estradiol was determined at 0.1 ng/ml (S/N=3) by using NH_2-WT-ESI-MS with the SRM mode at *m/z* 271 > *m/z* 145.

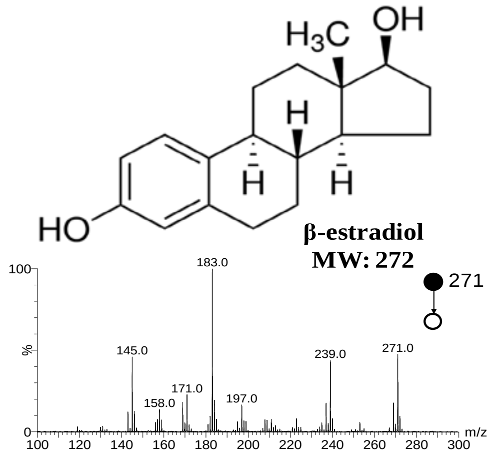

Figure 3-8 MS/MS spectrum of β-estradiol.

3.3.5 Detection of basic analytes

Cocaine, a typical alkaloid drug, continues to be a major drug of abuse worldwide. Recently, there has been a great deal of interest in using oral fluid as a potential biomatrix for drug testing.[153, 165] Oral fluid is much easier to collect than urine or blood, and thus can be used for field sampling.[166-168] Usually, rapid screening of oral fluid is performed with the immunological method.[153, 166]

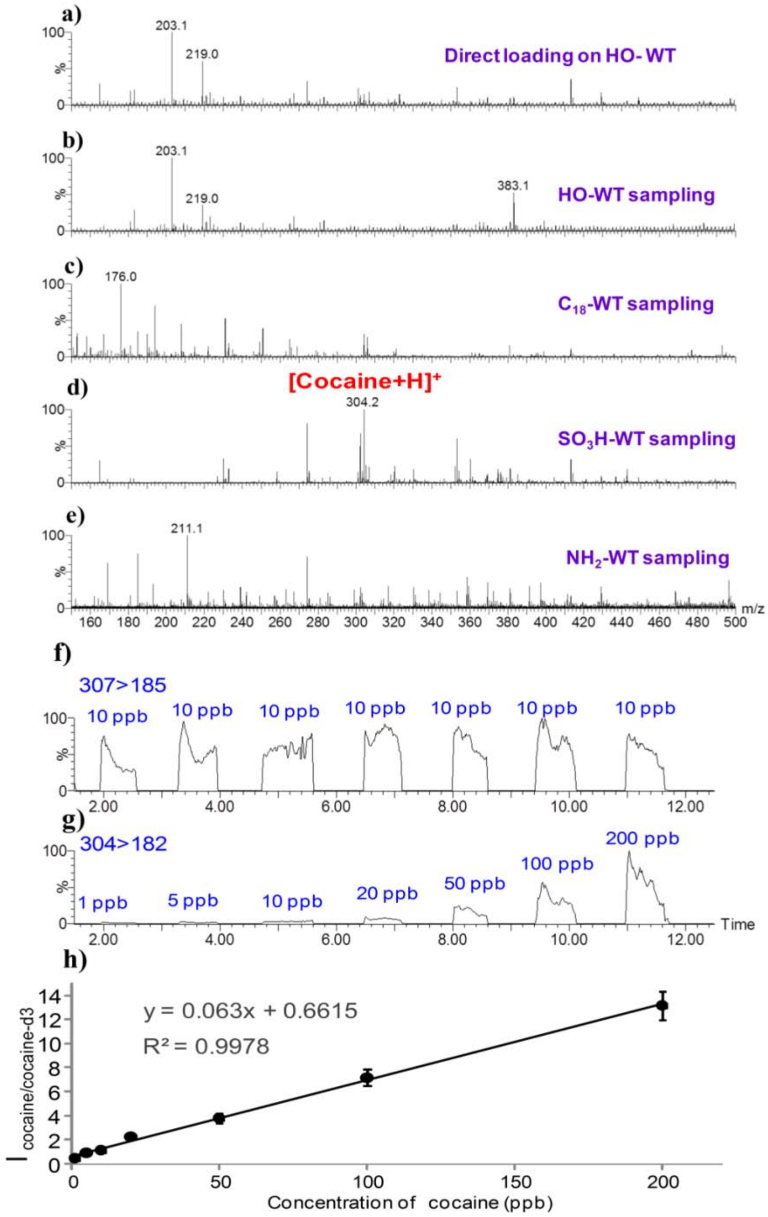

Figure 3-9 Quantitative detection of cocaine (0.1 µg/ml) in oral fluid: a) direct loading on normal wooden, b). C_{18}-WT extractive sampling, d) NH_2-WT extractive sampling, e) SO_3H-WT extractive sampling, f) SRM chromatograms of internal standard (10.0 µg/mL), g) SRM chromatograms of cocaine (1~200.0 µg/mL), calibration curve obtained for cocaine in oral fluid (1, 2, 5, 10, 20, 50, 100, and 200 ng/ml).

Nevertheless, immunological methods cannot be considered a reliable quantitative assay and the positive results are required to further confirm. Such confirmation is usually performed by using analytical techniques such as LC-MS and GC-MS, which usually require time-consuming sample pretreatments such as solid-phase extraction and liquid-liquid extraction, and chromatographic separation due to the high chemical complexity of oral fluid.[38, 39, 166] In this study, the applicability of the modified wooden tips for measurements of cocaine in oral fluid was investigated. As shown in Figure 3-9, the mass spectra displayed the response of cocaine in different WT-ESI-MS for detection of 0.1 µg/ml cocaine in raw oral fluid. As expected, it was found that the responses of cocaine at *m/z* 304 were significantly improved in SO_3H-WT-ESI-MS since it was alkaloid and thus could easily interact with SO_3H group in the extractive sampling of oral fluid. MS/MS experiment was performed to identify the peak at *m/z* 304 (Figure 3-10), which is good with the previous work.[169] In SRM mode, the typical ion pairs, *m/z* 304 > *m/z* 182 (cocaine) and *m/z* 307> *m/z* 185 (cocaine-d3, internal standard), were selected for quantification of cocaine in oral fluid. SRM chromatograms of oral fluid samples in different concentrations of the cocaine (1~200 ng/ml) and a fixed concentration of the internal standard (10 ng/ml) are shown in Figures 3-9f and h.

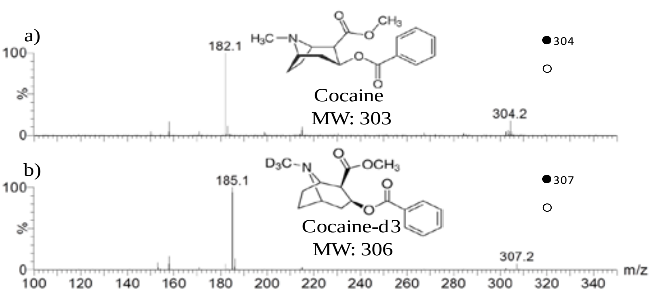

Figure 3-10 a) MS/MS spectrum of cocaine, b) MS/MS spectrum of cocaine-d3.

The calibration curves for cocaine showed excellent linearity (R^2 =0.9978) by the signal height of SRM chromatograms. The acceptable RSD was obtained at 9.4 %

(n=7) at 10 ng/ml. The LOD (S/N=3) and LOQ (S/N=10) of cocaine in oral fluid were determined at 0.01 ng/ml and 0.1 ng/ml (S/N=10), respectively, values that were significantly improved over GC-MS and LC-MS for detection of cocaine in oral fluid.[166, 170]

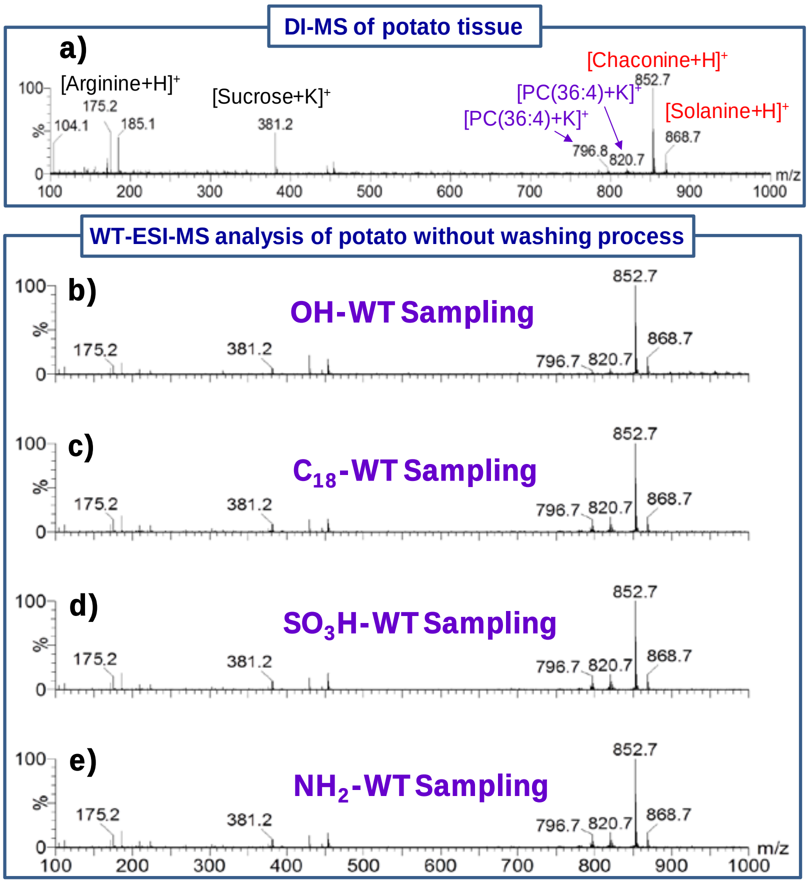

Figure 3-11 Analysis of potato using different methods in positive ion detection mode: a) DI-MS of potato tissue; sampling and ionization of potato with different wooden tips: b) normal WT; c) C_{18}-WT; d) NH_2-WT; e) SO_3H-WT.

3.3.6 Sampling and analysis of juicy solid sample.

Rapid analysis of living organisms at a molecular level is one of the most challenging topics in modern analytical chemistry.[119, 171] Direct ionization (DI) of living organisms for MS analysis has been demonstrated using living organism as

ESI emitter.[119, 156] Juice[172] and secretion[119] of living organisms also could be introduced by capillary for MS analysis. However, there is a great difference in the relative concentration of compounds between juice and secretion; resulting in the ion suppression effect in direct analysis of complex samples. To investigate the target of compounds from living organisms, the multiple-step sample pretreatments are usually required prior to MS analysis.[118, 173] In this study, analysis of fresh potato tube, which is an important part of the diet in many countries and contains many organic acids, fatty acid, lipids, sugars, glycoalkaloid and free amino acids,[174, 175] was demonstrated the selective detection of the target components using modified WT-ESI-MS.

In positive ion detection mode, the ions of arginine, sucrose, phosphatidylcholines (PC) and glycoalkaiods (i.e., solanine and chaconine), were observed in DI mass spectrum of potato tissue (Figure 3-11a) and were identified upon MS/MS experiments (Figure 3-12), in the MS/MS experiments, MS/MS spectrum of potassiated sucrose, the fragment ions at *m/z* 219 and 201 were generated by loss of anhydroglucose and anhydroglucose+H_2O; MS/MS spectrum of protonated arginine, generated ion fragments of *m/z* 158, 140, 130, 116, and 112 by the loss of NH_3, [NH_3+H_2O], [NH_3+CO], HN=C(NH_2)$_2$ and [NH_3+HCOOH], respectively; MS/MS spectrum of potassiated PC(34:2), the fragments of *m/z* 763, 613 by loss of $HPO_4CH_2CH_2N(CH_3)_3$ and $N(CH_3)_3$, respectively, and the *m/z* 163 is correspoanding to potassiated cyclophosphane; MS/MS spectrum of potassiated PC(36:4), the fragments of *m/z* 763, 637 by loss of $HPO_4CH_2CH_2N(CH_3)_3$ and $N(CH_3)_3$, respectively, and the *m/z* 163 is correspoanding to potassiated cyclophosphane; MS/MS spectrum of potassiated glucose, the peaks at *m/z* 201 is produced by loss of H_2O; MS/MS spectrum of protonated ascorbic acid, the fragment ions at *m/z* 141 and 129 was generated by loss of $2H_2O$, $CO+H_2O$; g) MS/MS spectrum of protonated chaconine, the fragment ions at *m/z* 722, 706,560 and by loss of L-rhamnose, D-glucose, [L-rhamnose + D-glucose], [L-rhamnose, 2D-glucose], respectively; MS/MS spectrum of protonated ascorbic acid, the fragment ions at *m/z* 141 and 129 was generated by loss of $2H_2O$, $CO+H_2O$; f) MS/MS spectrum of protonated chaconine, the fragment ions at *m/z* 722, 706, 560 and 398 by loss of L-rhamnose, D-glucose, [L-rhamnose + D-glucose], [L-rhamnose, 2D-glucose], respectively; MS/MS spectrum of protonated solanine, the fragment ions at *m/z* 706, 560 and 398 by loss of D-glucose, 2L-rhamnose + D-glucose], [2L-rhamnose, D-glucose], respectively.

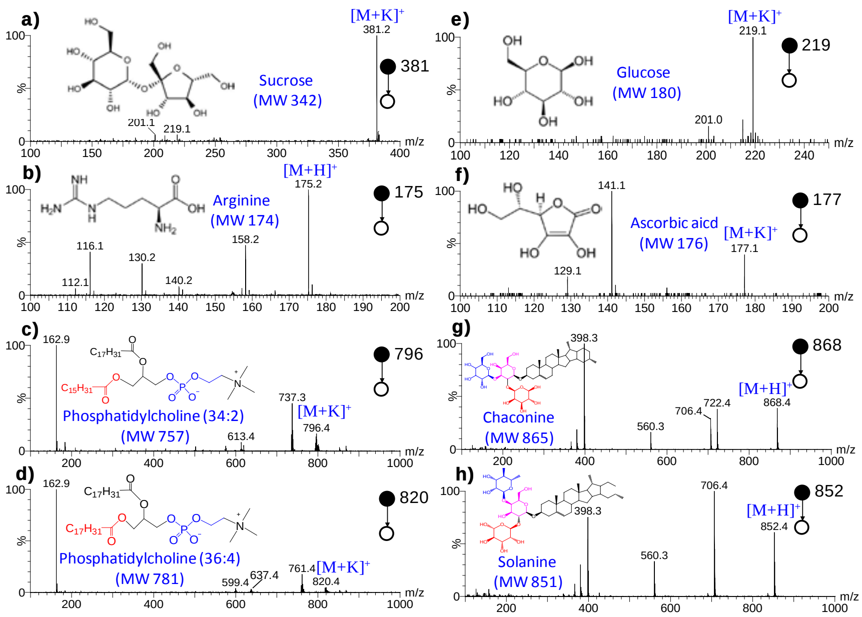

Figure 3-12 MS/MS spectra collected from potato using DI-MS and WT-ESI-MS in positive ion detection mode.

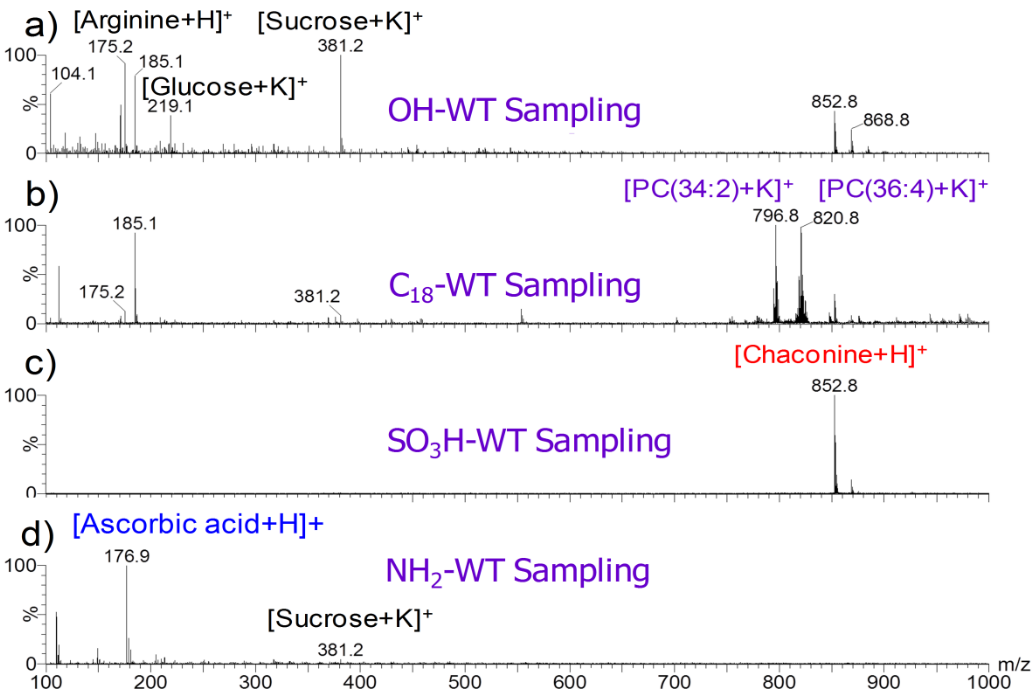

Figure 3-13 Sampling and analysis of potato using different wooden tips in positive ion detection mode: a) normal HO-WT extractive sampling, b) C_{18}-WT extractive sampling, c) NH_2-WT extractive sampling, d) SO_3H-WT extractive sampling.

It is found that there are no significant differences using different among the wooden tips in sampling potato juice, because the juice was easily absorbed in the porous wooden tips. Noted that the chaconine as the base peak were detected using these methods (Figure 3-11 a-e). When the wooden tips was washed with water then with hexane after extractive sampling, no significant changes were observed in the normal HO-WT (Figure 3-13a). However, it is clearly shown that modified wooden tips have significant selections of sampling of targets (Figure 6b-d). Particularly, in the C_{18}-WT-ESI-MS, the hydrophobic PCs were significantly improved to be base peaks (Figure 3-13b); in the SO_3-WT-ESI-MS, the basic chaconine was mainly maintained in the acidic SO_3-WT and other compounds were believed to be washed away (Figure 3-13c); on the contrary, the acidic compound, the ascorbic acid, was maintained in the basic NH_2-WT (Figure 3-13d) and was identified (Figure 3-12f).

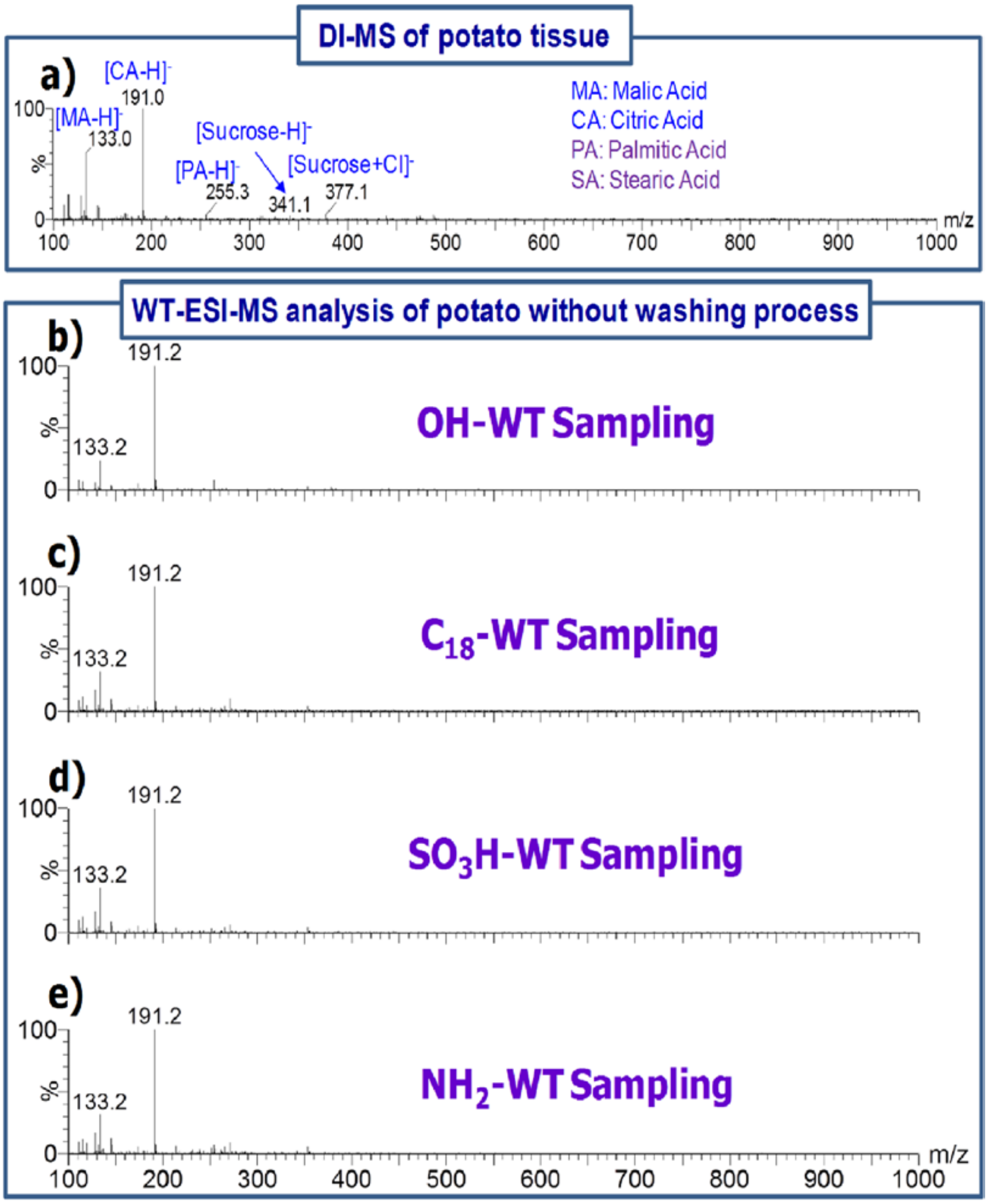

Figure 3-14 Analysis of potato using different methods in negative ion detection mode: a) DI-MS of potato tissue; sampling and ionization of potato with different wooden tips: b) normal WT; c) C_{18}-WT; d) NH_2-WT; e) SO_3H-WT.

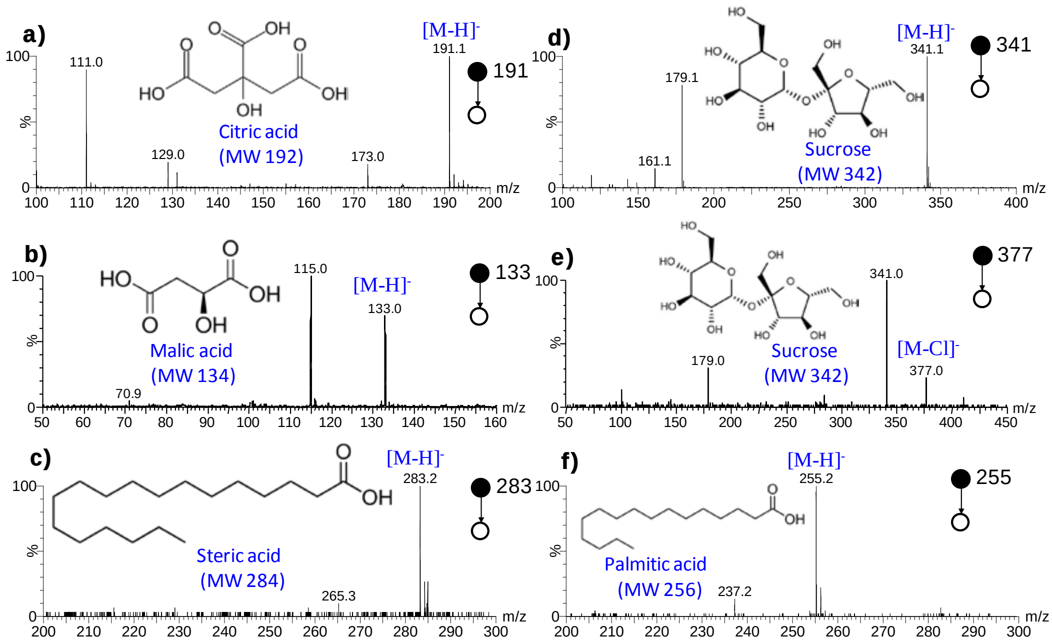

Figure 3-15 MS/MS spectra collected from potato using DI-MS and WT-ESI-MS in negative ion detection mode.

In negative ion detection mode, sucrose and organic acid such as malic acid, citric acid, and fatty acids were observed in the DI-MS spectrum of potato tissue (Figure 3-14a) and were identified upon MS/MS experiments (Figure 3-15). In the MS/MS experiments, MS/MS spectrum of deprotonated citric acid, the fragment ions at *m/z* 173, 129, and 111 were generated by loss of H_2O, CO_2+H_2O. CO_2+2H_2O, respectively; MS/MS spectrum of deprotonated malic acid, generated ion fragments of *m/z* 115 was produced by the loss of H_2O; MS/MS spectrum of deprotonated steric acid, the fragments of *m/z* 265 was generated by loss of CO; MS/MS spectrum of deprotonated sucrose, the fragments at *m/z* 179 and *m/z* 161 were produced by loss of anhydroglucose and anhydroglucose+H_2O, respectively; MS/MS spectrum of chloridized sucrose, the fragments at *m/z* 341 and *m/z* 179 were created *m/z* by loss of Cl and anhydroglucose, respectively; MS/MS spectrum of deprotonated palmitic acid, the fragments of *m/z* 237 was generated by loss of CO. Similarly, there are no significant changes in various wooden tips without washing process (Figure 3-15 b-e). As expected, the compounds were selectively maintained in the modified wooden tips after washing process (Figure 3-16).

For example, the signals of hydrophobic fatty acids were significantly improved in the C_{18}-WT-ESI-MS. These results demonstrated that the modified wooden tips not only could be used for selective sampling of liquid raw sample, but also for selective sampling of juicy solid samples.

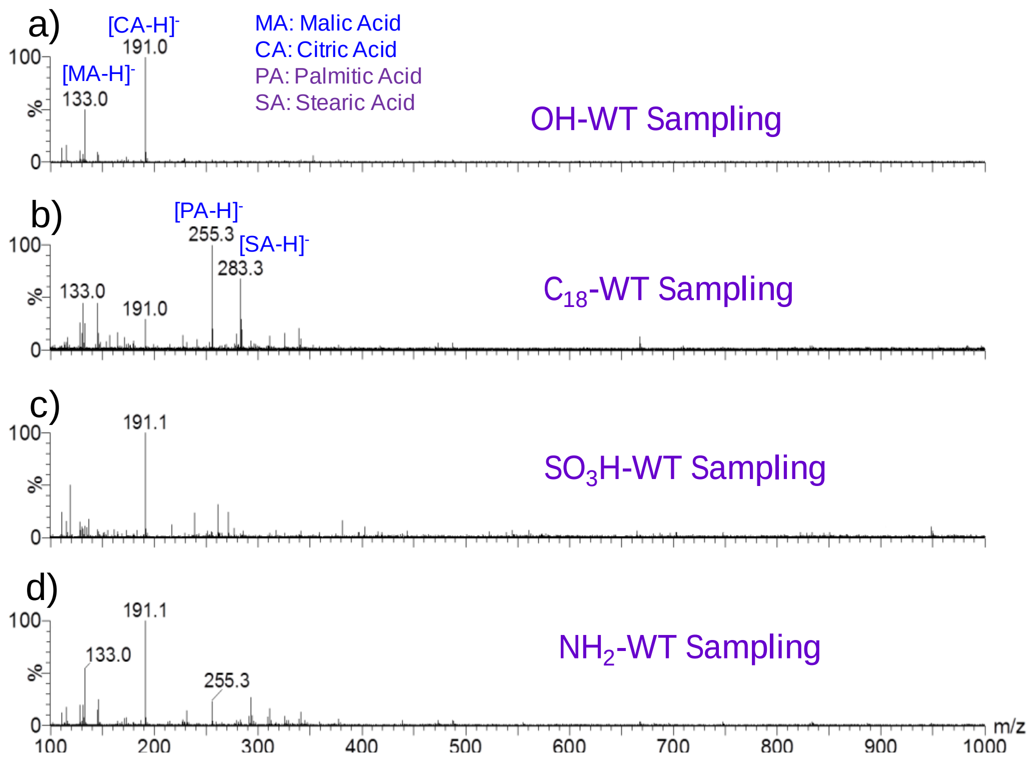

Figure 3-16 Sampling and analysis of potato using different wooden tips in negative ion detection mode: a) normal HO-WT extractive sampling, b). C_{18}-WT extractive sampling, c) NH_2-WT extractive sampling, d) SO_3H-WT extractive sampling.

3.4 Summary

Normal wooden tips (the disposable toothpick) had been modified with different functional groups for selective sampling and detection of analytes in complex samples, including liquid samples and juicy solid samples. The effect of surface on the sampling and ionization for WT-ESI-MS analysis was investigated in positive/negative ion detection modes, these results showed that the interaction of surface and analyte played a crucial role in sampling and ionization. This study revealed that the mechanism of selective sampling/ionization is the interaction of surface-analyte. High sensitivity, high specificity and quantitative analysis have been demonstrated for some important applications, including forensic analysis, environmental analysis, bioanalysis, and analysis of juicy solid sample. The successful modification of wood tip with C_{18}-WT, SO_3H and NH_2-WT not only provided a simple method of modification of wooden tips, but also gives insights into the modifications of other materials and other functional groups in the feature work.

Chapter 4: Mobility of Proteins in Porous Substrates under ESI Conditions

4.1 Introduction

Proteins play critical roles in life activities and are of hot topics in many fundamental and applied researches in chemistry and biology. Characterization of proteins provides information crucial to understanding the structures and functions of proteins. Techniques such as liquid chromatography[176, 177] and gel electrophoresis[178, 179] are commonly used for separation of proteins, and mass spectrometry (MS),[7, 73, 124, 180, 181] nuclear magnetic resonance spectroscopy[182-184] and X-ray crystallograpgy[185, 186] are widely used for investigation of protein structures, conformations and interactions. Among these techniques, MS is superior to others in speed, sensitivity and specificity for characterization of proteins. Combination of MS with hydrogen/deuterium exchange[187, 188] and ion mobility (IM)[189-193] have allowed mapping and measurement of different conformations of proteins. ESI is one of the most powerful ionization techniques for MS characterization of proteins.[124, 194] Under ESI conditions, a protein is typically detected as a series of multiply charged ions that can be correlated to the protein conformations. These multiply charged ions of proteins could be selectively deposited on a surface using the soft-landing method.[195]

Conventional ESI uses capillary for sample loading and ionization. Upon application of a high voltage, sample solution sprays out of the capillary to form fine droplets. Solvent evaporation and repeated droplet fission ultimately lead to production of gas-phase ions of analytes. ESI is a very gentle ionization technique and allows maintenance and investigation of protein conformations and interactions. Non-capillary ESI has also been developed and solid substrates such as metal needles, paper and wood tips have been successfully applied for ESI analysis of various samples.[89, 91, 128] Use of solid substrates for sample loading and ionization much facilitates sample introduction, avoids sample clogging and allows analysis of samples of various forms. Moreover, new ionization features were found with ESI on solid substrates. For example, proteins could be ionized separately from salts or detergents in the solutions with ESI using stainless steel needles,[94, 95] presumably because the small amount of solution loaded on the hard and smooth tip end could be considered as a large droplet and proteins, the more hydrophobic components of the solutions were easier to accumulate on the surface to be ionized.

Herein we report that ESI with porous substrates could allow separation of proteins based on their sizes and shapes. The separated proteins can be monitored using the ESI spectra or by analysis of the protein remained on the substrates. This new observation may imply a new strategy for analysis and studies of proteins.

4.2 Experimental section

4.2.1 Materials

The proteins and organic solvents used in this study were purchased from Sigma (St. Louis, MO, USA).Water was Milli-Q water generated from a Synthesis A10 Milli-Q water generator (Millipore, Billerica, MA, USA). Polyester tips were purchased from Deqing Huihua Pen Co., Ltd. (Deqing, China). Wooden tips (wooden toothpicks) were purchased from ParknShop (Hong Kong).The mixture of insulin and alpha-lactalbumin and the mixture of alpha-lactalbumin and lysozyme were prepared in methanol/water/formic acid (1/1/0.1%, v/v/v), with 4μM for alpha-lactalbumin each in the solutions and 1.5 μM for insulin or lysozyme in the solutions. Different concentrations of proteins in each mixture were used to generate similar intensities of the peaks for comparison. The cytochrome *c* solution was 5×10^{-4} M in acetonitrile/water/acetic acid (25/75/0.01, v/v/v) and myoglobin 20 μM in 12.5 % acetonitrile.

4.2.2 Extraction of proteins remained on the porous tips

After the spray ionization was completed, the tips were cut into two parts at the 1.0 mm position from the sharp end. Each part was soaked with50 μL solvent and votexed for 1 min. The extracts were analyzed using conventional ESI-MS. Methanol/water/formic acid (1/1/0.1% v/v/v) was used as the extraction solvent, except that 20 mM ammonium acetate was used for the myoglobin study.

4.2.3 Mass spectrometry

Mass spectra were acquired on a quadruple time-of-flight mass spectrometer (Q-TOF2, Micromass, Manchester, UK). The high voltage and cone voltage were set at 3.5 kV and 30 V, respectively. The ion source temperature was set at 80ºC, and the scan time was 0.2 sec.

4.2.4 Porous-tip ESI

As shown in Figure 4-1, polyester tips or wooden tips were connected to the high voltage supply of the mass spectrometers via a metal clip. The tips were parallel to the mass spectrometer inlet with the sharp end ~ 8.0 mm away from the inlet. Aliquots of 10 μL sample solutions were loaded onto the tips at 5 mm position from the sharp end. Other settings were similar to those previously described for the wooden-tip ESI.[91, 150]

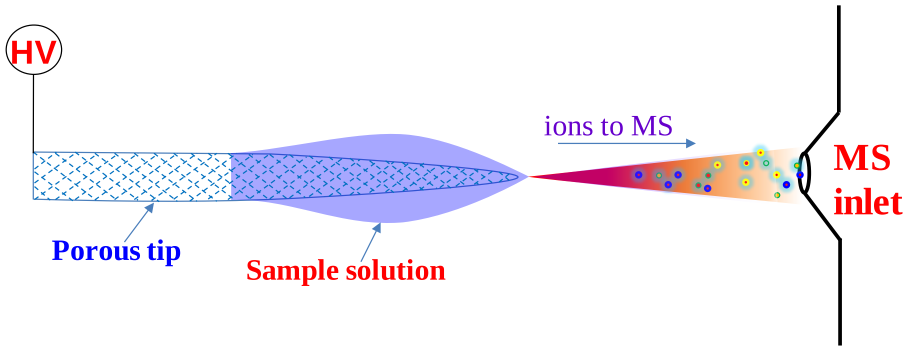

Figure 4-1 Schematic diagram of porous-tip ESI-MS.

4.2.5 Conventional ESI

Sample solutions were introduced to a capillary via transfer line by a syringe pump (KD Scientific Inc., Holliston, MA, USA). The flow rate of the sample was 5 μL/min. Nitrogen was used as the sheath gas, cone gas and desolvation gas with a flow rate of 15 L/h, 75 L/h, 250 L/h, respectively.

4.2.6 Scanning electron microscope

Scanning electron microscope (SEM) (JSM-6490, JEOL Ltd. Tokyo, Janpan) was used to characterize the surface of polyester tip.

4.3 Results and discussion

4.3.1 Separation of proteins with porous-tip ESI-MS.

The settings of solid-substrate ESI-MS is shown in Figure 4-1. When a sample solution is loaded onto the solid-substrate tip, with the application of a high voltage, the solution can be sprayed out and spectra of the sample can be obtained. In this study, an aliquot of 10 μL solution containing two proteins, alpha-lactalbumin (14.2 kDa) and insulin (5.7 kDa), was loaded onto a porous polyester tip at the 0.5 cm position from the sharp end, and was analyzed with the technique. The spraying lasted for about 2.5 min and the obtained selected ion chromatograms (SICs) of the two proteins are shown in Figure 4-2a. Two obviously different stages were observed during the sample analysis. At time 1 (T1), both proteins were abundantly observed in the mass spectrum (Figure 4-2b), which was very similar to the spectrum obtained by conventional ESI (Figure 4-2d). At time 2 (T2), however, insulin, the smaller protein, was almost exclusively detected in the spectrum (Figure 4-2c). Such two-stage ionization was believed to be related to the use of porous substrates for ionization (Figure 4-3). When a solution was loaded onto a porous tip for electrospray ionization, initially the pores on the tip were overwhelmed by the solution, and the movement and ionization of the sample solution was a bulk solution behaviour, in which larger and smaller protein molecules sprayed out together, leading to spectra similar with those of capillary-based ESI-MS. The second stage occurred when the solution was consumed to a low level. At this stage, protein molecules migrated mainly along the microchannels of the porous substrate and smaller molecules had higher mobilities and were easier to be sprayed out to be detected, leading to a spectrum predominated with the smaller protein (Figure 4-2c).Existence of microchannels in the porous tip could be supported by the scanning electron microscope (SEM) photos of the tip (Figure 4-4).

To verify the separation effect of the porous tip to the proteins and detect the remaining proteins on the tip, after the ESI process, the front 1.0 mm of the sharp end was cut out and extracted with solvent. The spectrum obtained by analysis of the extract using conventional ESI-MS analysis showed that only insulin, the smaller protein, was in this most front part (Figure 4-5a). Analysis of the other part of the tip revealed that both alpha-lactalbumin and insulin existed (Figure 4-5b), with alpha-lactalbumin predominated, in this back part, indicating that alpha-lactalbumin, the larger protein, and was more retained in the porous substrate. These results were consistent with those obtained by polyester-tip ESI-MS, and suggested that under the ESI conditions, porous substrates could allow separation of proteins based on their sizes with smaller proteins moving faster to be separately detected.

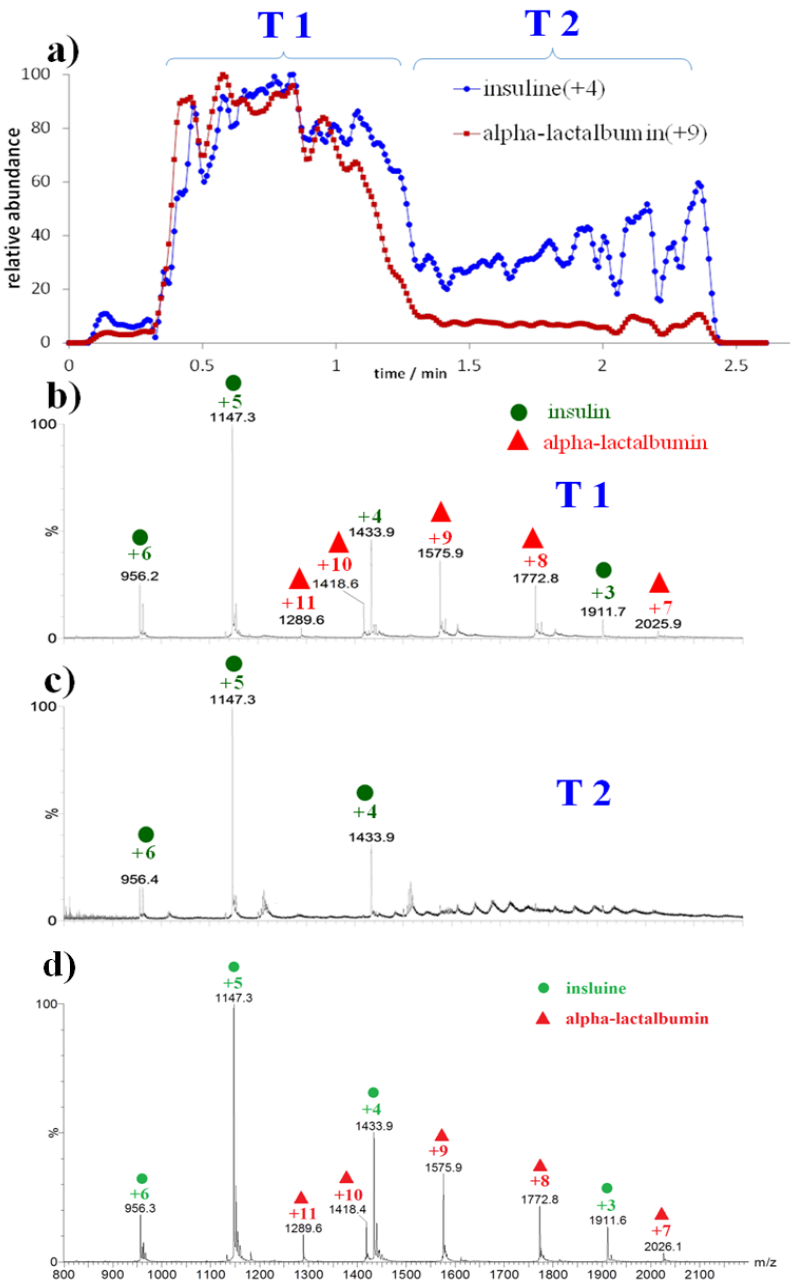

Figure 4-2 Analysisof the mixture of insulin and alpha-lactalbumin (4 μM and 1.5 μM respectively in methanol/water/formic acid (1/1/0.1%, v/v/v)). a) SICs of charge 4+ ion of insulin and charge 9+ ion of alpha-lactalbumin; b) mass spectrum for duration T1; c) mass spectrum for duration T2; d) mass spectrum obtained by using conventional ESI-MS. Different concentrations of insulin and alpha-lactalbumin were used to allow the two compared ions to have similar intensities in conventional ESI-MS.

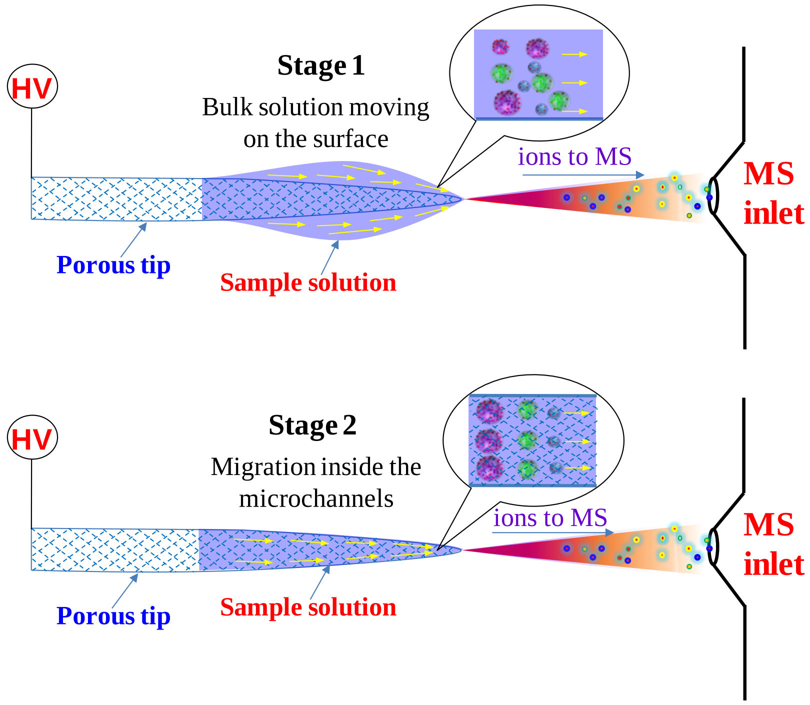

Figure 4-3 Solution movement on porous substrates under ESI conditions. At stage 1, bulk solution movement is mainly involved and larger and smaller molecules are sprayed out together. At stage 2, migration inside the microchannels is the main form of solution movement and smaller molecules are easier to be sprayed out of the porous tips.

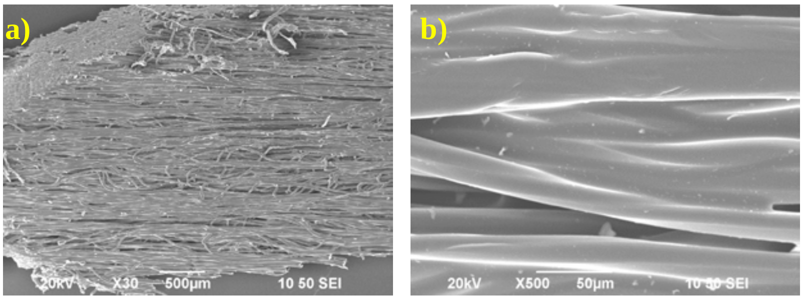

Figure 4-4 SEM images at magnifications of a) x30, b) x500 of polyester surface.

Such separate effect was further confirmed by using polyester-tip ESI-MS for analyzing a mixture containing alpha-lactalbumin (14.2 kDa) and lysozyme (14.3 kDa), which have similar amino acid sequences and molecular weights but different sizes.[196] The SICs of these two proteins at charge +9 were showed in

Figure 4-6a. Again, at time 1 (T1), both of the proteins were observed, and the spectrum obtained (Figure 4-6b) was very similar to that obtained with conventional ESI-MS (Figure 4-6d). At time 2 (T2), lysozyme was almost exclusively detected in the spectrum (Figure 4-6c). Although the mass of lysozyme (14.3 kDa) is slightly larger than that of alpha-lactalbumin (14.2 kDa), measurements with IM-MS showed that lysozyme has a smaller size than alpha-lactalbumin (~1460 Å^2 vs. ~1500 Å^2 for +9 charge state).[197] These results again demonstrated the two-stage ionization in porous-tip ESI-MS and the size-dependent ionization in the second stage. Notably, lysozyme and alpha-lactalbumin, the two proteins with very smaller size difference could be separately detected with this technique.

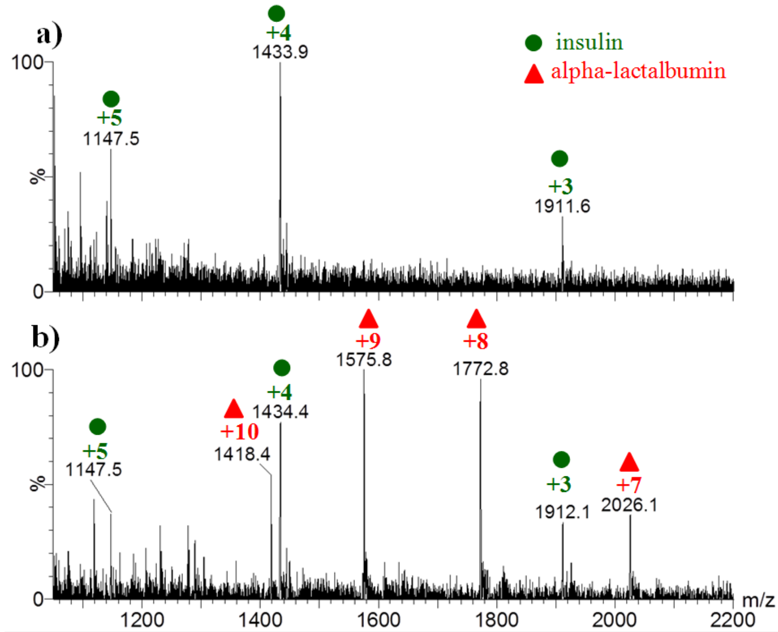

Figure 4-5 Analysis of insulin/alpha-lactalbumin residues on the tip. a) Spectrum obtained by analysis of the proteins remained on the front 1.0 mm part of the polyester tip after the polyester-tip ESI-MS analysis. b) Spectrum obtained by analysis of the proteins remained on the other part of the polyester tip after the polyester-tip ESI-MS analysis.

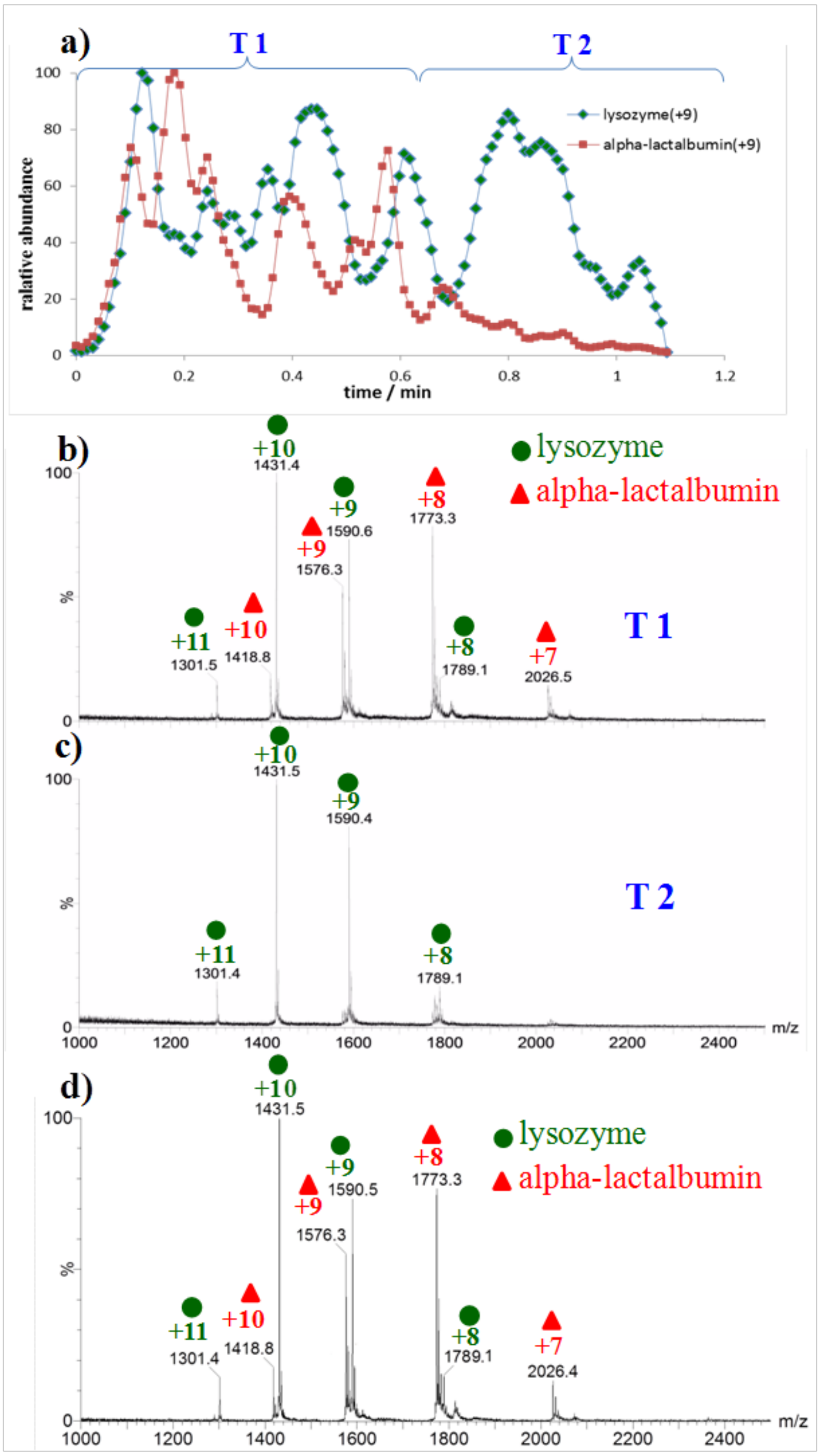

Figure 4-6 Analysis of the mixture of alpha-lactalbumin and lysozyme (1.5 µM and 4 µM respectively in methanol/water/formic acid, 1/1/0.1% v/v/v) using polyester-tip ESI-MS: a) SICs of lysozyme (+9) and alpha-lactalbumin (+9); b) mass spectrum for duration T1; c) mass spectrum for duration T2. d) Spectrum obtained by using conventional ESI-MS.

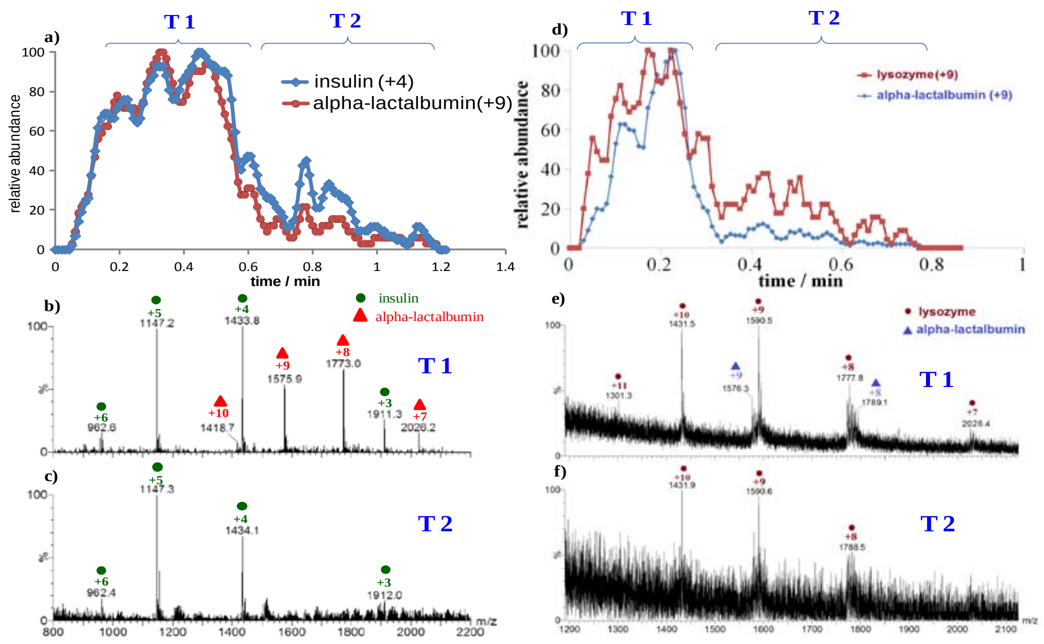

Figure 4-7 Analysis of the protein mixtures using wooden-tip ESI-MS. For analysis of the insulin/ alpha-lactalbumin mixture: a) selected ion chromatograms of insulin (+4) and alpha-lactalbumin (+9); b) mass spectrum obtained for duration T1 in a); and c) mass spectrum obtained for duration T2 in a). For analysis of the lysozyme/alpha-lactalbumin mixture: d) Selected ion chromatograms of lysozyme (+9) and alpha-lactalbumin (+9); e) mass spectrum obtained for duration T1 in d); and f) mass spectrum obtained for duration T2 in d).

To better understand the mechanism for the separation and investigate the effect of substrate material to the separation, wooden tips, which have hydrophilic surfaces that are very different from the hydrophobic surfaces of polyester tips, were used for analysis of the alpha-lactalbumin/insulin and alpha-lactalbumin/lysozyme samples. In general, the signals obtained by using wooden tips for these two protein samples were not as good as those obtained by using polyester tips. However, the spectral results (Figure 4-7) clearly revealed similar two-stage ionization and the same elution and ionization order as for the polyester tips. These results indicated that under the present conditions, few interactions occurred between protein molecules and functional groups on the substrate surface, and the separation was dependent on the mobility of proteins along the microchannels.

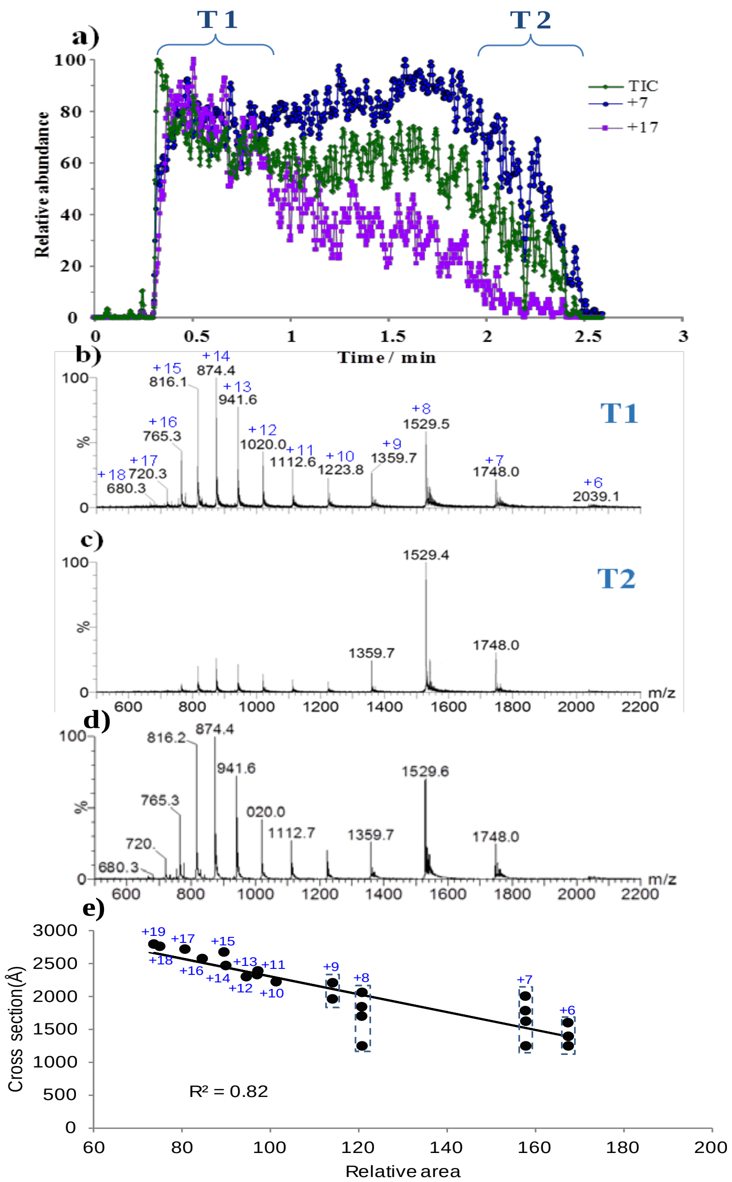

Figure 4-8 Analysis of cytochrome *c*(5×10^{-4} M in acetonitrile/water/acetic acid, 25/75/0.01, v/v/v). a) TIC of and SICsof ions at charges 7+ and 17+. b) Mass spectrum for duration T1. c) Mass spectrum for duration T2. d) mass spectrum obtained by conventional ESI-MS. e) Plot of CCS of ion against its relative integral area for different charged states (+6 ~ +19).

4.3.2 Relationship between the observed mobility of proteins and their collision cross sections

To gain more insight into the effect of protein size and shape on its mobility in the porous substrate, a solution of bovine cytochrome *c* was investigated with polyester-tip ESI-MS. Bovine cytochrome *c* has been well studied with conventional ESI-MS and it was revealed that the larger charge state of the protein observed in the ESI-MS spectrum had larger collision cross section (CCS).[198]The total ion chromatogram (TIC) obtained with polyester-tip ESI-MS analysis of the sample is shown in Figure4-8a, together with the SICs of ions with charges +7 and +17. It can be seen that in the first minute (T1), ions with +7 and +17 charges had very similar intensities and generated a spectrum (Figure 4-8b) very similar to what obtained by conventional ESI-MS (Figure 4-8d), indicating that these two charge states had similar tendency to be eluted and ionized. The spectra obtained indicated that under the present conditions, the protein existed in two conformations in the solution with charge centered at +8 and +14 respectively in the spectra. For the remaining spraying time (T2), however, the signal of +17 ion was significantly reduced, while the signal of +7 ion maintained relatively abundant (Figure 4-8c), leading to lower charge states much predominated in the spectrum (Figure 4-8c) and indicating that protein species with higher charge states, i.e., with larger CCSs, were more retained by the porous substrate. Area of SIC of each protein ion was proportional to the eluted and ionized amount of the species and can be used as an indicator for the mobility or elution tendency of the charged protein in the substrate. The relative integral area (RIA) of SIC of each protein ion was calculated (Figure 4-9) and plotted against the CCS value of protein ion at different charge states.[198] Although these CCS values were measured for the protein ions in the gas phase, they were for the same protein with different charges and are believed to be proportional to the values of the species in the solution phase A good linear correlation (R^2=0.82) was obtained (Figure 4-8e), indicating that the mobility of the protein was size and shape dependent and the present technique could be used to measure the cross sections of protein in solution and to study protein size, shape and conformations.

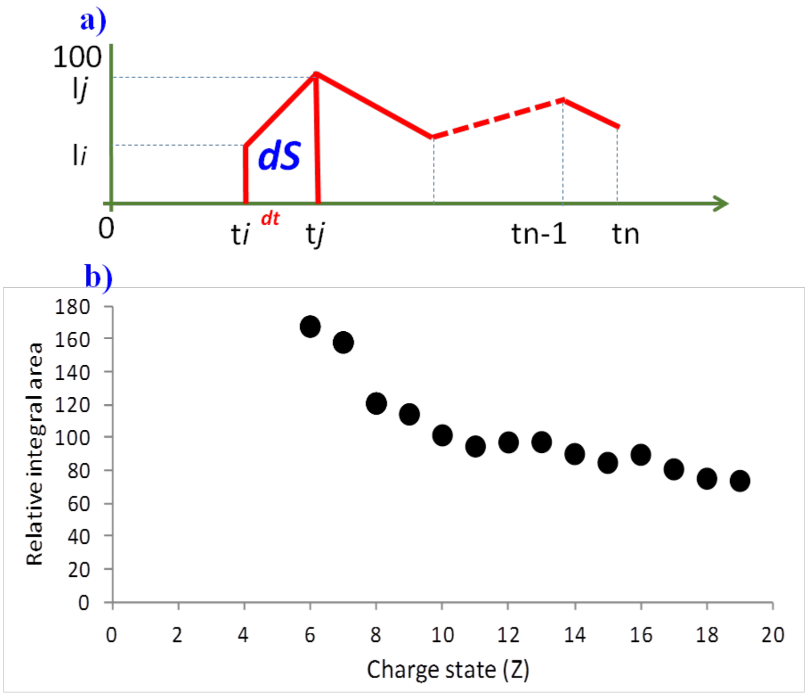

Figure 4-9 a) Calculation of relative integral area (RIA) using definite integration by S = ∫ f(*t*) d*t*, where *dt* is the time between twoneighboringscans (*ti* and *tj*), i.e., the scan time (0.2 sec), f(*t*) is signal of relative abundance between twoneighboringscans (Ii and Ij), and dS is relative integral area at one scan time. Thus, the indefinite integration area (S) can be calculated by S = ∫ f(*t*) d*t* = 1/2∑ [(ti-tj)(Ii+Ij)]. b) Plot of RIA against charged state (+6 ~ +19) of cytochrome *c*. The RIA of each protein ion was obtained by 5 individual measurements and the overall relative standard deviation of the measurements was 13.4 %.

4.3.3 Preparative separation of proteins

Preparative separation of proteins on porous tips can be achieved using this technique. In addition to preparative separation of smaller proteins from larger proteins as demonstrated with polyester tips (see Figure 4-3), separation of different conformations of the same protein can also be achieved. To illustrate this, a myoglobin solution (20 µM in 12.5% acetonitrile) was analyzed. Analysis of this solution by conventional ESI-MS showed that myoglobin was partially denatured in the solution, with both intact holo-myoglobin and denatured apo-myoglobin along with heme were observed in the spectrum (Figure 4-10a). Similar to the study with insulin/ alpha-lactalbumin, an aliquot of 10 µL myoglobin solution was analyzed using polyester-tip ESI-MS. After the electrospray, the polyester tip was cut to two parts at the 1.0 mm position (from the sharp end). Each part was extracted with 50 µL 20 mM ammonium acetate and the extracts were analyzed with conventional ESI-MS. As shown in Figure 4-10b, only a narrow charge state distribution in high mass range was observed in the spectrum obtained for the front 1 mm tip, indicating that only native holo-myoglobin with compact conformation was contained in this part. The spectrum obtained for the remainder part of the tip (Figure 4-10c) was predominated with heme and broad charge state distribution, indicating that this part mainly contained denatured apo-myoglobin with loosen conformations. These results demonstrated that proteins could be separated based on size and shape in the porous substrates and the technique could be used for preparative separation of proteins, e.g., isolation of native proteins from partially denatured proteins.

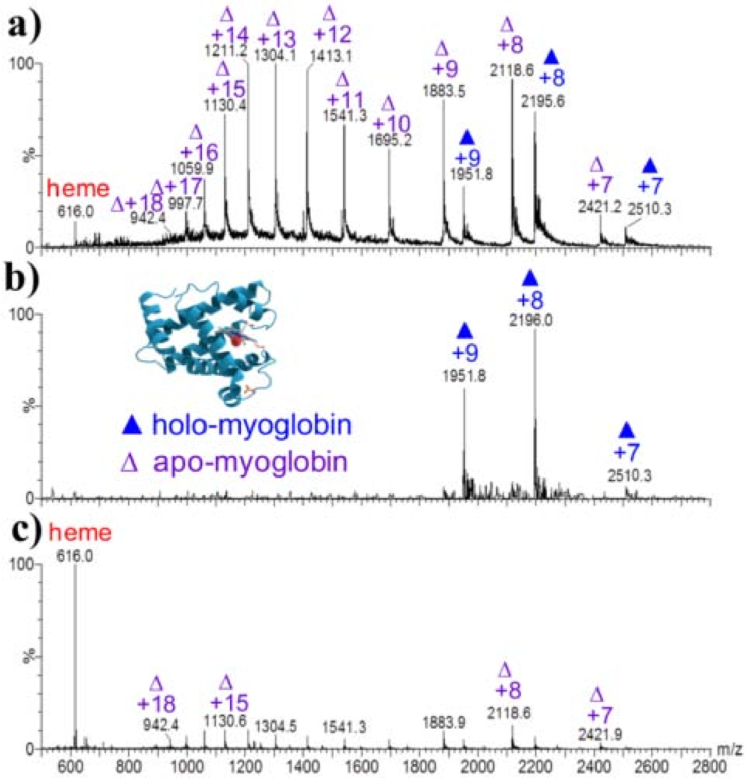

Figure 4-10 Analysis of myoglobin (20 µM in 12.5% acetonitrile). a) Spectrum obtained by conventional ESI-MS. Apo-myoglobin ions are labeled with hollow triangle (Δ) and holo-myoglobin ions are labeled with filled triangle (▲). b) Spectrum obtained by analysis of the proteins remained on the front 1.0 mm part of the polyester tip after the polyester-tip ESI-MS analysis. c) Spectrum obtained by analysis of the proteins remained on the other part of the polyester tip after the polyester-tip ESI-MS analysis.

4.4 Summary

In this study, we demonstrated that porous-tip ESI-MS could allow separation of proteins on porous substrates, followed by ESI-MS detection. The separation was size and shape dependent, suggesting that the technique could be used for separating proteins and for studying protein structures and conformations. Compared to IM-MS that can also separate and detect molecules based on their sizes and shapes, the protein separation in the present technique is caused by the mobility differences of proteins in microchannels of substrates in solution phase, while the separation in IM-MS is caused by the differences in collision with drift gas in the gas phase. Therefore, the information obtained by the present technique is directly about the proteins in the solution phase. Compared to gel electrophoresis and size-exclusion chromatography that are commonly used for protein separation, the present technique is directly linked to ESI-MS, and the high speed, high sensitivity and high specificity of mass spectrometry enable the technique to efficiently separate and detect proteins within a short time. Our results showed that the technique could also be used to measure cross sections of proteins and preparatively separate proteins, indicating the high potential of the technique for protein investigation. Further investigation on effects of various factors and developing the technique for more efficient protein analysis and studies is on-going.

Chapter 5: Solid-substrate ESI with Aluminium Foil

5. 1 Introduction

In conventional ESI and nano-ESI, sample solution is introduced through a capillary, to which a high voltage is applied. The electric field, usually with the assistance of heat and sheathing gas, causes the sample solution to spray out of the capillary and eventually leads to the formation of analyte ions.[26, 199] Due to the typical small bore diameter of the sample introducing capillary, particular precautions should be taken to prevent clogging of the capillary.[87-89, 93, 127, 150, 200, 201] In the past two decades, non-capillary emitters have been used for sampling and ionization in ESI to avoid the clogging problem and allow more convenient sampling.[87-89, 93, 127, 150, 200, 201] Study by Fenn demonstrated the use of a wicking element, e.g., cotton wire, for generation of ESI and the possibility of ionization from paper and thin layer chromatography plates.[202] In later years, various materials, e.g., copper wire and stainless steel needle,[89, 93, 126, 200, 203] optical fibre wired with metal coil [86, 127], polymer microchips [204], surface-modified glass rod,[87] nanostructured tungsten oxide,[88] and fabric materials such as polyester and polyethylene[149] had been successfully applied as substrates for sample loading and ionization of ESI. Recently, the use of paper for ESI, termed as paper spray, was developed and demonstrated to be applicable in analysis of a wide range of compounds and direct analysis of complex samples.[90, 128, 131, 134, 205-209] In recent years, our research group has advocated for development of solid-substrate ESI. An ESI method making use of wooden tips (wooden toothpicks) for loading and ionization of samples was first developed[91]. This method, termed as wooden-tip

ESI herein, was shown to allow on-tip chromatographic separation of samples, separated detection of samples, e.g., proteins and salts, direct detection and quantitation of analytes in complex mixtures, and analysis of slurry samples and powder samples that are less accessible to capillary-based ESI.[91, 146, 150] Our further studies demonstrated that other solid substrates, such as bamboo, fabrics, and sponges, could be used as emitters for ESI and analytes from solid biological tissues, e.g., plant and animal tissues, could be directly ionized by application of electric voltage.[150, 210] These studies indicated that solid-substrate ESI could allow not only convenient sampling but also new features. Unlike capillary-based ESI, ionization on solid-substrate ESI occurs on surface and it has been found that solid-substrate ESI could be more tolerant to salts and detergents,[93, 94] allowed sequential and exhaustive ionization of analytes[95] and direct analysis of raw samples.[91, 128, 146, 150] Compared with other ambient ionization techniques that also allow direct analysis of samples with little or no sample preparation,[6, 156, 211] solid-substrate ESI does not require additional solvent beam, plasma or laser to assist desorption and ionization, and is thus simpler and easier to be widely used.

In this study, we developed a novel ESI method using household aluminum foil (Al foil) as a solid substrate for loading and ionization of samples. Al foil is widely used for many household applications, e.g., packaging of foods, and readily available in low price worldwide. A previous study demonstrated that household Al foil could be applied for laser desorption/ionization (LDI) of biomolecules without the use of matrix, providing a simple and cost-effective mean for LDI mass spectrometric analysis.[212] As a substrate for ESI, Al foil has several remarkable characteristics compared with cellulose-based solid substrates, i.e., paper and wooden tip. First, Al foil is intrinsically of excellent conductivity,[213] and thus can be readily used as emitter for ESI without the need to use solvents to improve conductivity. Second, Al foil is flexible and can be easily folded into desired configuration for holding samples for effective and durable ionization. Third, Al foil has a relatively hydrophobic and impermeable surface, which provides a desired medium for sample pretreatment, e.g., extraction or desalting, prior to ESI-MS analysis. Furthermore, Al foil is an excellent heat conductor and is highly heat-tolerant, [213] thus enables on-target heating of samples or monitoring of thermal reactions. In the present study, the properties of Al foil as an ESI emitter were investigated and different applications of this novel method were demonstrated.

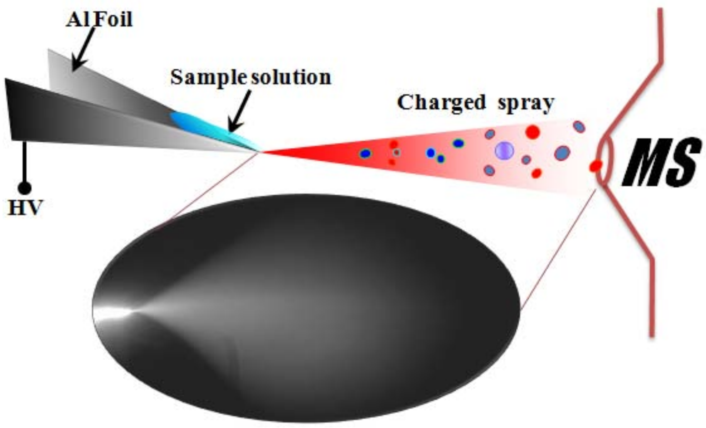

Figure 5-1 Schematic diagram for the setup of Al-foil ESI

5.2 Experimental section

5.2.1 Materials

Aluminium foil used in this study was manufactured by Reynolds Wrap (Vernon, IL) and purchased from PARKnSHOP supermarket in Hong Kong. Reserpine, βestardiol, vanadium (III) acetylacetonate, aspartic acid, ammonium acetate,formic acid, melamine, sodium chloride (NaCl), octyl β-D glucopyranside (OG), horse heart myoglobin, lysozyme, and cytrochorme *c* were purchased form Sigma. Gramincidin D (from bacillus subtilis) was purchased from International Laboratory U.S.A. (San Francisco, CA). Gly-Val-Phe was purchased from Bachem Bionscience Inc (PA, USA). Energy beverage (Red bull), coffee powder (Maxwell House), porcine muscle, skin care cream (Nivea), and wooden tips (BESTbuy brand toothpicks) were purchased from PARKnSHOP supermarket in Hong Kong. Herbal medicines, American ginseng and Fuzi (*Radix aconiti carmichaeli*), were purchased from authentic local traditional Chinese medicine pharmacies. Terbinafine hydrochloride cream was purchased from Dihon pharmaceutical Co. Ltd. (Kunming, China). Papers used for paper spray experiments were grade 1 filter papers purchased from GE Healthcare-Whatman (Uppsala, Sweden), the same as the model used in the previous studies.[128]

5.2.2 Sample preparation for ESI with aluminium foil

The Al foil used in this study was a typical model for household use, having a "shiny" side, which is coated with a layer of organic lubricant during manufacturing, and a "matte" side, which has no coating and is oxidized in air to generate an oxide layer.[212] The thickness of the Al foil was 13 μm, as measured by a micrometer calliper. The "shiny" side was applied for loading of samples, unless specified elsewhere. For sample loading, the Al foil was cut into a symmetrical triangle with typical dimensions of 20 mm width of base and 30 mm height, and folded symmetrically to form a mini-reservoir for sample solution (Figure 5-1).

Rolling up of the Al foil allows it to contain sample solution in bulk for longer duration of ion signals, and increases its rigidity to avoid bending of the substrate upon application of samples, particularly viscous and solid samples that are of considerable weight.

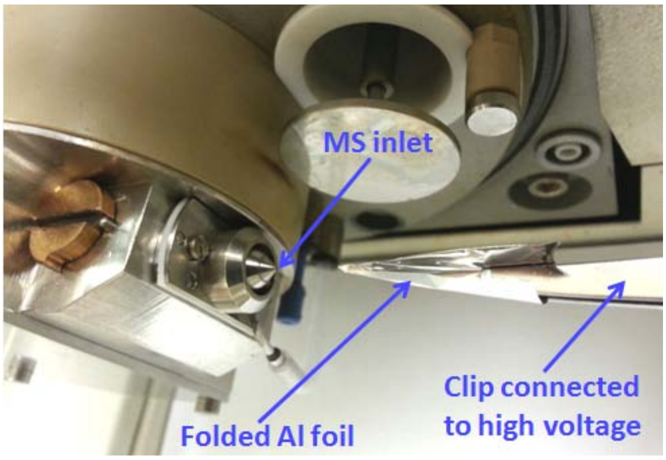

Figure 5-2 A photo for the instrumental setup of Al-foil ESI

The instrumental setup of Al-foil ESI is shown in Figure 5-1 and Figure 5-2. The Al foil cut and folded into desired configuration was held and positioned parallel to the inlet of the mass spectrometer with a clip, which is connected to the high voltage supply of the mass spectrometer (Micromass, Q-TOF2). The tip of the Al foil was positioned ~ 10 mm from the mass spectrometer inlet, from which optimal ion signals could be obtained. The Al foil tip should not be positioned too close to the mass spectrometer inlet to prevent electrical discharge and loss of sample solution in bulk due to the potential difference between the two regions. Unless specified, 10 μL of solution or viscous sample was loaded onto the Al foil tip by pipetting for analysis. For analysis of solid samples, desired amount, typically 1-5 mg, of solid sample was loaded onto the Al foil tip, and subsequently 10 μL of solvent was added for analysis. During data acquisition, the capillary voltage was adjusted in the range of 2.5 to 3.5 kV until optimal ion signals could be acquired. The cone voltage was set at 30 V. The ion source temperature was set at 80 ^{o}C and all desolvation gases were turned off. Mass spectra were obtained by averaging the spectral data obtained in a time window of at least 20 s (one mass spectrum acquired in one second), unless specified.

5.3.3 On-target clean-up of protein and peptide samples

A raw protein/peptide sample with contaminants, e.g., salts or detergents (i.e., octyl β-D glucopyranside), was first deposited onto a piece of Al foil and allowed to dry. Subsequently, an aliquot, typically 2 μL, of cold distilled water is applied onto the sample spot. The water droplet was allowed to stand for ~30 s, during which the water droplet was pipetted in and out for several times to enhance the efficiency of sample clean-up. Finally, the water droplet was pipetted off and 10 μL of spray and ionization solvent (water/methanol/formic acid, 50/50/0.1%, v/v/v) was added for analysis. Both sides of the Al foil were tested for the efficiency of sample clean-up in this study.

5.2.3 Direct monitoring of thermal reactions

For direct monitoring of thermal reactions, an electric heater was placed under the Al foil and a thermometer was located on the Al foil surface for monitoring the temperature. The heating temperature was controlled by the distance between the electric heater and the Al foil. Further refinement are required to achieve fine control of the heating temperature, yet the present setup allowed a demonstration of the applicability of Al-foil ESI in direct monitoring of thermal reactions.

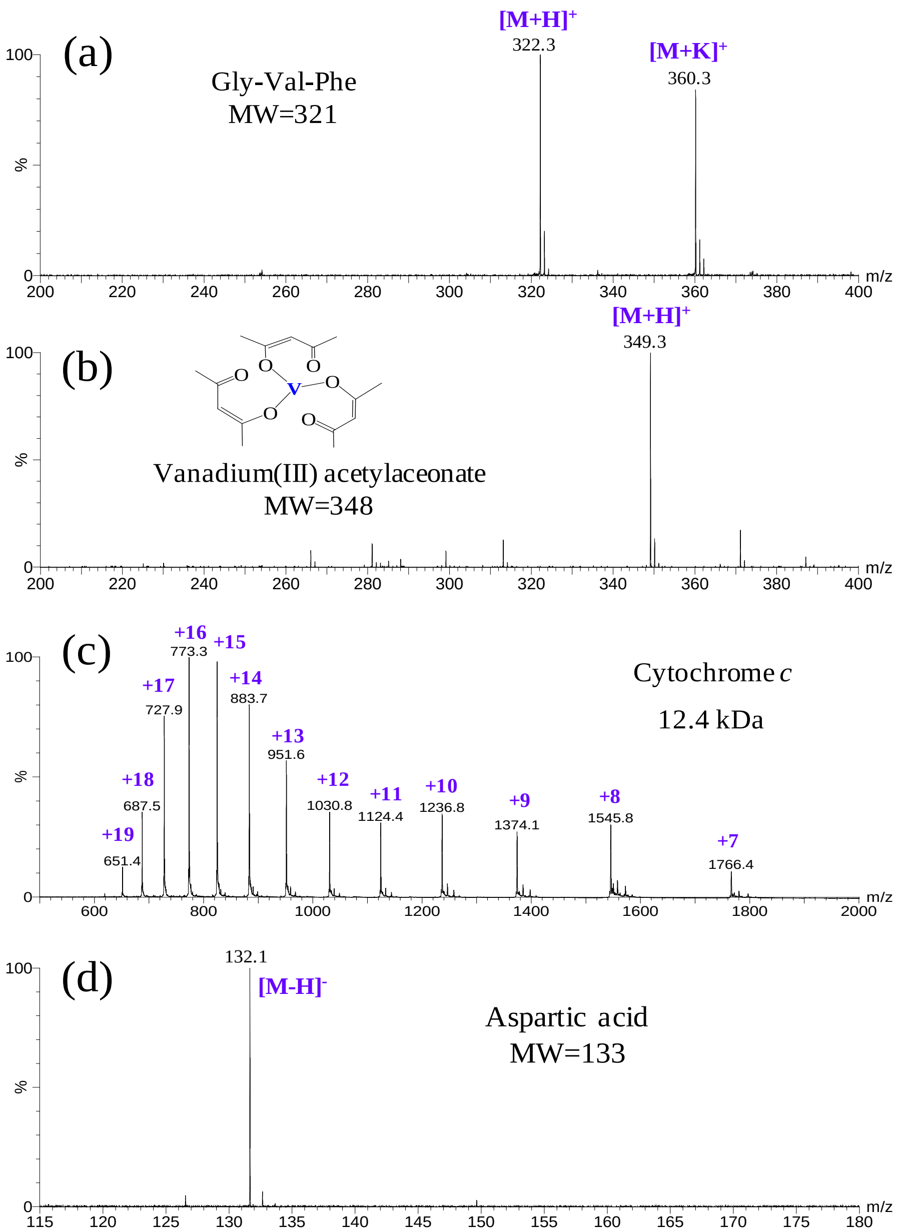

Figure 5-3 Mass spectra obtained for a) 1 ppm Gly-Val-Phe in water/methanol/formic acid (50/50/0.1%, v/v/v); b) 1 ppm vanduium(III) acetylacetonate in water/methanol (50/50, v/v); c) 10 μM cytochrome *c* in water/methanol/formic acid (50/50/0.1%, v/v/v); and d) 1 ppm aspartic acid in water/methanol (50/50, v/v)using Al-foil ESI with sample solutions loaded onto the "shiny" side of the Al foil. Spectra were obtained in the positive ion mode except for aspartic acid in negative ion mode. Very similar spectra with similar intensities were obtained with the "matte" side of the Al foil.

5.3 Results and discussion

5.3.1 Analytical properties of ESI with aluminium foil

Common household Al foils have two sides, the "shiny" side and "matte" side. To investigate the applicability of Al foil as an emitter for ESI and the effect of the two sides of Al foil on the ion signals, solutions of various compounds were loaded to the both sides of the Al foil and analyzed in both positive and negative ion modes. Upon application of high voltage, electrospray could be generated from the tip of either side of the Al foil (Figure 5-1). In general, quality mass spectra with distinct peaks of interest could be obtained (see Figure 5-3 for some of the spectra), and similar mass spectra were obtained from both sides, indicating that samples can be readily ionized from Al foil in both positive and negative ion modes and the different surface properties of two sides of Al foil exert no significant effect on the ion signals.

The mass spectra acquired for 5 µL of 0.1 ppm reserpine in methanol with 0.1% formic acid using Al-foil ESI, paper spray and wooden-tip ESI are shown in Figure 3. In terms of the ion intensity of reserpine, the signals obtained by Al-foil ESI were higher than paper spray and wooden-tip ESI by 2~3 folds (Figure 5-4). However, the noise level of the mass spectrum obtained by Al-foil ESI was significantly lower than those of paper spray and wooden-tip ESI, resulting in a signal-to-noise ratio ~ 20 folds higher than those of the other two techniques (Figure 5-4). The smooth and hydrophobic surface of Al foil is clean and thus allows enhanced signal-to-noise ratio in ESI analysis. Our results also showed that the variation in signal intensity obtained from five individual pieces of Al foils was ~ 40% for the reserpine sample, mostly likely due to the differences in morphology of different Al foil tips and alignment of the tip positions. Internal standard could be used to compensate the variation in signal intensity [146] if quantitative analysis is persuade.

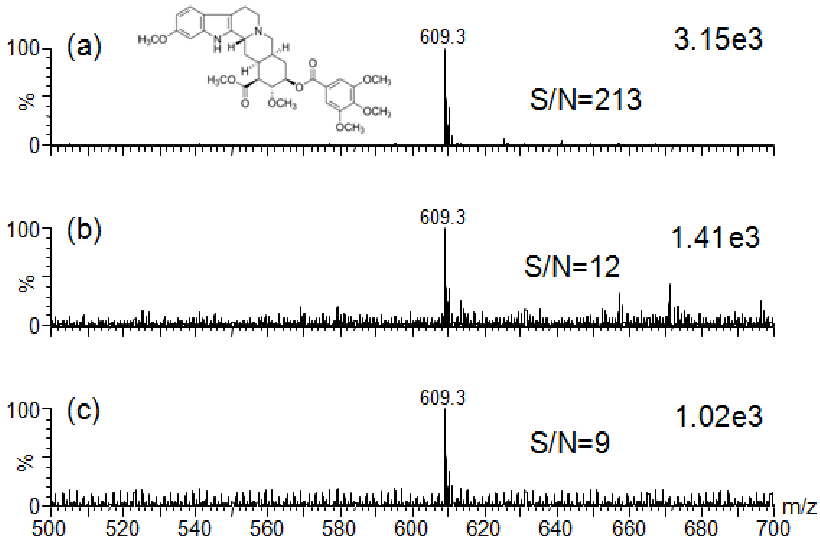

Figure 5-4 Mass spectra acquired for 5 μL of 0.1 ppm reserpine in methanol with 0.1% formic acid using a) Al-foil ESI, b) paper spray, and c) wooden-tip ESI.

The duration of ion signals obtained by Al-foil ESI was investigated using reserpine. Upon application of 10 μL of reserpine in methanol with 0.1% formic acid, signal duration of ~ 2 min could be obtained with Al-foil ESI (Figure 5-5a). In general, the signal duration exhibited a positive correlation with the volume of sample applied and the results obtained for both sides of Al foil were similar (Figure 5-5b). Al-foil ESI showed longer signal duration than paper spray for the same volume of sample solutions, presumably because the folded Al foil effectively held the sample solution to reduce exposure and evaporation. The signal duration of Al-foil ESI was generally shorter than that of wooden-tip ESI when relatively small volume, i.e., < 25 μL, of sample solution was applied

(Figure 5-5b).

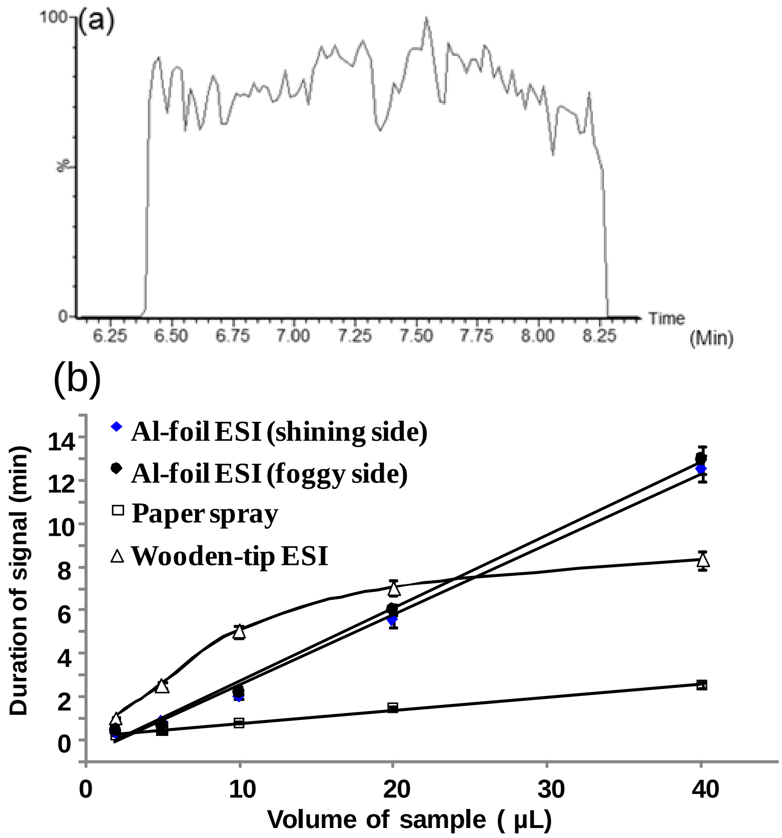

Figure 5-5 a) Ion chromatogram obtained for 10 μL of 0.1 ppm reserpine solution using Al-foil ESI; b) Plot of duration of ion signals against volume of sample solution for Al-foil ESI, paper spray, and wooden-tip ESI.

However, the Al-foil ESI was able to generate more durable ion signals than wooden-tip ESI for larger sample volume, i.e., > 25 μL, probably because the sample solution on the wooden-tip tended to dry out after prolonged exposure to an open atmosphere. In addition, Al foil is more readily to accommodate a large volume of sample solution than wooden tip, for which extra care was required to prevent gravity-driven falling of the liquid droplet when applying a large volume of sample solution.

5.3.2 Direct analysis of raw samples and viscous samples

The mass spectrum obtained for direct analysis of a well-known energy drink, Red Bull, by Al-foil ESI is shown in Figure 5-6a. Protonated molecules of various components, such as nicotinamide (*m/z* 123), vitamin B6 (*m/z* 170), caffeine (*m/z* 195), glucose (*m/z* 203), and sucrose (*m/z* 365), were clearly observed. The present technique was also applied in direct analysis of urine spiked with 1.0 ppm of melamine, which is the major source of fatal kidney stones in the melamine incident in 2008.[214] As shown in Figure 5-7, protonated molecules of melamine could be clearly detected, indicating the applicability of this technique in diagnosis of melamine-mediated kidney failure.

Direct analysis of viscous samples is a challenging task in mass spectrometry and is less readily accessible in conventional ESI, because of the difficulty in generation of electrospray from viscous samples through the small bore of capillary. Extractive electrospray ionization (EESI) approach was developed for direct analysis of complex viscous samples, and different samples, such as olive oil and honey, were successfully analyzed.[215] However, a relatively large amount of samples is required for this EESI approach as samples had to be placed in a large reservoir for microjetting sampling.[215] An ESI technique with a tiny stainless steel tip as an emitter was also applied to direct analysis of viscous samples in our recent study.[150]

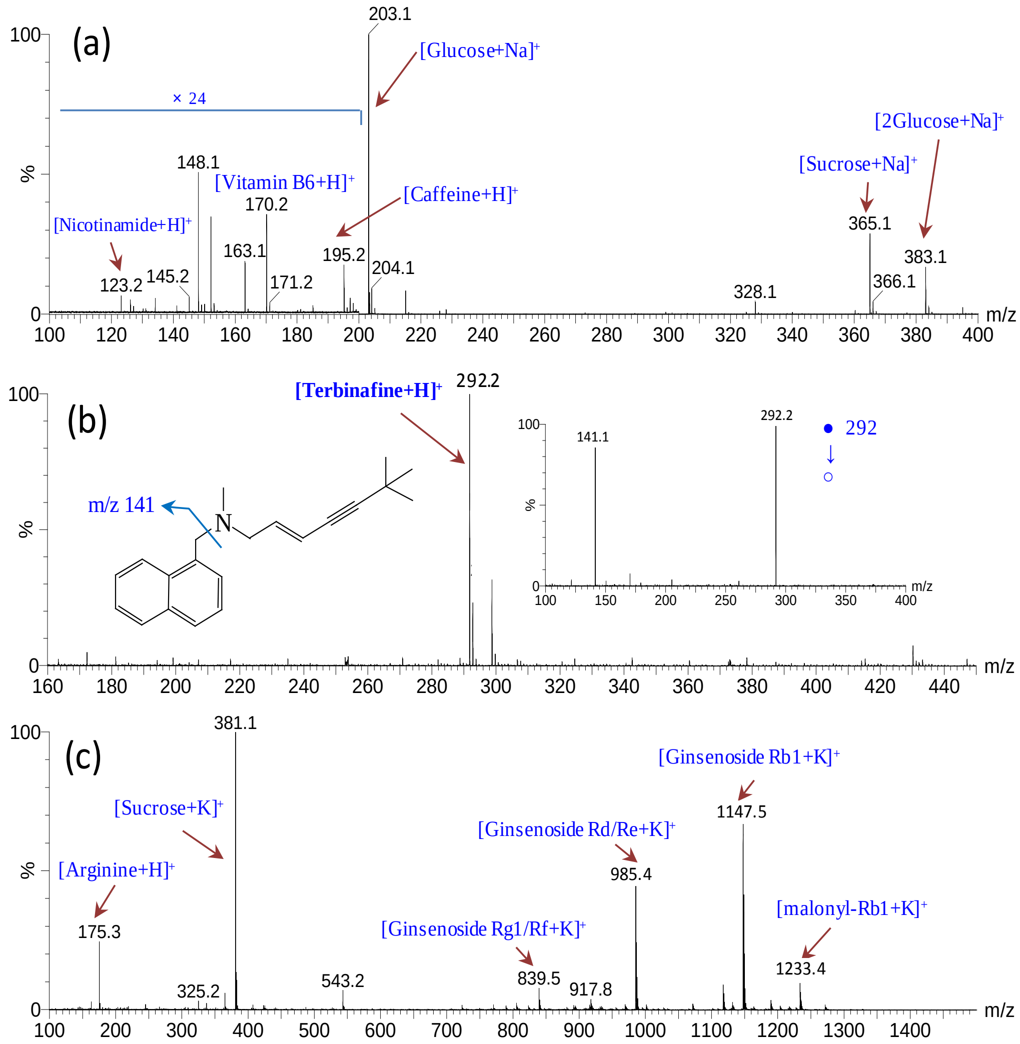

Figure 5-6 Mass spectra obtained by direct analysis of a) energy drink (Red Bull), b) terbinafine cream, and c) American ginseng using Al-foil ESI. The "●" and "○" in the inset of b) stand for the protonated precursor ion and its daughter ions, respectively.

Herein, the capability of Al-foil ESI in direct analysis of viscous samples was investigated. By applying 10 μL of terbinafine hydrochloride cream, a common medicinal cream for treating fungal infection,[216] onto the Al foil without addition of solvent, protonated molecules of terbinafine (*m/z* 292), the active component, could be predominately detected and further confirmed by tandem mass spectrometry (Figure 5-6b) [217]. This technique was also applied to direct analysis of skin care cream spiked with 1 ppm β-estradiol ($[M+H]^+$: *m/z* 273), a restricted estrogen hormone,[218] and the ion signals of the target analyte could be unambiguously detected (Figure 5-7b). These results showed that Al-foil ESI could be conveniently used for direct analysis of complex viscous samples. Our preliminary results showed that samples with higher viscosity produced smaller Taylor cones during the Al-foil ESI, as demonstrated by generation of electrospray from samples with different compositions of glycerol and methanol (Figure 5-8). The detailed mechanism of electrospray ionization of viscous samples needs further investigation.

5.3.3 On-target extraction

The impermeability, inertness and relative rigidity of Al foil enable it to serve as a desirable medium for on-target sample processing, e.g., solvent extraction, prior to ESI-MS analysis. The Al foil can be readily folded to hold extraction solvent in bulk and allows the solid samples to completely soak in the extraction solvent for effective extraction of samples. The mass spectrum obtained for powder of American ginseng, a commonly used herbal medicine,[219] by this approach is shown in Figure 5-6c. Using 10 μL of methanol as the extraction and spray solvent for 5 mg of the herbal powder, various ginsenosides, active ingredients of ginseng herbs, e.g., Rg1/Rf (*m/z* 839), Rd/Re (*m/z* 985), Rb1 (*m/z* 1147), and malonyl-Rb1 (*m/z* 1233) could be predominately detected in their potassiated forms.

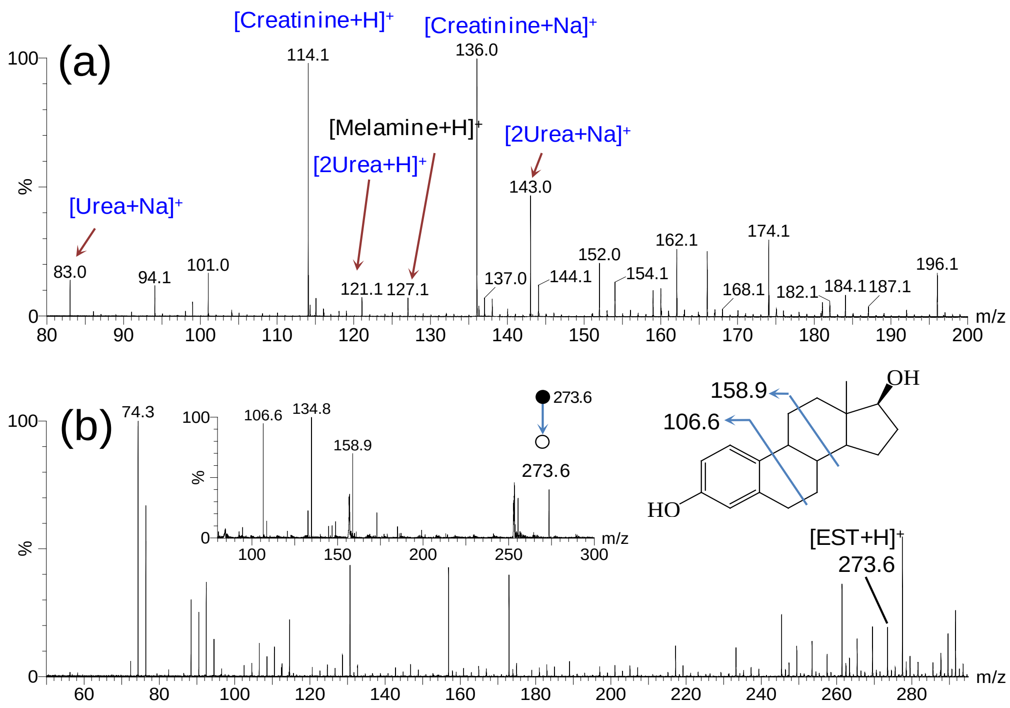

Figure 5-7 Mass spectra obtained for direct analysis of a) melamine in raw urine and b) β-estradiol in skincare cream using Al-foil ESI. The "●" and "○" in the inset of (b) stand for the protonated precursor ion and its daughter ions, respectively.

Similar ginsenosides were observed in our previous LC-MS analysis for a brief solvent extraction of American ginseng powder.[219] On-target extraction followed by ESI-MS analysis with Al foil was also applied to other samples. Spectral results for Fuzi (*Radix aconiti carmichaeli*), a common herbal medicine, coffee powder and biological tissue (porcine muscle) are shown in Figure 5-9.

5.3.4 On-target clean-up of protein and peptide samples

ESI-MS analysis of proteins and peptides is highly susceptible to interferences from salts and detergents.[158] Different methods, e.g., liquid chromatography, solid-phase extraction, ultrafitration, and ion exchange, have been applied to remove the interfering compounds prior to ESI-MS analysis.[158, 160] However, these methods are usually time-consuming and laborious; therefore it is highly desirable to develop simple and convenient techniques for rapid pre-clean-up of samples for ESI-MS analysis.[158, 160]

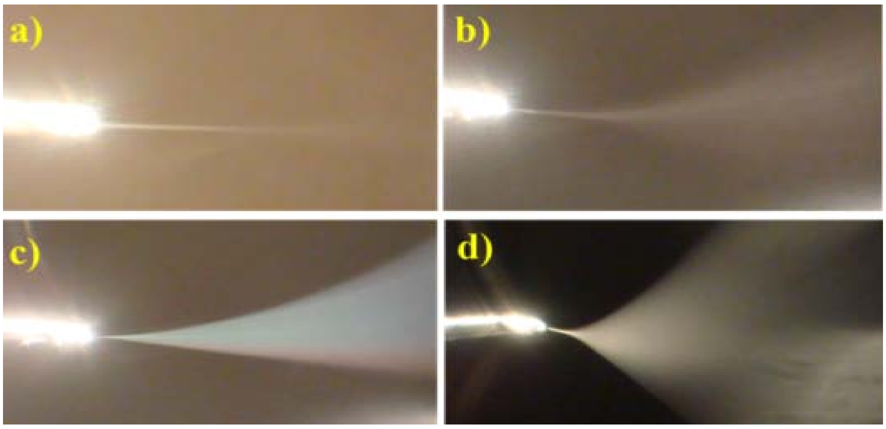

Figure 5-8 Photos showing the electrospray plumes generated from samples with different compositions of glycerol and methanol using Al-foil ESI: a) 100% glycerol, b) glycerol:methanol 3:1 (v/v); c) glycerol:methanol 1:1 (v/v); d) 100% methanol. The ionization voltage was set at 2.5 kV for all the samples.

Taking the idea from matrix-assisted laser desorption/ionization MS that on-target washing of samples on MALDI plate has been a successful strategy to remove salts and other contaminants,[220] we explored similar on-target sample washing approach with Al foil. The detailed experimental procedure of the on-target sample clean-up approach is described in Experimental Section. The mass spectral results obtained for a sample solution containing 10 μM lysozyme and a high concentration (2.5 M) of NaCl deposited on the "shiny" side of the Al foil are shown in Figure 5-10. Without on-target clean-up, only the NaCl clusters were observed and no ion signal for the interested protein was obtained (Figure 5-10a). After the on-target clean-up process, the ion signals of the protein of interest could be clearly observed, while the salt cluster ions became less significant (Figure 5-

10b). Repeating the cleaning procedure for several more cycles, i.e., three cycles, was found to enhance the desalting efficiency when the concentrations of salts were high. This sample clean-up approach was determined to be effective for samples containing up to 4 M NaCl. The present sample clean-up approach was also demonstrated to be effective in removal of NaCl from other proteins, e.g., cyotchrome *c* and myoglobin (Figure 5-11). With ammonium acetate buffer as the extraction and spray solvent after the on-surface clean-up procedure, the native structures of proteins could be maintained and detected, as demonstrated with myoglobin in this study (Figure 5-11c).

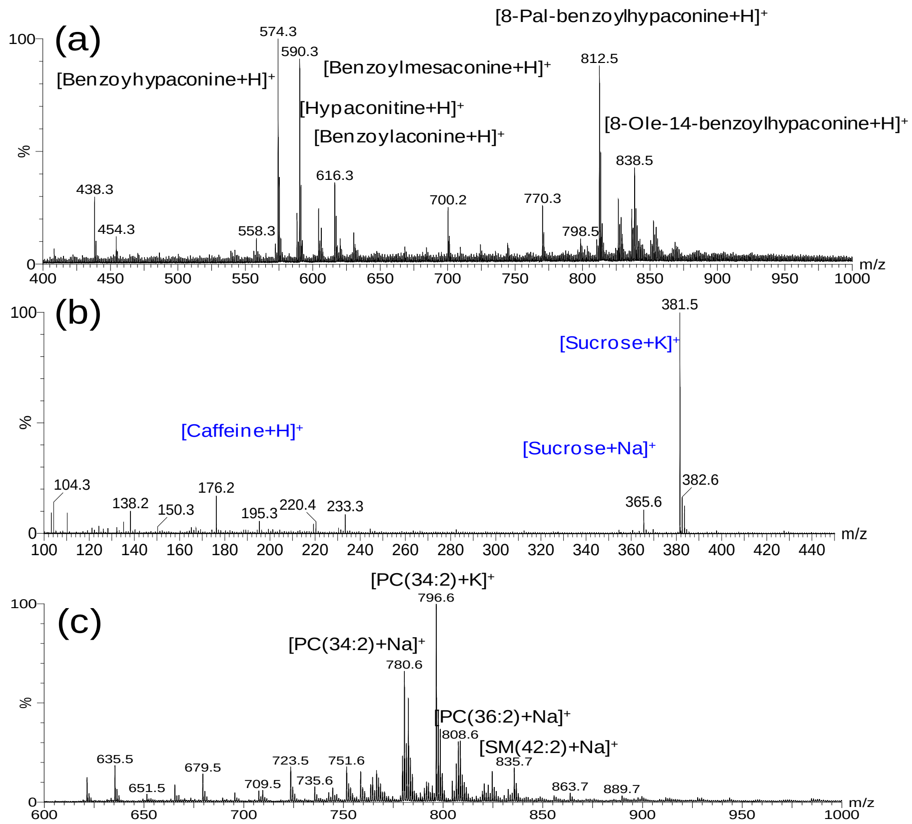

Figure 5-9 Mass spectra obtained for (a) herbal medicine Fuzi, (b) coffee powder, and (c) porcine muscle (PC: Glycerophosphocholine; SM: Sphingomyelin) using Al-foil ESI with on-target extraction

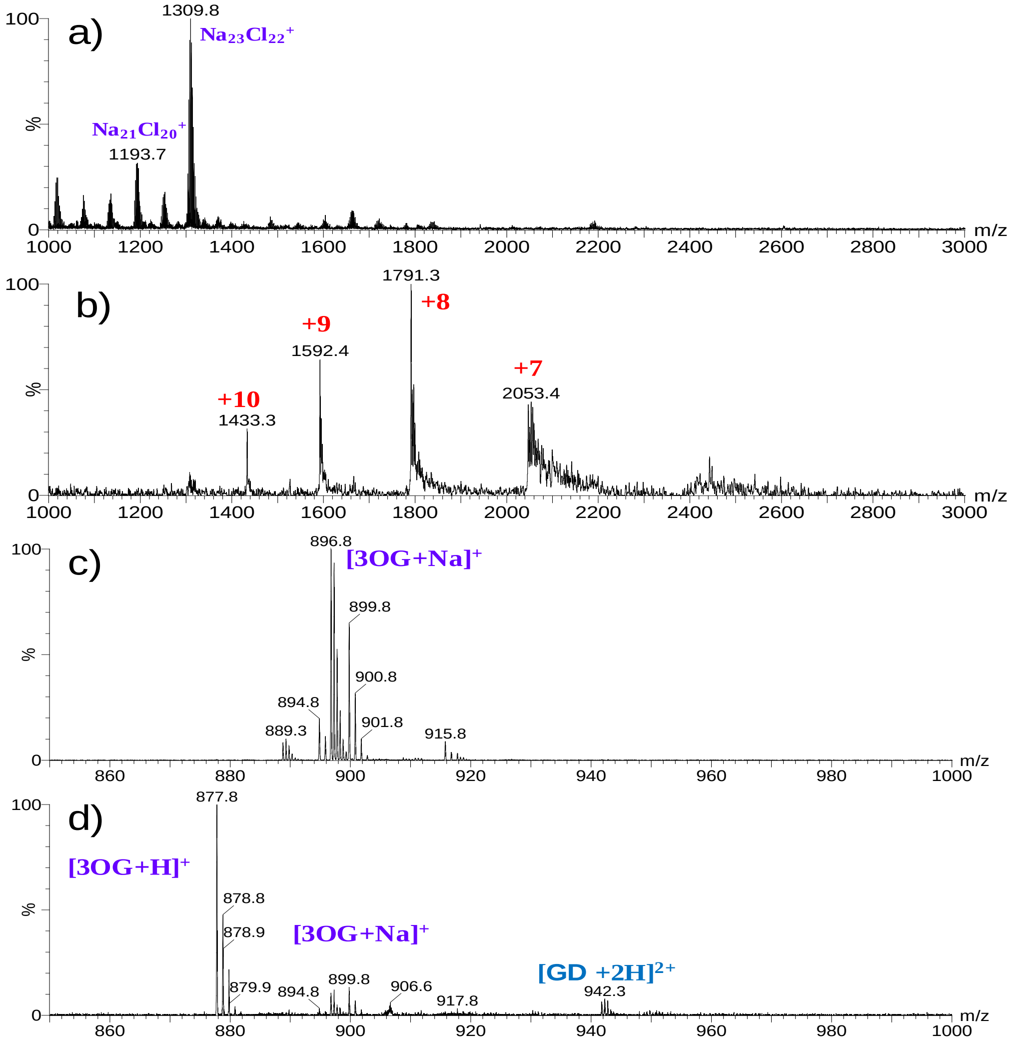

Figure 5-10 a) and b): Mass spectra obtained for a sample solution containing 10 µM lyzosome and 2.5 M NaCl before a) and after b) the on-target sample clean-up; c) & d): mass spectra obtained for a sample solution containing 10 µM gramicidin D and 5% (w/w) octly β-D glucopyranside before c) and after d) the on-target sample clean-up.

The on-target sample clean-up approach was also applied for protein and peptide samples containing detergents, which are often used in extraction of membrane peptides and proteins.[161] The mass spectra obtained for a solution containing 10 µM gramicidin D, a membrane peptide, and 5% (w/w) octly β-D glucopyranside (OG) deposited on the "shiny" side of Al foil are shown in Figures 5c and d.

Without the washing process, predominant sodiated ions of the detergent and no ion signal of the interested peptide could be observed (Figure 5-10c). After the on-target washing, the mass peak of the interested peptide could be clearly observed, although the spectrum remained dominant with the detergent signals (Figure 5-10d). This result is similar to that obtained in a previous study using a microfluidic electrocapture-assisted approach for removal of OG from gramicidin [161].

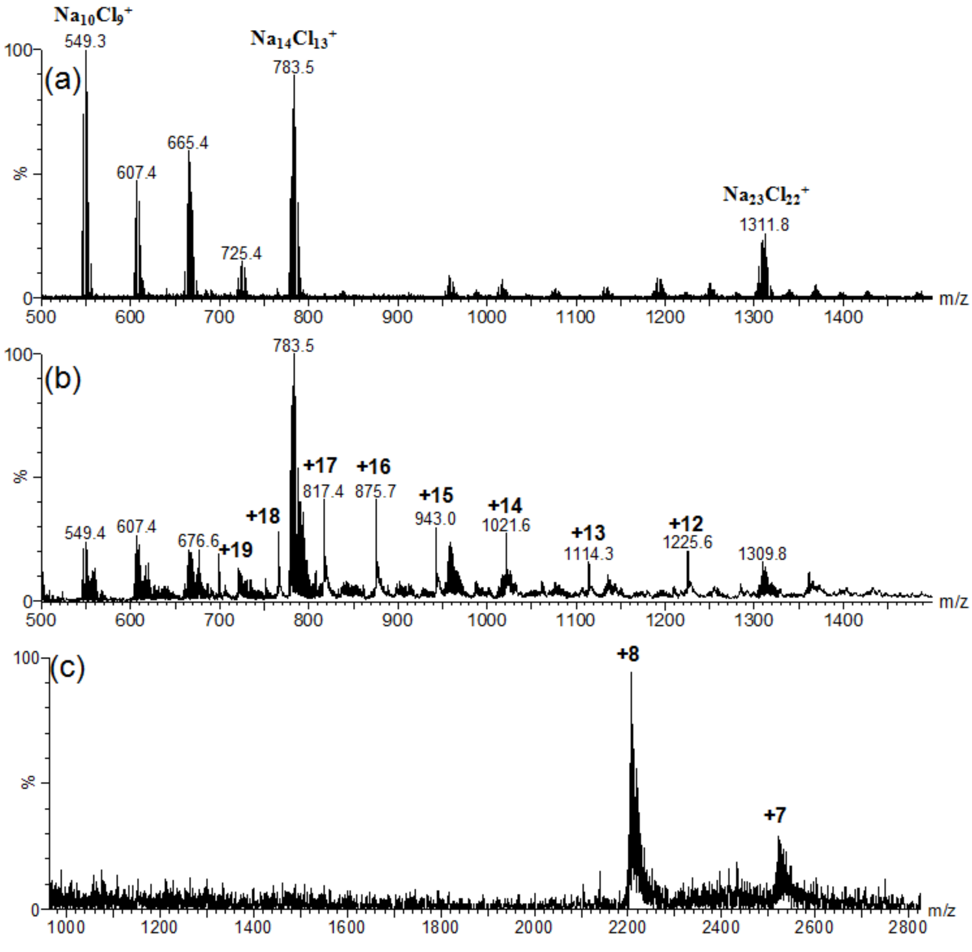

Figure 5-11 a) and b): Mass spectra obtained for a sample solution containing 10 μM cytochrome *c* and 2.5 M NaCl before a) and after b) on-target sample clean-up; c) mass spectrum obtained for a sample solution containing 10 μM myoglobin and 2.5 M NaCl after on-target sample clean-up using 20 mM ammonium acetate buffer as the extraction and spray solvent.

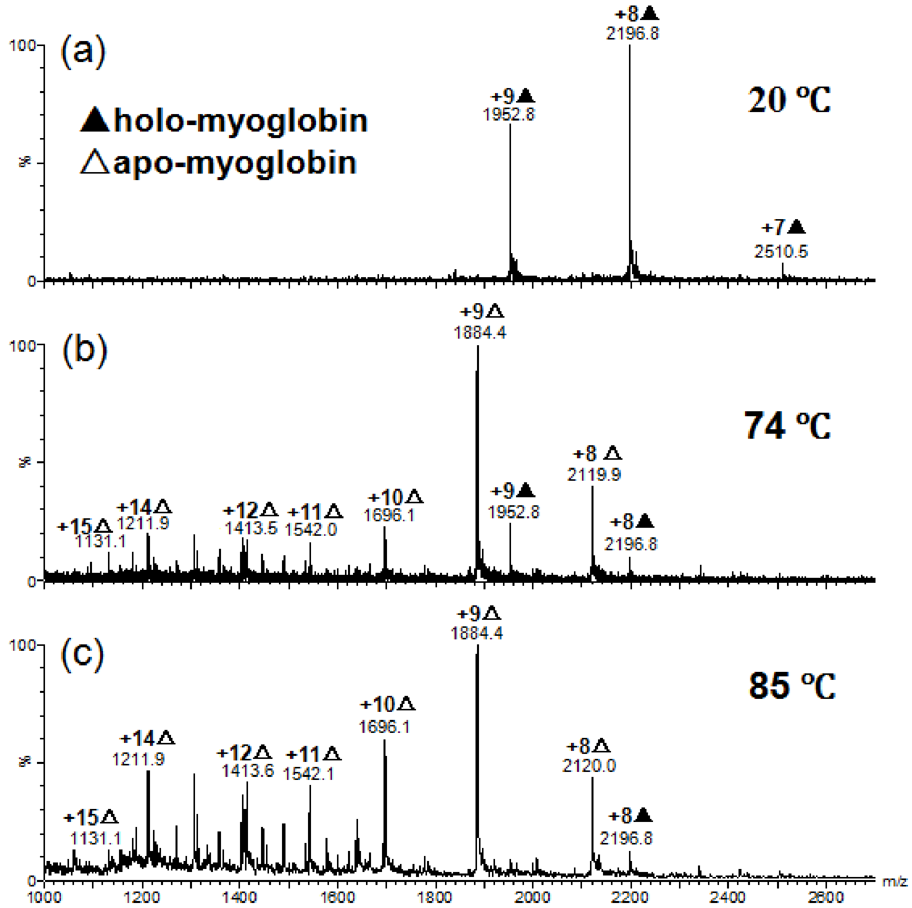

Figure 5-12 Mass spectra obtained for 10 μL of 10 μM myoglobin in 20 mM ammonium acetate using Al-foil ESI at different temperature: (a) 20 °C, (b) 74 °C and (c) 85 °C.

Attempt was also made to apply the "matte" side of Al foil for the sample clean-up of proteins and peptides, yet no improved signal of proteins and peptides could be obtained after the washing process. A plausible explanation could be that unlike the "shiny" side, for which the layer of organic lubricant might play a role as a hydrophobic surface for absorption of proteins and peptides, the sample molecules might not bind effectively to the aluminium oxide layer of the "matte" side, therefore were washed away during the clean-up process. Our investigation also showed that paper and wooden-tip were not applicable for this on-target washing approach. For these two substrates, the washing solvent appeared to spread over the substrate surface and could not be effectively pipetted off, resulting in poor washing efficiency. Moreover, the salts and detergents tend to bind to the hydrophilic surface of the substrates, and proteins and peptides cannot be effectively retained by the hydrophilic surface.

5.3.5 Direct monitoring of thermal reactions

Another desirable feature of Al foil as ESI substrate is that Al foil is heat-tolerant and a good conductor of heat. This feature renders Al-foil ESI a particular capability in direct monitoring of thermal reactions. Thermal denaturation is a common approach to investigate stability of proteins and protein complexes. Nano-ESI-MS has been applied to study thermal-denaturating behavior of proteins and protein complexes by heating the sample solutions in the nano-ESI probe [221] and more recently laser spray was developed for investigating thermal stability of proteins. [222] In this study, thermal denaturation of myoglobin was investigated using Al-foil ESI. As shown in Figure 5-12a, the ESI-MS spectrum acquired for myoglobin at room temperature (20 ^{o}C) displayed a narrow charge state distribution ranging from +7 to +9, similar to that obtained by wooden-tip ESI and EESI. [91, 157] The deconvoluted molecular mass obtained was 17,566 Da, which is consistent with the noncovalent complex between the apo-protein (16,951 Da) and the heme (615 Da). When the sample solution on the Al foil was heated to 74 ^{o}C, predominant multiply charged ion peaks of the apo-protein were observed and the signal intensity for the noncovalently bound complex significantly reduced (Figure 5-12b). In addition, the charged state distribution became much wider, indicating that the protein became denatured upon heating.

When the sample was further heated to 85 ^oC, the intensities of the peaks with higher charge states became more significant and the signals for the noncovalent complex were further lower, revealing further denaturation of the protein (Figure 5-12c).

Similar approach was also applied to investigate the thermal denautration of another protein, lysozyome (Figure 5-13). From 20 ^oC to 40 ^oC, the charge state distribution showed an average charge state of +8 (Figure 5-13c), which is similar to that obtained by conventional ESI [223] and laser spray when moderate laser energy was applied.[222] The average charge state increased gradually with the temperature, reached a maximum of +9.6 at 65 ^oC, and eventually levelled off beyond 65 ^oC (Figure 5-13c). The temperature-dependent charge state distribution allows estimation of melting temperature (T_m), which reflects thermal stability of proteins.[224-226] The T_m of lysozyme estimated herein was ~ 60 ^oC which is ~ 10 ^oC less than that obtained for the aqueous solution of lysozyme by near-UV and far-UV circular dichroism.[227] The discrepancy between these results could be due to variation in solvent and/or the fact that ESI charge state distribution reveals the overall conformational properties while the near-UV and far-UV circular dichroism are more correlated to local environments or secondary structures of proteins.[224-227] Nevertheless, these results demonstrated that Al-foil ESI could be an alternative method in monitoring thermal denaturation of proteins and protein complexes and can be extended to studies of other thermal reactions

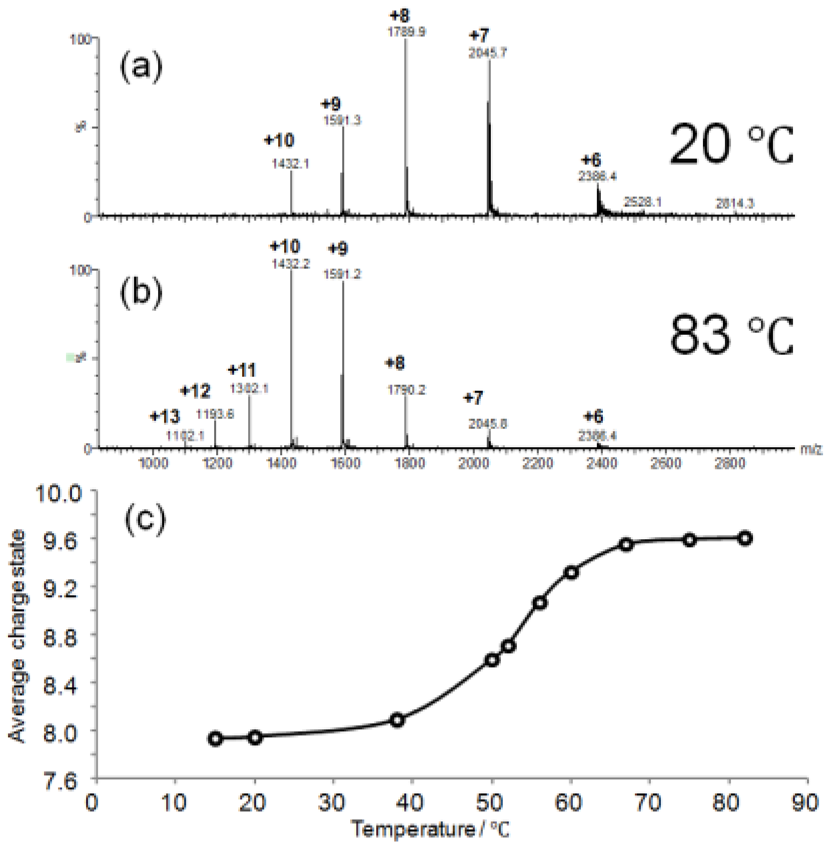

Figure 5-13 a) and b): Mass spectra obtained for 10 μM lysozyme using Al-foil ESI with Al foil maintained at 20 °C and 83 °C respectively. c) Plot of average charge state of the protein against temperature of the Al foil surface.

5.4 Summary

In this study, we developed an ESI method using household Al foil for loading and ionization of samples. This technique was found to be able to generate desired ion signals for a wide range of compounds with high signal-to-noise ratios, and applicable in direct analysis of viscous samples and solid samples that are less accessible in conventional capillary-based ESI. Al foil has an inert and impermeable surface and can be readily folded into desired configuration for holding (wrapping) of sample solution in bulk. These allow effective on-target extraction of solid samples prior to ESI-MS analysis and thus could generate quality mass spectra for the samples. The “shiny” side of Al foil has a relatively hydrophobic surface, rendering it the capability in absorbing proteins and peptides and enabling on-target clean-up of protein and peptide samples, i.e., removal of salts and detergents. Al foil is an excellent heat-conductor and highly tolerant to heat, permitting direct monitoring of thermal reactions, e.g., thermal denaturation of proteins. To conclude, the present study showed that Al-foil ESI further extended the capability of ESI and could be an economical and versatile method for mass spectrometric analysis.

Chapter 6: Thin Layer Chromatography Coupled with ESI-MS for Direct Analysis of Raw Samples

6.1 Introduction

In the past decades, development of sampling and ionization methods for direct analysis of raw samples has been an important research area in MS. For instance, the development of various ambient ionization techniques, e.g., DART,[228] DESI,[37] and ELDI,[74] which are characterized by that samples are ionized under atmospheric pressure and no or only little sample preparation is involved, significantly reduces the time and cost required for analysis of raw samples.[6, 73, 211] In addition, the techniques of solid-substrate ESI, in which sample solution is loaded and ionized on the open surface of a solid substrate, e.g., paper,[128] wooden tips,[91, 146] Al foil,[92] and metal needles,[89, 127, 200] were also developed to facilitate mass spectrometric analysis of complex samples by avoiding the problem of clogging potentially encountered in conventional capillary-based ESI, allowing more convenient sampling and reducing matrix interferences.[88, 89, 229] The techniques of ESI with paper (paper spray) and wooden tips (WT-ESI) were successfully applied in direct detection of analytes in raw samples, e.g., blood, urine and oral fluid samples, and the separation effects of filter paper and wooden-tip surface were believed to be related to their capabilities for direct sample analysis. [90, 91, 146, 150, 208, 230]

Thin layer chromatography (TLC) is a simple, rapid and more efficient separation method, and thus coupling of TLC with MS has good potential for direct analysis of raw samples. Coupling of TLC with MS has attracted much attention,[138, 231] and various ionization methods, including fast atom bombardment (FAB),[232, 233] matrix-assisted laser desorption/ionization (MALDI),[234] electrospray ionization (ESI),[139, 235] and ambient ionization methods such as liquid extraction junction,[236-238] desorption electrospray ionization (DESI),[239, 240] direct analysis in real time (DART),[241-243] easy ambient sonic-spray ionization (EASI) [121] and electrospray laser desorption ionization (ELDI),[244] has been employed for the coupling. However, the major purposes of these coupling studies were to facilitate identification of the separated TLC spots by MS. In this study, we mainly aimed to employ TLC plate as a solid substrate medium to reduce matrix interference and enhance detection of target analytes in raw samples by ESI-MS. Separation and identification of components in mixture samples were also investigated in this study. Our results indicated that TLC coupled with ESI-MS (TLC-ESI-MS) could be a simple, rapid and efficient method for detection, including quantitative measurements, of analytes in raw samples.

6.2 Experimental section

6.2.1 Materials

Acetone, ligroin, sodium chloride (NaCl), sodium dodecyl sulfonate (SDS), octly β-D-glucopyranside (OG), myoglobin and lysozyme were purchased from Sigma (St. Louis, MO). Gramincidin D (from *bacillus breris*) was purchased from International Lab (USA). Ketamine was purchased from Alfasan (Woerden, Holland). Methanol, acetonitrile and chloroform were purchased from Tedia (Fairfield, OH). Ketamine-D4 was purchased from Cerilliant (Round Rock, Texas). Petroleum was purchased from Fisher Scientific (Pittsburgh, PA). Formic acid was purchased from Fluka (Buchs, Germany). Ethyl acetate was purchased from Acros (Fairlawn, NJ). Ginsenoside Rc was purchased from Shanghai Tauto Biotech (Shanghai, China). The urine sample was collected from a healthy male volunteer. Herbal medicines Fruit of *Schisandra sphenanthera* (FSS) and Fruit of *Schisandra chinensis* (FSC) were purchased from pharmacy stores in Hong Kong. Fresh spinach leaves were purchased from a supermarket in Hong Kong. Extracts of spinach leaves, FSS and FSC, were prepared as described previously.[102] TLC plates (Model: Alugram Sil G/UV254) were purchased from Macherey-Nagel (Düren, Germany).

6.2.2 Thin layer chromatography-electrospray ionization mass spectrometry (TLC-ESI-MS)

6.2.2.1 TLC plates

The TLC plates used in this study have an aluminum base coated with silica gel stationary phase particles. Prior to sample loading, the TLC plates were cut into a dimension of 20 ×15 mm (L × W) and one end of each plate was cut into V-shape with a tip angle of ~ 60^{o}.

6.2.2.2 TLC-ESI-MS

TLC-ESI-MS experiments were performed on a quadrupole-time-of-flight mass spectrometer (Q-TOF2, Waters-Micromass) or a triple quadrupole mass spectrometer (Quattro Ultima, Waters-Micromass). The experimental setup was similar to the direct coupling method reported previously.[139] Briefly, a TLC plate, which has an aluminium base for conduction of electric current, was cut into triangle shape and positioned ~ 8 mm from the inlet of the mass spectrometer with a clip connected with the high voltage supply of the mass spectrometer (Figure 6-1). During data acquisition, the capillary voltage and cone voltage were set at 3.5 kV and 30 V, respectively. The ion source temperature was set at 80 ^{o}C and all desolvation gases were turned off. Nano-ESI experiments were carried as similar to our previous study.[91]

6.2.3 Sampling methods

6.2.3.1 Sampling for analysis of samples with interfering contaminants and raw urine

Sample solution, typically 2 µl, was applied onto the TLC plate at a position ~ 8.0 mm away from the sharp tip end. The sample solution appeared to dry quickly after the sample loading. The TLC plate with loaded sample was first mounted in front of the MS inlet and high voltage was applied to the plate. Ten microliter of elution solvent was then applied around the sample spot position for extraction of the sample. Electrospray could be generated when the elution solvent reached the tip end of the plate, i.e., methanol for small organic molecules and ACN/H_2O 50:50, 0.5% formic acid for peptides and proteins, reached the tip end of the plate.

6.2.3.2 Analysis of plant extracts: offline and online samplings

For offline sampling, the plant extract was pre-separated on a TLC plate and the sample spots were located with an UV lamp (Spectronics Corp, NY). The plate regions with sample spots of interest were cut out into a triangle shape with a size covering the entire sample spot, typically with ~6 mm base and tip angel of ~ 45 ^{o}, which were then mounted in front of the MS inlet with a clip connected with high voltage. Ten microliters of elution solvent, i.e., methanol, was added onto the plate for elution of the sample for MS analysis.

For online sampling, sample solution was loaded ~ 20 mm from the tip end of the TLC plate and stood until dryness. The plate with sample was then mounted in front of the MS inlet and connected with high voltage. Ten microliter of developing solvent (acetone/ligroin, 1/9, v/v) was applied onto the sample spot for several repeated cycles until the pigments were seen to travel and separate along the TLC plate to the tip end.

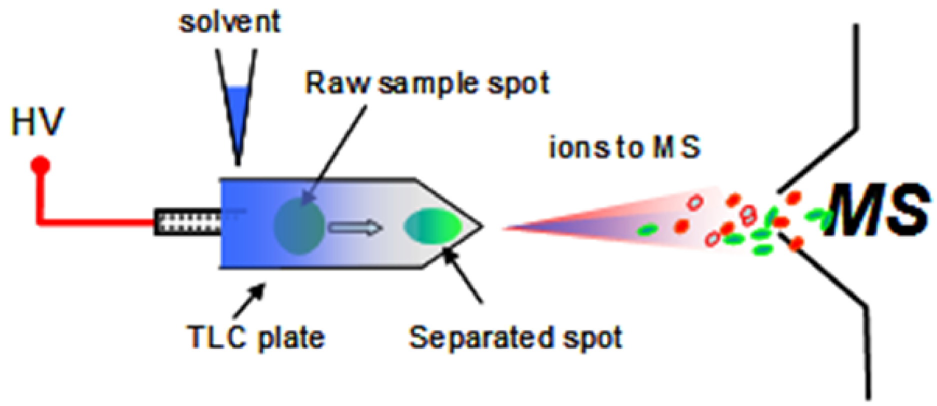

Figure 6-1 Schematic diagram of the TLC-ESI-MS setup.

6.3 Results and discussion

6.3.1 Analysis of samples containing salts and detergents

Salts and detergents are common interference compounds that can significantly suppress signals of analytes in ESI-MS. The capability of TLC-ESI-MS in analysis of samples containing salts and detergents was investigated in this study.

First, methanoic solutions of 1 μM of ginsenoside Rc (GRc), an active component of ginseng herbs,[219] containing different concentrations of NaCl, were analyzed with TLC-ESI-MS. For samples containing 1 to 100 mM of NaCl, sodiated ion of GRc at *m/z* 1101 was predominately observed and no interfering ion signal from cluster ions of NaCl was detected (Figures 6-2a-c). When the concentration of NaCl was increased to as high as 1 M, the interfering signals from NaCl cluster ions were observed, yet the sodiated ion of GRc remained the major detected peak (Figure 6-2d). As a comparison, when NaCl-containing samples were analyzed by nano-ESI, which has a higher salt tolerance than normal ESI, interfering signals of NaCl cluster ions were detected even at low salt contents, i.e., 1 mM, and completely masked the signals of GRc at 1M NaCl (Figures 6-2f-i). These observations could lead to an argument that the less salt interference observed for TLC-ESI than nano-ESI was due to the dilution of salt content by elution solvent in TLC-ESI. To address this argument, an aliquot of 2 μL a solution containing 1 μM GRc and 2 M NaCl was diluted with 10 μL methanol and analyzed with nano-ESI. In parallel, TLC-ESI-MS analysis with the same dilution conditions, i.e., applying 2 μL of the same solution onto a TLC plate followed by addition of 10 μL of methanol as elution solvent, was carried out and the results obtained with the two conditions were compared. The results showed that the ion signals for GRc could be clearly detected with TLC-ESI (Figures 6-2e), while no ion signal of GRc and only significant ion signals from the salt clusters could be detected for nano-ESI (Figure 6-2j). These observations revealed that salt interference could be reduced with the present TLC-ESI-MS method, but not by simple dilution of salt content. These results indicated that the present TLC-ESI approach allowed rapid detection of analytes in high salt content with reduced salt interference, which could be achieved by retention of salts by the polar silica gel stationary phase and selective elution of the more nonpolar analyte for MS detection.

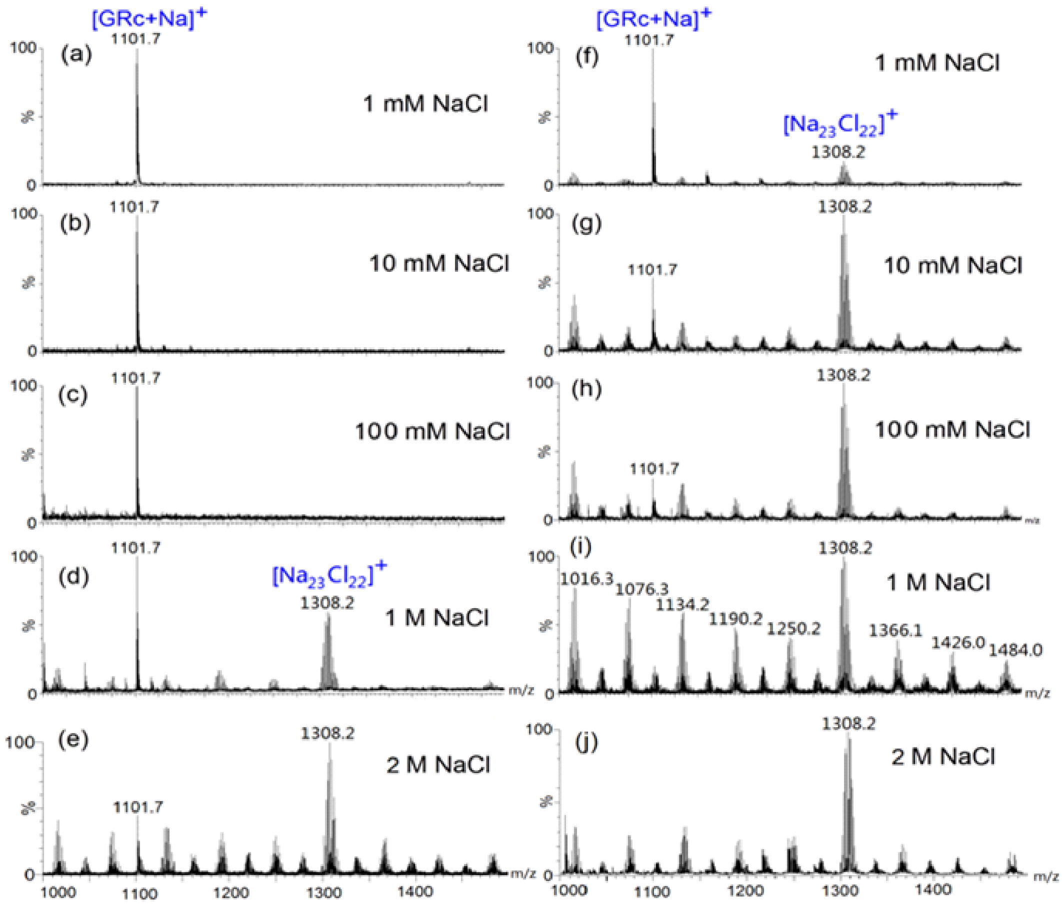

Figure 6-2 Mass spectra of 1 μM ginsenoside Rc (GRc) (MW=1079) in different concentrations of NaCl obtained by (a-e) TLC-ESI-MS with 10 μL of methanol as the elution solvent and (f-i) nanoESI-MS. j) A Nano-ESI-MS mass spectrum for 2 μL of solution with 1 μM GRc and 2 M NaCl diluted with 10 μL methanol.

The salt tolerance of the present TLC-ESI-MS method for analysis of proteins with high salt contents was also investigated with horse heart myoglobin. As shown in Figure 6-3a, the mass spectrum obtained with TLC-ESI-MS showed predominant protein ion signals with no interfering signal from NaCl ion clusters. When the same sample was analyzed with nano-ESI, however, the interfering signals from NaCl cluster ions were predominately detected (Figure 6-3b). These results further demonstrated that TLC-ESI-MS could be effectively applied in analysis of samples containing high contents of salts.

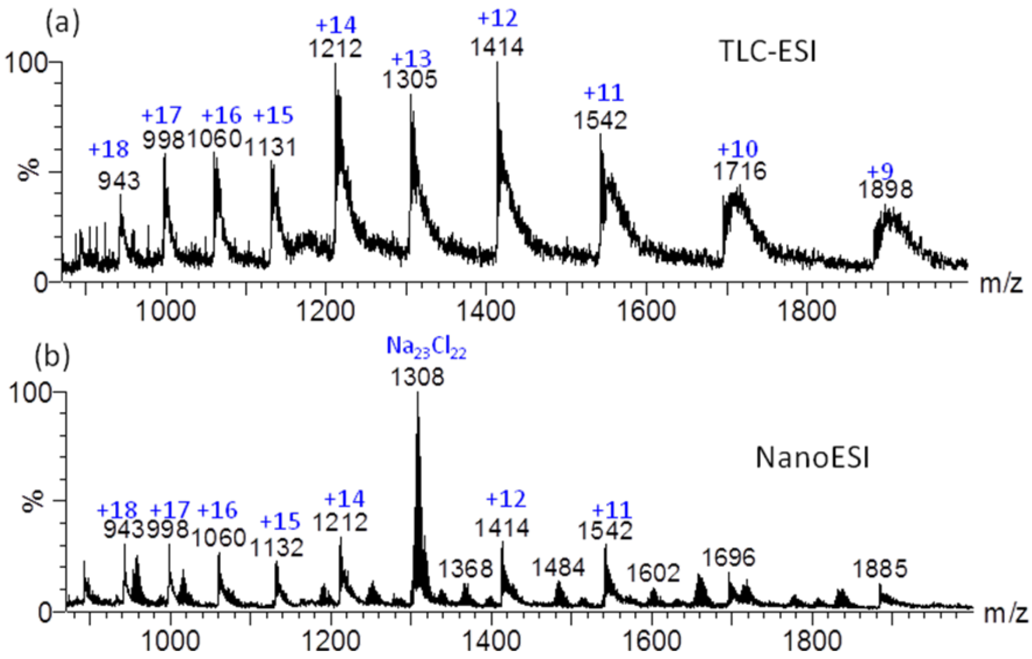

Figure 6-3 Mass spectra of 0.1 mM myoglobin in ACN:water (50:50) containing 0.5% formic acid and 1 mM NaCl obtained by a) TLC-ESI-MS and b) nano-ESI-MS.

Detergents are commonly used to improve the solubilities of membrane peptides and proteins, and analysis of membrane peptides and proteins has been a challenging problem in analytical chemistry and protein studies.[161, 245] In this study, the TLC-ESI-MS method was applied to analyze membrane peptide samples containing detergents. The mass spectrum obtained for 50 µM of gramicidin D, a membrane peptide, containing 5 % (w/w) of octyl β-D-glucopyranside (OG) is shown in Figure 4a. The doubly charged sodiated ion ([M

+ 2Na]$^{2+}$) of gramicidin D could be unambiguously detected with the TLC-ESI-MS approach, although the detergent signals were also observed.

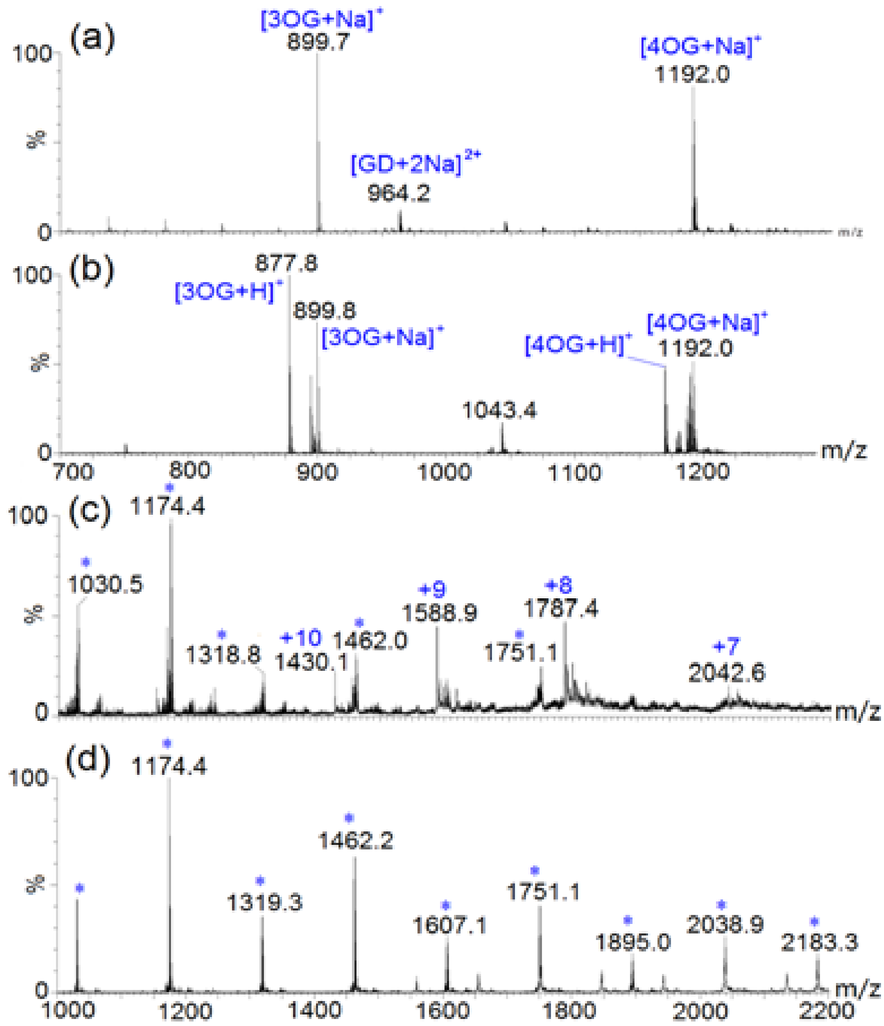

Figure 6-4 a) & b) Mass spectra of 50 μM gramicidin D in methanol containing 0.1% formic acid and octly β-D-glucopyranside (5% w/w) obtained by a) TLC-ESI-MS with methanol containing 0.1% formic acid as the elution solvent and b) nano-ESI-MS. c) & d) Mass spectra of lysozyme in ACN:water (50:50) containing 0.5% formic acid and 5 mM SDS obtained using c) TLC-ESI-MS and d) nano-ESI-MS. Peaks originated from SDS are denoted with "*".

However, no ion signal of gramicidin D and only predominate detergent signals were detected when the same sample was analyzed with nano-ESI (Figure 6-4b).

Similar observations were obtained for a protein sample, lysozyme, containing SDS, a commonly interfering detergent in proteomics studies. The analyte of interest could be clearly detected with the TLC-ESI-MS approach, but not with nano-ESI (Figures 6-4c and d). These data reveal the applicability of the present TLC-ESI-MS method in analysis of samples containing detergents.

6.3.2 Detection and quantitation of analytes in raw urine

Detection and quantitation of analytes in raw biological fluids, e.g., urine, usually requires sample pre-treatment and HPLC or GC separation. In this study, the capability of TLC-ESI-MS in rapid detection and quantitation of ketamine, a commonly recreational drug, in raw urine was investigated. The quantitation settings were similar to what we used previously for wooden-tip ESI.[146] Upon loading of the raw urine samples spiked with ketamine (0.1, 0.2, 1, 2, 5 and 10 ppm) and ketamine-d4 internal standard (1.0 ppm) onto the TLC plate and monitoring the analyte and internal standard by selected reaction monitoring (SRM), distinctive SRM signals of ketamine (*m/z* 238> 125) (Figure 5a) and ketamine-d4 (*m/z* 242 > 129) (Figure 5b) could be obtained. The peak intensity (peak height) ratio of ketamine and ketamine-d4 internal standard ($I_{ketamine}/I_{ketamine\text{-}d4}$) exhibited a good linear relationship over the concentration range of 0.1 – 10 ppm (Figure 5c), which was comparable to the results obtained by wooden-tip ESI and within the quantity range commonly found in drug abusers.[146] The limit-of-detection (LOD) and limit of-quantitation (LOQ) were determined by comparing the $I_{ketamine}/I_{ketamine\text{-}d4}$ of the spiked samples, i.e., samples spiked with both analyte and internal standard, with that of the blank sample, i.e., the sample spiked with internal standard only [($I_{ketamine}/I_{ketamine\text{-}d4})_{spiked}/(I_{ketamine}/I_{ketamine\text{-}d4})_{blank}$]. The LOD and LOQ are defined as the quantity of analyte that could achieve a [($I_{ketamine}/I_{ketamine\text{-}d4})_{spiked}/(I_{ketamine}/I_{ketamine\text{-}d4})_{blank}$] of three and ten, respectively. With the present TLC-ESI-MS method, the LOD and LOQ were determined as 30 ppb and 90 ppb, respectively, which are acceptable for real applications. Multiple analysis of a urine sample spiked with 100 ppb of ketamine and 1.0 ppm of ketamine-d4 internal standard (n = 9) showed a relative standard deviation (RSD) of 9.5 %, indicating an acceptable level of reproducibility. The above results indicated that TLC-ESI-MS could be used for selective and quantitative detection of analytes in raw samples.

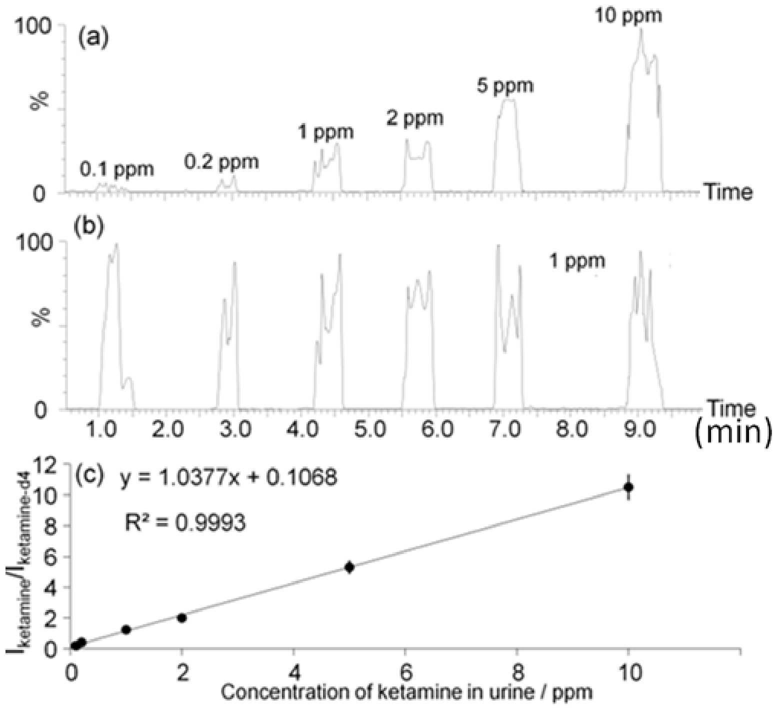

Figure 6-5 (a) & (b) Selected reaction monitoring (SRM) chromatograms of (a) ketamine (*m/z* 238>125) and (b) ketamine-d4 (*m/z* 242>129) in raw urine samples obtained by TLC-ESI-MS. (c) A calibration curve obtained for quantitation of ketamine in raw urine with TLC-ESI-MS.

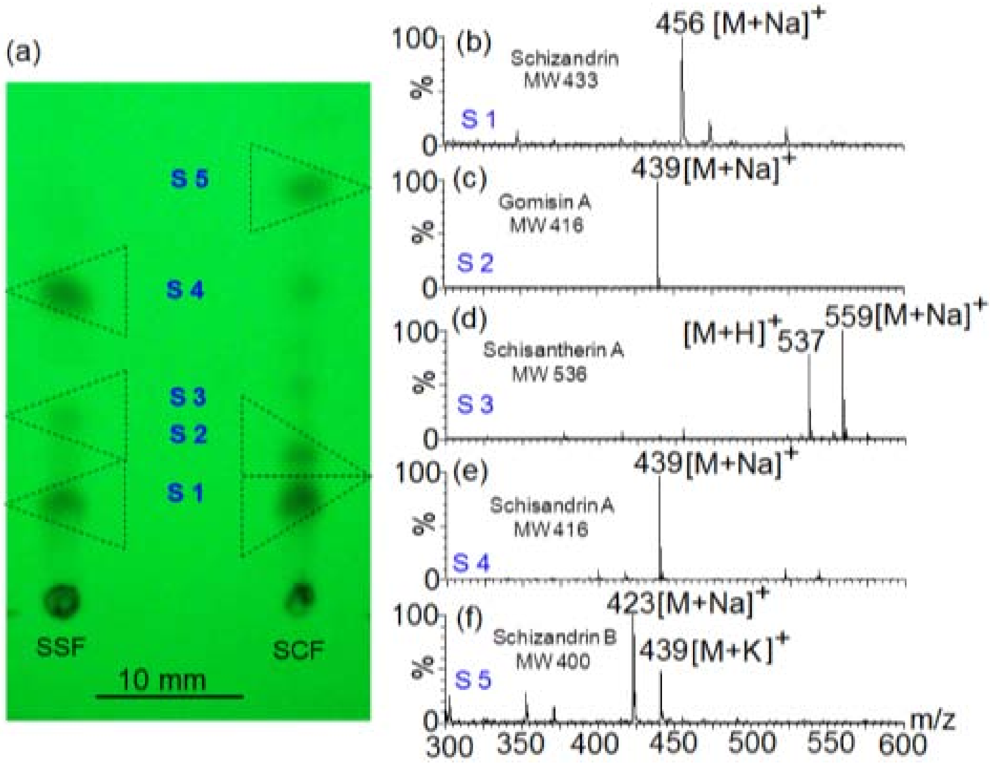

Figure 6-6 a) TLC profiles of extracts of Fruit of *Schisandras phenanthera* (FSS) and Fruit of *Schisandr achinensis* (FSC) under a 254 nm UV lamp. The spot areas denoted with triangles with dotted lines were cut out for TLC-ESI-MS analysis. The mass spectra obtained for each spot are shown in b) – f).

6.3.3 Characterization of different components in mixture samples

Two sampling approaches, offline sampling and online sampling, were applied for rapid analysis of mixture samples by TLC-ESI-MS. The offline sampling approach was applied for differentiation of Fruit of *Schisandra sphenanthera* (FSS) and Fruit of *Schisandra chinensis* (FSC), two common Traditional Chinese Medicines distributed in southern China and northen China, respectively.[246] The FSC is considered to be of higher quality than FSS.[246] The methanolic extracts of the two herbs were separated on a TLC plate and the sample spots were identified under an UV lamp. The TLC profile of FSC showed five major spots and that of

FSS showed three major spots (Figure 6-6a). By cutting out each spot for TLC-ESI-MS analysis, ion signals for various lignans could be clearly detected.

Our results showed that gomisin A and schizandrin B were detected only for FSC, and other lignans, including schizandrin, schisantherin A and schisandrin A, were detected for both FSC and FSS (Figures 6b-f). These results were similar to the previous report [246] and allowed differentiation of the two closely related herbal species. These data indicated the capability of the TLC-ESI-MS method in rapid identification of chemical components in plant extracts with high reliability.

An online sampling method, which does not require separation of samples in an external container, for analysis of mixture samples was also investigated. An extract of spinach was analyzed with the online TLC-ESI-MS approach and the results are shown in Figure 6-7. When the developing solvent was loaded onto the sample spot, a yellow-orange component was first separated out, and a predominate peak corresponding to carotene ($[M+H]^+$: *m/z* 537), a major pigment in spinach,[247] could be detected (Figure 6-7a). Following the separation and detection of the yellow carotene component, a grey-green component appeared to separate out and a major peak corresponding to pheophytin A ($[M+H]^+$: *m/z* 871), another well known pigment in spinach,[247] was observed (Figure 7b). These data indicated the potential of TLC-ESI-MS for rapid online separation and detection of components in mixture samples. However, care should be taken to allow solvent migration and component separation compatible with spray rate and ESI-MS detection.

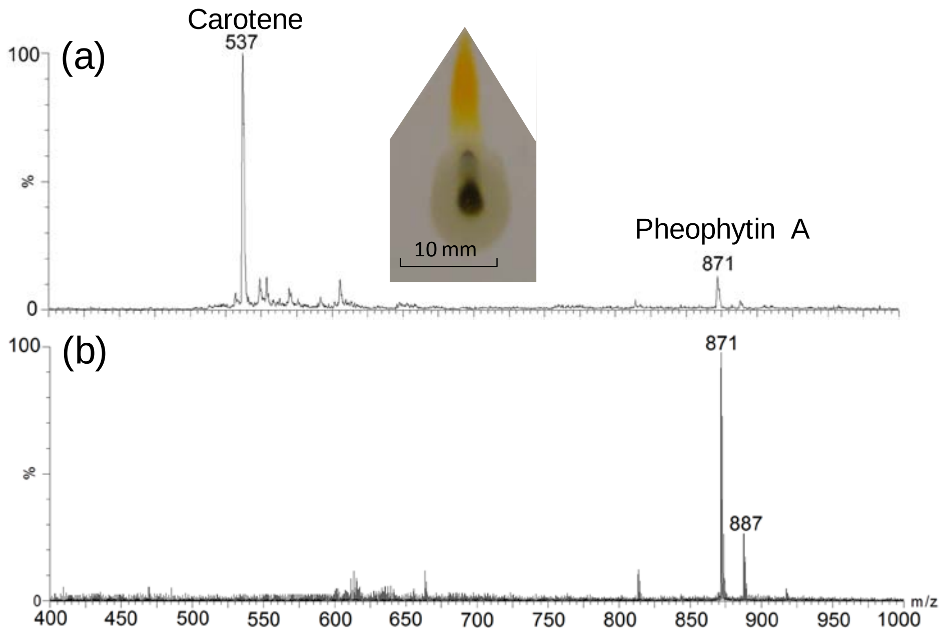

Figure 6-7 Online TLC-ESI-MS analysis of a spinach extract. Mass spectra obtained for pigments a) carotene and b) pheophytin.

6.4 Summary

This study presented a TLC-ESI-MS method for direct analysis of raw samples. Our results showed that the TLC plate could serve as a medium for absorbing interfering substances, allowing target analytes to be detected with reduced matrix interference. This mode of separation of analytes and interfering substances avoided the need of laborious and time-consuming sample pretreatment and HPLC or GC separation in analysis of complex samples. This method was shown to be effective in direct analysis of samples with interfering substances, e.g., salts and detergents, and rapid detection and quantitation of target analyte in raw biological fluids. In addition, the TLC-ESI-MS method was also demonstrated to be an effective method for rapid identification of different components in mixture samples. In view of the capability in rapid analysis of raw samples demonstrated here, the availability of TLC plates with various sorbent materials and the features of simplicity, economy and easy operation, TLC-ESI-MS could be a decent analytical technique potentially extended to a widely range of chemical and biological applications.

Chapter 7: Direct Coupling of Solid Phase Microextraction with ESI-MS: A Case Study for Detection of Ketamine in Urine

7.1 Introduction

In recent years, use of solid substrates for sample loading and ionization has attracted a lot of attentions.[89, 200, 248] Solid-substrate ESI-MS avoids the clogging problem that can easily be encountered in capillary-based ESI-MS, allows convenient sample loading and direct analysis of various forms of samples,[118, 150, 153, 249] including viscous liquids, powders and bulky solids, as well as quantitation.[150, 153] Moreover, *in vivo* and real-time detection[250] using solid-substrate ESI-MS has been reported to provide transient and dynamic information about the metabolic processes, with no or minimal sample preparation. Unlike other ambient ionization techniques that typically require additional settings such as solvent beam, plasma or laser, the assembling and operation of solid-substrate ESI-MS are usually quite simple and straightforward. The minimal modification of mass spectrometers and the use of easily accessible objects enable this technique easy to use, especially to the increasing non-expert MS users in the booming fields of proteomics, metabonomics, and molecular diagnostics.

Although solid-substrate ESI-MS allows direct analysis of real samples, the presence of other compounds and complex matrices in the directly loaded samples can severely suppress the signals of target compounds. Solid substrates that allow selective enrichment and ionization would thus be very helpful for improving analysis of solid-substrate ESI-MS. Solid phase microextraction (SPME) may be a good choice for this purpose. SPME is a simple and efficient technique for sample pretreatment. The materials coated on the surface of SPME fibres allow selective extraction and enrichment of analytes from raw samples. The interfering substances and matrix can be washed out, while analytes are retained due to their affinities to the surface. Since its introduction in 1990, SPME has been widely used for various analytical purposes and SPME devices with various coating materials have been commercially available.[251, 252] SPME is typically coupled with gas chromatography-MS (GC-MS)[253] and liquid chromatography-MS (LC-MS)[254] for detection of the extracted and enriched analytes. Direct detection of analytes on SPME fibres by mass spectrometry has been also explored,[255] and ionization techniques such as matrix-assisted laser desorption ionization (MALDI),[256,257] direct analysis in real-time (DART),[258,259] desorption electrospray ionization (DESI)[260] have been attempted. SPME fibres have also been coupled with direct electrospray probe (DEP) that was made of copper coil,[127] thermal desorption technique,[261] and designed holder for direct ionization.[262] Very recently, electrospray ionization from surface-modified blades[263] and wooden tips[264] were reported as well.

Detection and quantitation of drugs-of-abuse in body fluid is a critical step for control of drug abuse that is a severe problem worldwide nowadays. Conventionally, detection and quantitation of drugs-of-abuse in body fluid require intensive sample pretreatments and time-consuming chromatographic separation prior to MS analysis.[264-269] Due to the high analytical demand, development of rapid and reliable methods for analysis of drugs-of-abuse is highly desirable. We previously developed a wooden-tip ESI-MS (WT-ESI-MS) method[154] for direct analysis of drugs-of-abuse in urine and oral fluid.[153] Our results demonstrated that

WT-ESI-MS was an efficient method for rapid detection and quantitation of ketamine and its metabolite norketamine in urine and oral fluid. However, in general, the performance of WT- ESI-MS for analysis of the drug in body fluids was not as good as those obtained by conventional methods, mainly due to the interferences and signal suppression of the co-existing matrix and other compounds during the ionization process. Replacing the wooden tip with a SPME fibre, which can selectively extract and enrich drug analytes and rule out interference compounds, for sampling and ionization is expected to improve the performance for drug analysis.

In this study, we explored the direct coupling of SPME with ESI-MS (termed SPME-ESI-MS), and applied this technique for analysis of ketamine in urine. The commercial SPME fibres were directly mounted onto a common nano-ESI ion source without any hardware modification, and desorption and elution solvent was introduced by a spray emitter for SPME-ESI-MS analysis (see Figure 7-1). By employing a C_{18}/benzenesulfonic acid coated SPME fibre, SPME-ESI-MS showed analytical performance much better than that of WT-ESI-MS and even better than that of most conventional methods. This study demonstrated that SPME could be conveniently coupled with ESI-MS for rapid and sensitive detection of analytes in raw samples.

7.2 Experimental section

7.2.1 Reagents and materials

Ketamine and internal standard d_4-ketamine were purchased from Alfasan

(Woerden, Holland). Methanol and acetonitrile were the products of Tedia (Fairfield, OH, USA). Other chemicals were purchased from Sigma-Aldrich (St. Louis, MO, USA). Five types of SPME fibres tested in this study are the products of Supelco (Bellefonte, PA, USA). Three of them are fused silica fibres with PDMS (polydimethylsiloxane, 100 μm), PDMS/DVB (polydimethylsiloxane/ divinylbenzene, 60 μm), and polyacrylate (85 μm) as the coating, respectively. The other two are silica metal alloy with C_{18} (octadecyl, 50 μm) and mixed mode (C_{18}/ benzenesulfonic acid, 45 μm)[252] as the coating, respectively. The urine samples were collected from healthy volunteers and used without dilution. A urine sample spiked with 1 μg/ml ketamine was used to optimize the analytical procedures. Wooden toothpicks (BestBuy brand) were purchased from the PARKnSHOP supermarket in Hong Kong.

7.2.2 Mass spectrometry

The qualitative analysis of ketamine was carried on a Q-TOF2 mass spectrometer (Waters, Milford, MA, USA), while the quantitative data were acquired with multiple reaction monitoring (MRM) mode on a triple quadrupole mass spectrometer (Waters Quattro Ultima, Milford, MA, USA). The high voltage applied onto the SPME fibres for ionization was 3.5 kV; Cone voltage was 30 V; Source temperature was set to 80°C. For the MRM experiments, ion pairs *m/z* 238 > *m/z* 125 and *m/z* 242> *m/z* 129, for ketamine and d_4-ketamine (internal standard) respectively, were chosen for quantitation as in the previous study.[10] The collision energy was 25 eV, and the dwelling time was set at 0.2 second.

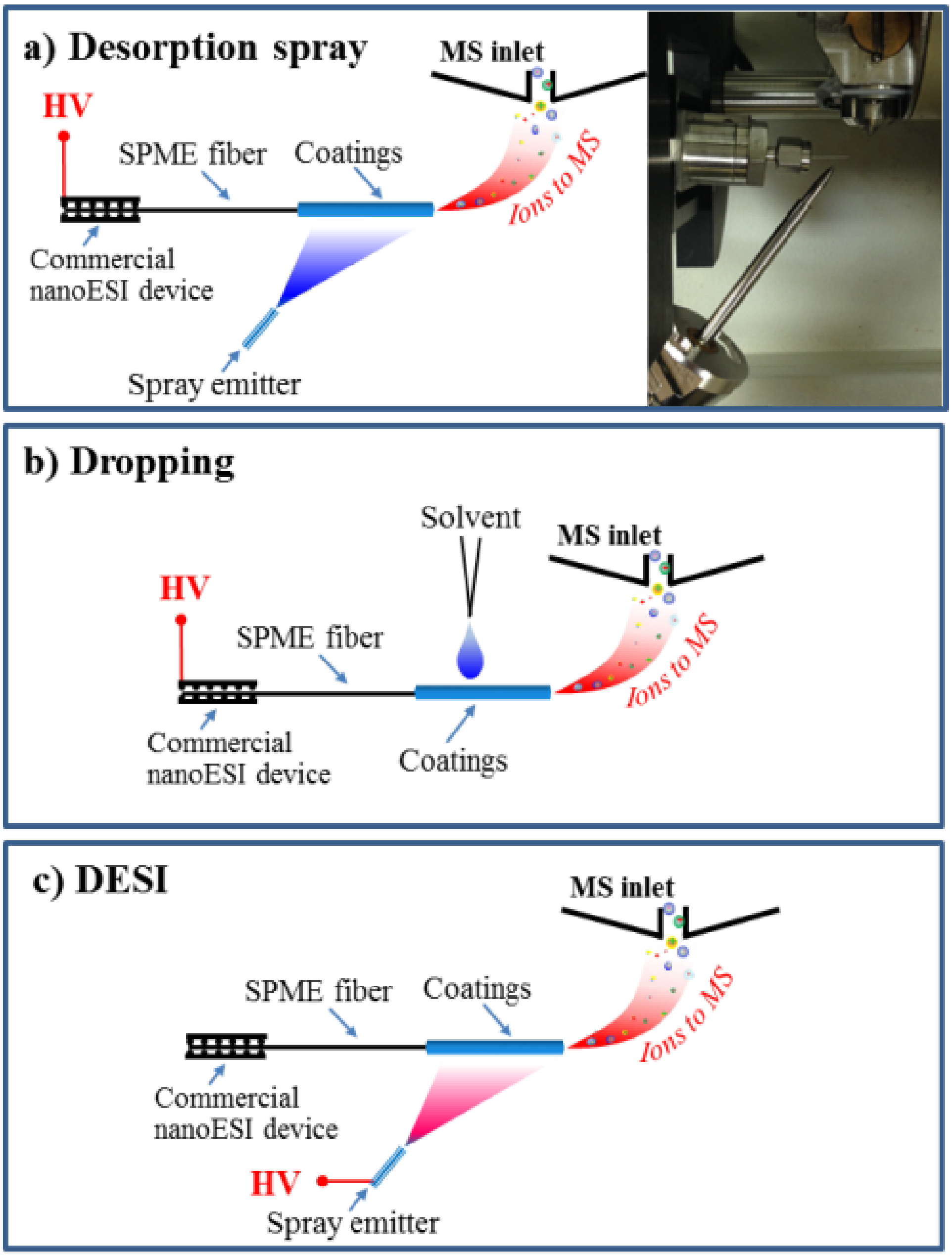

Figure 7-1 Schematic diagram of SPME-ESI-MS with different settings for applying elution and ionization solvent: a) desorption spray (with a top view photo), b) dripping, and c) DESI.

7.2.3 SPME-ESI-MS

The whole procedure of SPME-ESI-MS analysis, including cleaning/conditioning, sampling/extraction, washing, and elution/ionization, is shown in Figure 7-2.

Briefly speaking, the SPME fibres were immersed into 1 mL solvent (water/methanol, 1/1, v/v) for 5 min with vortex to remove the residue compounds and condition the solid phase coating. Then, the SPME fibres were immersed into 300 μL urine sample with 10 minutes vortex for the extraction. After washing in pure water for 50 seconds, the SPME fibres were directly mounted onto the nano-ESI ion source for detection.

As shown in Figure 7-1, the setup of SPME-ESI-MS was simple and convenient, without any hardware modification. SPME fibres could be amounted onto a commercial nano-ESI ion source (Waters, Milford, MA, USA) by an inner screw and with the aid of a conductive elastomer, similar to the setup of nano-ESI needles for nano-ESI analysis. SPME fibre's direction and distance from the MS inlet could be readily adjusted by using the settings of the nano-ESI ion source. Three different approaches, desorption spray, dripping and desorption electrospray (DESI), were attempted for desorption and ionization of the retained analytes. As shown in Figure 1a, in the Desorption Spray approach, a high voltage was applied onto the SPME fibre and desorption solvent (1/1/0.001 for water/methanol/formic acid, v/v/v) was sprayed out from a spray emitter towards the fibre. The distance between the SPME fibre and MS inlet (d1) was 6 mm and the angle between them was 90°. The spray emitter was placed 3 mm (d2) away from the SPME fibre with an angle of 45°. The flow rate of desorption solvent was 5.0 μL/min and the nitrogen gas pressure rate was 10 psi. In the Dripping approach (see Figure 7-1b), desorption solvent was directly dripped onto the SPME fibre by a pipette. In the DESI approach (see Figure 7-1c), all the settings were identical with those of the Desorption Spray approach, except that the high voltage was applied onto the spray emitter instead of the SPME fibre.

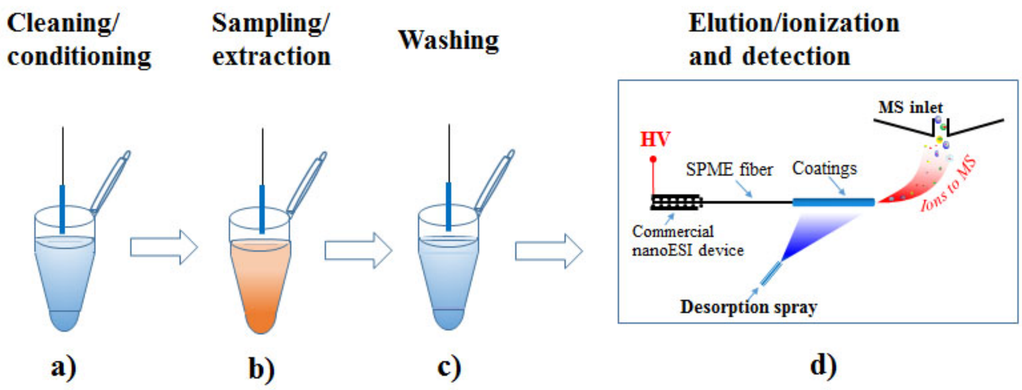

Figure 7-2 Procedures for SPME-ESI-MS analysis: a) cleaning/ conditioning, b) sampling/ extraction,c) washing, and d) elution/ ionization and detection.

7.2.4 WT-ESI-MS

The procedures of WT-ESI-MS analysis were similar to those of the SPME-ESI-MS analysis, except using wooden toothpicks instead of SPME fibres. Mass spectrometric settings of WT-ESI-MS were similar to those in the previous study.[10]

7.3 Results and discussion

7.3.1 Optimization of SPME-ESI-MS

Figure 7-3a is a typical ESI-MS spectrum of ketamine obtained by SPME-ESI-MS analysis of 1 μg/ml ketamine aqueous solution. Ketamine was observed as the protonated molecule at *m/z* 238 in the spectrum. In this study, a urine sample spiked with 1.0μg/mlketamine was used to optimize the conditions and establish the analytical protocol. Different types of SPME fibres and various experimental settings were attempted and their influences to the ketamine signals were investigated and evaluated.

Five commercially available SPME fibres with different supporting materials (fused silica and silica metal alloy) and coating materials of different polarities (PDMS, PDMS/DVB, polyacrylate, C_{18} and C_{18}/benzenesulfonic acid) were used to analyze the urine sample and their performances were shown in Figure 7-4a. For the three fibres coated with PDMS, PDMS/DVB and polyacrylate, poor signal intensities of ketamine were obtained. In addition to the nonpolar properties of the coating materials that might not effectively extract ketamine from the urine samples, the poor performance of these fibres was believed to mainly arise from the poor electrical conductivity of fused silica, the supporting material of these fibres. It was found that electrospray fume was difficult to generate from these tips even if higher voltages were applied. For the C_{18}-coated and C_{18}/benzenesulfonic acid-coated SPME fibres that used electrically conductive silica metal alloy as the supporting material, electrospray could be readily generated and good ketamine signals were obtained. Particularly, much higherketamine signals were obtained using the C_{18}/benzenesulfonic acid-coatedmixed-mode fibre than the C_{18}-coated fibre. The mixed-mode fibre contained C_{18} and benzenesulfonic acid on the fibre surface and could have both hydrophobic and electrostatic interactions with ketamine that contains phenyl and amino groups. This was believed to contribute to its better extraction and enrichment with ketamine in urine, compared to the C_{18} fibre that could have only hydrophobic interaction with ketamine. The above results indicated that for direct coupling with ESI-MS and selective detection of target analytes in raw samples, SPME fibres with electrically conductive supporting materials and suitable coating material should be used. In this study, the mixed-mode SPME fibre was selected for the further experiments.

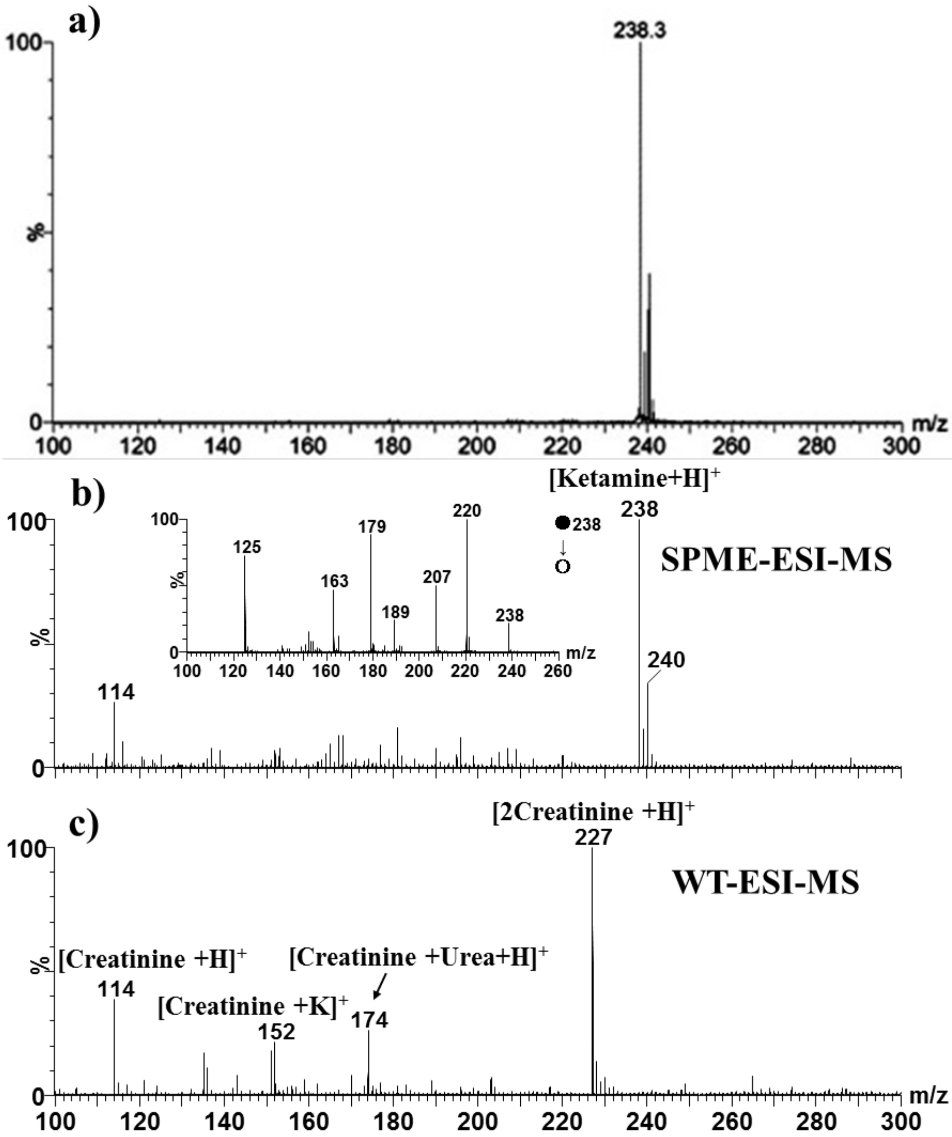

Figure 7-3 Mass spectra obtained by a)analysis of 1 μg/ml ketamine aqueous solution with SPME-ESI-MS, analysis of a urine sample spiked with 100 ng/ml ketamine using b) SPME-ESI-MS and c) WT-ESI-MS. The insert is the MS/MS spectrum of ion at *m/z* 238.

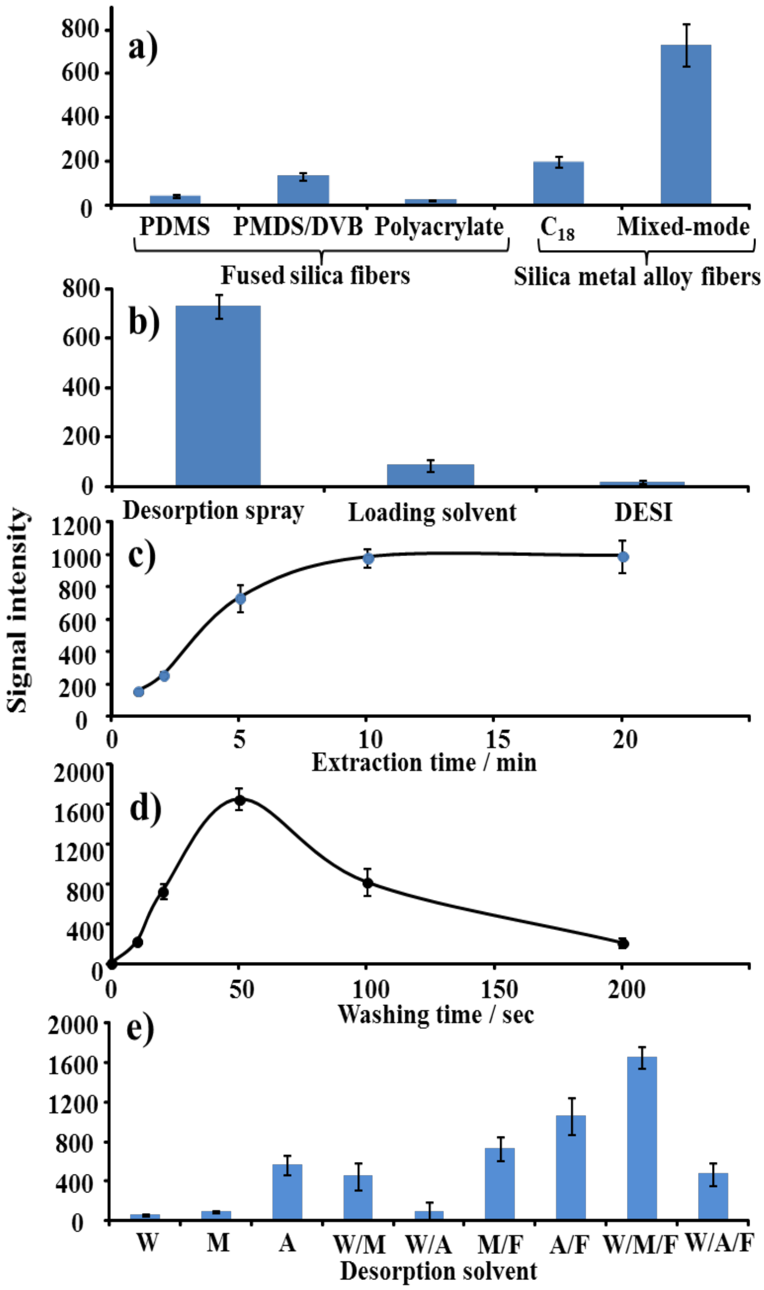

Figure 7-4 Effects of various factors on SPME-ESI-MS signal of ketamine: (a) SPME fibre, (b) application approach for elution and ionization solvent, (c) extraction time, (d) washing time, and (e) desorption solvents (W: Water, M: Methanol, A: Acetonitrile, W/M: Water/Methanol (1/1, v/v), W/A: Water/Acetonitrile (1/ 1, v/v), M/F: Methanol/Formic acid (1/0.001, v/v), A/F: Acetonitrile/Formic acid (1/0.001, v/v), W/M/F: Water/Methanol/Formic acid (1/1/0.001, v/v/v), and W/A/F: Water/Acetonitrile/Formic acid (1/1/0.001, v/v/v)). Each data point was the average of three individual measurements, where the error bar represented the standard deviation.

Three approaches (see Figure 7-1), i.e., desorption spray, dripping, and DESI, as described in the experimental section, were attempted for elution and ionization of the analytes retained on the SPME fibres with water/methanol/ formic acid (1/1/0.001, v/ v/ v) as the elution and ionization solvent. Desorption Spray showed much better performance than the other two in terms of the signal intensity (see Figure 7-4b) and signal duration. For the dripping approach that could be conveniently used, the solvent was added manually using a pipette and typically only a limited volume (typically 1 μL) of solvent could be held by the fibre for analyte elution and ionization, leading to poorer signal intensities and duration. For the DESI approach, charged droplets were generated and sprayed onto the surface of the SPME fibre for the elution and ionization of the retained analytes; the results revealed that desorption and ionization efficiency was quite low with this approach. In the Desorption Spray approach, the solvent was spray out onto the SPME fibre surface in a continuous and uniform manner, which provided sufficient solvent volume and duration for analyte elution, and the high voltage was directly applied onto the fibre to more efficiently induce spray ionization, leading to good signal response and signal duration. The Desorption Spray approach was thus employed for the SPME-ESI-MS analysis.

Extraction time, washing time as well as elution and ionization solvent were also optimized for SPME-ESI-MS analysis of ketamine in urine using the mixed-mode fibre. As shown in Figures 7-4c and d, the extraction was found to reach equilibrium at 10 minutes, and the optimal washing time was determined as 50 seconds with water as the washing solvent for removing polar interference compounds. Various solvents were attempted to elute and ionize the ketamine analyte retained on the SPME fibre, and water/methanol/formic acid (1/1/0.001, v/v/v) was found to be the most efficient one as indicated by the detected ketamine signals (Figure 7-4e).

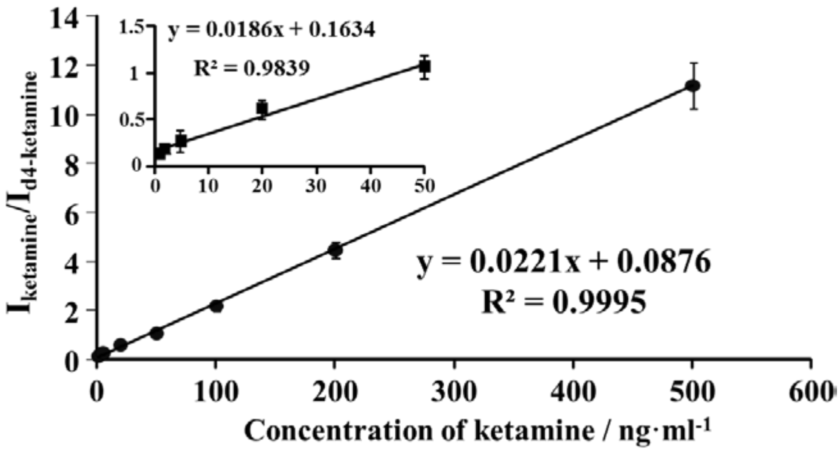

Figure 7-5 The calibration curve for ketamine in urine. Samples analyzed contained different concentrations of the analyte (1.0, 2.0, 5.0, 20.0, 50.0, 100.0, 200.0 and 500.0 ng/ml) and a fixed amount of internal standard (50.0 ng/ml). The insert plot was the calibration curve in low concentration range (1.0-50.0 ng/ml). Each data point was the average of three individual measurements, where the error bar represented the standard deviation.

7.3.2 Performance for analysis of ketamine in urine using SPME-ESI-MS

Figure 7-5b is a spectrum obtained by analyzing a urine sample spiked with 100 ng/ml ketamine with SPME-ESI-MS using the optimized conditions as shown above. Protonated molecule of ketamine, confirmed by the mass and the insert MS/MS spectrum,[10] was predominately observed in the spectrum with very few interference signals. The same urine sample was also analyzed using WT-ESI-MS for comparison. As shown in Figure 7-5c, ketamine signals could hardly be observed in the mass spectrum obtained with WT-ESI-MS, and the spectrum was predominated with peaks corresponding to $[2\text{Creatinine} + H]^+$ and $[\text{Creatinine} + \text{Urea} + H]^+$, revealing that these polar interference compounds were retained on the hydrophilic surface of wooden toothpicks and suppressed the signals of ketamine. In the previous study with WT-ESI-MS, ketamine in urine could only be detected using MRM mode.[10] The above results indicated that the SPME-ESI-MS protocol could efficiently enrich ketamine in urine, significantly reduce matrix effect and enhance the signal of ketamine.

The limit of detection (LOD) and limit of quantitation (LOQ) of the SPME-ESI-MS method for analysis of ketamine in urine were measured. The signal/noise ratio (S/N) was obtained by comparing the intensity of the "spiked" sample to that of the "blank" one, i.e., $(I_{analyte}/I_{IS})_{spiked}/(I_{analyte}/I_{IS})_{blank}$, where $I_{analyte}$ and I_{IS} were the MRM chromatogram intensities (peak areas) of analyte and internal standard (i.e., deuterated ketamine), respectively, and the subscripts "blank" and "spiked" represented the samples spiked with the internal standard only and spiked with both analyte and internal standard, respectively. As shown in Table 7-1, the LOD (S/N=3) and LOQ (S/N=10) of the present method for detection of ketamine in urine were 0.1 and 1 ng/ml, respectively. These results indicated that the LOD and LOQ of SPME-ESI-MS for detection of ketamine in urine could be at least 200 folds better than those obtained with WT-ESI-MS (LOD of 20 ng/ml and LOQ of 50 ng/ml)[10] and even better than those obtained with most conventional methods (LOD of 0.5-25 ng/ml and LOQ of 1.5-50 ng/ml for using GC-MS,[268,270, 271] and LOD of 0.03-5 ng/ml and LOQ of 0.1-3.17 ng/ml for using LC-MS[272-274]).

Table 7-1 Response of ketamine at low concentrations in urine by SPME-ESI-MS

Concentration / ng/ml	S/N (n=3)
0.1	3.9 ± 0.7
0.2	7.2 ± 1.0
0.5	8.9 ± 0.9
1.0	17.4 ± 1.4
2.0	45.7 ± 4.2

The calibration curve for quantitation was constructed with eight sets of experimental data. Each set of data was obtained by detecting samples containing different levels of the analyte (1.0, 2.0, 5.0, 20.0, 50.0, 100.0, 200.0 and 500.0 ng/ml) and a fixed amount of internal standard (50.0 ng/ml). As shown in Figure 7-5, excellent linearity (R^2=0.9995) was obtained over the concentration range from 1.0 to 500.0 ng/ml. In low concentration range (1.0-50.0 ng/ml), satisfactory linearity for ketamine was also obtained with a linear coefficient of R^2 = 0.9839. Such low concentration could not be achieved for quantitation with the WT-ESI-MS method[153] and the conventional methods.[268, 270-272, 274]

The precision and accuracy of the SPME-ESI-MS approach were evaluated by quantitation of ketamine in urine samples spiked with ketamine of low, middle and high concentrations. Each sample was analyzed five times and the data were averaged for comparison. For 4.0-400.0 ng/ml spiked amounts, the recoveries were in the range of 91.7-109.9 % with RSD 5.9-7.7 % (Table 7-2). These data revealed that accuracy and precision obtained by the SPME-ESI-MS was well acceptable for sample analysis.

Table 7-2 Precision and accuracy for detection of ketamine in urine by SPME-ESI-MS

Spiked amount (ng/ml)	Detected amount (ng/ml)	Recovery (%)	RSD (%, n=5)
4.0	4.4 ± 0.3	109.1	7.7
40.0	44.0 ± 3.3	109.9	8.3
400.0	366.9 ± 23.7	91.7	5.9

Typically, the whole procedure of SPME-ESI-MS analysis took around 18 minutes for analyzing one single sample, and this could be further reduced to around 13 minutes with pre-conditioned SPME fibres. This analysis time is much shorter than the analysis time of several hours per sample required with the conventional methods, although it is still longer than the analysis time of 1 minute per sample with the WT-ESI-MS method.[154] SPME technique could be applied in a high-throughput manner to accelerate the analysis speed. For example, by using a simple 4×4 matrix automated extraction system, the average analytical time per sample with SPME-ESI-MS could be reduced to a couple of minutes, which would be particularly useful for analysis of large batch of samples. Another notable feature of SPME fibres is their re-usability. Our experimental results showed that a SPME fibre could be reused over 200 times with no obvious degradation of performance. This allows much reduced costs and good reproducibility of results since the same fibre can be used for the experiments.

7.4 Summary

Direct coupling of SPME with ESI-MS has been developed in this study. The setup of SPME-ESI-MS was optimized and Desorption Spray was employed to provide extensive and durable signals of attained analytes. Using detection of ketamine in urine as an example, we demonstrated that SPME-ESI-MS could provide analytical performance much better than wooden-tip ESI-MS and better than most conventional methods. These results indicated that SPME-ESI-MS could be a simple and efficient method for rapid, sensitive, and selective detection of analytes in mixture samples. Given the popularity of both SPME and ESI-MS and the easy setup of the coupling technique, SPME-ESI-MS is expected to have great potentials for sample analysis in various areas.

Chapter 8: Facilitated Sample Analysis with Field-induced ESI-MS

8.1 Introduction

Recently, development in ambient ionization techniques, e.g., DESI,[37] DART,[228] and other techniques[69, 73, 275, 276] has been introduced for direct sample analysis with little or no sample preparation, greatly facilitating the sample analysis with mass spectrometry. More recently, the high-voltage free ionization techniques have attracted great attention duo to their unique features such as sonic-spray ionization (SSI),[120] easy ambient sonic-spray ionization (EASI),[121] matrix assisted ionization vacuum (MAIV),[122] and the zero-volt paper spray ionization.[123] Interestingly, those high-voltage free ionization are suggested to similar the ESI for formation of ions.

ESI was first introduced for soft ionization of biomacromolecules in the late 1980s by Fenn et al [277] and has been widely used in MS analysis in various fields.[278, 279] In general, ESI is generated by the application of a high voltage directly on a capillary emitter where sample solution sprays out to form a Taylor cone and produce fine droplets and subsequent gas-phase ions for mass spectrometric analysis. Moreover, there is new development in ESI on non-capillary emitter for sample loading and ionization such as metal needle,[89] paper,[90] wooden tip,[91] and aluminum foil[92] for mass spectrometric analysis of complex samples with little or no sample pre-treatment. Alternatively, an opposite high voltage can also be applied on the MS inlet for creation of a strong electric field between the emitter (which is grounded) and the MS inlet to induce the ESI for mass spectrometric analysis.[114] In general, ambient ionization techniques based on ESI mechanism with direct sample analysis strategy can be operated with field-induced mode, and a few methods have been recently reported. In our previous work, field-induced ESI was reported for *in vivo* and real-time monitoring of secondary metabolites of living organisms,[280] by placing a sample or sample-preloaded substrate (generally grounded) in front of the MS inlet and applying of an opposite high voltage on the MS inlet to induce ESI for mass spectrometric analysis. A similar technique polarization induced ESI-MS[281] was also reported for direct analysis of liquid droplet under ambient condition.

In this study, we furthered the previous studies[280] and demonstrated direct ionization ambient methods operated with field-induced mode systematically and comprehensively. Several representative ambient ionization methods based on the ESI mechanism with analytical strategy of direct sample analysis were applied for field-induced mass spectrometric analysis, and their analytical strategies and applications were discussed.

8.2 Experimental section

8.2.1 Reagent, materials and sample preparation

Trptophan, dimethoate, reserpine, dithylhexly phalste, and lysozyme were purchased from Sigma (St. Louis, MO, USA); Val-His-Leu-Thr-Pro was purchased from Bachem Bioscience Inc (King of Prussia, PA, USA). Ketamine was purchased from Alfasan (Woerden, Holland).All standard samples were prepared in 1.0 mg/L in water/methanol (1/1, v/v) solution. Methanol was purchased from Tedia (Fairfield, OH, USA). Amoxicillin capsule were purchased from The United Laboratories Ltd (Hong Kong). Salbutamol aerosol and Terbinafine hydrochloride cream were purchased from Sino-American Tianjin Smith Kline & French Laboratories Ltd (Tianjin, China). Aluminium foil was purchased from Reynolds Wrap (Vernon, IL, USA). All living organisms, biological tissue, and herbal medicines were supplied by College of Pharmacy, Jinan University (Guangzhou, China). The surfaces of living organism were washed with MilliQ water before use. All living plant were pot plants (30 ~ 70 cmhigh). Glass capillaries (ID: 0.5 mm; OD: 1.0 mm; length: 10 mm) were purchased from Huaxi Medical University Instrument Factory (Chengdu, China). Metal capillaries (ID: 0.4 mm; OD: 0.8 mm; length: 20 mm) were purchased from Samk Wang Ind Corp (Korea). Wooden tips (BESTbuy brand toothpicks) were purchased from PARKnSHOP supermarket (Hong Kong). Filter paper was purchased from Macherey-Nagel Gmbh & Co. Kg (Düren, Germany). Metal needles were purchased from Tongyong Office Co. Ltd. (Shanghai, China), the needle tip size was modified to 0.1 mm.

For preparation of the living tumorous rats, the freeze-stored H22 cells were inoculated in the abdominal cavity of the rats. After 7 days, the ascites (0.1 mL) of rats were taken out and the cell number was adjusted to 10^7/mL for vaccinating the normal rats at their right armpit subcutaneous spaces. After another 6 days, the tumorous rats were prepared. All rats were anesthetized by intraperitoneal injection of 0.1 mL 10% hydration chlorine aldehyde (m/v) before this study. All experiments on animals have been approved by Jinan University (Guangzhou, China).

8.2.2 Sample loading, sampling and field-induced ESI-MS analysis

For direct sample loading with non-capillary ESI emitters, the samples were loaded on the tip-end by pipetting (liquid and viscous samples) or by weighing paper (powder sample). For the sampling with glass capillary, liquid sample (1.0 mL) was deposited on an Eppendorf tube to introduce with the capillary action under strong electric field to induce spray ionization. For field-induced ESI of dry biological tissue, the biological tissue was cut into a V-shape. All ESI emitters (i.e., wooden tips, paper, needle, Al foil, biological tissue) were mounted on a clip as previous work,[91, 92, 150, 156] and the tip-ends were placed on the MS inlet, and then the spray solvent (methanol) was loaded onto the tip-end to induce spray ionization except for the biological tissue that contained water (i.e. tomato tissue) as spray solvent was not required. For the field-induced direct ionization of living organisms, the living plants were placed in front of the MS, with a leaf pointing to the MS inlet. The spray solvent (5.0 μL methanol) was added on the leaf surface for extraction and ionization of analytes. In addition, the leaf was cut in a sharp tip-end and the solvent (methanol) was added on the leaf tip to induce spray ionization of its internal compositions. For localization analysis of internal compositions of bulk plants (i.e. potato and cactus), glass capillary containing methanol was used to sample the internal tissue. For the field-induced ESI of the surface fluid of fish, the fishtail was closed to the MS inlet, and the surface fluid was induced to create spray ionization without any spray solvent and ground connection. For sampling of the body fluid of fish and the rats, the skin (2.0 mm of depth) was pierced by a metal capillary (syringe needle), and the body fluid was introduced and was sprayed out to the MS inlet for direct analysis.

All mass spectrometric experiments were carried out by using a high resolution TOF mass spectrometer (6210, Agilent Technologies, Santa Clara, CA, USA) unless otherwise specified using a QQQ mass spectrometer (6400, Agilent Technologies, Santa Clara, CA, USA). A high voltage (3.5 kV) was applied to the MS inlet. It should be noted that the setting of high voltage applied to the MS inlet is fixedly designed at some commercial mass spectrometers. The tips such as metal needle, paper tip, wooden tip, pipette tip and biological tissue are grounded, while it is not necessary to connect the bulk sample (i.e., Eppendorf tube, stone, wet biological tissue, and living organism) to the ground. The distance between the tip-end of emitter and the MS inlet is typically 8.0 mm, and a spray solvent could be optionally added on the tip. To perform the normal non-capillary ESI, in which the high voltage was applied to the ESI emitter rather than the MS inlet, a high voltage (3.5 kV) was applied to the ESI emitter by a high-voltage power supply (produced by Tianjin Hengbo High-voltage Power Supply Factory, Tianjin, China), and the voltage of MS inlet was set at 0.

8.3 Results and discussion

8.3.1 Facilitated routine analysis using various materials for sample loading

Various materials such as steel needle, paper tip, and wooden tip, pipette tip, capillary and stone were placed in the front of the MS inlet as the ESI emitter for sample loading and field-induced ionization, respectively (Figure 8-1). It was found that various compounds, including small molecules and protein, are could be induced spray ionization. The reproducibility of these signals (Figure 8-1) were also investigated, providing the acceptable relative standard deviation (RSD) of 9.6~15.7 % (n=5) for these six experiments. Interestingly, noted that ground connection is unnecessary for bulk sample such as the Eppendorf tube containing 1.0 ml liquid sample and the stone (~ 0.5 kg) spiked with the 5.0 μL liquid droplets, probably a current circuit was formed in the conductive big-bulk sample when it was placed close enough to the MS inlet.

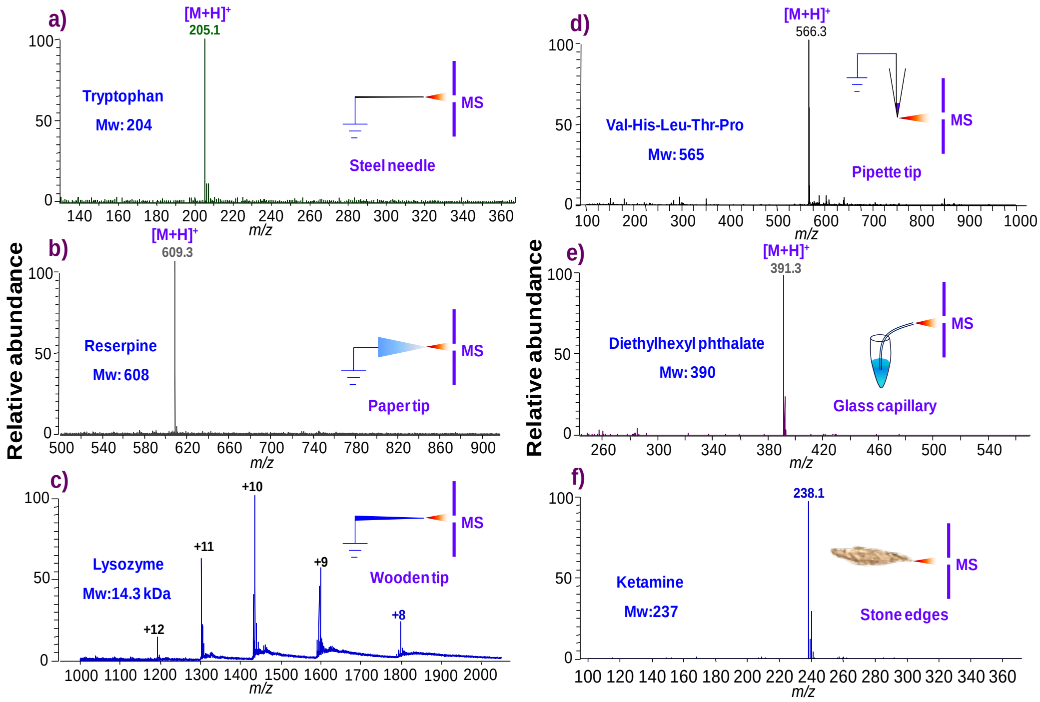

Figure 8-1 Mass spectra obtained from various ESI emitters: a) grounded steel needle with spiked 2.0 μL tryptophan, b) grounded paper with 5.0 μL reserpine, c) grounded wooden tip with 5.0 μL lysozym, d) grounded pipette with 2.0 μL Val-His-Leu-Thr-Pro solution, e) 2.0 μL diethylhexyl phthalate in an Eppendorf tube with a glass capillary, f) an edge of the store with spiked 5.0 μL ketamine. All standard samples were prepared at the concentration of 1.0 μg/mL in water/methanol (1/1, v/v) solution.

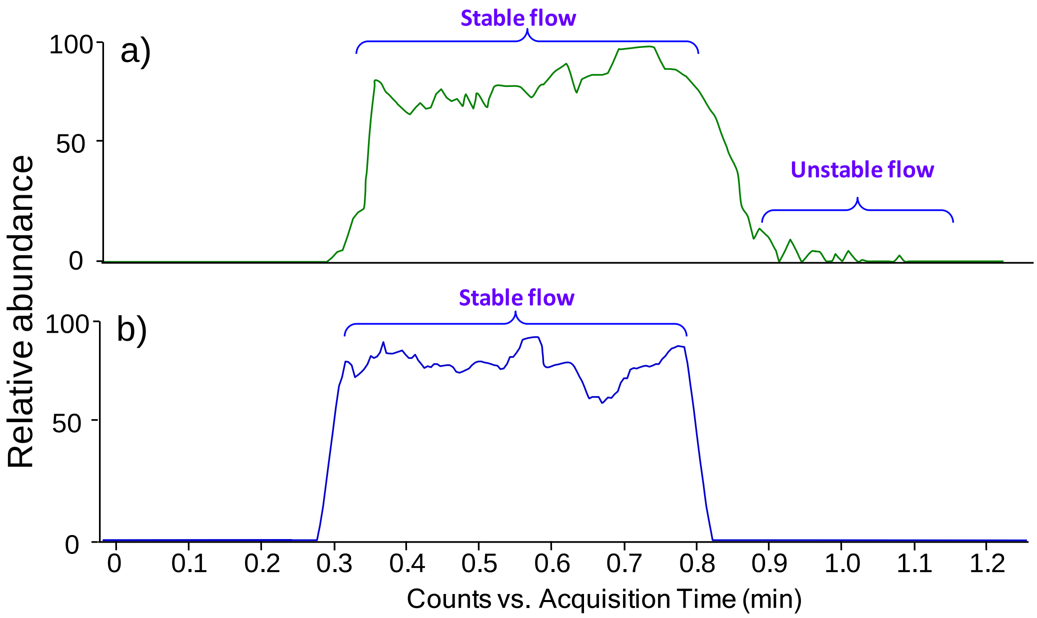

Figure 8-2 Total ion chromatorgram obtained from a) normal wooden-tip ESI-MS and b) field-induced wooden-tip ESI-MS. All spectra were acquired on a QQQ mass spectrometer.

By comparing the normal wooden-tip ESI (high voltage was applied to wooden tip) with the field-induced wooden-tip ESI, it was found that the ion current gradually was decreased and the flow was unstable when solvent was approached exhausted in normal wooden-tip ESI (Figure 8-2a), which is in good agreement with the previous studies on ESI with porous tip.[91, 149, 205] The observation of unstable ion currents was believed to be due to changes in flow rate.[205] The ion currents on field-induced wooden-tip ESI were reposefully run and declined sharply in the last few seconds (Fig. Figure 8-2b) probably because the Taylor cone could not be formed at the wooden tips (tip size: 0.2 mm) and the electric field (3.5 kV, 0.8 cm) was not strong enough to bring about any field ionization[117, 205] after significant solvent depletion at the few last seconds. The rectangular response of ion current in field mode should be beneficial to the quantitative analysis of analytes using field-induced ESI technique since the quantitation is the remain a remain a challenge by ambient MS.[275]

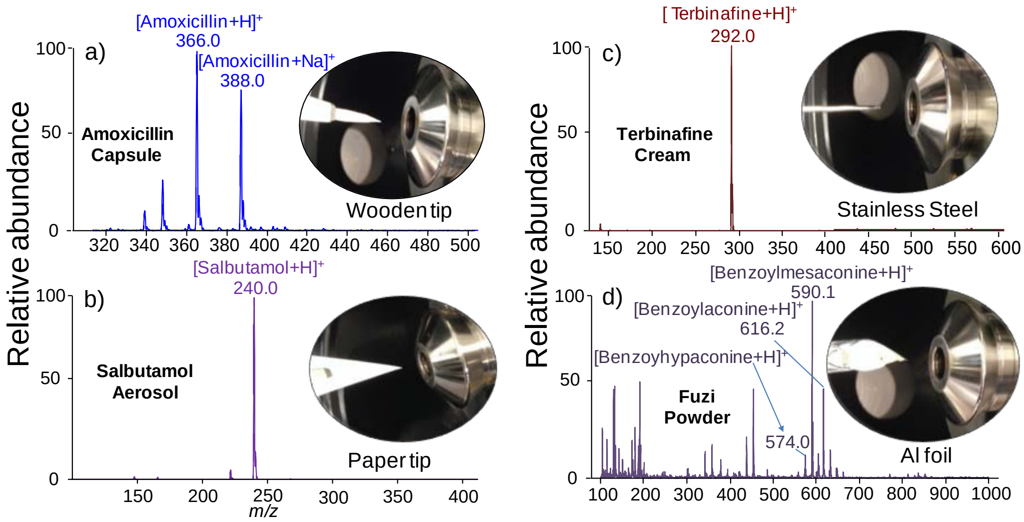

Figure 8-3 Pharmaceutical analysis using field-induced ESI-MS with various ESI emitter: a) wooden tip, b) paper tip, c) steel needle, and d) Al foil; inset show the experimental photographs. All spectra acquired on a QQQ mass spectrometer.

8.3.2 Facilitated complex sample analysis using various materials for sampling

To investigate the applications of field-induced ESI-MS, typical ESI emitters such as paper, wooden tip, steel needle and aluminium foil, which have been well studied in previous work,[89-92] have been demonstrated for pharmaceutical analysis, which is one of the most challenging topics in analytical chemistry.[282, 283]

The first example of analysis of complex sample is a demonstration of analysis of powder samples. The powders of amoxicillin capsule (~ 0.1 mg) were directly sampled by a wet wooden tip,[147] a high voltage (3.5 kV) was applied to the MS inlet and the wooden tip was placed in the front of the far away ~ 8.0 mm (inset of Figure 8-3a), when 5.0 μL of methanol was pipetted onto the wooden tip as spray solvent to induce spray ionization, the mass spectrum obtained is shown in Figure 8-3a. It is clearly shown that the peaks of amoxicillin ions were dominated the spectrum and lower background ions than the analysis of amoxicillin using plasma desorption/ionization MS,[284] showing the sensitive detection of amoxicillin using wooden-tip sampling and ionization.

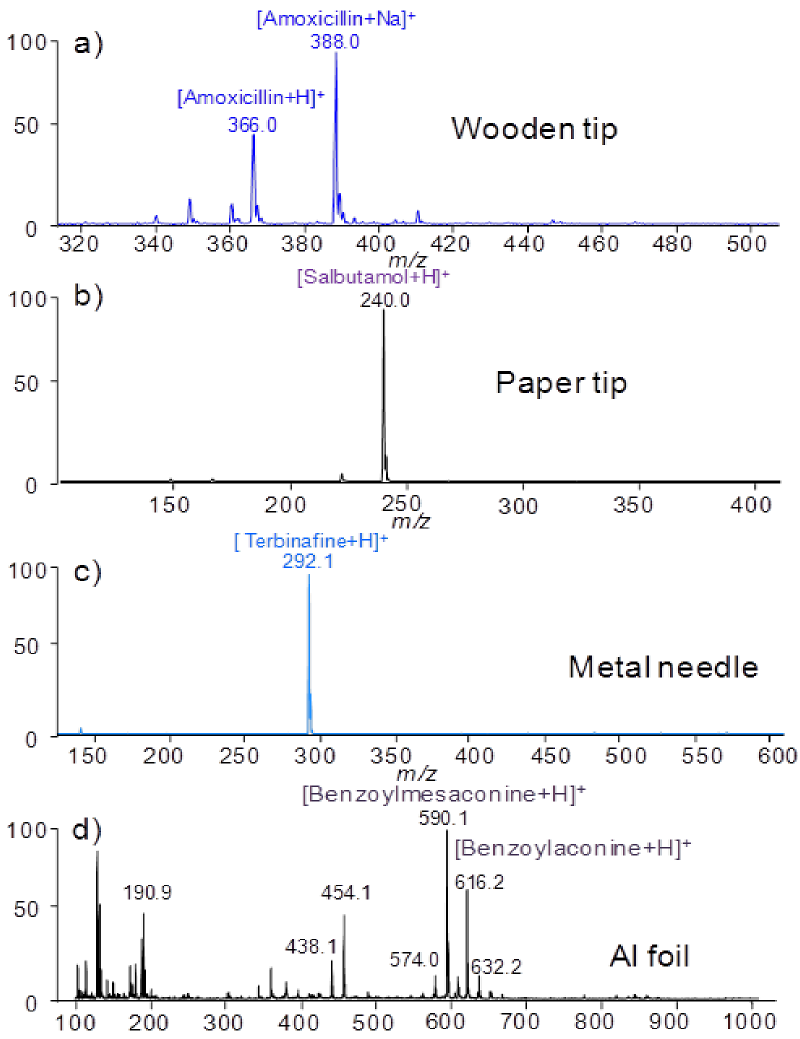

Figure 8-4 Pharmaceutical analysis using normal ESI-MS with various emitters: a) wooden tip, b) paper tip, c) steel needle, and d) Al foil. All spectra were acquired on a QQQ mass spectrometer.

Filter paper was used for development of field-induced paper spray ionization (PSI). In this study, the aerosol drug was firstly analyzed by field-induced PSI-MS.

Salbutamol aerosol (0.1 mg) was sprayed to surface paper triangle (inset of Figure 8-3b), then 5.0 μL of methanol was pipetted onto the paper surface for field-induced ionization. The protonated salbutamol (*m/z* 240) was recorded as the base peak, showing field-induced PSI-MS is rapid, simple and efficient method for analysis of aerosol drug. There was no condensation reaction through sulbutamol and ethanol (i.e., the solvent of sulbutamol) or methanol (i.e., the spray solvent), which reaction was reported in extractive ESI processes. [67, 68]

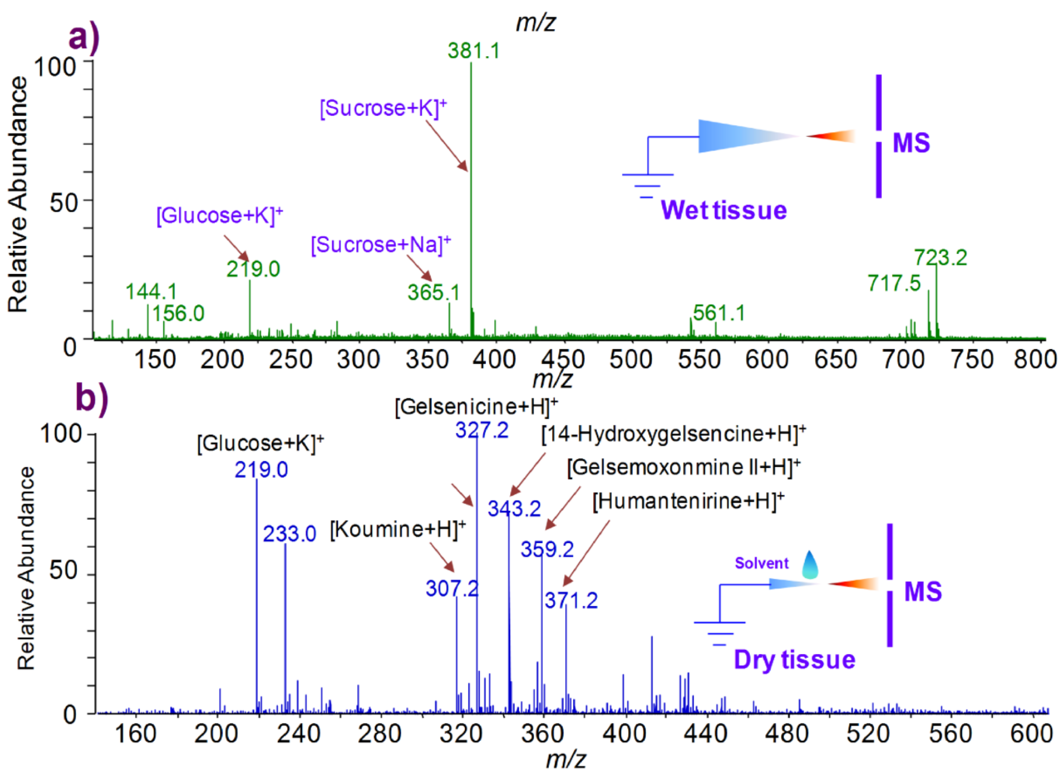

Figure 8-5 Mass spectra obtained from a) tissue of fresh tomato and b) dry leaf of *Gelsmium elegans*.

Another important pharmaceutical analysis is analysis of viscous drugs. The steel needle (as probe) is reliable to sample the viscous ointment (inset of Figure 8-3c); Figure 8-3c displays a field-induced probe ESI mass spectrum obtained from terbinafine cream (~ 0.05 mg). The protonated terbinafine was observed as the base peak, showing the trace amounts of viscous samples could be held by metal needles and analyzed by field-induced probe ESI-MS.

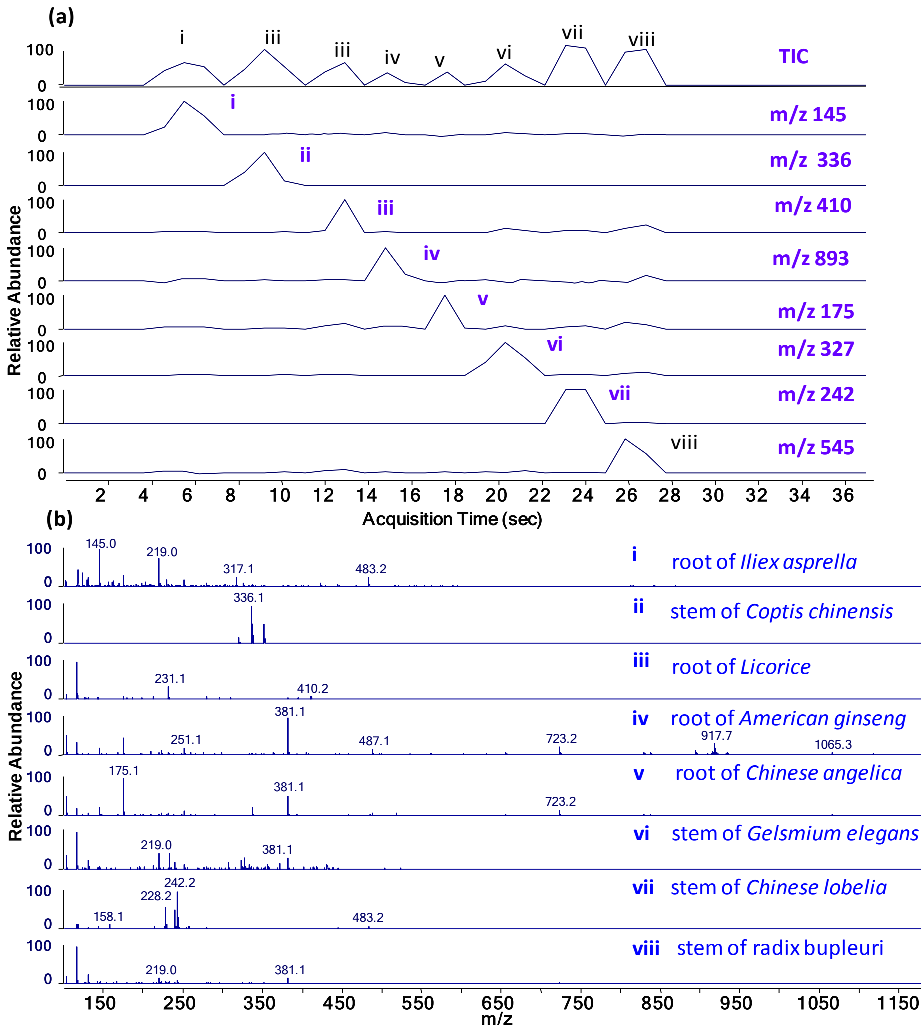

Figure 8-6 High throughput analysis of 8 herbal medicines: a) total ion chromatogram (TIC) and selected ion chromatogram (SIC) of different characteristic compounds detected at (i) *m/z* 145, (ii) *m/z* 336, (iii) *m/z* 410, (iv) 893, (v) *m/z* 175, (vi) *m/z* 327, (vii) *m/z* 242 and (viii) *m/z* 545 from (i) root of*Iliex asprella*; (ii) stem of *Coptis chinensis*; (iii) root of *Licorice*; (iv) root of *American ginseng*; (v) root of *Chinese angelica*; (vi) stem of *Gelsmium elegans*; (vii) stem of *Chinese lobelia*;and (viii) stem of *Radix bupleuri*, respectively; b) mass spectra of 8 hebal medicines.

Finally, experiments were performed to analyze herbal medicine using Al foil for sample loading and ionization (inset of Figure 8-3d). For example, as shown in Figure 8-3d, several alkaloids peaks were observed in the mass spectrum from the Fuzi powder (10.0 mg) by adding 10.0 µL of methanol, shown that field-induced Al-foil ESI-MS suitable for large amount of powder.

It is noted that the spectra of pharmaceutical analysis mentioned above by field-induced ESI-MS are very similar to the spectra obtained by solid-substrate ESI-MS (Figure 8-4), suggesting that the ionization mechanism is same to the conventional ESI. In general, the experimental results demonstrated field-induced solid-substrate ESI greatly facilitate the analysis of real sample, providing the unique features such as the low background, no ion-molecule reaction and compatible various substrates.

8.3.3 Facilitated rapid analysis of biological tissues

Figure 8-5a displays an obtained FI-ESI spectrum of fresh tomato tissue when it was close to the MS inlet without adding solvent, and the peaks at *m/z* 381, 365 and 219 correspond to $[sucrose+K]^+$, $[sucrose+Na]^+$ and $[glucose+K]^+$, respectively. Dry tissue could be directly ionized by adding solvent. For instance, leaf of dry *Glsmium elegans* with added methanol; the ions of alkaloids were observed (Figure 8-5b). Furthermore, a movable metal belt, which was connecting to the ground, was established for high-throughput analysis of herbal medicines. Herbal medicines were directly placed on a grounded belt; all tips of the herbal medicines were horizontally directing to the MS inlet and methanol (5.0 µL) was added for extraction and spray ionization of analytes. Selected ion chromatograms

(SIC) of characteristic ions of each herbal medicine and TIC were observed (Figure 8-6a) when eight herbal medicines were sequentially moved to the MS inlet within 24 sec. The FI-ESI-MS spectrum of each herbal medicine was obtained as well (Figure 8-6b), provided a high-throughput analysis at ~3 sec per sample and no significant carryover effect between two neighboring samples.

8.3.4 Facilitated in vivo analysis of living organisms

8.4.4.1 *In vivo* analysis of surface compositions of living organisms

Rapid detection of pesticide residues in fresh vegetables is of paramount important for public safety.[285, 286] In routine detection of pesticides, methods of sample preparation and separation such as the solid-phase extraction (SPE), liquid-phase extraction (LPE), gas chromatography (GC) and liquid chromatography (LC) are usually used before MS detection.[285, 286] Recently, ambient mass spectrometry techniques including desorption electrospray ionization (DESI),[38] direct analysis in real time (DART),[287] thermal desorption electrospray ionization (TD-ESI),[288] and surface desorption atmospheric chemical ionization (DAPCI)[289] are available for the detection of trace amounts of analyst on plant leaf surface without any sample pretreatment. More recently, direct detection of pesticide residues on leaf surface by leaf spray mass spectrometry was introduced,[290] in which high voltage was directly applied to the leaf to generate spray ionization. Compared with the ionization methods mentioned above, the field-induced ESI allows experiment to be performed with the whole of living plant pot, which makes it attractive for in situ, in vivo analysis of living vegetable samples. Figure 8-7a shows a mass spectrum recorded from an intact living spinach leaf spiked with 5.0 ng of dimethoate. The base peak at *m/z* 230 corresponded to the protonated dimethoate. Dimethoate on the leaf surface was detected only because no damage was done to the leaf surface, providing a simple and rapid method for sensitive detection of substances on the surface of living plants.

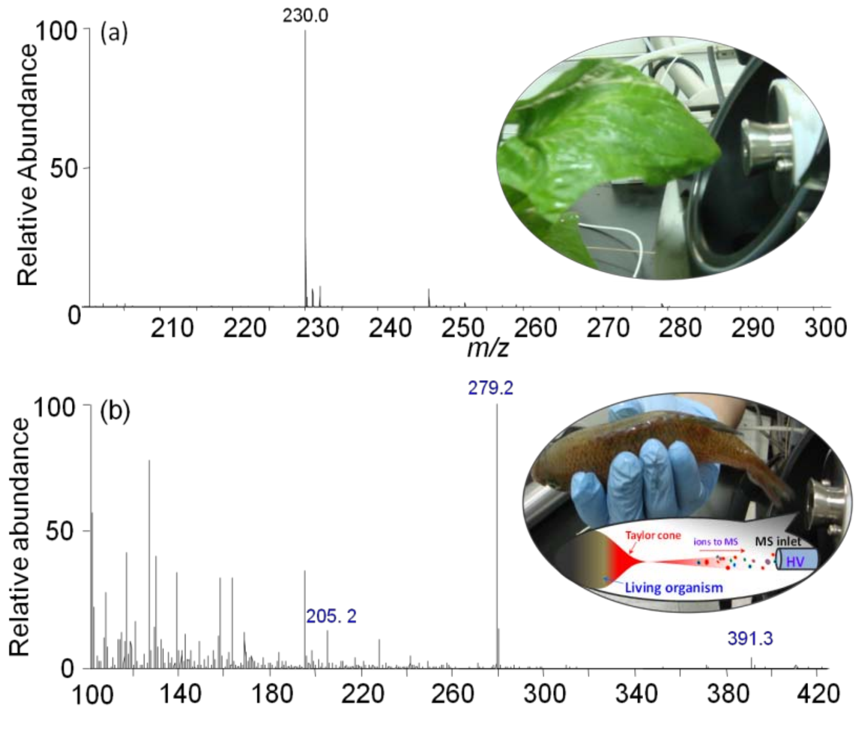

Figure 8-7 *In vivo* analysis of surface compositions of living organisms: a) analysis of leaf surface of spinach with spiked 5.0 μg dimethoate; b) analysis of surface mucus of fish. Insets in a) and b) show the experimental operation of spinach and fish, respectively.

Furthermore, field-induced ESI-MS also allows direct analysis of animal surface. For example, fish skin mucus, which plays an important role in the ecological functions,[291] was investigated for ecological risk assessments of environmental contaminant analyzed by liquid chromatography-mass spectrometry,[292] which analytical process is time-consuming and complicated. In this study, the mass spectrum was immediately obtained (Fig. 8-7b) from skin mucus crucian (*Carassius carassius*), because spray ionization was induced when the tail of fish was close to the MS inlet (inset of Figure 8-7b). The peak at *m/z* 205 corresponded to the tryptophan, which is one of the typical free amino acids of fish mucus.[293] It is not surprising to observe the peaks of dibutyl phthalate (*m/z* 279) and diethylhexyl phthalate (*m/z* 391) in the high resolution mass spectrum due to plasticizers in the body of fish,[294] suggesting that plasticizers not only could be found in the brain, muscle, heart, or kidney of fish,[295-297] but also in mucus due to water pollution in China.[298, 299] There were no obvious side-effects observed in the week after the rapid analysis of the living fish in this study due to the weak current (nA level) and short time (< 1.0 min) of the ionization process. These results demonstrated that the field-induced ESI-MS could be used for rapid analysis of the surface of living organisms, including plants and animals.

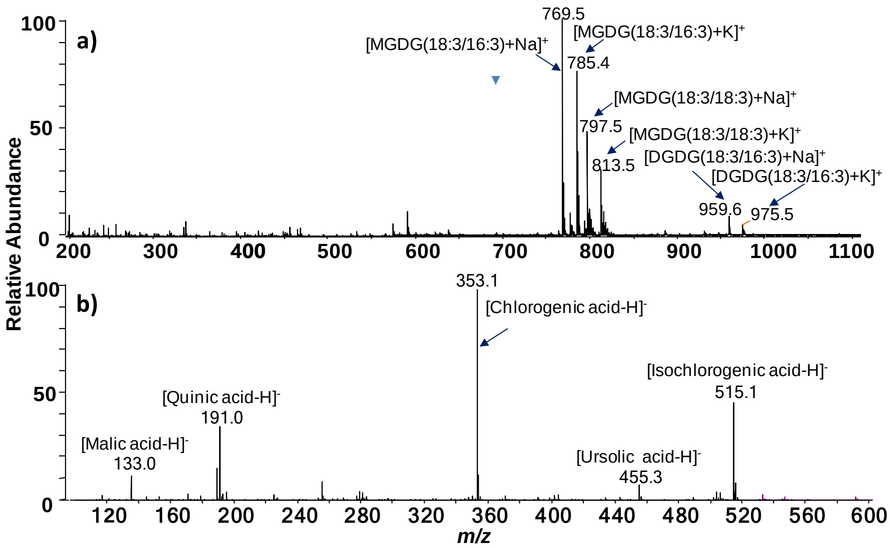

Figure 8-8 Mass spectra obtained from internal compositions of leaves: a) living spinach in positive ion mode, MGDG: monogalactosyl diacylglycerol, DGDG: digalactosyl diacylglycerol; b) living *I. latifolia* in negative ion mode. Methanol was used as the added solvent.

8.4.4.2 *In vivo* analysis of internal compositions of plant leaves

The internal compositions of living plants also could be rapid analyzed. For example, the leaf of living spinach was cut in a V-shape as in the previous work of the direct ionization of fresh spinach leaf.[156] The mass spectrum (Figure 8-8a) obtained was dominated by peaks of monogalactosyl diacylglycerol (MGDG) and digalactosyl diacylglycerol (DGDG) in positive ion detection mode, a spectrum that is similar to the spectrum obtained by analysis of a fresh tissue of spinach leaf.[156] In the negative ion detection mode, a leaf of a living Ilex latifolia was cut into a V-shape and analyzed under field-induced ESI conditions (Figure 8-8b) and deprotonated malic acid (*m/z* 133), quinic acid (*m/z* 191), chlorogenic acid (*m/z* 353), and isochlorogenic acid (*m/z* 515) were observed. Rapid in vivo analysis of various living plants was carried out by using the field-induced ESI-MS (Figure 8-9). Results demonstrated that the field-induced ESI method could be used for rapid analysis of various plants. It is noted that analytical capability could be only performed for living plant pots that can entry to laboratory.

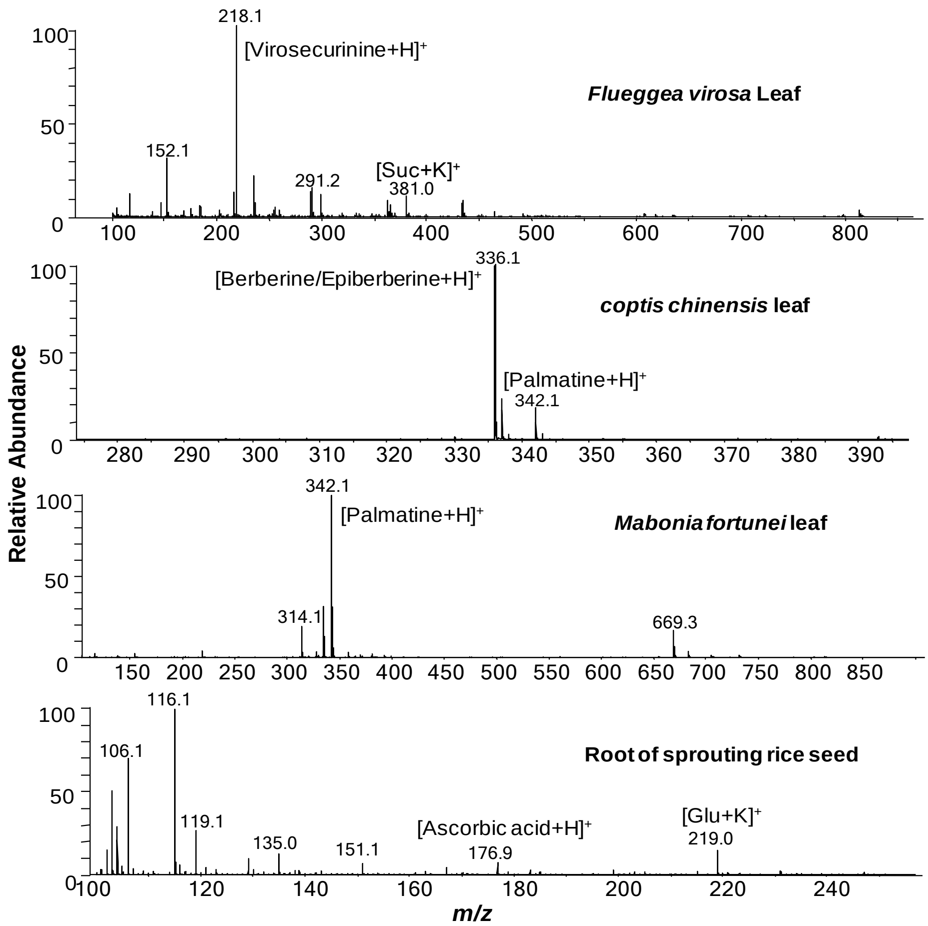

Figure 8-9 Field-induced ESI-MS spectra obtained from various living plants. Methanol was used as the spray solvent.

8.4.4.3 *In vivo* analysis of internal compositions of bulk plants

To sample the bulk plant, a novel sampling and ionization of juicy plant with a capillary was developed by Peng et al,[172] in which the plant was inserted a capillary for introduce of the juicy analyst. However, anther wire is required to connect the plant for connection of high voltage. Therefore, it is hard to avoid the influence of high-voltage stimulation on living plant, especially in real-time monitoring.

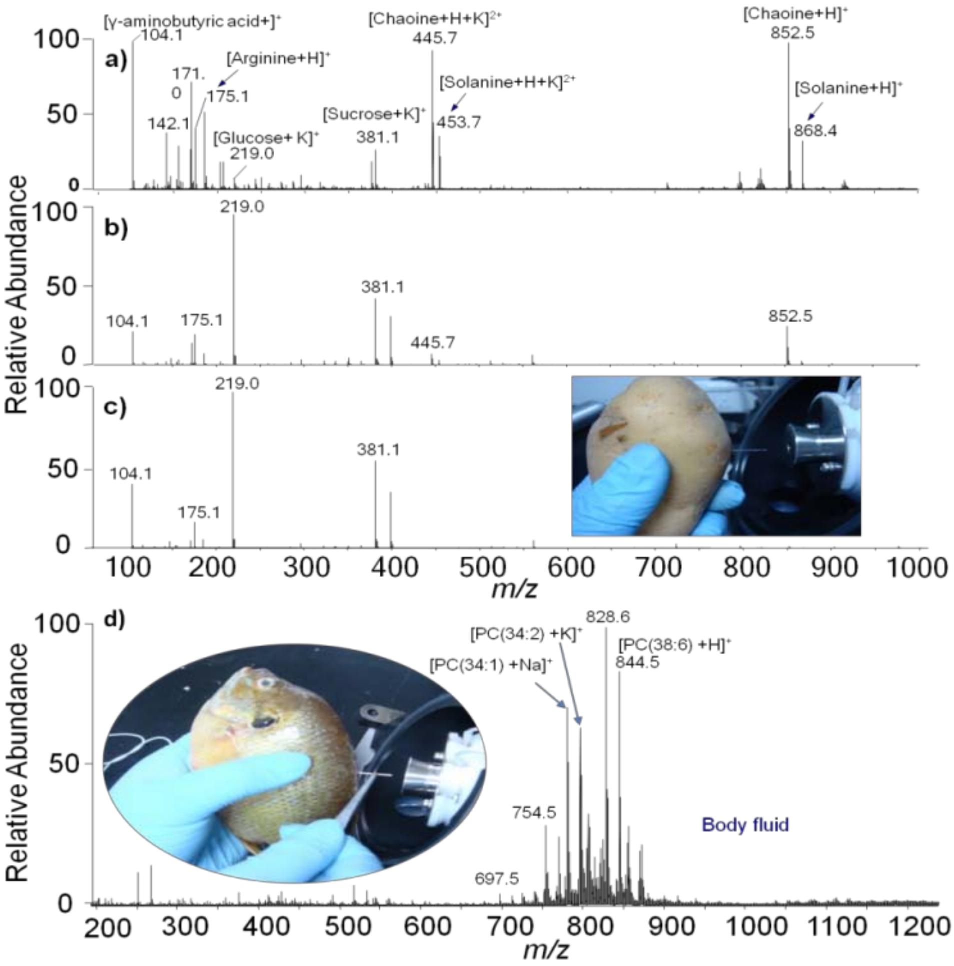

Figure 8-10 a-c) Mass spectra of a sprouting potato tuber obtained by sampling with a glass capillary at the depth of a) 2 mm b) 4 mm and c) 14 mm, respectively; d) Mass spectrum obtained from the fish with capillary sampling, insets of c) and d) show the experimental operation of potato and fish, respectively. PC: phosphatidylcholine.

In this study, field-induced ESI is also useful for location analysis of bulk plants with capillary. As a demonstration of investigation of glycoalkaloids distributions in potato using a capillary sampling (inset of Figure 8-8a), it was found that glycoalkaloids were distributed mainly in the cortex. At a depth of 2.0 mm below the peel of potato in the sprouting area of a sprouted potato tuber, a very high level of glycoalkaloids (solanine, $[M+H]^+$, *m/z* 868; chaconne, $[M+H]^+$, *m/z* 853) was observed (Figure 8-10a). The relative abundance of the two glycoalkaloids were significantly reduced at 4.0 mm below the peel (Figure 8-10b) and completely disappeared at 14.0 mm below the peel (Figure 8-10c). Glycoalkaloids are a natural defense strategy of potato against viral and microbial diseases and enemies,[300] but they are toxic to human.[301, 302] Glycoalkaloids are s also used as pesticide in agriculture and was as medicines to treat some diseases.[303] Therefore, understanding of the depth distributions of glycoalkaloids in potato is benefit for removing it in food industry and collecting it in pharmaceutical industry.

8.4.4.4 *In vivo* analysis of internal compositions of living animals

The body fluid of living animals was also could be introduced with metal capillary (syringe needle) for rapid analysis, which is important in disease diagnosis and metabolism.[304, 305] For example, the metabolic biofluid of living crucian (*Carassius carassius*) was introduced by capillary (inset of Figure 8-10d) for field-induced ESI-MS analysis and the FI-ESI spectrum was rapidly obtained (Figure 8-10d), showing abundant of lipids in crucian biofluid. The lipids species were also obtained from rats by metal-capillary sampling and field-induced ESI-MS analysis. The predominant peaks of normal rat in the spectrum (Figure 8-12a) were at *m/z* 756, 780, 796, and 808, which corresponding to $[SM(30:6)+Na]^+$, $[PC(34:1)+Na]^+$, $[PC(36:8)+Na]^+$, and $[PC(36:2)+Na]^+$, respectively. However, in the mass spectrum of tumorous rat of hepatoma H22 cells, the lipids such as peak at *m/z* 853.7, 869.7, 879.7 and 895.7 were significantly raised, while the lipids such as the peak at *m/z* 796 $[PC(36:8)+Na]^+$ disappeared(Figure 8-11b). The significant changes of lipids compositions were believed to be the rat canceration. This finding could help identification of the hepatoma H22 cells.[306] Lipids have important biological functions in signalling, energy storage and construction of cells, and have been shown as biomarkers in some diseases.[307] This result shown field-induced ESI with capillary sampling is rapid detection of lipids from animals, indicated this method is a powerful tool for seeking biomarkers and thus is expected to analyze drug target in living animals.

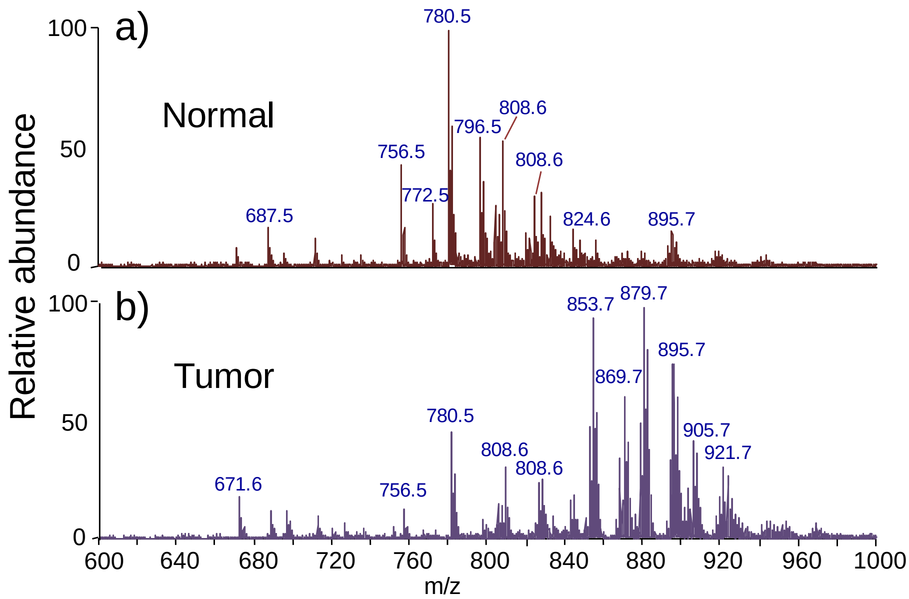

Figure 8-11 Mass spectra obtained from of a) normal rat and b) tumorous rat; the sampling points is at itsright armpit subcutaneous spaces with metal capillary.

8.4 Summary

The high voltage was applied to the MS inlet has been used to induce spray ionization of various solid substrate, which is just placed in the front of the MS inlet. Consequently, a field-induced ESI has been developed for rapid analysis of various samples with litter sample pretreatment. Field-induced ESI source was examined by various solid substrates such as metal needle, paper tip, wooden tip pipette tip, capillary, and stone. And the applications of pharmaceutical analysis have been demonstrated by rapid analysis of powders, aerosol, and cream. High throughput, sensitivity, and specificity analysis has been demonstrated for a wide range of applications, including in situ analysis of biological tissue, rapid analysis of herbal medicines, *in vivo* analysis of plant, location analysis, real-time monitoring, and biomarker discovery. In addition, field-induced ESI allows performance of experiments without any high-voltage connection, sheath gas, or laser beam, the instrument and the operation are significantly simplified.

Chapter 9: *In Vivo* and Real-time Monitoring of Secondary Metabolites of Living Organisms by Field-induced Direct Ionization Mass Spectrometry

9.1 Introduction

Secondary metabolites play important roles in the survival and propagation of animals and plants. For example, some animals and plants produce chemicals to resist environmental stress, and some secret toxic chemicals for defense purpose upon attack. Understanding these processes at the molecular level is an integral part of our exploration of nature. Conventional approaches for detection of secondary metabolites typically involve homogenization and extraction followed by analysis of extracts using techniques such as chromatography coupled with mass spectrometry.[308, 309] These approaches are not only time-consuming and labor-intensive; they are also unable to provide transient and dynamic information about the metabolic processes. Techniques for *in vivo* and real-time detection of secondary metabolites from biological systems are thus highly desirable.

MS is a rapid and sensitive tool for analysis of metabolites from biological samples.[8, 310, 311] The development of ambient ionization techniques in recent years allows direct analysis of samples with little or no sample preparation.[6, 312-314] DESI,[37, 315] DART,[316] and neutral desorption coupled with extractive electrospray ionization (EESI)[317, 318] enable direct detection of analytes on surfaces of living organisms at atmospheric pressure. Endogenous compounds of plant tissue could be directly analyzed using techniques such as direct ionization (DI),[156] leaf spray[105] and tissue spray.[103] In these techniques, tissue sample is cut into V-shape and connected to a high voltage. With addition of some solvents to the tissue sample if necessary, spray ionization is induced from the tip of the plant tissue and a spectrum is obtained. Leaf spray has been attempted to detect chemical response of cabbage leaf upon punching,[105] and DI has also been used for direct analysis of animal tissue.[156] These techniques, however, require connection of samples to a high voltage (typically 3-5 kV), which can be a strong stimulus to living plants and animals, and thus may have limited application in *in vivo* stimulation studies of living organisms.

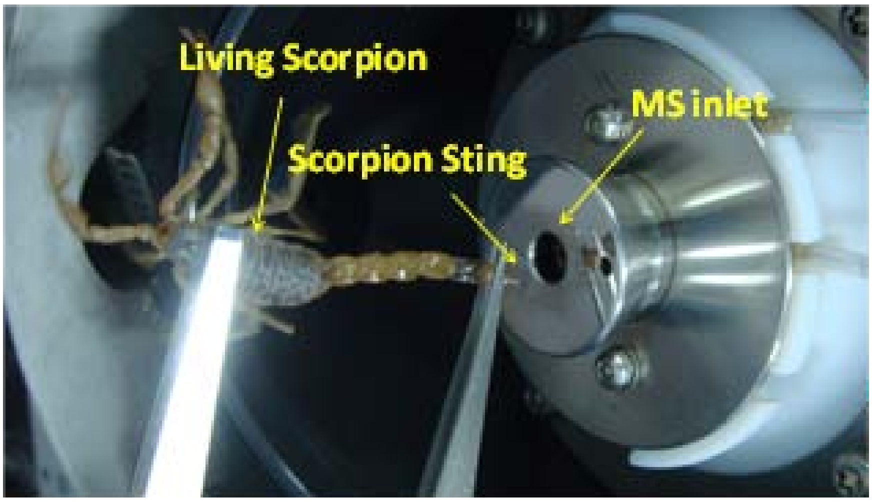

Figure 9-1 Photo of *in vivo* analysis of a living scorpion by field-induced DI-MS.

We herein further develop DI-MS for *in vivo* studies of living organisms. In this study, the high voltage required for inducing ESI is applied to the MS inlet, and the studied animal or plant sample is grounded and placed in front of the MS inlet. To differentiate from the previous DI technique that requires connection of the sample to a high voltage,[156] the current technique is termed field-induced DI in this study. We demonstrated that *in vivo* and real-time detection of secondary metabolites released from both living animals and plants upon stimulation could be easily achieved by using field-induced DI-MS.

9.2 Experimental section

9.2.1 Materials

Living scorpion (Buthus martensii Karsch; origin: Guanzhou, China), toad (*Bufo bufo gargarizans Cantor*; origin: Guanzhou, China), and *Catharanthus roseus* were supplied by College of Pharmacy, JinanUniversity (Guangzhou, China). Glass capillaries (ID: 1.0 mm, length: 10 mm) were purchased from Huaxi Medical University Instrument Factory (Chengdu, China). Methanol was purchased from Tedia (Fairfield, OH, USA). The experiments related to animalswere approved by the Ethical Committee of Jinan University.

Figure 9-2 Procedures for *in vivo* detection of toad's secretion upon stimulation by field inducedDI-MS.

Figure 9-3 Photo of *in vivo* analysis of a leaf of a living *C. rosues* by field induced DI-MS.

9.2.2 Field-induced DI-MS Measurements

Mass spectra were acquired on an Agilent 6210 TOF-MS instrument (Agilent Technologies, Santa Clara, CA, USA) in positive ion mode. The analyzed biological samples were directly placed in front of the MS inlet with a tip pointing to the inlet. Note that when handling living animals such as scorpion and toad, gloves, mask and goggles should be worn for protection purpose. For FI-DI-MS analysis of scorpion, a living scorpion was held by tweezers and its sting was directly used as the spraying tip (Figure 9-1); for field-induced DI-MS analysis of toad, a living toad was held by hand and a glass capillary, which is commonly used for thin layer chromatography (TLC) spotting, was employed as the spraying tip as described in Results and Discussion and shown in Figure 9-2; for field-induced DI-MS analysis of the potted *C. roseus*, the analyzed leaf was held by hand and the spraying tip was formed by cutting the leaf into a small V-shape

(Figure 9-3). The distance between the tip and MS inlet was typically ~ 8 mm. The capillary voltage (Vcap), that is the voltage applied to the MS inlet, was set at –5 kV for positive ion mode. Drying gas was used with a temperature of 100ºC and a flow rate of 4.0 L/min. ESI-MS and MS/MS experiments were performed on a QToF II mass spectrometer (Waters, Milford, MA) for confirmation of peak assignments. Some of the mass spectral results and the experimental details are shown in Figure 9-4 and Figure 9-5.

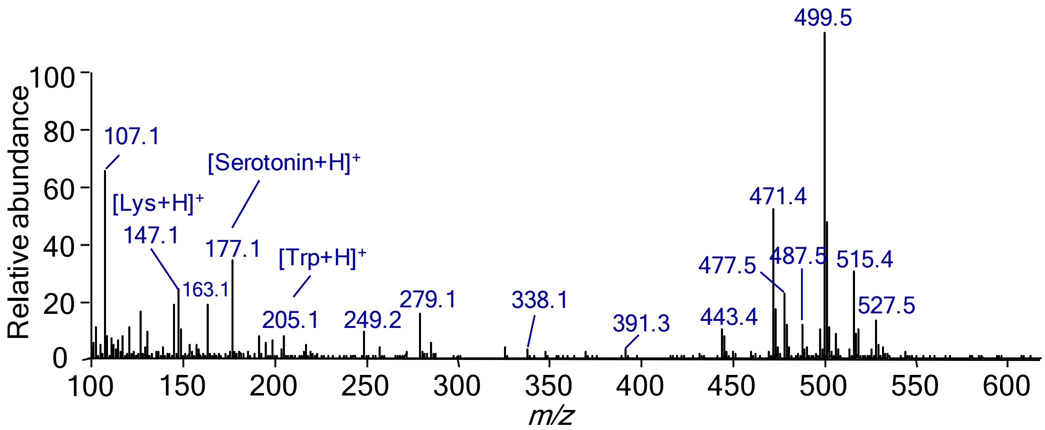

Figure 9-4 Mass spectrum obtained by field-induced DI-MS analysis of the secretion released from a living scorpion upon stimulation.

9.3 Results and discussion

9.3.1 *In vivo* monitoring of scorpion

Scorpion uses its sting on the tail as weapon and can release toxic substances from the sting upon attack. Using field-induced DI, spray ionization can be directly induced from such a sharp sting when the sting is placed close and pointing to the MS inlet, and secretion can be conveniently detected (Figure 9-1). In this study, a living scorpion was stimulated at its tail with a pair of tweezers, clear liquid secretion was soon released from the sting, and subsequently spray ionization was induced from the sting and a mass spectrum was obtained (Figure 9-6). Compound such as lysine (*m/z* 147), tryptophan (*m/z* 205) and serotonin (*m/z* 177) could be identified from the spectrum based on their masses and literature searching.[319-323] Serotonin is one of the toxins in scorpion to produce pain,[322] and lysine and tryptophan have been reported to play important roles in the biological activity of the scorpion toxin.[320] Peaks of high abundances were observed in the range of *m/z* 400-600. These peaks have a mass interval of 28 Da (e.g., *m/z* 443.4, 471.4, 499.5 and 527.5) or 16 Da (e.g., *m/z* 471.4 and 487.5, and *m/z* 499.5 and 515.4). No masses of reported scorpion toxins matched these peaks. When the venom was collected and dissolved in common solvents such as methanol, white precipitate was formed. No significant peaks were observed in the spectra obtained by analysis of the supernatant using conventional ESI-MS.

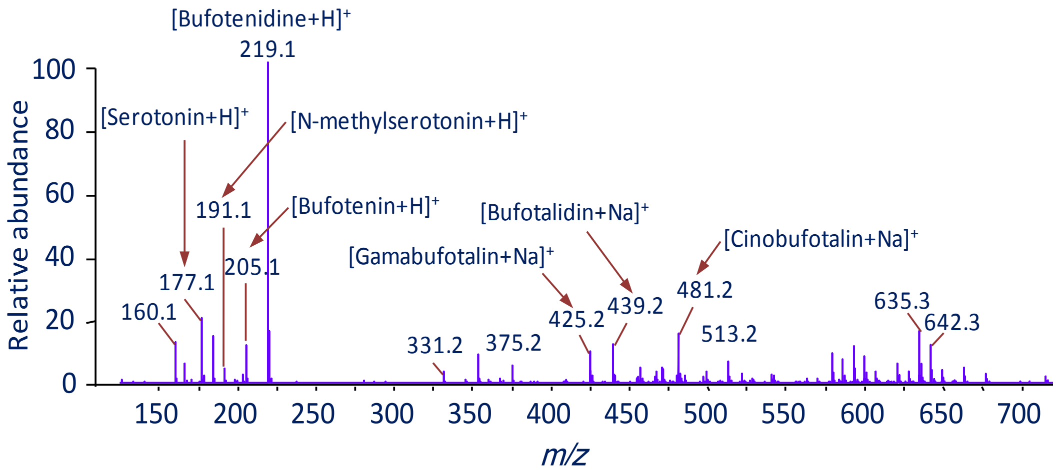

Figure 9-5 Mass spectrum obtained by field-induced DI-MS analysis of the secretion released from a living toad upon stimulation.

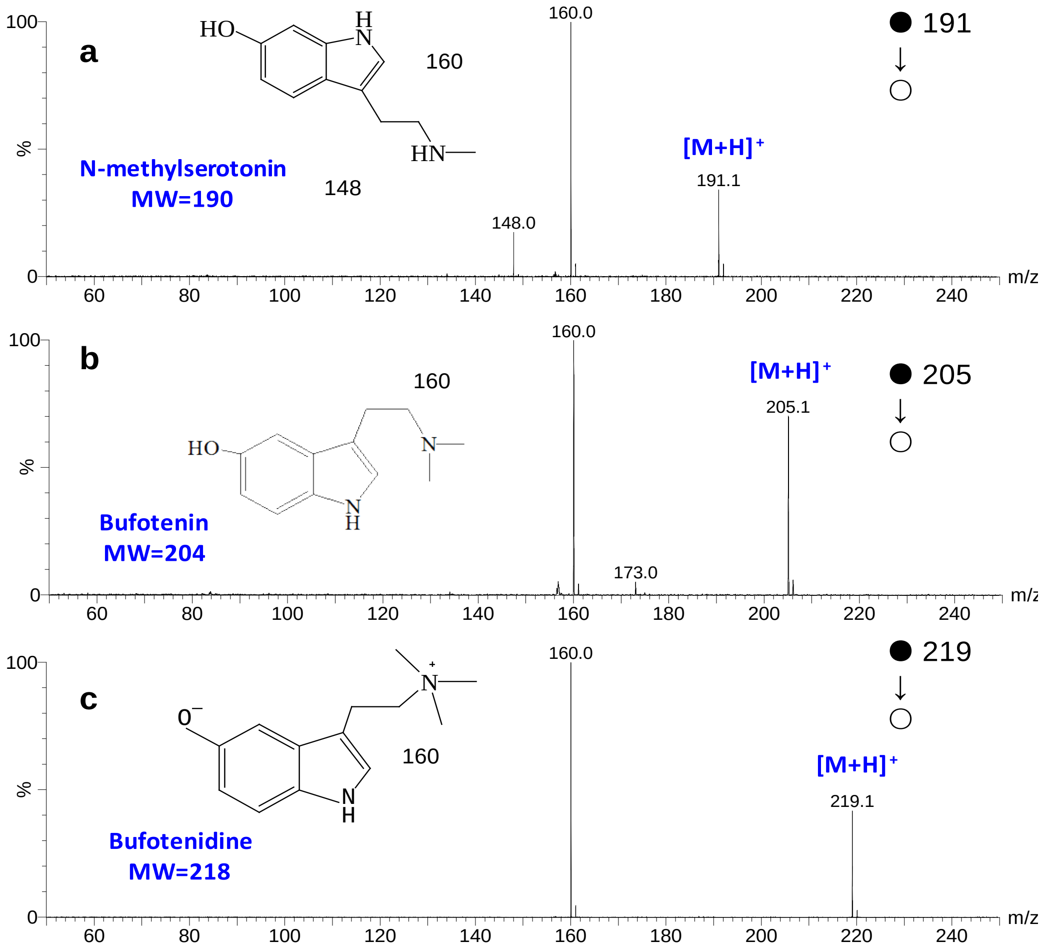

Figure 9-6 ESI-MS/MS analysis of venom solution using the QToF II mass spectrometer. a) MS/MS spectrum of *m/z* 191. b) MS/MS spectrum of *m/z* 205. c) MS/MS spectrum of *m/z* 219. The venom solution was obtained by dilution of 2 μL toad secretion into 1 mL methanol.

9.3.2 *In vivo* monitoring of toad

Toad can release venom from its skin for defense purpose upon attack.[324] In this study, a toad was stimulated at its postauricular gland with a needle (without breaking the skin), and release of white and milky venom was soon observed. A glass capillary containing some methanol was used with one side putting into the secretion and the other side pointing to the MS inlet (Figure 9-2). Spray ionization was observed from the capillary tip pointing to the MS inlet and a spectrum was obtained (Figure 9-7). Recently, Peng et al[325] used a capillary for localized analysis of plants. In their method, a quartz capillary was inserted into a plant for sampling and a stainless needle was inserted into the plant to provide a high voltage to induce spray ionization from the capillary. In this study, with a high voltage applied to the MS inlet, spray ionization was conveniently induced from a glass capillary linked to the sample, and no additional power supply device connected to the sample was required. This is particularly important and useful for *in vivo* study of living animals since connection to a high voltage in such cases is very inconvenient and can cause adverse effects to the animals and severe perturbation to the stimulation studies

Table 9-1 Proposed species detected from the scorpion toxin in the positive ion mode (*m/z* 400-700). The high resolution masses of the ions were compared with data in an affiliated database of MassHunter software (Agilent Technologies,Santa Clara, CA, USA).

Proposed formula	Ion species	Measured *m/z*	Calculation *m/z*	Mass error (ppm)	Mass Match
$C_{23}H_{50}N_8O_2$	$[M+H]^+$	471.41231	471.41295	0.83	99.3
$C_{29}H_{58}Cl_2$	$[M+H]^+$	487.40050	477.39883	-11.56	24.95
$C_{29}H_{50}N_4O_2$	$[M+H]^+$	477.46002	487.40065	-0.26	99.93
$C_{29}H_{58}N_2O_4$	$[M+H]^+$	499.44693	499.44693	0.27	99.92
$C_{27}H_{50}N_{10}$	$[M+H]^+$	515.42917	515.42927	-0.32	99.89
$C_{30}H_{62}N_4OS$	$[M+H]^+$	527.47321	527.47171	-2.93	90.78

As shown in Figure 9-7, compounds including serotonin ($[M+H]^+$, *m/z* 177.1), N-methylserotonin ($[M+H]^+$, *m/z* 191.1), bufotenin ($[M+H]^+$, *m/z* 205.1), bufotenindine ($[M+H]^+$, *m/z* 219.1), gamabufotalin ($[M+Na]^+$, *m/z* 425.2), bufotalidin ($[M+H]^+$, *m/z* 439.2) and cinobufotalin ($[M+H]^+$, *m/z* 481.2) could be identified based on masses, MS/MS studies and literature searching (Figure 9-4). These compounds were reported as the major compositions of the toad venom, and some of the compounds had been found to have antitumor activity.[326, 327] The secretion became solidified and brown after exposure in the air for about 3 min, indicating some chemical reactions occurred to the secretion. No signals were then observed after solidification of the secretion. It was also noticed that a cluster of ions were observed in the *m/z* range of 550-700 Da. Venom compositions with masses corresponding to these peaks had not been reported in previous studies, which were mainly based on analysis of dried toad venoms.[327] Identification of these compounds and studies of their biological functions will be carried out.

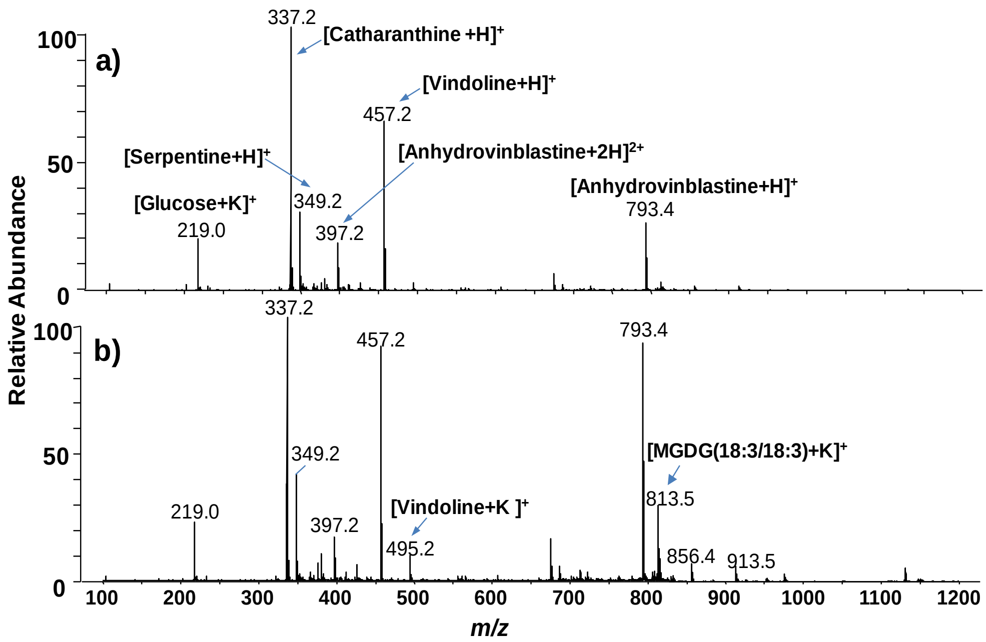

Figure 9-7 Field-induced DI spectra obtained for a leaf of a living *C. rosues* a) before and b) after heating. Methanol (2 μL) was used as the added solvent.

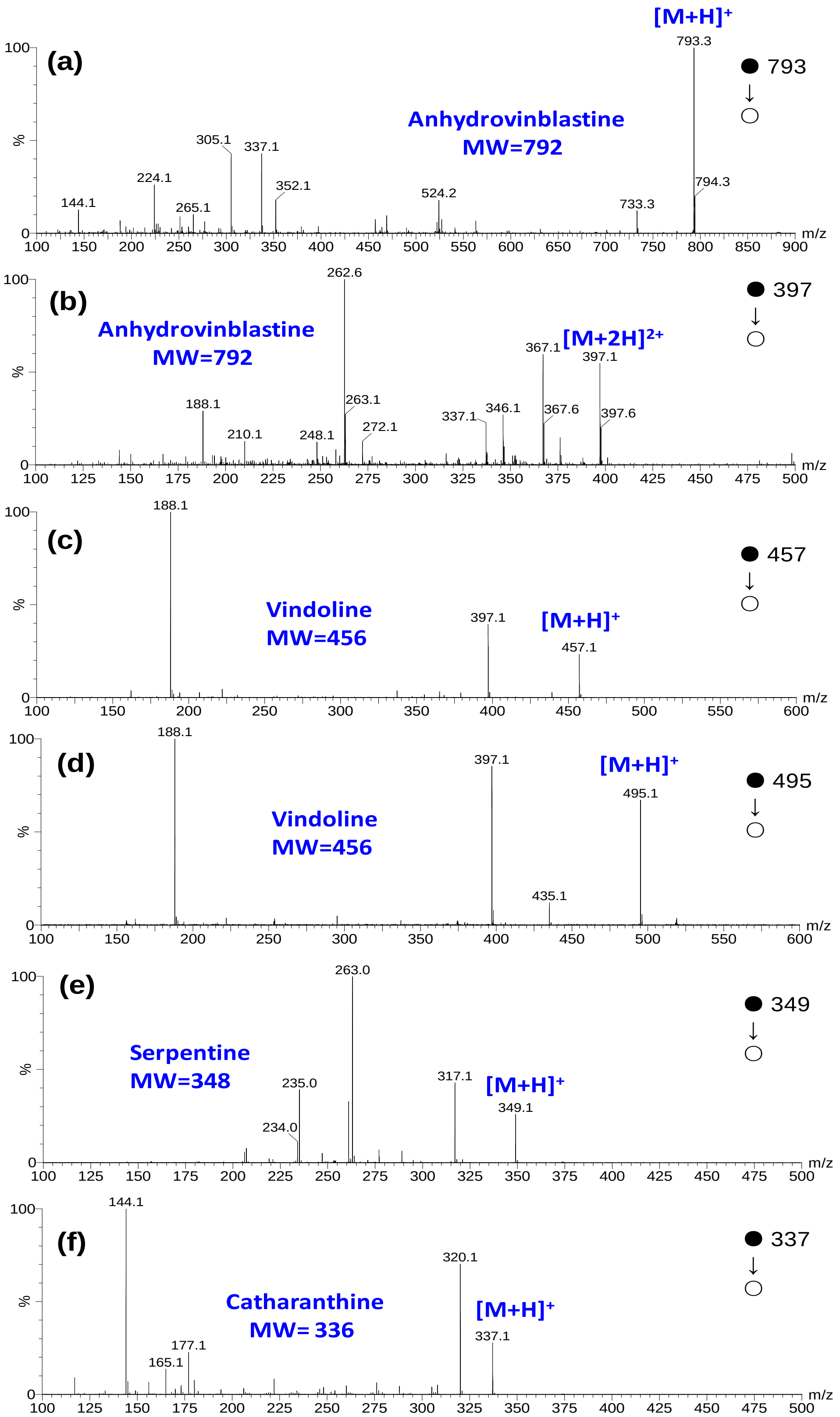

Figure 9-8 MS/MS spectra for major peaks in Figure 9-7, obtained from a *C. rosues* leaf using DI method on the QToF II mass spectrometer.

9.3.3 *In vivo* monitoring of *Catharanthus roseus*

Dimeric vinca alkaloids such as vinblastine are powerful anticancer drugs derived from *Catharanthus roseus*. These drugs are produced within the plant by coupling catharanthine with vindoline. Detailed mechanisms for the assembly are still unclear.[309] These dimeric drugs are found at very low levels in the plant and their commercial productions are very expensive.[328] A recent study by Roepke et al[308] revealed that catharanthine and vindoline are spatially separated in the leaves of *C. roseus*. It was hypothesized that when the plant is wounded or attacked, catharanthine and vindoline are brought together to allow the formation of the dimers that can prevent cell division.

A living *C. roseus* was analyzed using the field-induced DI technique and the spectrum obtained from a leaf of the plant is shown in Figure 9-8a. Catharanthine, vindoline and anhydrovinblastine were predominantly observed in the spectrum. When heating at 450 ºC was induced for 3 sec at ~ 30 mm away from the spray tip of the leaf, anhydrovinblastine was found to significantly increase, as shown in Figure 10-8b.Repeated measurements with three different leaves revealed that abundance of anhydrovinblastine increased from 25.0 ± 4.1% in the spectrum before heating to 93.6 ± 5.2% after heating. Increased abundances of potassium adducts were also observed in the spectrum obtained after the heating, along with the appearance of monogalactosyldiacylglycerol (MGDG). Accumulation of potassium has been reported as an osmotic adjustment strategy for plants to maintain water,[329] and MGDG was previously reported to accumulate when plants were wounded.[330] As such, the increase of extracted ions of anhydrovinblastine and MGDG were also observed in their selected ion chromatogram after heating stimulation (Figure 9-9). Our results confirmed the hypothesis proposed by Roepke et al,[308] and suggested that production of anhydrovinblastine in *C. roseus* might be increased by changing the living environment of the plant.

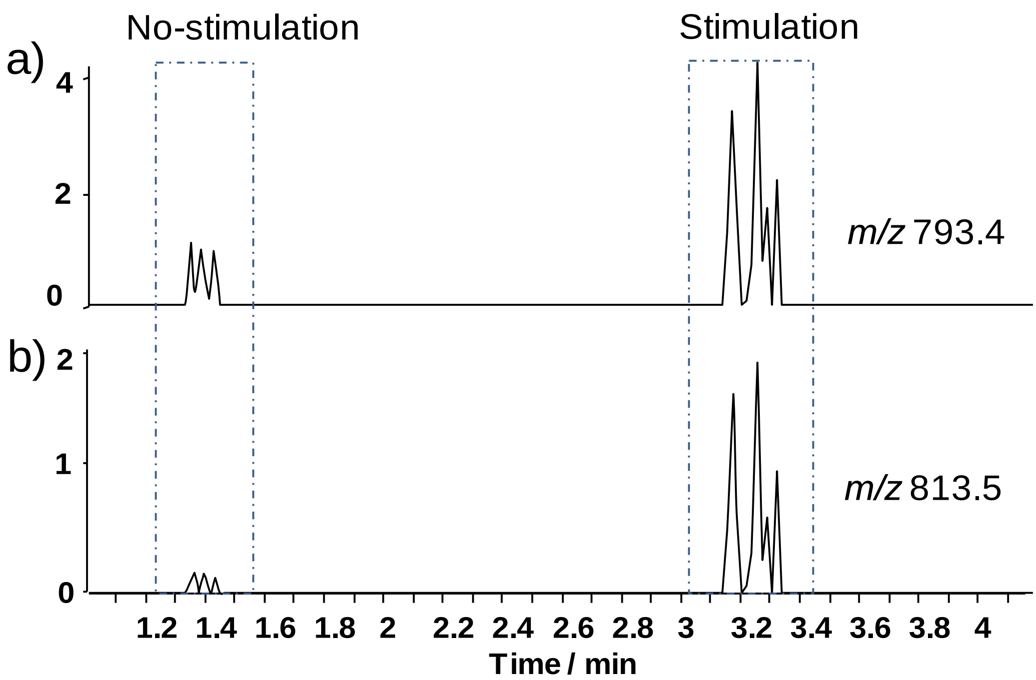

Figure 9-9 SIC of a) anhydrovinblastine ($[M+H]^+$, 793.4) and b) MGDG($[M+Na]^+$, 813.5).

9.3.4 *In vivo* monitoring of Cactus

Field-induced ESI with capillary sampling could also be used for real-time monitoring of metabolites of bulk pants due to minimization the effects of the high voltage wire. For example, a glass capillary was inserted into cactus at a depth of 10.0mm for sampling and ionization. An abundance of information on metabolites, which are corresponding to amino acid and sugars, were obtained (Figure 9-11a). After heating at 450 ºC for 3 seconds at ~10 mm away from the insertion location of the capillary, monogalactosyldiacylglycerol (MGDG) was detected as the base peak in the spectrum (Figure 9-11b). The release of MGDG was believed to be a response when plant was wounded,[330] while raised of potassium adducts along with the appearance of MGDG could be due to stimulation of osmotic adjustment of plants.[329] Our results indicated that field-induced ESI could be a simple and rapid method for investigation of plant growth, development, and ecological functions.

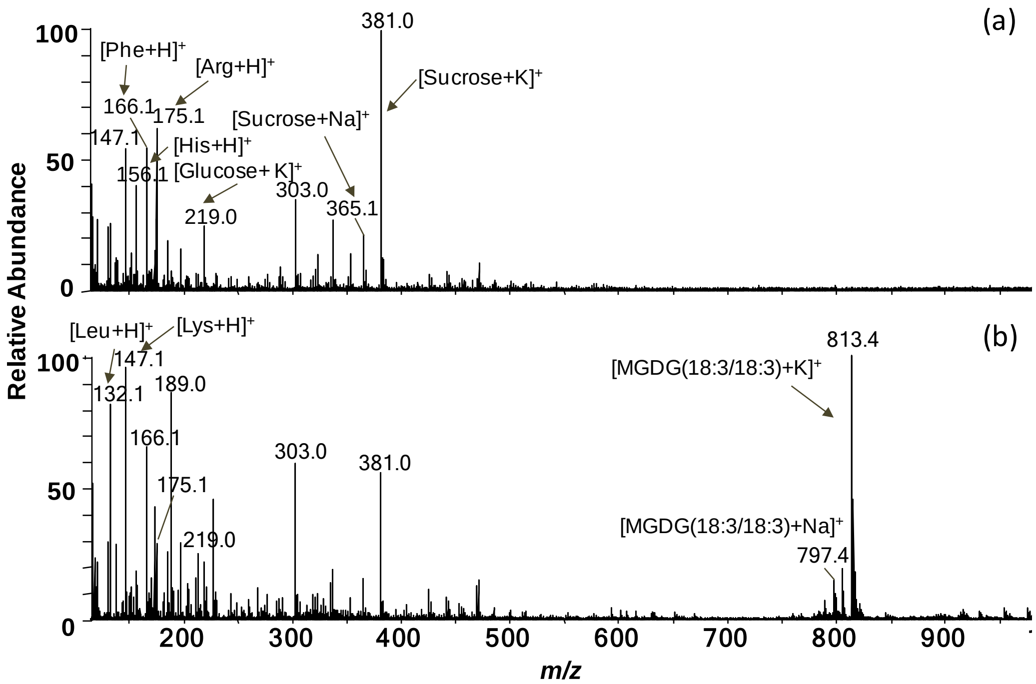

Figure 9-10 Mass spectra obtained from cactus (a) before and (b) after heating.

9.4 Summary

In this study, we presented proof-of-principle for *in vivo* and real-time monitoring of secondary metabolites released from living organisms using field-induced DI-MS. Without to connect the analyzed samples to a high voltage not only avoids perturbation to the stimulation studies and potential hazard to living organisms, but also greatly facilitates sampling from living organisms. Spray ionization could be conveniently induced from a tip of the studied living organism when the tip is moved close and pointing to the MS inlet. Such a tip can be intrinsic to own by the living organism (e.g., the sting of a scorpion), or become available by introducing a capillary to the sample (e.g., in the case for *in vivo* study of toad), or formed by modifying the sample (e.g., cutting the analyzed leaf to a small V-shape). We demonstrated that this technique could be readily used for *in vivo* and real-time monitoring of secondary metabolites released from both living animals and plants upon stimulation. Such *in vivo* and real-time monitoring is simple, rapid, and able to detect compounds that could not be observed by analysis using conventional approaches. Further development of field-induced DI-MS, including allowing observation of more compounds of interest in a single spectrum, coupling it to a high resolution tandem mass spectrometer to improve identification of compounds, and applying it for studying other interesting biosystems and clinical studies, are on-going.

Chapter 10: Overall Conclusions and Prospects

The research in this thesis involves the development, mechanistic studies, and applications of new ESI using solid substrates. The effects of various factors of solid-substrate ESI, including substrate surface, spray solvent, application mode of ionization voltage, and properties of analytes, were systematically investigated. Different functional materials such as Al foil, TLC plate, and SPME fibre, were successfully developed as ESI emitters with additional functions of on-target sample preteatment, sample separation, and sample extraction and enrichment. Moreover, *in vivo* analysis and real-time monitoring of metabolites from living organisms, including plants and animals, were also investigated with the solid-substrate ESI-MS technique.

Unlike capillary-based ESI, solid-substrate ESI sample is ionized on the surface of a solid substrate. Therefore, the analytical performance might be influenced by interactions between analytes and the solid-substrate surface. It is thus important to understand such interactions and their effects to the signals and to improve solid-substrate ESI for various applications. This work revealed that solid-substrate ESI with different functional groups allowed selective detection of analytes from complex samples with reduced matrix interferences. Metal substrates, e.g., Al foil, enabled not only on-target sample extraction, cleanup of analytes, but also monitoring of thermal reaction. Development of functional ESI techniques extended ESI emitters from sample loading and ionization to sample pretreatments.

Compared with conventional ESI, solid-substrate ESI does not have the problem of no sample clogging and allows for more convenient sampling. With replacement of capillary with solid substrates in conventional ESI, a wide range of materials can be selected for sampling and analysis, leading to many new possibilities for further developments and applications. Solid-substrate ESI also significantly facilitates mass spectrometric analysis in various applications, e.g., food safety analysis, drug quality control, environmental analysis, forensic analysis, clinical diagnosis, therapeutic monitoring, and *in vivo* studies.

Solid-substrate ESI can be used for *in situ*, real-time, *in vivo*, high-throughput analysis with high-sensitivity, high selectivity, low cost, and requires no or litter sample preparation. These render solid-substrate ESI very suitable for on-site analysis. Therefore, development and coupling miniature mass spectrometers with solid-substrate ESI may be a good direction for future study. Meanwhile, given that single cell analysis is an emerging hot research topic, the current study indicated that solid-substrate ESI-MS could become an important tool for single cell analysis in the future, if the technique could become faster, more convenient, more sensitive, and more selective. In addition, quantitative detection of solid-substrate ESI-MS still requires great improvement to ensure the accuracy of the results for complex sample analysis.

References

1. Thomson, J. J., Bakerian Lecture: Rays of Positive Electricity. *Proc Royal Soc A* **1913,** *89*, 1-20.

2. Berlin, K. P. A. d. W. z., *Monatsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin*. The Academy1876.

3. du Bois, H., Ueber magnetische Schirmwirkung. *Annalen der Physik* **1898,** *301*, 1-37.

4. Anonymous, Proceedings of the American Physical Society. *Phys Rev* **1946,** *69*, 674-674.

5. Domon, B.; Aebersold, R., Review - mass spectrometry and protein analysis. *Science* **2006,** *312*, 212-217.

6. Cooks, R. G.; Ouyang, Z.; Takats, Z.; Wiseman, J. M., Ambient mass spectrometry. *Science* **2006,** *311*, 1566-1570.

7. Aebersold, R.; Mann, M., Mass spectrometry-based proteomics. *Nature* **2003,** *422*, 198-207.

8. Glinski, M.; Weckwerth, W., The role of mass spectrometry in plant systems biology. *Mass Spectrom Rev* **2006,** *25*, 173-214.

9. DeCarlo, P. F.; Kimmel, J. R.; Trimborn, A.; Northway, M. J.; Jayne, J. T.; Aiken, A. C.; Gonin, M.; Fuhrer, K.; Horvath, T.; Docherty, K. S.; Worsnop, D. R.; Jimenez, J. L., Field-deployable, high-resolution, time-of-

flight aerosol mass spectrometer. *Anal Chem* **2006,** *78*, 8281-8289.

10. Canagaratna, M. R.; Jayne, J. T.; Jimenez, J. L.; Allan, J. D.; Alfarra, M. R.; Zhang, Q.; Onasch, T. B.; Drewnick, F.; Coe, H.; Middlebrook, A.; Delia, A.; Williams, L. R.; Trimborn, A. M.; Northway, M. J.; DeCarlo, P. F.; Kolb, C. E.; Davidovits, P.; Worsnop, D. R., Chemical and microphysical characterization of ambient aerosols with the aerodyne aerosol mass spectrometer. *Mass Spectrom Rev* **2007,** *26*, 185-222.

11. Bantscheff, M.; Schirle, M.; Sweetman, G.; Rick, J.; Kuster, B., Quantitative mass spectrometry in proteomics: a critical review. *Anal Bioanal Chem* **2007,** *389*, 1017-1031.

12. Yang, M.; Kim, T. Y.; Hwang, H. C.; Yi, S. K.; Kim, D. H., Development of a palm portable mass spectrometer. *J Am Soc Mass Spectrom* **2008,** *19*, 1442-1448.

13. Keil, A.; Hernandez-Soto, H.; Noll, R. J.; Fico, M.; Gao, L.; Ouyang, Z.; Cooks, R. G., Monitoring of toxic compounds in air using a handheld rectilinear ion trap mass spectrometer. *Anal Chem* **2008,** *80*, 734-741.

14. Gao, L.; Song, Q. Y.; Patterson, G. E.; Cooks, R. G.; Ouyang, Z., Handheld rectilinear ion trap mass spectrometer. *Anal Chem* **2006,** *78*, 5994-6002.

15. Keil, A.; Talaty, N.; Janfelt, C.; Noll, R. J.; Gao, L.; Ouyang, Z.; Cooks, R. G., Ambient mass spectrometry with a handheld mass spectrometer at high pressure. *Anal Chem* **2007,** *79*, 7734-7739.

16. Liang, G.; Sugiarto, A.; Harper, J. D.; Cooks, R. G.; Zheng, O. Y., Design and characterization of a multisource hand-held tandem mass spectrometer. *Anal Chem* **2008,** *80,* 7198-7205.

17. Li, L. F.; Chen, T. C.; Ren, Y.; Hendricks, P. I.; Cooks, R. G.; Ouyang, Z., Mini 12, miniature mass spectrometer for clinical and other applications-introduction and characterization. *Anal Chem* **2014,** *86*, 2909-2916.

18. Covey, T. R.; Thomson, B. A.; Schneider, B. B., Atmospheric pressure ion sources. *Mass Spectrom Rev* **2009,** *28,* 870-897.

19. Chait, E. M., Ionization sources in mass-spectrometry. *Anal Chem* **1972,** *44,* 77A -91A.

20. El-Aneed, A.; Cohen, A.; Banoub, J., Mass spectrometry, review of the basics: electrospray, maldi, and commonly used mass analyzers. *Appl Spectro Rev* **2009,** *44*, 210-230.

21. Middleton, R., A review of ion sources for accelerator mass-spectrometry. *Nucl Instrum Meth B* **1984,** *5*, 193-199.

22. Zeleny, J., Instability of electrified liquid surfaces. *Phys Rev* **1917,** *10*, 1-6.

23. Taylor, G., *Electrically Driven Jets*. **1969,** 313, 453-475.

24. Taylor, G., Disintegration of water drops in an electric field. *Proc Royal Soc A.* **1964,** *280*, 383-397.

25. Dole, M.; Mack, L. L.; Hines, R. L.; Mobley, R. C.; Ferguson, L. D.; Alice,

M. B., Molecular beams of macroions. *J Chem Phys* **1968,** *49*, 2240-2249.

26. Fenn, J. B.; Mann, M.; Meng, C. K.; Wong, S. F.; Whitehouse, C. M., Electrospray ionization for mass-spectrometry of large biomolecules. *Science* **1989,** *246*, 64-71.

27. Yamashita, M.; Fenn, J. B., Negative-ion production with the electrospray ion-source. *J Phys Chem* **1984,** *88*, 4671-4675.

28. de la Mora, J. F., Electrospray ionization of large multiply charged species proceeds via Dole's charged residue mechanism. *Anal Chim Acta* **2000,** *406*, 93-104.

29. Iavarone, A. T.; Williams, E. R., Mechanism of charging and supercharging molecules in electrospray ionization. *J Am Chem Soc* **2003,** *125*, 2319-2327.

30. Konermann, L.; Ahadi, E.; Rodriguez, A. D.; Vahidi, S., Unraveling the mechanism of electrospray ionization. *Anal Chem* **2013,** *85*, 2-9.

31. Blades, A. T.; Ikonomou, M. G.; Kebarle, P., Mechanism of electrospray mass-spectrometry - electrospray as an electrolysis cell. *Anal Chem* **1991,** *63*, 2109-2114.

32. Kebarle, P., A brief overview of the present status of the mechanisms involved in electrospray mass spectrometry. *J Mass Spectrom* **2000,** *35*, 804-817.

33. Kebarle, P.; Verkerk, U. H., Electrospray: from ions in solution to ions in

the gas phase, what we know now. *Mass Spectrom Rev* **2009,** *28*, 898-917.

34. Loeb, L. B.; Kip, A. F.; Hudson, G. G.; Bennett, W. H., Pulses in negative point-to-plane corona. *Phys Rev* **1941,** *60*, 714-722.

35. Gaskell, S. J., Electrospray: principles and practice. *J Mass Spectrom* **1997,** *32*, 677-688.

36. Banerjee, S.; Mazumdar, S., Electrospray ionization mass spectrometry: a technique to access the information beyond the molecular weight of the analyte. *Int J Anal Chem* **2012,** *2012,* 282574

37. Takats, Z.; Wiseman, J. M.; Gologan, B.; Cooks, R. G., Mass spectrometry sampling under ambient conditions with desorption electrospray ionization. *Science* **2004,** *306*, 471-473.

38. Garcia-Reyes, J. F.; Jackson, A. U.; Molina-Diaz, A.; Cooks, R. G., Desorption electrospray ionization mass spectrometry for trace analysis of agrochemicals in food. *Anal Chem* **2009,** *81*, 820-829.

39. Gerbig, S.; Takats, Z., Analysis of triglycerides in food items by desorption electrospray ionization mass spectrometry. *Rapid Commun Mass Spectrom* **2010,** *24*, 2186-2192.

40. Leuthold, L. A.; Mandscheff, J. F.; Fathi, M.; Giroud, C.; Augsburger, M.; Varesio, E.; Hopfgartner, G., Desorption electrospray ionization mass spectrometry: direct toxicological screening and analysis of illicit Ecstasy tablets. *Rapid Commun Mass Spectrom* **2006,** *20*, 103-110.

41. Leuthold, L. A.; Mandscheff, J. F.; Fathi, M.; Giroud, C.; Augsburger, M.; Varesio, E.; Hopfgartner, G., Direct ambient analysis of pharmaceutical and ecstasy tablets. *Chimia* **2006,** *60*, 190-194.

42. Hemalatha, R. G.; Pradeep, T., Understanding the molecular signatures in leaves and flowers by desorption electrospray ionization mass spectrometry (DESI MS) imaging. *J Agri Food Chem* **2013,** *61*, 7477-7487.

43. Morelato, M.; Beavis, A.; Kirkbride, P.; Roux, C., Forensic applications of desorption electrospray ionisation mass spectrometry (DESI-MS). *Forensic Sci Int* **2013,** *226*, 10-21.

44. Stojanovska, N.; Tahtouh, M.; Kelly, T.; Beavis, A.; Fu, S. L., Qualitative analysis of seized cocaine samples using desorption electrospray ionization- mass spectrometry (DESI-MS). *Drug Test Anal* **2015,** *7*, 393-400.

45. Pan, Z. Z.; Gu, H. W.; Talaty, N.; Chen, H. W.; Shanaiah, N.; Hainline, B. E.; Cooks, R. G.; Raftery, D., Principal component analysis of urine metabolites detected by NMR and DESI-MS in patients with inborn errors of metabolism. *Anal Bioanal Chem* **2007,** *387*, 539-549.

46. Wiseman, J. M.; Ifa, D. R.; Zhu, Y. X.; Kissinger, C. B.; Manicke, N. E.; Kissinger, P. T.; Cooks, R. G., Mass spectrometry across the sciences special feature: desorption electrospray ionization mass spectrometry: imaging drugs and metabolites in tissues. *Pro Natl Acad Sci USA* **2009,** *106*, 6022-6022.

47. Wiseman, J. M.; Ifa, D. R.; Zhu, Y. X.; Kissinger, C. B.; Manicke, N. E.; Kissinger, P. T.; Cooks, R. G., Desorption electrospray ionization mass spectrometry: Imaging drugs and metabolites in tissues. *Pro Natl Acad Sci USA* **2008,** *105*, 18120-18125.

48. Ifa, D. R.; Manicke, N. E.; Dill, A. L.; Cooks, G., Latent fingerprint chemical imaging by mass spectrometry. *Science* **2008,** *321*, 805-805.

49. Lostun, D.; Perez, C. J.; Licence, P.; Barrett, D. A.; Ifa, D. R., Reactive DESI-MS Imaging of Biological Tissues with Dicationic Ion-Pairing Compounds. *Anal Chem* **2015,** *87*, 3286-3293.

50. Chen, H. W.; Venter, A.; Cooks, R. G., Extractive electrospray ionization for direct analysis of undiluted urine, milk and other complex mixtures without sample preparation. *Chem Commun* **2006**, 2042-2044.

51. Chipuk, J. E.; Brodbelt, J. S., Transmission Mode Desorption Electrospray Ionization. *J Am Soc Mass Spectrom.* **2008,** *19*, 1612-1620.

52. Nyadong, L.; Galhena, A. S.; Fernandez, F. M., Desorption Electrospray/Metastable-Induced Ionization: A Flexible Multimode Ambient Ion Generation Technique. *Anal Chem* **2009,** *81*, 7788-7794.

53. Ozdemir, A.; Chen, C. H., Electrode-assisted desorption electrospray ionization mass spectrometry. *J Mass Spectrom* **2010,** *45*, 1203-1211.

54. Roach, P. J.; Laskin, J.; Laskin, A., Nanospray desorption electrospray ionization: an ambient method for liquid-extraction surface sampling in

mass spectrometry. *Analyst* **2010,** *135*, 2233-2236.

55. Forbes, T. P.; Brewer, T. M.; Gillen, G., Desorption electro-flow focusing ionization of explosives and narcotics for ambient pressure mass spectrometry. *Analyst* **2013,** *138*, 5665-5673.

56. Luo, Z. G.; He, J. M.; Chen, Y.; He, J. J.; Gong, T.; Tang, F.; Wang, X. H.; Zhang, R. P.; Huang, L.; Zhang, L. F.; Lv, H. N.; Ma, S. G.; Fu, Z. D.; Chen, X. G.; Yu, S. S.; Abliz, Z., Air flow-assisted ionization imaging mass spectrometry method for easy whole-body molecular imaging under ambient conditions. *Anal Chem* **2013,** *85*, 2977-2982.

57. Wu, C.; Siems, W. F.; Hill, H. H., Secondary electrospray ionization ion mobility spectrometry/mass spectrometry of illicit drugs. *Anal Chem* **2000,** *72*, 396-403.

58. Tam, M.; Hill, H. H., Secondary electrospray ionization-ion mobility spectrometry for explosive vapor detection. *Anal Chem* **2004,** *76*, 2741-2747.

59. Chang, D. Y.; Lee, C. C.; Shiea, J., Detecting large biomolecules from high-salt solutions by fused-droplet electrospray ionization mass spectrometry. *Anal Chem* **2002,** *74*, 2465-2469.

60. Hsu, H. J.; Kuo, T. L.; Wu, S. H.; Oung, J. H.; Shiea, J., Characterization of synthetic polymers by electrospray-assisted pyrolysis ionization-mass spectrometry. *Anal Chem* **2005,** *77*, 7744-7749.

61. Hsu, H. J.; Oung, J. N.; Kuo, T. L.; Wu, S. H.; Shiea, J., Using electrospray-assisted pyrolysis ionization/mass spectrometry for the rapid characterization of trace polar components in crude oil, amber, humic substances, and rubber samples. *Rapid Commun Mass Spectrom* **2007,** *21*, 375-384.

62. Wang, R.; Grohn, A. J.; Zhu, L.; Dietiker, R.; Wegner, K.; Gunther, D.; Zenobi, R., On the mechanism of extractive electrospray ionization (EESI) in the dual-spray configuration. *Anal Bioanal Chem* **2012,** *402*, 2633-2643.

63. Law, W. S.; Wang, R.; Hu, B.; Berchtold, C.; Meier, L.; Chen, H. W.; Zenobi, R., On the mechanism of extractive electrospray ionization. *Anal Chem* **2010,** *82*, 4494-4500.

64. Chen, H. W.; Wortmann, A.; Zenobi, R., Neutral desorption sampling coupled to extractive electrospray ionization mass spectrometry for rapid differentiation of bilosamples by metabolomic fingerprinting. *J Mass Spectrom* **2007,** *42*, 1123-1135.

65. Chen, H. W.; Wortmann, A.; Zhang, W. H.; Zenobi, R., Rapid *in vivo* fingerprinting of nonvolatile compounds in breath by extractive electrospray ionization quadrupole time-of-flight mass spectrometry. *Angew Chem Int Ed* **2007,** *46*, 580-583.

66. Chen, H. W.; Hu, B.; Hu, Y.; Huan, Y. F.; Zhou, Z. Q.; Qiao, X. F., Neutral desorption using a sealed enclosure to sample explosives on human skin for rapid detection by EESI-MS. *J Am Soc Mass Spectrom* **2009,** *20*, 719-

722.

67. Li, M.; Hu, B.; Li, J. Q.; Chen, R.; Zhang, X.; Chen, H. W., Extractive electrospray ionization mass spectrometry toward in situ analysis without sample pretreatment. *Anal Chem* **2009,** *81*, 7724-7731.

68. Gu, H. W.; Hu, B.; Li, J. Q.; Yang, S. P.; Han, J.; Chen, H. W., Rapid analysis of aerosol drugs using nano extractive electrospray ionization tandem mass spectrometry. *Analyst* **2010,** *135*, 1259-1267.

69. Harris, G. A.; Galhena, A. S.; Fernandez, F. M., Ambient sampling/ionization mass spectrometry: applications and current trends. *Anal Chem* **2011,** *83*, 4508-4538.

70. Van Berkel, G. J.; Pasilis, S. P.; Ovchinnikova, O., Established and emerging atmospheric pressure surface sampling/ionization techniques for mass spectrometry. *J Mass Spectrom* **2008,** *43*, 1161-1180.

71. Ifa, D. R.; Wu, C. P.; Ouyang, Z.; Cooks, R. G., Desorption electrospray ionization and other ambient ionization methods: current progress and preview. *Analyst* **2010,** *135*, 669-681.

72. Weston, D. J., Ambient ionization mass spectrometry: current understanding of mechanistic theory; analytical performance and application areas. *Analyst* **2010,** *135*, 661-668.

73. Yao, Z. P., Characterization of proteins by ambient mass spectrometry. *Mass Spectrom Rev* **2012,** *31*, 437-447.

74. Shiea, J.; Huang, M. Z.; HSu, H. J.; Lee, C. Y.; Yuan, C. H.; Beech, I.; Sunner, J., Electrospray-assisted laser desorption/ionization mass spectrometry for direct ambient analysis of solids. *Rapid Commun Mass Spectrom* **2005,** *19*, 3701-3704.

75. Sampson, J. S.; Hawkridge, A. M.; Muddiman, D. C., Generation and detection of multiply-charged peptides and proteins by matrix-assisted laser desorption electrospray ionization (MALDESI) Fourier transform ion cyclotron resonance mass spectrometry. *J Am Soc Mass Spectrom* **2006,** *17*, 1712-1716.

76. Nemes, P.; Barton, A. A.; Li, Y.; Vertes, A., Ambient molecular imaging and depth profiling of live tissue by infrared laser ablation electrospray ionization mass spectrometry. *Anal Chem* **2008,** *80*, 4575-4582.

77. Rezenom, Y. H.; Dong, J.; Murray, K. K., Infrared laser-assisted desorption electrospray ionization mass spectrometry. *Analyst* **2008,** *133*, 226-232.

78. Sampson, J. S.; Muddiman, D. C., Atmospheric pressure infrared (10.6 mu m) laser desorption electrospray ionization (IR-LDESI) coupled to a LTQ Fourier transform ion cyclotron resonance mass spectrometer. *Rapid Commun Mass Spectrom* **2009,** *23*, 1989-1992.

79. Cheng, S. C.; Cheng, T. L.; Chang, H. C.; Shiea, J., Using Laser-induced acoustic desorption/electrospray ionization mass spectrometry to characterize small organic and large biological compounds in the solid state and in solution under ambient conditions. *Anal Chem* **2009,** *81*, 868-

874.

80. Liu, J.; Qiu, B.; Luo, H., Fingerprinting of yogurt products by laser desorption spray post-ionization mass spectrometry. *Rapid Commun Mass Spectrom* **2010,** *24*, 1365-1370.

81. Brady, J. J.; Judge, E. J.; Levis, R. J., Identification of explosives and explosive formulations using laser electrospray mass spectrometry. *Rapid Commun Mass Spectrom* **2010,** *24*, 1659-1664.

82. McEwen, C. N.; Larsen, B. S.; Trimpin, S., Laserspray ionization on a commercial atmospheric pressure-maldi mass spectrometer ion source: selecting singly or multiply charged ions. *Anal Chem* **2010,** *82*, 4998-5001.

83. Fenn, J. B., *US Patent 6297499* **1998**.

84. Hong, C. M.; Lee, C. T.; Lee, Y. M.; Kuo, C. P.; Yuan, C. H.; Shiea, J., Generating electrospray from solutions predeposited on a copper wire. *Rapid Commun Mass Spectrom* **1999,** *13*, 21-25.

85. Hiraoka, K.; Nishidate, K.; Mori, K.; Asakawa, D.; Suzuki, S., Development of probe electrospray using a solid needle. *Rapid Commun Mass Spectrom* **2007,** *21*, 3139-3144.

86. Kuo, C. P.; Yuan, C. H.; Shiea, J., Generation of electrospray from a solution predeposited on optical fibres coiled with a platinum wire. *J Am Soc Mass Spectrom* **2000,** *11*, 464-467.

87. Jeng, J. Y.; Shiea, J., Electrospray ionization from a droplet deposited on a

surface-modified glass rod. *Rapid Commun Mass Spectrom* **2003,** *17*, 1709-1713.

88. Jeng, J. Y.; Lin, C. H.; Shiea, J., Electrospray from nanostructured tungsten oxide surfaces with ultralow sample volume. *Anal Chem* **2005,** *77*, 8170-8173.

89. Hiraoka, K.; Nishidate, K.; Mori, K.; Asakawa, D.; Suzuki, S., Development of probe electrospray using a solid needle. *Rapid Commun Mass Spectrom* **2007,** *21*, 3139-3144.

90. Liu, J. J.; Wang, H.; Manicke, N. E.; Lin, J. M.; Cooks, R. G.; Ouyang, Z., Development, characterization, and application of paper spray ionization. *Anal Chem* **2010,** *82*, 2463-2471.

91. Hu, B.; So, P. K.; Chen, H. W.; Yao, Z. P., Electrospray ionization using wooden tips. *Anal Chem* **2011,** *83,* 8201-8207.

92. Hu, B.; So, P. K.; Yao, Z. P., Electrospray ionization with aluminum foil: a versatile mass spectrometric technique. *Anal Chim Acta* **2014,** *817*, 1-8.

93. Mandal, M. K.; Chen, L. C.; Hashimoto, Y.; Yu, Z.; Hiraoka, K., Detection of biomolecules from solutions with high concentration of salts using probe electrospray and nano-electrospray ionization mass spectrometry. *Anal Methods* **2010,** *2*, 1905-1912.

94. Mandal, M. K.; Chen, L. C.; Yu, Z.; Nonami, H.; Erra-Balsells, R.; Hiraoka, K., Detection of protein from detergent solutions by probe

electrospray ionization mass spectrometry (PESI-MS). *J Mass Spectrom* **2011,** *46*, 967-975.

95. Mandal, M. K.; Chen, L. C.; Hiraoka, K., Sequential and exhaustive ionization of analytes with different surface activity by probe electrospray ionization. *J Am Soc Mass Spectrom* **2011,** *22*, 1493-1500.

96. Liu, J. J.; Wang, H.; Manicke, N. E.; Lin, J. M.; Cooks, R. G.; Ouyang, Z., Development, Characterization, and Application of Paper Spray Ionization. *Anal Chem* **2010,** *82*, 2463-2471.

97. Wang, H.; Liu, J. J.; Cooks, R. G.; Ouyang, Z., Paper spray for direct analysis of complex mixtures using mass spectrometry. *Angew Chem Int Ed* **2010,** *49*, 877-880.

98. Hu, B.; So, P. K.; Chen, H. W.; Yao, Z. P., Electrospray ionization using wooden tips. *Anal Chem* **2011,** *83*, 8201-8207.

99. Hu, B.; So, P. K.; Yao, Z. P., Analytical properties of solid-substrate electrospray ionization mass spectrometry. *J Am Soc Mass Spectrom* **2013,** *24*, 57-65.

100. So, P. K.; Ng, T. T.; Wang, H. X.; Hu, B.; Yao, Z. P., Rapid detection and quantitation of ketamine and norketamine in urine and oral fluid by wooden-tip electrospray ionization mass spectrometry. *Analyst* **2013,** *138*, 2239-2243.

101. Yao, Z. P., Mass Spectrometry: Tip spray and direct analysis of biological

tissues. In *Proceedings of the NSFC meeting "Analytical Techniques in Life Science: Opportunities and Challenges"*: Hong Kong, 20-21 May 2011.

102. Hu, B.; Lai, Y. H.; So, P. K.; Chen, H. W.; Yao, Z. P., Direct ionization of biological tissue for mass spectrometric Analysis. *Analyst* **2012,** *137*, 3613-3619.

103. Chan, S. L. F.; Wong, M. Y. M.; Tang, H. W.; Che, C. M.; Ng, K. M., Tissue-spray ionization mass spectrometry for raw herb analysis. *Rapid Commun Mass Spectrom* **2011,** *25*, 2837-2843.

104. Liu, J. J.; Wang, H.; Cooks, R. G.; Ouyang, Z., Leaf Spray: Direct Ambient Ionization for Plant Tissue Analysis. In *Proceedings of the 59th ASMS Annual Meeting on Mass Spectrometry and Allied Topics*: Denver, USA, 6-9 June 2011.

105. Liu, J. J.; Wang, H.; Cooks, R. G.; Ouyang, Z., Leaf Spray: Direct chemical analysis of plant material and living plants by mass spectrometry. *Anal Chem* **2011,** *83*, 7608-7613.

106. Zeleny, J., The electrical discharge from liquid points, and a hydrostatic method of measuring the electric intensity at their surfaces. *Phys Rev* **1914,** *3*, 69-91.

107. Iribarne, J. V.; Thomson, B. A., On the evaporation of small ions from charged droplets. *J Chem Phys* **1976,** *64*, 2287-2294.

108. Hirabayashi, A.; Sakairi, M.; Koizumi, H., Sonic spray mass-spectrometry. *Anal Chem* **1995,** *67*, 2878-2882.

109. Chetwani, N.; Cassou, C. A.; Go, D. B.; Chang, H. C., High-frequency AC electrospray ionization source for mass spectrometry of biomolecules. *J Am Soc Mass Spectrom* **2010,** *21*, 1852-1856.

110. Lu, Y.; Zhou, F.; Shui, W. Q.; Bian, L. P.; Yang, P. Y., Pulsed electrospray for mass spectrometry. *Anal Chem* **2001,** *73*, 4748-4753.

111. Qiao, L.; Sartor, R.; Gasilova, N.; Lu, Y.; Tobolkina, E.; Liu, B. H.; Girault, H. H., Electrostatic-spray ionization mass spectrometry. *Anal Chem* **2012,** *84*, 7422-7430.

112. Ozdemir, A.; Lin, J. L.; Gillig, K. J.; Chen, C. H., Triboelectric spray ionization. *J Mass Spectrom* **2013,** *48*, 154-163.

113. Robert, C., Electrospray ionization mass spectrometry. In *Applied Electrospray Mass Spectrometry*, CRC Press 2002.

114. Bertsch, J. L.; Henry, K. D. US Patent 5838003, 1998.

115. Hsieh, C. H.; Chang, C. H.; Urban, P. L.; Chen, Y. C., Capillary action-supported contact less atmospheric pressure ionization for the combined sampling and mass spectrometric analysis of biomolecules. *Anal Chem* **2011,** *83*, 2866-2869.

116. Hsieh, C. H.; Chao, C. S.; Mong, K. K. T.; Chen, Y. C., Online monitoring of chemical reactions by contactless atmospheric pressure ionization mass

spectrometry. *J Mass Spectrom* **2012,** *47*, 586-590.

117. Pirkl, A.; Dreisewerd, K.; Yew, J. Y.; Konig, S., Field-based ion generation from microscale emitters on natural and artificial objects for atmospheric pressure mass spectrometry. *Anal Bioanal Chem* **2010,** *397*, 3311-3316.

118. So, P. K.; Hu, B.; Yao, Z. P., Mass spectrometry: towards in vivo analysis of biological systems. *Mol Biosys* **2013,** *9*, 915-929.

119. Hu, B.; Wang, L.; Ye, W. C.; Yao, Z. P., *In vivo* and real-time monitoring of secondary metabolites of living organisms by mass spectrometry. *Sci Rep* **2013,** *3*, 2104.

120. Hirabayashi, A.; Sakairi, M.; Koizumi, H., Sonic spray ionization method for atmospheric-pressure ionization mass-spectrometry. *Anal Chem* **1994,** *66*, 4557-4559.

121. Haddad, R.; Milagre, H. M. S.; Catharino, R. R.; Eberlin, M. N., Easy ambient sonic-spray ionization mass spectrometry combined with thin-layer chromatography. *Anal Chem* **2008,** *80*, 2744-2750.

122. Inutan, E. D.; Trimpin, S., Matrix assisted ionization vacuum (MAIV), a new ionization method for biological materials analysis using mass spectrometry. *Mol Cell Proteomics* **2013,** *12*, 792-796.

123. Wleklinski, M.; Li, Y.; Bag, S.; Sarkar, D.; Narayanan, R.; Pradeep, T.; Cooks, R. G., Zero volt paper spray ionization and its mechanism. *Anal Chem* **2015**, *87*, 6786-6789.

124. Fenn, J. B.; Mann, M.; Meng, C. K.; Wong, S. F.; Whitehouse, C. M., Electrospray ionization for mass spectrometry of large biomolecules. *Science* **1989,** *246*, 64-71.

125. Wilm, M.; Mann, M., Analytical properties of the nanoelectrospray ion source. *Anal Chem* **1996,** *68*, 1-8.

126. Chen, L. C.; Nishidate, K.; Saito, Y.; Mori, K.; Asakawa, D.; Takeda, S.; Kubota, T.; Hori, H.; Hiraoka, K., Characteristics of probe electrospray generated from a solid needle. *J Phys Chem B* **2008,** *112*, 11164-11170.

127. Kuo, C. P.; Shiea, J., Application of direct electrospray probe to analyze biological compounds and to couple to solid-phase microextraction to detect trace surfactants in aqueous solution. *Anal Chem* **1999,** *71*, 4413-4417.

128. Wang, H.; Liu, J. J.; Cooks, R. G.; Ouyang, Z., Paper spray for direct analysis of complex mixtures using mass spectrometry. *Angew Chem Int Ed* **2010,** *49*, 877-880.

129. Liu, J. J.; Cooks, R. G.; Ouyang, Z., Biological tissue diagnostics using needle biopsy and spray ionization mass spectrometry. *Anal Chem* **2011,** *83*, 9221-9225.

130. Zheng, O. Y.; Wang, H.; Manicke, N. E.; Yang, Q. A.; Zheng, L. X.; Shi, R. Y.; Cooks, R. G., Direct analysis of biological tissue by paper spray mass spectrometry. *Anal Chem* **2011,** *83*, 1197-1201.

131. Manicke, N. E.; Yang, Q. A.; Wang, H.; Oradu, S.; Ouyang, Z.; Cooks, R. G., Assessment of paper spray ionization for quantitation of pharmaceuticals in blood spots. *Int J Mass Spectrom* **2011,** *300*, 123-129.

132. Li, A. Y.; Wang, H.; Ouyang, Z.; Cooks, R. G., Paper spray ionization of polar analytes using non-polar solvents. *Chem Commun* **2011,** *47*, 2811-2813.

133. Manicke, N. E.; Abu-Rabie, P.; Spooner, N.; Ouyang, Z.; Cooks, R. G., Quantitative analysis of therapeutic drugs in dried blood spot samples by paper spray mass spectrometry: an avenue to therapeutic drug monitoring. *J Am Soc Mass Spectrom* **2011,** *22*, 1501-1507.

134. Li, A. Y.; Wang, H.; Ouyang, Z.; Cooks, R. G., Paper spray ionization of polar analytes using non-polar solvents. *Chem Commun* **2011,** *47*, 2811-2813.

135. Todd, D., *Experimental Organic Chemistry*. Prentice-Hall: New Jersey, 1979.

136. MacKenzie, C. A., *Experimental Organic Chemistry*. 4th ed.; Prentice-Hall: New Jersey, 1971.

137. Mendiola, J. A.; Marin, F. R.; Hernandez, S. F.; Arredondo, B. O.; Senorans, F. J.; Ibanez, E.; Reglero, G., Characterization via liquid chromatography coupled to diode array detector and tandem mass spectrometry of supercritical fluid antioxidant extracts of Spirulina platensis microalga. *J Sep Sci* **2005,** *28*, 1031-1038.

138. Cheng, S. C.; Huang, M. Z.; Shiea, J., Thin layer chromatography/mass spectrometry. *J Chromatogr A* **2011,** *1218*, 2700-2711.

139. Hsu, F. L.; Chen, C. H.; Yuan, C. H.; Shiea, J., Interfaces to connect thin-layer chromatography with electrospray ionization mass spectrometry. *Anal Chem* **2003,** *75*, 2493-2498.

140. Li, X.; Hu, B.; Ding, J. H.; Chen, H. W., Rapid characterization of complex viscous samples at molecular levels by neutral desorption extractive electrospray ionization mass spectrometry. *Nat Protocols* **2011,** *6*, 1010-1025.

141. Low, A. S.; Wangboonskul, J., An HPLC assay for the determination of ketoconazole in common pharmaceutical preparations. *Analyst* **1999,** *124*, 1589-1593.

142. Yang, B.C.; Wang, F.; Deng, W.; Zou, Y.; Liu, F. Y.; Wan, X. D.; Yang, X.; Liu, H.; Huang, O.-p., Wooden-tip electrospray ionization mass spectrometry for trace analysis of toxic and hazardous compounds in food samples. *Anal Methods* **2015**, *7*, 5886-5890.

143. Yang, Y. Y.; Deng, J. W.; Yao, Z. P., Pharmaceutical analysis by solid-substrate electrospray ionization mass spectrometry with wooden tips. *J Am Soc Mass Spectrom* **2014,** *25*, 37-47.

144. Yang, B. C.; Liu, F. Y.; Wang, L.Q.; Zou, Y.; Wang, F.; Deng, W.; Wan, X. D; Yang, X.; He, M.; Huang, O. P., Serum metabolic profiling study of endometriosis by using wooden-tip electrospray ionization mass

spectrometry. *Anal Methods* **2015**, *5*, 6125-6132

145. Yang, B. C.; Liu, F. Y.; Guo, J. B.; Wan, L.; Wu, J.; Wang, F.; Liu, H.; Huang, O. P., Rapid assay of neopterin and biopterin in urine by wooden-tip electrospray ionization mass spectrometry. *Anal Methods* **2015,** *7*, 2913-2916.

146. So, P. K.; Ng, T.-T.; Wang, H.; Hu, B.; Yao, Z. P., Rapid detection and quantitation of ketamine and norketamine in urine and oral fluid by wooden-tip electrospray ionization mass spectrometry. *Analyst* **2013,** *138*, 2239-2243.

147. Xin, G. Z.; Hu, B.; Shi, Z. Q.; Lam, Y. C.; Dong, T. T. X.; Li, P.; Yao, Z. P.; Tsim, K. W. K., Rapid identification of plant materials by wooden-tip electrospray ionization mass spectrometry and a strategy to differentiate the bulbs of Fritillaria. *Anal Chim Acta* **2014,** *820*, 84-91.

148. Deng, J. W.; Yang, Y. Y.; Fang, L.; Lin, L.; Zhou, H. Y.; Luan, T. G., Coupling solid-phase microextraction with ambient mass spectrometry using surface coated wooden-tip probe for rapid analysis of ultra trace perfluorinated compounds in complex samples. *Anal Chem* **2014,** *86*, 11159-11166.

149. Wong, M. Y. M.; Tang, H. W.; Man, S. H.; Lam, C. W.; Che, C. M.; Ng, K. M., Electrospray ionization on porous spraying tips for direct sample analysis by mass spectrometry: enhanced detection sensitivity and selectivity using hydrophobic/hydrophilic materials as spraying tips. *Rapid*

Commun Mass Spectrom **2013,** *27*, 713-721.

150. Hu, B.; So, P. K.; Yao, Z. P., Analytical properties of solid-substrate electrospray ionization mass spectrometry. *J Am Soc Mass Spectrom* **2013,** *24*, 57-65.

151. Wong, M. Y. M.; Man, S. H.; Che, C. M.; Lau, K. C.; Ng, K. M., Negative electrospray ionization on porous supporting tips for mass spectrometric analysis: electrostatic charging effect on detection sensitivity and its application to explosive detection. *Analyst* **2014,** *139*, 1482-1491.

152. Li, A.; Hollerbach, A.; Luo, Q.; Cooks, R. G., On-demand ambient ionization of picoliter samples using charge pulses. *Angew Chem Int Ed* **2015,** *54*, 6893-6895.

153. So, P. K.; Ng, T. T.; Wang, H. X.; Hu, B.; Yao, Z. P., Rapid detection and quantitation of ketamine and norketamine in urine and oral fluid by wooden-tip electrospray ionization mass spectrometry. *Analyst* **2013,** *138*, 2239-2243.

154. Hu, B.; So, P. K.; Chen, H.; Yao, Z. P., Electrospray ionization using wooden tips. *Anal Chem* **2011,** *83*, 8201-8207.

155. Zhang, Z. P.; Xu, W.; Manicke, N. E.; Cooks, R. G.; Ouyang, Z., Silica Coated paper substrate for paper-spray analysis of therapeutic drugs in dried blood spots. *Anal Chem* **2012,** *84*, 931-938.

156. Hu, B.; Lai, Y. H.; So, P. K.; Chen, H.; Yao, Z. P., Direct ionization of

biological tissue for mass spectrometric analysis. *Analyst* **2012,** *137*, 3613-3619.

157. Chen, H. W.; Yang, S. P.; Li, M.; Hu, B.; Li, J. Q.; Wang, J., Sensitive detection of native proteins using extractive electrospray ionization mass spectrometry. *Angew Chem Int Ed* **2010,** *49*, 3053-3056.

158. Lion, N.; Gellon, J. O.; Jensen, H.; Girault, H. H., On-chip protein sample desalting and preparation for direct coupling with electrospray ionization mass spectrometry. *J Chromatogr A* **2003,** *1003*, 11-19.

159. Hu, B.; Yang, S. P.; Li, M.; Gu, H. W.; Chen, H. W., Direct detection of native proteins in biological matrices using extractive electrospray ionization mass spectrometry. *Analyst* **2011,** *136*, 3599-3601.

160. Gilar, M.; Belenky, A.; Wang, B. H., High-throughput biopolymer desalting by solid-phase extraction prior to mass spectrometric analysis. *J Chromatogr A* **2001,** *921*, 3-13.

161. Shariatgorji, M.; Astorga-Wells, J.; Jornvall, H.; Ilag, L. L., Microfluidic electrocapture-assisted mass spectrometry of membrane-associated polypeptides. *Anal Chem* **2008,** *80*, 7116-7120.

162. Laganowsky, A.; Reading, E.; Hopper, J. T. S.; Robinson, C. V., Mass spectrometry of intact membrane protein complexes. *Nat Protocols* **2013,** *8*, 639-651.

163. McLachlan, J. A., Environmental signaling: What embryos and evolution

teach us about endocrine disrupting chemicals. *Endocr Rev* **2001,** *22*, 319-341.

164. Briciu, R. D.; Kot-Wasik, A.; Namiesnik, J., Analytical challenges and recent advances in the determination of estrogens in water environments. *J Chromatogr Sci* **2009,** *47*, 127-139.

165. Pirro, V.; Jarmusch, A. K.; Vincenti, M.; Cooks, R. G., Direct drug analysis from oral fluid using medical swab touch spray mass spectrometry. *Anal Chim Acta* **2015,** *861*, 47-54.

166. Clauwaert, K.; Decalestecker, T.; Mortier, K.; Lambert, W.; Deforce, D.; Van Peteghem, C.; Van Bocxlaer, J., The determination of cocaine, benzoylecgonine, and cocaethylene in small-volume oral fluid samples by liquid chromatography-quadrupole-time-of-flight mass spectrometry. *J Anal Toxicol* **2004,** *28*, 655-659.

167. Samyn, N.; De Boeck, G.; Verstraete, A. G., The use of oral fluid and sweat wipes for the detection of drugs of abuse in drivers. *J Forensic Sci* **2002,** *47*, 1380-1387.

168. Caplan, Y. H.; Goldberger, B. A., Alternative specimens for workplace drug testing. *J Anal Toxicol* **2001,** *25*, 396-399.

169. Hu, B.; Peng, X. J.; Yang, S. P.; Gu, H. W.; Chen, H. W.; Huan, Y. F.; Zhang, T. T.; Qiao, X. L., Fast quantitative detection of cocaine in beverages using nanoextractive electrospray ionization tandem mass spectrometry. *J Am Soc Mass Spectrom* **2010,** *21*, 290-293.

170. Kala, S. V.; Harris, S. E.; Freijo, T. D.; Gerlich, S., Validation of analysis of amphetamines, opiates, phencyclidine, cocaine, and benzoylecgonine in oral fluids by liquid chromatography-tandem mass spectrometry. *J Anal Toxicol* **2008,** *32*, 605-611.

171. Cross, L. M.; Cook, M. A.; Lin, S.; Chen, J. N.; Rubinstein, A. L., Rapid analysis of angiogenesis drugs in a live fluorescent zebrafish assay. *Arteriosclerosis Thrombosis and Vascular Biology* **2003,** *23*, 911-912.

172. Peng, Y. E.; Zhang, S. C.; Wen, F.; Ma, X. X.; Yang, C. D.; Zhang, X. R., *In vivo* nanoelectrospray for the localization of bioactive molecules in plants by mass spectrometry. *Anal Chem* **2012,** *84*, 3058-3062.

173. Jones-Lepp, T. L.; Momplaisir, G. M., New applications of LC-MS and LC-MS2 toward understanding the environmental fate of organometallics. *Trac-Trends Anal Chem* **2005,** *24*, 590-595.

174. Cotrufo, C.; Lunsetter, P., The fatty acids of potato tubers (Solanum tuberosum). *Am J Pot Res* **1964,** *41*, 18-22.

175. Putz, B., Lisinska, G., and W. Leszczynski: Potato Science and Technology. Elsevier Applied Science Publishers Ltd., London, New York 1989. 391 pages, 164 figures and 54 schedules. Hardcover, 58. *Starch - Stärke* **1990,** *42*, 326-326.

176. Bouvier, E. S. P.; Koza, S. M., Advances in size-exclusion separations of proteins and polymers by UHPLC. *Trac-Trend Anal Chem* **2014,** *63*, 85-94.

177. Yates, J. R., 3rd; Carmack, E.; Hays, L.; Link, A. J.; Eng, J. K., Automated protein identification using microcolumn liquid chromatography-tandem mass spectrometry. *Methods Mol Bio* **1999,** *112*, 553-69.

178. Guttman, A., Capillary sodium dodecyl sulfate-gel electrophoresis of proteins. *Electrophoresis* **1996,** *17*, 1333-41.

179. Rabilloud, T.; Lelong, C., Two-dimensional gel electrophoresis in proteomics: a tutorial. *J Proteomics* **2011,** *74*, 1829-1841.

180. Benesch, J. L. P.; Ruotolo, B. T.; Simmons, D. A.; Robinson, C. V., Protein complexes in the gas phase: Technology for structural genomics and proteomics. *Chem Rev* **2007,** *107*, 3544-3567.

181. Heck, A. J. R., Native mass spectrometry: a bridge between interactomics and structural biology. *Nat Methods* **2008,** *5*, 927-933.

182. Baker, L. A.; Baldus, M., Characterization of membrane protein function by solid-state NMR spectroscopy. *Curr Opin Struc Biol* **2014,** *27*, 48-55.

183. Marion, D., An introduction to biological NMR spectroscopy. *Mol Cell Proteomics* **2013,** *12*, 3006-3025.

184. Vuister, G. W.; Fogh, R. H.; Hendrickx, P. M. S.; Doreleijers, J. F.; Gutmanas, A., An overview of tools for the validation of protein NMR structures. *J Biomol NMR* **2014,** *58*, 259-285.

185. Kornreich, M.; Avinery, R.; Beck, R., Modern X-ray scattering studies of complex biological systems. *Curr Opin Biotech* **2013,** *24*, 716-723.

186. Richardson, J. S.; Richardson, D. C., Biophysical highlights from 54 years of macromolecular crystallography. *Biophys J* **2014,** *106*, 510-525.

187. Hoofnagle, A. N.; Resing, K. A.; Ahn, N. G., Protein analysis by hydrogen exchange mass spectrometry. *Annu Rev Bioph Biom* **2003,** *32*, 1-25.

188. Yao, Z. P.; Tito, P.; Robinson, C. V., Site-specific hydrogen exchange of proteins: insights into the structures of amyloidogenic intermediates. *Method Enzymol* **2005,** *402*, 389-402.

189. Laganowsky, A.; Reading, E.; Allison, T. M.; Ulmschneider, M. B.; Degiacomi, M. T.; Baldwin, A. J.; Robinson, C. V., Membrane proteins bind lipids selectively to modulate their structure and function. *Nature* **2014,** *510*, 172-172.

190. Lanucara, F.; Holman, S. W.; Gray, C. J.; Eyers, C. E., The power of ion mobility-mass spectrometry for structural characterization and the study of conformational dynamics. *Nat Chem* **2014,** *6*, 281-294.

191. Uetrecht, C.; Rose, R. J.; van Duijn, E.; Lorenzen, K.; Heck, A. J. R., Ion mobility mass spectrometry of proteins and protein assemblies. *Chem Soc Rev* **2010,** *39*, 1633-1655.

192. Williams, D. M.; Pukala, T. L., Novel insights into protein misfolding diseases revealed by ion mobility-mass spectrometry. *Mass Spectrom Rev* **2013,** *32*, 169-187.

193. Zhou, M.; Politis, A.; Davies, R. B.; Liko, I.; Wu, K. J.; Stewart, A. G.;

Stock, D.; Robinson, C. V., Ion mobility-mass spectrometry of a rotary ATPase reveals ATP-induced reduction in conformational flexibility. *Nat Chem* **2014,** *6*, 208-215.

194. Mehmood, S.; Allison, T. M.; Robinson, C. V., Mass spectroscopy in protein complexes: from origins to applications. *Annu Rev Phys Chem* **2015,** *66*.

195. Ouyang, Z.; Takats, Z.; Blake, T. A.; Gologan, B.; Guymon, A. J.; Wiseman, J. M.; Oliver, J. C.; Davisson, V. J.; Cooks, R. G., Preparing protein microarrays by soft-landing of mass-selected ions. *Science* **2003,** *301*, 1351-1354.

196. Nitta, K.; Sugai, S., The Evolution of lysozyme and alpha-lactalbumin. *Eur J Biochem* **1989,** *182*, 111-118.

197. Jurneczko, E.; Barran, P. E., How useful is ion mobility mass spectrometry for structural biology? The relationship between protein crystal structures and their collision cross sections in the gas phase. *Analyst* **2011,** *136*, 20-28.

198. Shelimov, K. B.; Clemmer, D. E.; Hudgins, R. R.; Jarrold, M. F., Protein structure in vacuo: Gas-phase confirmations of BPTI and cytochrome *c*. *J Am Chem Soc* **1997,** *119*, 2240-2248.

199. Fenn, J. B.; Mann, M.; Meng, C. K.; Wong, S. F.; Whitehouse, C. M., Electrospray ionization-principles and practice. *Mass Spectrom Rev* **1990,** *9*, 37-70.

200. Hong, C. M.; Lee, C. T.; Lee, Y. M.; Kuo, C. P.; Yuan, C. H.; Shiea, J., Generating electrospray from solutions predeposited on a copper wire. *Rapid Commun Mass Spectrom* **1999,** *13,* 21-25.

201. Monge, M. E.; Harris, G. A.; Dwivedi, P.; Fernández, F. M., Mass spectrometry: recent advances in direct open air surface sampling/ionization. *Chem Rev* **2013**, *113*, 2269-2308

202. Fenn, J. B. Method and apparatus for electrospray ionization. U.S. Patent 6297499. October 2, 2001.

203. Yu, Z.; Chen, L. C.; Erra-Balsells, R.; Nonami, H.; Hiraoka, K., Real-time reaction monitoring by probe electrospray ionization mass spectrometry. *Rapid Commun Mass Spectrom* **2010,** *24,* 1507-1513.

204. Svedberg, M.; Pettersson, A.; Nilsson, S.; Bergquist, J.; Nyholm, L.; Nikolajeff, F.; Markides, K., Sheathless electrospray from polymer microchips. *Anal Chem* **2003,** *75*, 3934-3940.

205. Espy, R. D.; Muliadi, A. R.; Ouyang, Z.; Cooks, R. G., Spray mechanism in paper spray ionization. *Int J Mass Spectrom* **2012,** *325*, 167-171.

206. Espy, R. D.; Manicke, N. E.; Ouyang, Z.; Cooks, R. G., Rapid analysis of whole blood by paper spray mass spectrometry for point-of-care therapeutic drug monitoring. *Analyst* **2012,** *137*, 2344-2349.

207. Wang, H.; Manicke, N. E.; Yang, Q. A.; Zheng, L. X.; Shi, R. Y.; Cooks, R. G.; Zheng, O. Y., Direct analysis of biological tissue by paper spray mass

spectrometry. *Anal Chem* **2011,** *83*, 1197-1201.

208. Manicke, N. E.; Abu-Rabie, P.; Spooner, N.; Ouyang, Z.; Cooks, R. G., Quantitative analysis of therapeutic drugs in dried blood spot samples by paper spray mass spectrometry: an avenue to therapeutic drug monitoring. *J Am Soc Mass Spectrom* **2011,** *22*, 1501-1507.

209. Cooks, R. G.; Manicke, N. E.; Dill, A. L.; Ifa, D. R.; Eberlin, L. S.; Costa, A. B.; Wang, H.; Huang, G. M.; Zheng, O. Y., New ionization methods and miniature mass spectrometers for biomedicine: DESI imaging for cancer diagnostics and paper spray ionization for therapeutic drug monitoring. *Faraday Discuss* **2011,** *149*, 247-267.

210. Hu, B.; Lai, Y. H.; So, P. K.; Chen, H.; Yao, Z. P., Direct ionization of biological tissue for mass spectrometric analysis. *Analyst* **2012,** *137*, 3613-3619.

211. Huang, M. Z.; Yuan, C. H.; Cheng, S. C.; Cho, Y. T.; Shiea, J., Ambient ionization mass spectrometry. *Annu Rev Anal Chem,* **2011**; *3,* 43-65.

212. Hsu, N. F.; Tseng, S. Y.; Wu, C. Y.; Ren, C. T.; Lee, Y. C.; Wong, C. H.; Chen, C. H., Desorption ionization of biomolecules on metals. *Anal Chem* **2008,** *80*, 5203-5210.

213. Turhan, S., Aluminium contents in baked meats wrapped in aluminium foil. *Meat Sci* **2006,** *74*, 644-647.

214. Liu, C. C.; Wu, C. F.; Shiea, J.; Cho, Y. T.; Hsieh, T. J.; Chou, Y. H.; Chen,

B. H.; Huang, S. P.; Wu, W. J.; Shen, J. T.; Chang, M. Y.; Huang, C. H.; Chang, A. W.; Wu, M. T., Detection of melamine in a human renal uric acid stone by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS). *Clin Chim Acta* **2012,** *413*, 1689-1695.

215. Law, W. S.; Chen, H. W.; Ding, J. H.; Yang, S. P.; Zhu, L.; Gamez, G.; Chingin, K.; Ren, Y. L.; Zenobi, R., Rapid characterization of complex viscous liquids at the molecular level. *Angew Chem Int Ed* **2009,** *48*, 8277-8280.

216. Cardoso, S. G.; Schapoval, E. E. S., Microbiological assay for terbinafine hydrochloride in tablets and creams. *Int J Pharm* **2000,** *203*, 109-113.

217. Lim, A. Y.; Seviour, J., Doped silica nanoparticles for the detection of pharmaceutical terbinafine in latent fingerprints by mass spectrometry. *Anal Methods* **2012,** *4*, 1983-1988.

218. Yang, S.; Zhang, D.; Xu, Y.; Wang, X.; Liu, X.; Wang, S.; Wang, J.; Wu, M.; He, Z.; Zhao, J.; Yuan, H., Discriminating the endogenous and exogenous urinary estrogens in human by isotopic ratio mass spectrometry and its potential clinical value. *Steroids* **2013,** *78*, 297-303.

219. Lai, Y. H.; So, P. K.; Lo, S. C. L.; Ng, E. W. Y.; Poon, T. C. W.; Yao, Z. P., Rapid differentiation of Panax ginseng and Panax quinguefolius by matrix-assisted laser desorption/ionization mass spectrometry. *Anal Chim Acta* **2012,** *753*, 73-81.

220. Cohen, S. L.; Chait, B. T., Influence of matrix solution conditions on the MALDI-MS analysis of peptides and proteins. *Anal Chem* **1996,** *68*, 31-37.

221. Benesch, J. L. P.; Sobott, F.; Robinson, C. V., Thermal dissociation of multimeric protein complexes by using nanoelectrospray mass spectrometry. *Anal Chem* **2003,** *75*, 2208-2214.

222. Takamizawa, A.; Fujimaki, S.; Sunner, J.; Hiraoka, K., Denaturation of lysozyme and myoglobin in laser spray. *J Am Soc Mass Spectrom.* **2005,** *16*, 860-868.

223. Halgand, F.; Laprevote, O., Mean charge state and charge state distribution of proteins as structural probes. an electrospray ionisation mass spectrometry study of lysozyme and ribonuclease A. *Eur J Mass Spectrom* **2001,** *7*, 433-439.

224. Yan, X. G.; Watson, J.; Ho, P. S.; Deinzer, M. L., Mass spectrometric approaches using electrospray ionization charge states and hydrogen-deuterium exchange for determining protein structures and their conformational changes. *Mol Cell Proteomics* **2004,** *3*, 10-23.

225. Kim, M. Y.; Maier, C. S.; Reed, D. J.; Deinzer, M. L., Conformational changes in chemically modified Escherichia coli thioredoxin monitored by H/D exchange and electrospray ionization mass spectrometry. *Protein Sci* **2002,** *11*, 1320-1329.

226. Maier, C. S.; Schimerlik, M. I.; Deinzer, M. L., Thermal denaturation of escherichia coli thioredoxin studied by hydrogen/deuterium exchange and

electrospray ionization mass spectrometry: monitoring a two-state protein unfolding transition. *Biochemistry* **1999,** *38*, 1136-1143.

227. Knubovets, T.; Osterhout, J. J.; Connolly, P. J.; Klibanov, A. M., Structure, thermostability, and conformational flexibility of hen egg-white lysozyme dissolved in glycerol. *Pro Natl Acad Sci USA* **1999,** *96*, 1262-1267.

228. Cody, R. B.; Laramee, J. A.; Durst, H. D., Versatile new ion source for the analysis of materials in open air under ambient conditions. *Anal Chem* **2005,** *77*, 2297-2302.

229. So, P. K.; Hu, B.; Yao, Z. P., Electrospray ionization on solid substrates. *Mass Spectrometry* **2014,** *3*, S0028.

230. Su, Y.; Wang, H.; Liu, J. J.; Wei, P.; Cooks, R. G.; Ouyang, Z., Quantitative paper spray mass spectrometry analysis of drugs of abuse. *Analyst* **2013,** *138*, 4443-4447.

231. Pasilis, S. P.; Van Berkel, G. J., Atmospheric pressure surface sampling/ionization techniques for direct coupling of planar separations with mass spectrometry. *J Chromatogr A* **2010,** *1217*, 3955-3965.

232. Chang, T. T.; Lay, J. O.; Francel, R. J., Direct analysis of thin-layer chromatography spots by fast atom bombardment mass-spectrometry. *Anal Chem* **1984,** *56*, 109-111.

233. Yao, Z. P.; Liu, G. Q.; Wen, H. H.; He, W. D.; Zhao, S. K.; Hu, W. M., Determination of the glycosyl sequence of glycosides by enzymatic-

hydrolysis followed by TLC-FABMS. *Planar Chromat* **1994,** *7*, 410-412.

234. Distler, U.; Hulsewig, M.; Souady, J.; Dreisewerd, K.; Haier, J.; Senninger, N.; Friedrich, A. W.; Karch, H.; Hillenkamp, F.; Berkenkamp, S.; Peter-Katalinic, J.; Muthing, J., Matching IR-MALDI-o-TOF mass spectrometry with the TLC overlay binding assay and its clinical application for tracing tumor-associated glycosphingolipids in hepatocellular and pancreatic cancer. *Anal Chem* **2008,** *80*, 1835-1846.

235. Himmelsbach, M.; Waser, M.; Klampfl, C., Thin layer chromatography-spray mass spectrometry: a method for easy identification of synthesis products and UV filters from TLC aluminum foils. *Anal Bioanal Chem* **2014,** *406*, 3647-3656.

236. Aranda, M.; Morlock, G., New method for caffeine quantification by planar chromatography coupled with electropray ionization mass spectrometry using stable isotope dilution analysis. *Rapid Commun Mass Spectrom* **2007,** *21*, 1297-1303.

237. Smrke, S.; Vovk, I., Comprehensive thin-layer chromatography mass spectrometry of flavanols from Juniperus communis L. and Punica granatum L. *J Chromatogr A* **2013,** *1289*, 119-126.

238. Jautz, U.; Morlock, G., Efficacy of planar chromatography coupled to (tandem) mass spectrometry for employment in trace analysis. *J Chromatogr A* **2006,** *1128*, 244-250.

239. Van Berkel, G. J.; Tomkins, B. A.; Kertesz, V., Thin-layer

chromatography/desorption electrospray ionization mass spectrometry: Investigation of goldenseal alkaloids. *Anal Chem* **2007,** *79*, 2778-2789.

240. Paglia, G.; Ifa, D. R.; Wu, C. P.; Corso, G.; Cooks, R. G., Desorption electrospray ionization mass spectrometry analysis of lipids after two-dimensional high-performance thin-layer chromatography partial separation. *Anal Chem* **2010,** *82*, 1744-1750.

241. Morlock, G.; Ueda, Y., New coupling of planar chromatography with direct analysis in real time mass spectrometry. *J Chromatogr A* **2007,** *1143*, 243-251.

242. Morlock, G.; Schwack, W., Determination of isopropylthioxanthone (ITX) in milk, yoghurt and fat by HPTLC-FLD, HPTLC-ESI/MS and HPTLC-DART/MS. *Anal Bioanal Chem* **2006,** *385*, 586-595.

243. Kim, H. J.; Oh, M. S.; Hong, J.; Jang, Y. P., Quantitative analysis of major dibenzocyclooctane lignans in schisandrae fructus by online TLC-DART-MS. *Phytochem Anal* **2010,** *22*, 258-262.

244. Lin, S. Y.; Huang, M. Z.; Chang, H. C.; Shiea, J., Using electrospray-assisted laser desorption/ionization mass spectrometry to characterize organic compounds separated on thin-layer chromatography plates. *Anal Chem* **2007,** *79*, 8789-8795.

245. Yeung, Y. G.; Nieves, E.; Angeletti, R. H.; Stanley, E. R., Removal of detergents from protein digests for mass spectrometry analysis. *Anal Biochem* **2008,** *382*, 135-137.

246. Lu, Y.; Chen, D. F., Analysis of Schisandra chinensis and Schisandra sphenanthera. *J Chromatogr A* **2009,** *1216*, 1980-1990.

247. Quach, H. T.; Steeper, R. L.; Griffin, G. W., An improved method for the extraction and thin-layer chromatography of chlorophyll A and B from spinach. *J Chem Educ* **2004,** *81*, 385-387.

248. Wang, H.; Liu, J.; Cooks, R. G.; Ouyang, Z., Paper spray for direct analysis of complex mixtures using mass spectrometry. *Angew Chem Int Ed* **2010,** *49*, 877-880.

249. Hu, B.; So, P. K.; Yao, Z. P., Electrospray ionization with aluminum foil: A versatile mass spectrometric technique. *Anal Chim Acta* **2014,** *817*, 1-8.

250. Hu, B.; Wang, L.; Ye, W. C.; Yao, Z. P., *In vivo* and real-time monitoring of secondary metabolites of living organisms by mass spectrometry. *Sci Rep* **2013,** *3*, 2104

251. Bojko, B.; Cudjoe, E.; Gomez-Rios, G. A.; Gorynski, K.; Jiang, R.; Reyes-Garces, N.; Risticevic, S.; Silva, E. A. S.; Togunde, O.; Vuckovic, D.; Pawliszyn, J., SPME - Quo vadis? *Anal Chim Acta* **2012,** *750*, 132-151.

252. Vuckovic, D.; Zhang, X.; Cudjoe, E.; Pawliszyn, J., Solid-phase microextraction in bioanalysis: new devices and directions. *J Chromatogr A* **2010,** *1217*, 4041-4060.

253. Hossain, S. M. Z.; Bojko, B.; Pawliszyn, J., Automated SPME-GC-MS monitoring of headspace metabolomic responses of E. coli to biologically

active components extracted by the coating. *Anal Chim Acta* **2013,** *776*, 41-49.

254. Du, F.; Alam, M. N.; Pawliszyn, J., Aptamer-functionalized solid phase microextraction-liquid chromatography/tandem mass spectrometry for selective enrichment and determination of thrombin. *Anal Chim Acta* **2014,** *845*, 45-52.

255. Deng, J.; Yang, Y.; Wang, X.; Luan, T., Strategies for coupling solid-phase microextraction with mass spectrometry. *Trac-Trends Anal Chem* **2014,** *55*, 55-67.

256. Tong, H.; Sze, N.; Thomson, B.; Nacson, S.; Pawliszyn, J., Solid phase microextraction with matrix assisted laser desorption/ionization introduction to mass spectrometry and ion mobility spectrometry. *Analyst* **2002,** *127*, 1207-1210.

257. Wang, Y.; Schneider, B. B.; Covey, T. R.; Pawliszyn, J., High-performance SPME/AP MALDI system for high-throughput sampling and determination of peptides. *Anal Chem* **2005,** *77*, 8095-8101.

258. Mirnaghi, F. S.; Pawliszyn, J., Reusable Solid-phase microextraction coating for direct immersion whole-blood analysis and extracted blood spot sampling coupled with liquid chromatography-tandem mass spectrometry and direct analysis in real-time tandem mass spectrometry. *Anal Chem* **2012,** *84*, 8301-8309.

259. Rodriguez-Lafuente, A.; Mirnaghi, F. S.; Pawliszyn, J., Determination of

cocaine and methadone in urine samples by thin-film solid-phase microextraction and direct analysis in real time (DART) coupled with tandem mass spectrometry. *Anal Bioanal Chem* **2013,** *405*, 9723-9727.

260. D'Agostino, P. A.; Chenier, C. L.; Hancock, J. R.; Lepage, C. R. J., Desorption electrospray ionisation mass spectrometric analysis of chemical warfare agents from solid-phase microextraction fibres. *Rapid Commun Mass Spectrom* **2007,** *21*, 543-549.

261. Chou, J. H.; Huang, M. Z.; Shiea, J. In *Development of solid-phase microextraction (SPME) coupled with ambient mass spectrometry for rapidly characterizing trace emerging contaminants in aqueous samples*, 4th Asia Oceania Mass Spectrometry Conference, 2013, 2013.

262. Ahmad, S.; Tucker, M.; Spooner, N.; Murnane, D.; Gerhard, U., Direct ionization of solid-phase microextraction fibres for quantitative drug bioanalysis: from peripheral circulation to mass spectrometry detection. *Anal Chem* **2015,** *87*, 754-759.

263. Gómez-Ríos, G. A.; Pawliszyn, J., Development of coated blade spray ionization mass spectrometry for the quantitation of target analytes present in complex matrices. *Angew Chemie Int Ed* **2014,** *53*, 14503-14507.

264. Deng, J.; Yang, Y.; Fang, L.; Lin, L.; Zhou, H.; Luan, T., Coupling solid-phase microextraction with ambient mass spectrometry using surface coated wooden-tip probe for rapid analysis of ultra trace perfluorinated compounds in complex samples. *Anal Chem* **2014,** *86*, 11159-11166.

265. Strano-Rossi, S.; Castrignanò, E.; Anzillotti, L.; Serpelloni, G.; Mollica, R.; Tagliaro, F.; Pascali, J. P.; di Stefano, D.; Sgalla, R.; Chiarotti, M., Evaluation of four oral fluid devices (DDS®, Drugtest 5000®, Drugwipe 5+® and RapidSTAT®) for on-site monitoring drugged driving in comparison with UHPLC–MS/MS analysis. *Forensic Sci Int* **2012,** *221*, 70-76.

266. Wille, S. M. R.; Samyn, N.; Ramírez-Fernández, M. d. M.; De Boeck, G., Evaluation of on-site oral fluid screening using Drugwipe-5+®, RapidSTAT® and Drug Test 5000® for the detection of drugs of abuse in drivers. *Forensic Science Int* **2010,** *198*, 2-6.

267. Huang, M. H.; Wu, M. Y.; Wu, C. H.; Tsai, J. L.; Lee, H. H.; Liu, R. H., Performance characteristics of ELISAs for monitoring ketamine exposure. *Clin Chim Acta* **2007,** *379*, 59-65.

268. Huang, M. K.; Liu, C.; Li, J. H.; Huang, S.-D., Quantitative detection of ketamine, norketamine, and dehydronorketamine in urine using chemical derivatization followed by gas chromatography–mass spectrometry. *J Chromatogr B* **2005,** *820*, 165-173.

269. Nosseir, N. S.; Michels, G.; Binder, P.; Wiesen, M. H. J.; Mueller, C., Simultaneous detection of ketamine, lorazepam, midazolam and sufentanil in human serum with liquid chromatography-tandem mass spectrometry for monitoring of analgosedation in critically ill patients. *J. Chromatogr B* **2014,** *973*, 133-141.

270. Cheng, P. S.; Fu, C. Y.; Lee, C. H.; Liu, C. R.; Chien, C. S., GC-MS quantification of ketamine, norketamine, and dehydronorketamine in urine specimens and comparative study using ELISA as the preliminary test methodology. *J. Chromatogr B* **2007,** *852*, 443-449.

271. Kim, E. M.; Lee, J. S.; Choi, S. K.; Lim, M. A.; Chung, H. S., Analysis of ketamine and norketamine in urine by automatic solid-phase extraction (SPE) and positive ion chemical ionization-gas chromatography-mass spectrometry (PCI-GC-MS). *Forensic Sci Int* **2008,** *174*, 197-202.

272. Chen, C. Y.; Lee, M. R.; Cheng, F. C.; Wu, G. J., Determination of ketamine and metabolites in urine by liquid chromatography-mass spectrometry. *Talanta* **2007,** *72*, 1217-1222.

273. Parkin, M. C.; Turfus, S. C.; Smith, N. W.; Halket, J. M.; Braithwaite, R. A.; Elliott, S. P.; Osselton, M. D.; Cowan, D. A.; Kicman, A. T., Detection of ketamine and its metabolites in urine by ultra high pressure liquid chromatography-tandem mass spectrometry. *J. Chromatogr B* **2008,** *876*, 137-142.

274. Cheng, J. Y. K.; Mok, V. K. K., Rapid determination of ketamine in urine by liquid chromatography-tandem mass spectrometry for a high throughput laboratory. *Forensic Sci Int* **2004,** *142*, 9-15.

275. Monge, M. E.; Harris, G. A.; Dwivedi, P.; Fernandez, F. M., Mass spectrometry: recent advances in direct open air surface sampling/ionization. *Chem Rev* **2013,** *113*, 2269-2308.

276. Huang, M. Z.; Yuan, C. H.; Cheng, S. C.; Cho, Y. T.; Shiea, J., Ambient ionization mass spectrometry. *Annu Rev Anal Chem* **2010,** *3*, 43-65.

277. Fenn, J. B.; Mann, M.; Meng, C. K.; Wong, S. F.; Whitehouse, C. M., Electrospray ionization for mass spectrometry of large biomolecules. *Science* **1989,** *6*, 64-71.

278. Smith, R. D.; Loo, J. A.; Edmonds, C. G.; Barinaga, C. J.; Udseth, H. R., New developments in biochemical mass-spectrometry - electrospray ionization. *Anal Chem* **1990,** *62*, 882-899.

279. Cech, N. B.; Enke, C. G., Practical implications of some recent studies in electrospray ionization fundamentals. *Mass Spectrom Rev* **2001,** *20*, 362-387.

280. Hu, B.; Wang, L.; Ye, W. C.; Yao, Z. P., *In vivo* and real-time monitoring of secondary metabolites of living organisms by mass spectrometry. *Sci Rep* **2013,** *3*, 2104.

281. Meher, A. K.; Chen, Y. C., Polarization induced electrospray ionization mass spectrometry for the analysis of liquid, viscous and solid samples. *J Mass Spectrom* **2015,** *50*, 444-450.

282. Lim, C. K.; Lord, G., Current developments in LC-MS for pharmaceutical analysis. *Bio Pharmac Bull* **2002,** *25*, 547-557.

283. Kataoka, H., Recent advances in solid-phase microextraction and related techniques for pharmaceutical and biomedical analysis. *Curr Pharmac*

Anal **2005,** *1*, 65-84.

284. Zhang, X. L.; Jia, B.; Huang, K. K.; Hu, B.; Chen, R.; Chen, H. W., Tracing origins of complex pharmaceutical preparations using surface desorption atmospheric pressure chemical ionization mass spectrometry. *Anal Chem* **2010,** *82*, 8060-8070.

285. Chowdhury, M. A. Z.; Fakhruddin, A. N. M.; Islam, M. N.; Moniruzzaman, M.; Gan, S. H.; Alam, M. K., Detection of the residues of nineteen pesticides in fresh vegetable samples using gas chromatography-mass spectrometry. *Food Control* **2013,** *34*, 457-465.

286. Vidal, J. L. M.; Plaza-Bolanos, P.; Romero-Gonzalez, R.; Frenich, A. G., Determination of pesticide transformation products: a review of extraction and detection methods. *J Chromatogr A* **2009,** *1216*, 6767-6788.

287. Gross, J. H., Direct analysis in real time-a critical review on DART-MS. *Anal Bioanal Chem* **2014,** *406*, 63-80.

288. Shiea, C.; Huang, Y. L.; Liu, D. L.; Chou, C. C.; Chou, J. H.; Chen, P. Y.; Shiea, J.; Huang, M. Z., Rapid screening of residual pesticides on fruits and vegetables using thermal desorption electrospray ionization mass spectrometry. *Rapid Commun Mass Spectrom* **2015,** *29*, 163-170.

289. Hu, B.; Zhang, X. L.; Li, M.; Peng, X. J.; Han, J.; Yang, S. P.; Ouyang, Y. Z.; Chen, H. W., Coupling corona discharge for ambient extractive ionization mass spectrometry. *Analyst* **2011,** *136*, 4977-4985.

290. Malaj, N.; Ouyang, Z.; Sindona, G.; Cooks, R. G., Analysis of pesticide residues by leaf spray mass spectrometry. *Anal Methods* **2012,** *4*, 1913-1919.

291. Forward, R. B.; Rittschof, D., Alteration of photoresponses involved in diel vertical migration of a crab larva by fish mucus and degradation products of mucopolysaccharides. *J Exp Marine Bio Eco* **2000,** *245*, 277-292.

292. Ekman, D. R.; Skelton, D. M.; Davis, J. M.; Villeneuve, D. L.; Cavallin, J. E.; Schroeder, A.; Jensen, K. M.; Ankley, G. T.; Collette, T. W., Metabolite profiling of fish skin mucus: a novel approach for minimally-invasive environmental exposure monitoring and surveillance. *Environ Sci Technol* **2015,** *49*, 3091-3100.

293. Chong, K.; Ying, T. S.; Foo, J.; Jin, L. T.; Chong, A., Characterisation of proteins in epidermal mucus of discus fish (Symphysodon spp.) during parental phase. *Aquaculture* **2005,** *249*, 469-476.

294. Adeniyi, A. A.; Okedeyi, O. O.; Yusuf, K. A., Flame ionization gas chromatographic determination of phthalate esters in water, surface sediments and fish species in the Ogun river catchments, Ketu, Lagos, Nigeria. *Environ Monit Ass* **2011,** *172*, 561-569.

295. Jee, J. H.; Koo, J. G.; Keum, Y. H.; Park, K. H.; Choi, S. H.; Kang, J. C., Effects of dibutyl phthalate and di-ethylhexyl phthalate on acetylcholinesterase activity in bagrid catfish, Pseudobagrus fulvidraco

(Richardson). *J Appl Ichthyol* **2009,** *25*, 771-775.

296. Zanotelli, V. R. T.; Neuhauss, S. C. F.; Ehrengruber, M. U., Long-term exposure to bis(2-ethylhexyl)phthalate (DEHP) inhibits growth of guppy fish (Poecilia reticulata). *J Appl Toxicol* **2010,** *30*, 29-33.

297. Szczerbik, P.; Socha, M.; Mikolajczyk, T.; Sokolowska-Mikolajczyk, M.; Chyb, J.; Epler, P., The effect of di(2-ethylheksyl) phthalate (DEHP) on LH secretion in Prussian carp (Carassius auratus gibelio B.) and common carp (Cyprinus carpio L.). *Cybium* **2008,** *32*, 52-53.

298. Shi, W.; Wang, X. Y.; Hu, G. J.; Hao, Y. Q.; Zhang, X. W.; Liu, H. L.; Wei, S.; Wang, X. R.; Yu, H. X., Bioanalytical and instrumental analysis of thyroid hormone disrupting compounds in water sources along the Yangtze River. *Environ Pollut* **2011,** *159*, 441-448.

299. Chi, J., Phthalate acid esters in potamogeton crispus l. from haihe river, china. *Chemosphere* **2009,** *77*, 48-52.

300. Haase, N. U., Glycoalkaloid concentration in potato tubers related to storage and consumer offering. *Potato Res* **2010,** *53*, 297-307.

301. Friedman, M.; McDonald, G. M., Potato glycoalkaloids: Chemistry, analysis, safety, and plant physiology. *Crit Rev Plant Sci* **1997,** *16*, 55-132.

302. Valkonen, J. P. T.; Keskitalo, M.; Vasara, T.; Pietila, L., Potato glycoalkaloids: a burden or a blessing? *Crit Rev Plant Sci* **1996,** *15*, 1-20.

303. Dorland, W. A. N.; Anderson, D. M., *Dorland's Illustrated Medical*

Dictionary 28th Edition. Elsevier Health Sciences: Philadelphia., 1994.

304. Schrier, R. W., Body-fluid volume regulation in health and disease - a unifying hypothesis. *Ann Intern Med* **1990,** *113*, 155-159.

305. Antunes-Rodrigues, J.; De Castro, M.; Elias, L. L. K.; Valenca, M. M.; McCann, S. M., Neuroendocrine control of body fluid metabolism. *Physiol Rev* **2004,** *84*, 169-208.

306. Wang, S. J.; Jia, L.; Zhou, H. M.; Shi, W.; Zhang, J. N., Knockdown of caveolin-1 by siRNA inhibits the transformation of mouse hepatoma h22 cells *in vitro* and *in vivo*. *Oligonucleotides* **2009,** *19*, 81-88.

307. Fernandis, A. Z.; Wenk, M. R., Lipid-based biomarkers for cancer. *J Chromatogr B* **2009,** *877*, 2830-2835.

308. Roepke, J.; Salim, V.; Wu, M.; Thamm, A. M. K.; Murata, J.; Ploss, K.; Boland, W.; De Luca, V., Vinca drug components accumulate exclusively in leaf exudates of Madagascar periwinkle. *Pro Natl Acad Sci USA* **2010,** *107*, 15287-15292.

309. Kessler, A.; Baldwin, I. T., Defensive function of herbivore-induced plant volatile emissions in nature. *Science* **2001,** *291*, 2141-2144.

310. Chughtai, K.; Heeren, R. M. A., Mass Spectrometric imaging for biomedical tissue analysis. *Chem Rev* **2010,** *110*, 3237-3277.

311. Dunn, W. B.; Broadhurst, D. I.; Atherton, H. J.; Goodacre, R.; Griffin, J. L., Systems level studies of mammalian metabolomes: the roles of mass

spectrometry and nuclear magnetic resonance spectroscopy. *Chem Soc Rev* **2011,** *40*, 387-426.

312. Alberici, R. M.; Simas, R. C.; Sanvido, G. B.; Romao, W.; Lalli, P. M.; Benassi, M.; Cunha, I. B. S.; Eberlin, M. N., Ambient mass spectrometry: bringing MS into the "real world". *Anal Bioanal Chem* **2010,** *398*, 265-294.

313. Huang, M. Z.; Yuan, C. H.; Cheng, S. C.; Cho, Y. T.; Shiea, J., Ambient ionization mass spectrometry. *Annu Rev Anal Chem* **2010,** *3*, 43-65.

314. Yao, Z. P., Characterization of proteins by ambient mass spectrometry. *Mass Spectrom. Rev* **2012,** *31*, 437-447.

315. Cooks, R. G.; Talaty, N.; Takats, Z., Rapid in situ detection of alkaloids in plant tissue under ambient conditions using desorption electrospray ionization. *Analyst* **2005,** *130*, 1624-1633.

316. Yew, J. Y.; Cody, R. B.; Kravitz, E. A., Cuticular hydrocarbon analysis of an awake behaving fly using direct analysis in real-time time-of-flight mass spectrometry. *Pro Natl Acad Sci USA* **2008,** *105*, 7135-7140.

317. Chen, H.; Yang, S.; Wortmann, A.; Zenobi, R., Neutral desorption sampling of living objects for rapid analysis by extractive electrospray ionization mass spectrometry. *Angew Chem Int Ed* **2007,** *46*, 7591-7594.

318. Chen, H. W.; Zenobi, R., Neutral desorption sampling of biological surfaces for rapid chemical characterization by extractive electrospray ionization mass spectrometry. *Nat Protocols* **2008,** *3*, 1467-1475.

319. RomiLebrun, R.; MartinEauclaire, M. F.; Escoubas, P.; Wu, F. Q.; Lebrun, B.; Hisada, M.; Nakajima, T., Characterization of four toxins from Buthus martensi scorpion venom, which act on apamin-sensitive Ca^{2+}-activated K^{+} channels. *Euro J Biochem* **1997,** *245*, 457-464.

320. Hassani, O.; Mansuelle, P.; Cestele, S.; Bourdeaux, M.; Rochat, H.; Sampieri, F., Role of lysine and tryptophan residues in the biological activity of toxin VII (Ts gamma) from the scorpion Tityus serrulatus. *Euro J Biochem* **1999,** *260*, 76-86.

321. Ali, S. A.; Stoeva, S.; Grossmann, J. G.; Abbasi, A.; Voelter, W., Purification, characterization, and primary structure of four depressant insect-selective neurotoxin analogs from scorpion (Buthus sindicus) venom. *Arch Biochem Biophys* **2001,** *391*, 197-206.

322. Weisel-Eichler, A.; Libersat, F., Venom effects on monoaminergic systems. *J Com Phys A* **2004,** *190*, 683-690.

323. Zeng, X. C.; Luo, F.; Li, W. X., Molecular dissection of venom from Chinese scorpion Mesobuthus martensii: identification and characterization of four novel disulfide-bridged venom peptides. *Peptides* **2006,** *27*, 1745-1754.

324. Delfino, G.; Brizzi, R.; Feri, L., Chemical skin defence in Bufo bufo: An ultrastructural study during ontogenesis. *Zool Anz* **1995,** *234*, 101-111.

325. Peng, Y. E.; Zhang, S. C.; Wen, F.; Ma, X. X.; Yang, C. D.; Zhang, X. R., *In vivo* nanoelectrospray for the localization of bioactive molecules in

plants by mass spectrometry. *Anal Chem* **2012,** *84*, 3058-3062.

326. Tian, H. Y.; Wang, L.; Zhang, X. Q.; Wang, Y.; Zhang, D. M.; Jiang, R. W.; Liu, Z.; Liu, J. S.; Li, Y. L.; Ye, W. C., Bufogargarizins A and B: two novel 19-norbufadienolides with unprecedented skeletons from the venom of bufo bufo gargarizans. *Chem Eur J* **2010,** *16*, 10989-10993.

327. Gao, H. M.; Popescu, R.; Kopp, B.; Wang, Z. M., Bufadienolides and their antitumor activity. *Nat Prod Rep* **2011,** *28*, 953-969.

328. Fukuyama, T.; Yokoshima, S.; Ueda, T.; Kobayashi, S.; Sato, A.; Kuboyama, T.; Tokuyama, H., Stereocontrolled total synthesis of (+)-vinblastine. *J Am Chem Soc* **2002,** *124*, 2137-2139.

329. Farooq, M.; Wahid, A.; Kobayashi, N.; Fujita, D.; Basra, S. M. A., Plant drought stress: effects, mechanisms and management. *Agron Sust Dev* **2009,** *29*, 185-212.

330. Ibrahim, A.; Schutz, A. L.; Galano, J. M.; Herrfurth, C.; Feussner, K.; Durand, T.; Brodhun, F.; Feussner, I., The alphabet of galactolipids in Arabidopsis thaliana. *Plant Sci* **2011,** *2*, 1-24.